ADULT CPR with AED

FOLLOWS 2010 ECC/ILCOR GUIDELINES

Updated with 2015 Guidelines



Welcome!

We're excited that you've decided to take CPR, and that you've chosen us to direct your learning experience.

Despite technological advances and new scientific discoveries, heart disease remains the number one cause of death globally, with 17.3 million deaths per year; 38.7% of cases of out of hospital cardiac arrests (OHCA) are witnessed by a bystander (AHA, 2015). Learning CPR can save lives- you can save a life.

In this course, you will learn:

- The Chain of Survival and your place within the chain
- How to provide high quality chest compressions for adults
- How to initiate early use of an AED (automated external defibrillator)
- How to provide rescue breathing for victims of respiratory arrest
- How to provide CPR with 2 rescuers
- How to assist adult victims of choking

Let's get started!

The Five Fears

Many bystanders are reluctant to perform CPR. There are many reasons why bystanders are reluctant to get involved. Let's take a look at some of these reasons so that you can understand why they have no real basis of support.

Fear of Disease: Use universal precautions when the possibility exists of coming into contact with bodily fluids. Use gloves, mask, and/or gown when you have them available. The benefit of initiating lifesaving resuscitation in a patient in cardiopulmonary arrest greatly outweighs the risk for secondary infection in the rescuer or the patient. Nevertheless, use of simple infection-control measures during CPR and CPR training can reduce a very low level of risk even further (Mejicano & Maki, 1998).



Fear of Lawsuits: These laws state that a person acting in good faith who is rendering reasonable first aid will not be held accountable for damages to the person to whom the aid is rendered, unless gross and willful misconduct are involved. Good Samaritan laws may differ for professional health care providers versus lay rescuers from state to state. Implied consent means that there is an assumption that if an unconscious person were able to request care, they would do so.



Fear of Uncertainty: People sometimes fear that they won't remember what to do. Remembering the correct number of compressions or the number of compressions to ventilations is not as important as the willingness to respond and to push hard and fast. Hands-only CPR is designed to provide simple life support. No mask, no worry- performing compressions alone has been shown to be effective at saving lives.



Fear of Harming the Victim: A victim in need or clinically dead can only be helped by your intervention efforts. You can't hurt someone who is dead, and any injuries you may unknowingly cause (such as injured ribs) can be dealt with in a surviving victim of cardiac arrest.



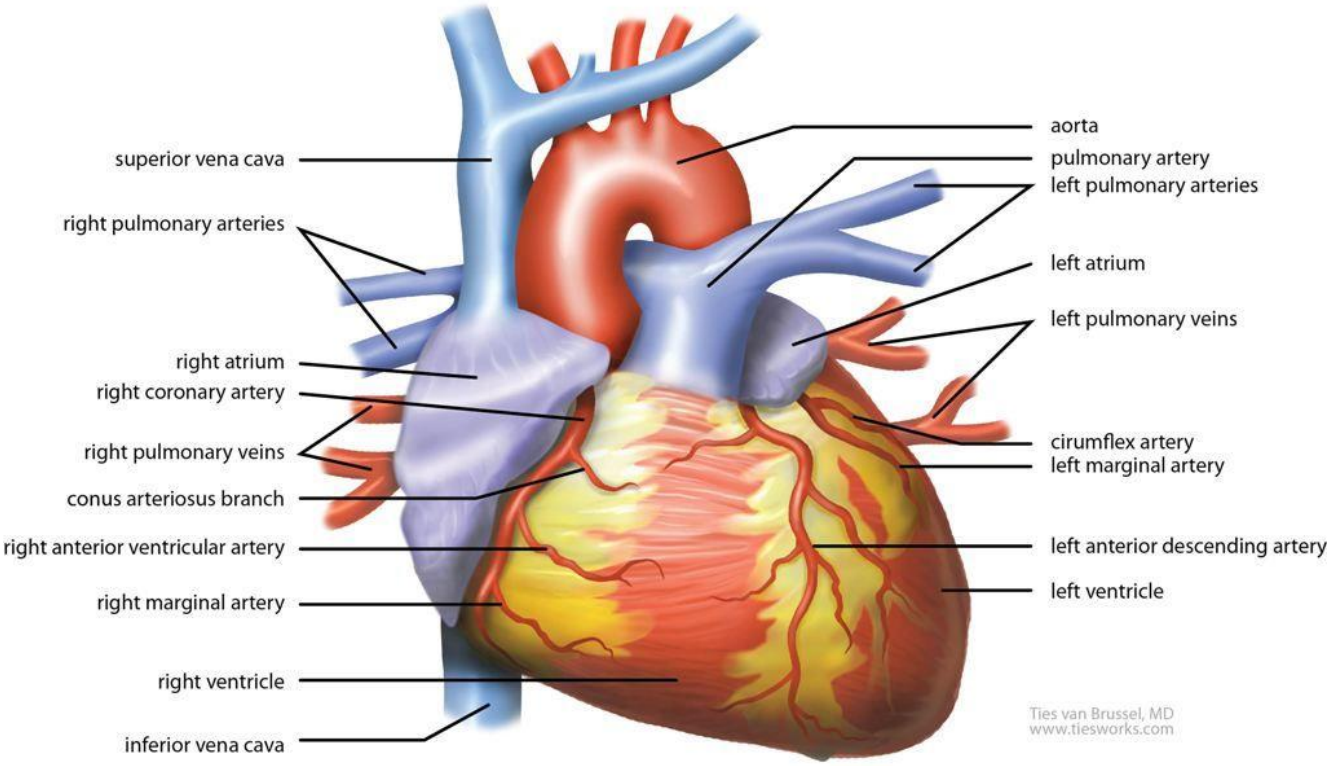
Fear of Injury to Self: As you will learn, scene safety is of utmost importance, and checking the scene for safety should be your first action when you come across an unconscious victim. If the scene is not safe for you to enter, you must not enter. This will only result in more victims. Under no circumstances should you risk becoming a victim. Instead, you should call for help (activate EMS).



BASIC ANATOMY AND PHYSIOLOGY

Before you learn how to perform CPR, it's important to understand how the heart, lungs, brain and cells perform. Here is a brief review of these systems.

THE HEART



The heart consists of four chambers, the right and left atria (singular: atrium) and the right and left ventricles. The atria are located above the ventricles, as can be seen in the above diagram. The heart is a muscular organ supplied by the coronary arteries. It is located below your breastbone (sternum) and, in an adult, is approximately the size of your fist.

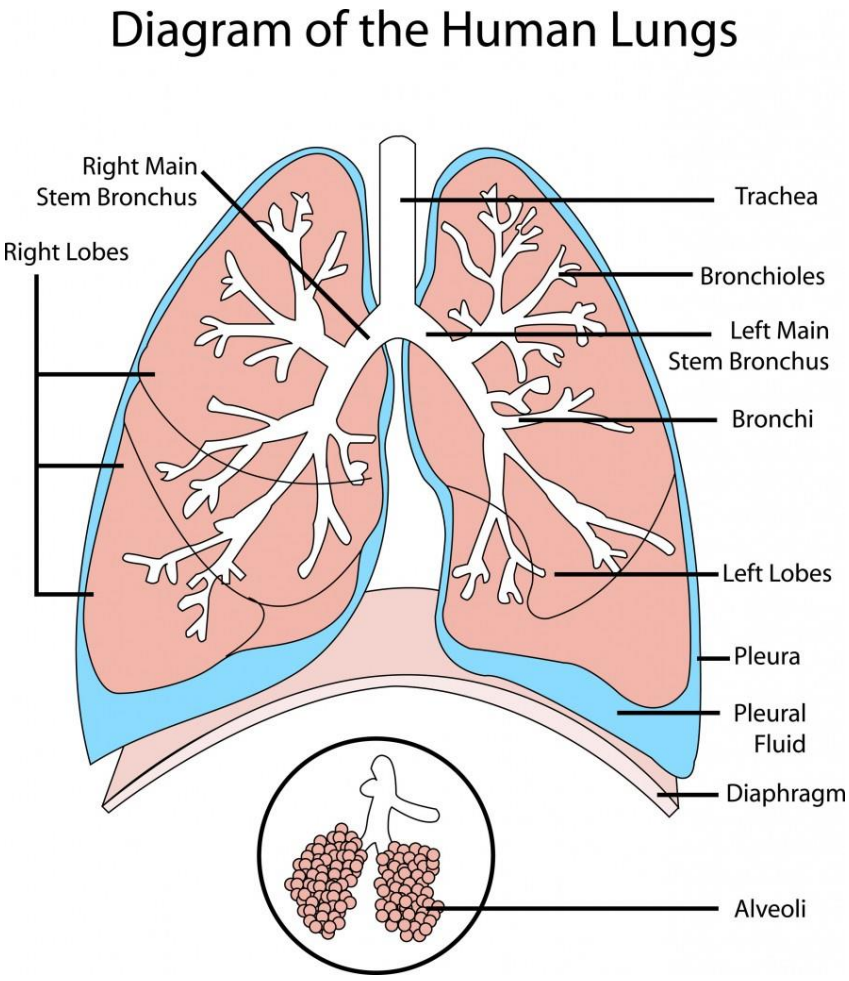
The heart pumps blood that has been deoxygenated after supplying the body's tissues into the lungs, and when that blood has been oxygenated again in the lungs, it exits the lungs to the left

side of the heart, where it is pumped out into the body once again, to provide oxygen to the vital organs.

Your body has about 5 liters of blood, which circulate through this system approximately 3 times per minute.

What is known as the cardiovascular system is composed of the heart, arteries, capillaries, and veins.

THE LUNGS



Your lungs are spongy, air-filled sacs, with one lung located on either side of the chest. The trachea, which is sometimes called the windpipe, conducts air down into the lungs through the

bronchi, which are smaller tubular branches. The bronchi then divide into smaller and smaller tubules called bronchioles. Air is exchanged in the alveoli, which are tiny sacs that allow oxygen and carbon dioxide to move between the lungs and the bloodstream via tiny capillaries.

Your lungs take in oxygen to supply your body's organs and tissues. They release carbon dioxide, a waste product, into the atmosphere when you exhale.

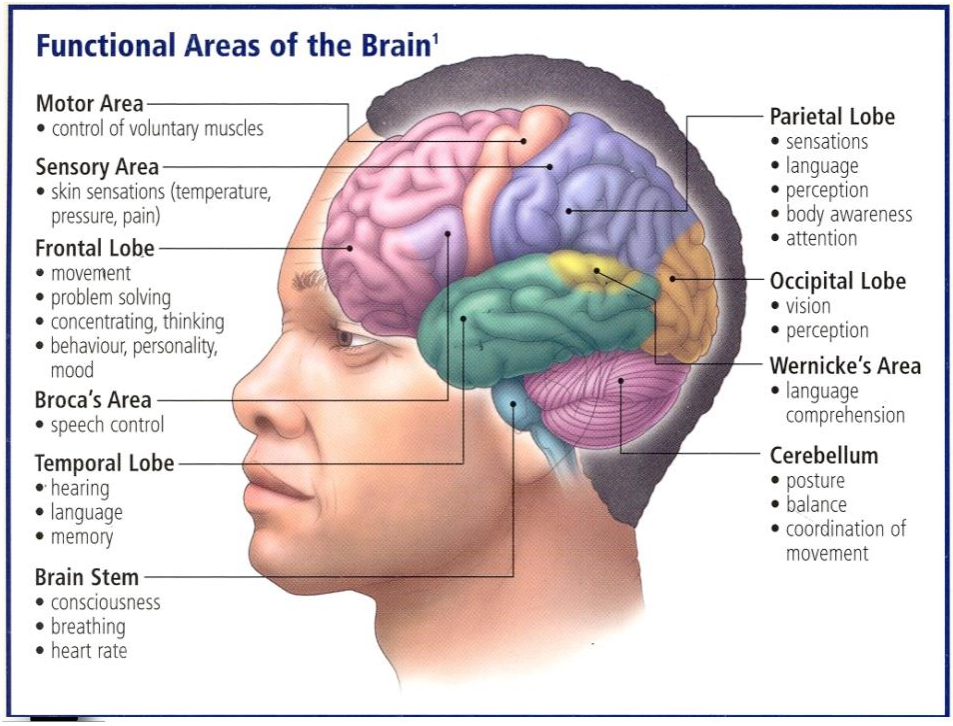
Room air at regular atmospheric pressure contains 21% oxygen. Our bodies utilize approximately 4-6% of that oxygen and release about 16-17% back into the atmosphere, along with carbon dioxide, when we exhale. This 16-17% that we exhale can provide just enough oxygen to sustain someone in cardiac arrest, at least for a short time.

THE BRAIN

Your brain needs a constant supply of oxygen. Without oxygen, brain cells begin to die in 4 to 6 minutes.

On average, the brain weighs three pounds, and uses a whopping 20% of the body's oxygen!

The medulla is located in the brain stem and controls automatic bodily functions, including consciousness and respiratory and cardiovascular function.



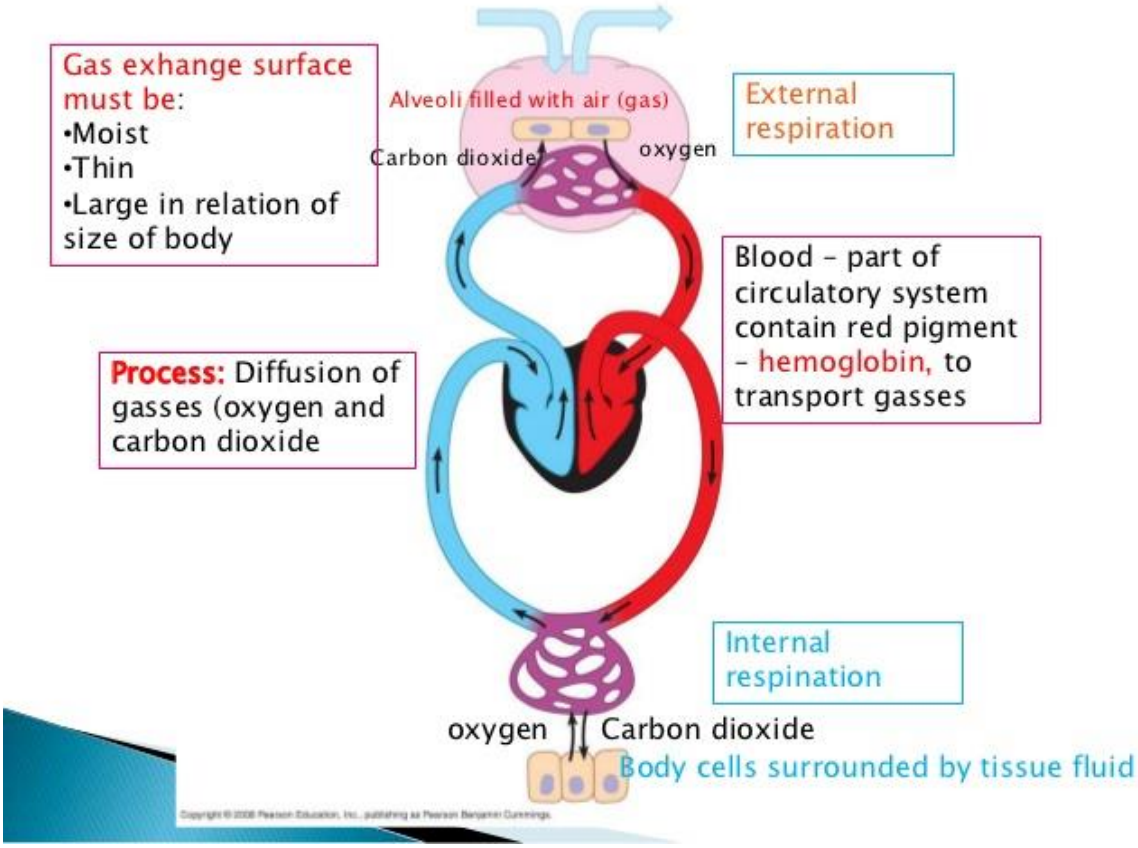
THE BODY'S CELLS

All of the body's tissues are made up of cells.

All cells require oxygen to carry out their normal functions. They also require nutrients. Oxygen deprivation, or hypoxia, will cause cells to die within a few short minutes.

The body's cells use oxygen and produce carbon dioxide as a waste product. Carbon dioxide must be eliminated via the lungs through ventilation.

Red blood cells transport oxygen throughout the body.



CARDIOVASCULAR DISEASE

As mentioned previously, cardiovascular disease is the number one cause of death worldwide. While it is important to learn what to do when cardiovascular disease leads to a heart attack or stroke, it is equally as important, if not more so, to understand how to prevent cardiovascular disease from occurring in the first place.

There are numerous factors that can increase an individual's risk of heart attack. Some of these factors can be controlled (modifiable risk factors), while others cannot (non-modifiable risk factors.

Modifiable risk factors, or factors that can be controlled to a certain degree by an individual, include:

- cigarette smoking
- hypertension
- obesity
- sedentary lifestyle (lack of exercise)
- high cholesterol
- high blood sugar (in diabetes)
- poor diet (diet high in sugar, fat)
- stress

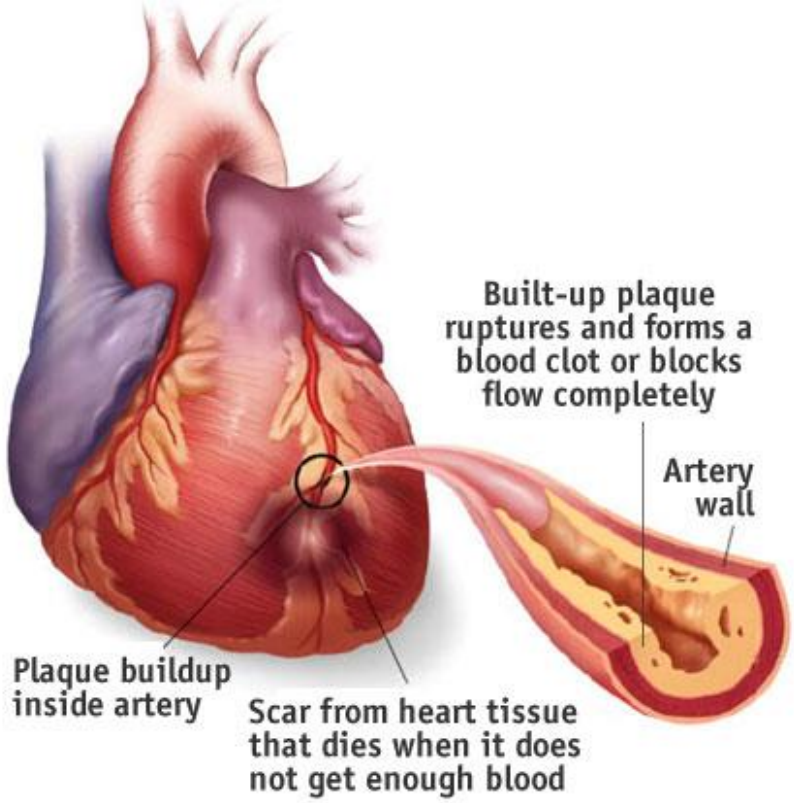
Non-modifiable risk factors, or risk factors that individuals cannot change, include:

- age
- sex
- genetics/hereditary factors (i.e. hereditary high cholesterol or hyperlipidemia)
- race (i.e., certain groups may be at higher risk for hypertension, or high blood pressure)

CARDIOVASCULAR DISEASE IS THE NUMBER ONE CAUSE OF DEATH IN THE UNITED STATES, ACCORDING TO THE CENTER FOR DISEASE CONTROL (CDC). THE CDC ESTIMATES THAT MORE THAN 650,000 AMERICANS DIE FROM CARDIOVASCULAR DISEASE (1 in 4 DEATHS) EACH YEAR IN THE UNITED STATES.

Cardiovascular disease damages the heart and blood vessels, and frequently causes heart attack and/or stroke. Americans of all ages should focus on prevention of cardiovascular disease by eating healthy foods, exercising 30 minutes (at least) each day, controlling weight, minimizing stress, consuming healthy fats and oils, and avoiding cigarette or cigar smoking.

Atherosclerosis, or hardening of the arteries, is a condition caused by the building up of plaque inside the body's arteries, the large blood vessels that carry blood away from the heart to the body's organs. Plaque is composed of fatty substances, cholesterol, fibrin (a clotting substance in the blood), calcium and cellular waste products. When plaque builds up, it can partially or totally block the flow of blood through an artery in the brain (which causes stroke), the heart (which causes a heart attack), the kidneys, the arms, the legs and other vital areas. Plaque may break off to block an artery, or a blood clot (thrombus) may form on the surface of the plaque- either of these two circumstances can lead to a heart attack or stroke.



HEART ATTACK

A heart attack often presents with one or more of the following symptoms:

- Chest tightness, pressure, or discomfort
- Nausea
- Sweating
- Shortness of breath
- Fatigue
- Weakness
- Pain in the jaw or arm
- Pallor (pale color of the skin)

Women and people with diabetes often present with atypical symptoms, such as nausea and vomiting or back pain. About one-third of patients report no chest pain at all.

WHAT CAN YOU DO?



- **Recognize** the symptoms and signs of a heart attack.
- **Call 911.**
- Don't let the patient drive him or herself to the hospital
- Keep the patient calm and seated in a position that is comfortable.
- **Offer 1 adult aspirin** or **2 baby aspirin**- aspirin should be chewed before swallowing to speed absorption (Note: do not give if the patient has already taken aspirin or if the patient has an allergy to aspirin)
- Be prepared to start CPR if it becomes necessary

STROKE

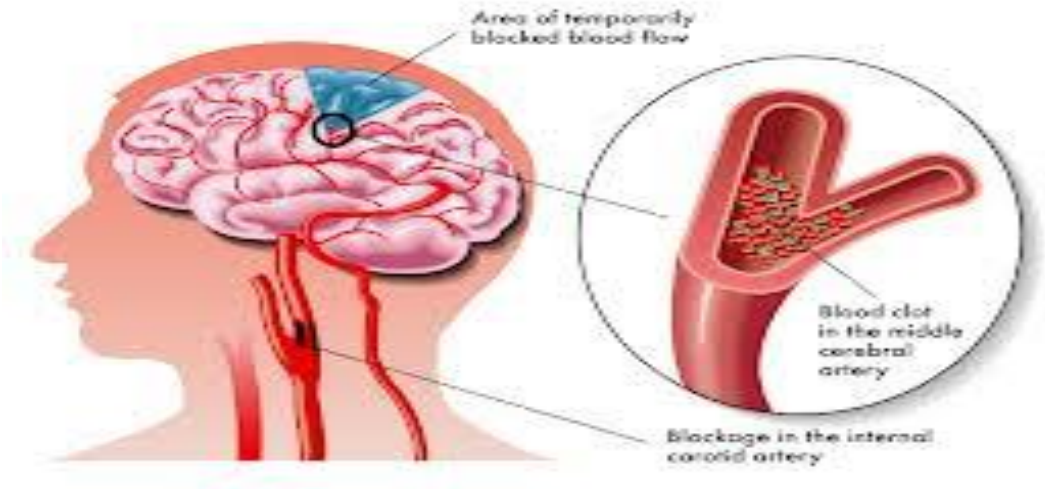
You should suspect stroke if a patient or victim presents with:

- Severe headache with no known cause
- Numbness or weakness of the face, arm or leg on one side of the body
- Drooping eyelid or mouth on one side
- Confusion or trouble understanding
- Difficulty speaking (slurred speech or difficulty finding words)
- Loss of balance or coordination
- Dizziness
- Trouble with vision (i.e. blurred or double vision, loss of vision in one eye)

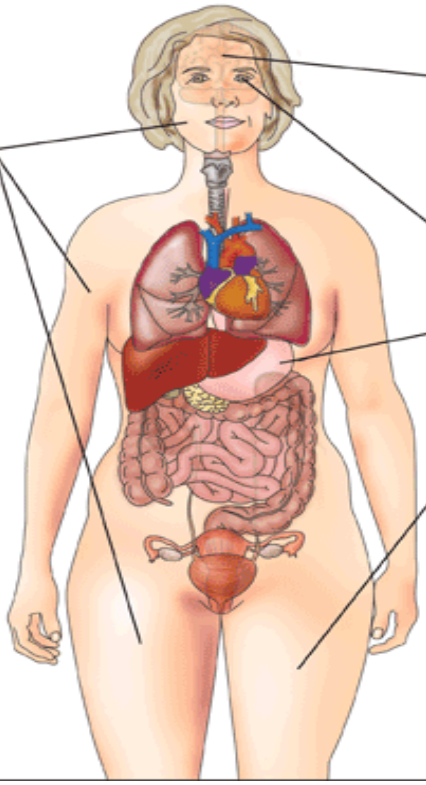
WHAT CAN YOU DO?

- **Know the signs and symptoms** of a stroke
- **Call 911**
- Don't give the patient food or drink
- Keep the patient calm and quiet
- Monitor the patient
- Be prepared to begin CPR if it becomes necessary

A stroke is caused by one of two mechanisms: it can be caused by a ruptured blood vessel in the brain (called a hemorrhagic stroke) or by a blockage in one of the arteries that causes loss of blood flow and oxygen to a part of the brain (called an ischemic stroke). Ischemic strokes are more common. Remember that "Time is brain," and act quickly. Stroke is the 3rd leading cause of death in the United States.



Signs of a Stroke



Face, Arm, or Leg
numbness or weakness (mainly on one side of the body)

Brain
confusion, trouble talking or understanding speech, dizziness, loss of balance, bad headache

Eyes
trouble seeing in one or both eyes

Stomach
throwing up (or urge to)

Body
feel tired

Legs
trouble walking

Guideline Changes

For anyone who has taken CPR prior to 2010, there have been a few key changes since the 2010 Guidelines were instituted. These changes include:

- Changing from the ABCs (Airway-Breathing-Circulation) to a C-A-B sequence of steps. This was one of the major changes that took place in 2010, when research showed that decreasing the delay in beginning chest compressions lead to a higher incidence of ROSC (return of spontaneous circulation). Rescuers are now trained to provide interventions in the following order: 1) Chest Compressions, 2) Airway, and 3) Breathing. Decreasing the delay in beginning compressions increases survival rates.
- The removal of "Look, listen and feel" for breathing. It is often hard for even trained providers to identify effective breathing. It is no longer recommended to look, listen and feel to determine whether a victim is breathing. Rather, if the victim is not responsive, has no pulse, and is not breathing or is breathing in an irregular fashion (i.e. only gasping), begin CPR immediately!
- Emphasis on high-quality CPR. High-quality CPR is defined as:
    - Compression rate of 100- 120 beats per minute on victims of all ages;
    - Compression depth of AT LEAST 2 inches in adults (but no greater than 2.4 inches). Allow the chest to completely expand (recoil) after each compression (do not lean on the chest between compressions);
    - Not interrupting CPR except to use an AED (keep interruptions in chest compressions to less than 10 seconds);
    - Avoiding excessive ventilations;
    - Above all else, PUSH HARD AND FAST!
- Since many responders (even professionals) are unable to feel or palpate a pulse quickly, the recommendation is to feel for a pulse for NO MORE THAN 10 seconds. If you can't feel a pulse or if you're not sure you can feel a pulse, begin CPR.
- It has been recognized that health care professionals should call for assistance when they come upon an unconscious victim, but they may also simultaneously assess breathing and check for a pulse before fully activating the emergency response system. This may encourage efficiency in assessment and response, rather than following a step-by-step response.
- Health care professionals may tailor their response to an unconscious victim by altering the response sequence to fit the situation or scenario (using an AED immediately when one is close by and the arrest is witnessed, or providing ventilations first when the cause of arrest is known to be an anoxic event (i.e. drowning).

- Rescuers should provide 1 breath every 6 seconds when an advanced airway is in place and compressions are ongoing continuously (rather than trying to remember a range of ventilations for adults, children and infants).

The Adult Chain of Survival



The Adult Chain of Survival represents a continuum of care, from early recognition of the victim in cardiac arrest to post-resuscitation care to provide the best chance of survival from cardiac arrest. The five links in the Adult Chain of Survival include:

- Early recognition of cardiac arrest
- Activation of the Emergency Response System
- Early CPR, with an emphasis on high-quality chest compressions
- Early defibrillation
- Comprehensive post- cardiac arrest care

Note that in the diagram above, the first two of the five steps (early recognition of cardiac arrest and activation of the emergency response system) have been combined to acknowledge the fact that these steps often occur simultaneously when multiple rescuers are present.

THE TEAM APPROACH

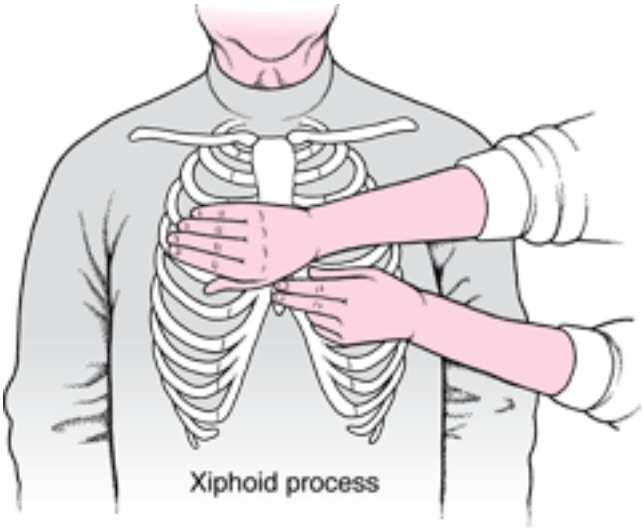
In many situations, there may be only one rescuer who will carry out the normal sequence of assessments and actions. However, in many situations, there is often more than one rescuer trained and willing to help.

This is when a team approach can be used. The first rescuer should take the role of team leader and delegate tasks. One rescuer can provide compressions, one can prepare to give breaths with a bag-valve mask, and one can prepare the AED. By working together, the most efficient care can be given to the patient.

Chest Compressions for Adults

Chest compressions are the most important component of CPR. Chest compressions are an attempt to mimic the normal activity of the heart. When a rescuer presses down on a victim's chest, blood is forced out of the heart and into the arteries. When pressure on the chest is released, blood is allowed to return to the heart. A small amount of oxygen will be present in the bloodstream for several minutes after the heart ceases to beat, just enough to keep the brain alive. Compressions can keep vital organs functioning until higher level care is available.

Adult Chest Compressions





To perform compressions on an adult, place the heel of your non-dominant hand on the victim's chest between the nipples. Then place your other hand on top of the first and interlace your fingers. The heel of your hand should be pressing on the bottom two-thirds of the sternum, avoiding the xiphoid process (the small bony prominence at the very bottom of the sternum (breastbone). You should be as close to the victim's side as possible, with your knees against the victim's side (this will help prevent back injury). Lock your elbows and press down hard, depressing the sternum at least 2 inches (5 cm), but no more than 2.4 inches (6 cm). Your shoulders should be positioned directly over your hands in a straight line. Push hard and fast 100 to 120 times per minute, counting out loud as you do so. It's important that you allow the chest to recoil (return to it's normal, relaxed position) in between compressions. If you do not allow the chest to recoil, the heart will not fill completely, which means that less blood (and therefore oxygen) will be pumped out of the heart to vital organs with the next compression.

Performing compressions is exhausting. Most people find that they become very tired after providing compressions for 2 or 3 minutes. When a person performing compressions becomes fatigued, there is a tendency to compress less firmly and more slowly; for this reason, it is recommended that rescuers trade off doing compressions every 2 minutes to prevent fatigue and optimize the quality of compressions. If you are alone, you will have to do the best you can- keep performing compressions until help arrives or you are physically so exhausted that you cannot continue.

One Rescuer Adult CPR

If you are alone and come across an individual who is down, follow the steps below. If someone else is in the area to assist, use the 'Two Rescuer' sequence.

Stay Safe: If you come upon an individual who may need CPR, look around and make sure you and the victim are in a safe place. For example, if the victim is in water or on a road, try to move the victim to a safer area. If you are in a safe area, do not try to move the victim as he/she may have other injuries (i.e., to the head or neck) that you cannot see. Simply roll him over onto his back. Make sure the victim is on a firm surface, in case compressions are needed.

Assess the Victim: To quickly assess the victim, shake his shoulder and yell at him. Check for breathing. If he/she is not breathing, or is not breathing normally (i.e., only gasping), you must summon help. Note: *Agonal breathing* is breathing that is very slow, shallow and/or gasping. Agonal breathing is a sign that the victim is dying. It is important not to mistake agonal breathing for normal breathing!

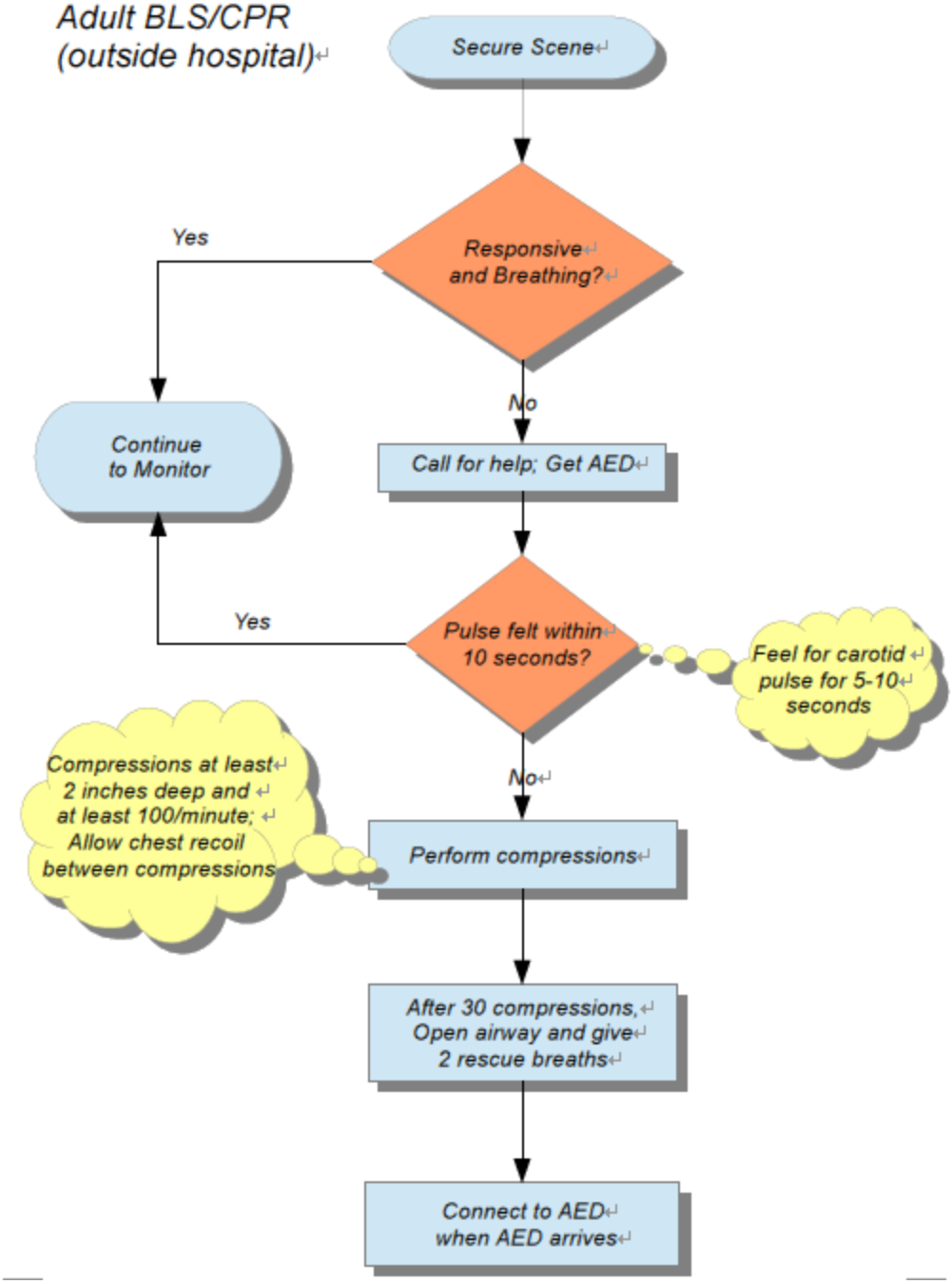
Activate the Emergency Response System and Find an AED: Yell for help. If someone responds, tell him/her to call for help by dialing 9-1-1. If you are in an area where an AED may be available, tell him to go find the AED. Make sure you tell the person to return to assist you as soon as possible. If you are alone, call for help by dialing 9-1-1 and run to get the AED if you know where one is nearby. If you do not know where an AED is, begin CPR immediately after calling for help. Mobile phones have made activating the emergency response system much easier- you can call 9-1-1 and put your phone on speaker while you start CPR.

Begin CPR

1. Check for a pulse on the side of the neck. Feel for a pulse for at least 5 seconds but NO MORE THAN 10 seconds. To check for a carotid pulse, slide 2 or 3 fingers into the groove between the windpipe and the neck muscles at the side of the neck.
2. If there is no pulse (or if you are unsure if there is a pulse), begin CPR starting with chest compressions. Provide 30 chest compressions, followed by two breaths. NOTE: If you are not comfortable giving rescue breathing and/or you do not have a mask available, do 'Compression Only' CPR.
    a. Use the heel of one hand on the lower half of the breastbone in the middle of the chest.
    b. Place the other hand on top of the first hand.
    c. Straighten your arms and lock your elbows so that your body weight is over your hands.

d. The most important part of CPR is to remember to push HARD and FAST. Each compression should be AT LEAST two inches deep (but no more than 2.4 inches) and the rate should be 100-120 compressions per minute.
e. Be sure to let up on the pressure on the sternum after each compression (chest recoil) so the chest can re-expand and blood can flow back into the heart. The purpose of CPR is to help the blood flow through the heart and into the rest of the vital organs; if you allow the chest to re-expand, more blood will flow into the heart and will be available to deliver to the rest of the body.
f. Count out loud as you do compressions. When you have done 30 compressions, try to open the victim's airway by doing a head tilt/chin lift. Note that if you are doing 'Compressions Only' CPR, you can skip this step.
    i. With your non-dominant hand, push on the victim's forehead to tilt the head back.
    ii. With your dominant hand, place your fingers under the bony part of the lower jaw and gently lift the jaw to bring the chin forward. Be sure you lift up on the bony part of the jaw and not the soft tissue under the jaw so you don't block the victim's airway. Do not use your thumb to lift the jaw. Allow the victim's mouth to remain slightly open.
    iii. If you think the person's neck may be injured, avoid the head tilt/chin lift. Use the jaw thrust maneuver if you have been trained to do so.
g. If you have a barrier device to use between your mouth and the victim's face, use it. Although the risk of infection from performing CPR is very, very low, it is expected that healthcare workers use a barrier device when providing CPR. This includes the use of face masks or bag-mask devices (see next section). Give each breath slowly – each breath should last one second. Make sure the chest rises with each breath. Repeat, giving a second breath.
h. Start another cycle of chest compressions. Remember, push HARD and FAST. Alternate chest compressions (30) and giving breaths (2) until help arrives.

Adult BLS/CPR (outside hospital)



Airway/ Breathing

Face masks provide a barrier between the rescuer and the victim. Some masks are equipped with a one-way valve that allows the rescuer's breaths to enter the victim's airway, but prevents the victim's expired air from entering the rescuer's airway. These masks also prevent contact with vomitus and blood, which could pose an infection risk to the rescuer. It takes practice to learn how to use these masks effectively to provide ventilations.

Using a Face Mask:

a) Position yourself at the victim's side. If you are a lone (single) rescuer, positioning yourself at the victim's side will allow you to provide both ventilations and compressions without having to move.
b) Position the mask on the victim's face. Masks are usually triangular in shape, and you will notice that the mask has a "pointy" end- this end goes over the bridge of the victim's nose.
c) Seal the mask against the victim's face. To do this, take the hand that is closest to the top of the victim's head and place it along the edge of the mask. Some people find it easier to form a 'C' with their index finger and thumb and use these digits to grasp the mask around the base of the mouthpiece. With the thumb of your other hand, apply pressure along the bottom edge of the mask. Then place the remaining fingers of your second hand along the bony edge of the jaw and lift the jaw upwards. Open the airway by performing a head-tilt chin-lift procedure. While you lift the jaw, ensure that you are sealing the mask all the way around the outside edge of the mask to obtain a good seal against the victim's face.
d) Deliver air over 1 second, ensuring that the victim's chest rises.
e) If the victim's chest does not rise, reposition the mask and try to get a better seal. Remember, you should be lifting the victim's jaw into the mask, rather than simply pushing the mask down onto the victim's face.
f) Provide 2 ventilations over 1 second each with the mask after every 30 compressions.
g) If the victim has a pulse but is *not breathing*, provide rescue breathing by providing 1 breath every 5 to 6 seconds (10-12 breaths/minute). Check for a pulse every 2 minutes- if there is no pulse, start chest compressions along with ventilations at a rate of 30:2.



Using a Bag-Mask Device:

A bag-mask device (or BVM, bag-valve mask) consists of a mask attached to a reservoir bag. They are commonly used to provide positive-pressure ventilations during CPR. They can be attached to an oxygen source to provide 100% oxygenation during resuscitation. As with the face mask, it takes practice to be able to use a bag-mask device. It can also be very difficult for one person to use a bag-mask device; therefore, it is recommended that use of a bag-mask device be used only when there are two rescuers available.

The steps to using a bag-mask device are as follows:

a) Position yourself at the top of the victim's head- this allows room for the second rescuer to provide compressions.
b) Place the mask of the bag-mask device on the victim's face, using the bridge of the patient's nose as a guide to correct positioning.
c) Use the E-C clamp technique to hold the mask in the correct position while you lift the jaw to obtain an open airway. To perform the E-C clamp technique, take the index finger and thumb of your non-dominant hand and form them into a 'C' around the top of the mask. Your other fingers of the same hand are used to lift the jaw (the 'E' part of the E-C clamp technique.
d) Squeeze the bag to deliver a breath- each breath should be delivered over 1 second. Watch for chest rise. If you do not observe chest rise, you do not have a tight seal. If this occurs, reposition the mask and try again. Be careful not to overinflate the lungs- each breath should result in visible and *natural* chest rise.
e) Provide 2 ventilations after every 30 compressions. If the patient has a pulse but is not breathing, provide one breath every 5-6 seconds (10-12 breaths/minute) and check for a

pulse every 2 minutes. If the victim loses their pulse, you will need to begin chest compressions.



The Jaw Thrust Maneuver

If you suspect that a victim may have a neck or spinal cord injury (i.e., the victim has fallen, been in a motor vehicle accident or suffered another mechanism of injury that could result in injury to the neck or spinal cord), you should not use the head tilt-chin lift maneuver to open the victim's airway. This could further damage the neck or spinal cord. Instead, you should use the jaw thrust maneuver to open and maintain the victim's airway.

To perform this maneuver:

a) Place your hands on either side of the victim's head. Rest your elbows on the surface that the victim is laying on.
b) Put the fingers of both your hands under the angle of the victim's lower jaw and lift so that the jaw slides forward.
c) Use your thumb to push the lower lip away from you if the victim's lips close.

Two Rescuer Adult CPR (without an AED)

If you come across a victim who is down and another person is available to help, send that person to activate the Emergency Response System and find an AED while you assess whether the victim needs CPR. When you have determined that the victim requires CPR, start CPR beginning with compressions. When the second person returns (without an AED in this case):

- You should continue chest compressions and count the compressions aloud each time you compress the victim's chest.
- The second rescuer should keep the person's airway open using the head tilt-chin lift maneuver. The second rescuer should give two rescue breaths after every 30 compressions, using a face mask or a bag-mask device.
- You and the second rescuer should switch positions/roles about every two minutes so neither one of you gets tired. If you are counting out loud, two minutes is about 5 cycles of thirty compressions and two breaths. You should change positions sooner if the person doing compressions becomes too tired to perform high-quality compressions.
- Continue switching roles as above until EMS arrives to relieve you or the victim regains consciousness.

Using an AED



An AED, or automated external defibrillator, is a device that has the ability to detect and treat, through electrical energy, the lethal arrhythmias known as ventricular fibrillation and ventricular tachycardia. These rhythms are a common cause of sudden cardiac arrest.

Ventricular fibrillation is a condition in which the lower chambers of the heart, the ventricles, quiver in an unorganized fashion, which renders them incapable of pumping blood to the rest of the body. Untreated, ventricular fibrillation rapidly causes cardiac arrest. Ventricular tachycardia is a rapid rhythm also originating in the ventricles. In ventricular tachycardia, the ventricles contract so quickly, albeit in a somewhat organized fashion, that inadequate blood flow is produced. Ventricular tachycardia often precedes ventricular fibrillation. Both rhythms are lethal if not treated.

An AED sends electrical energy (a ‘shock’) through the heart, which stuns the heart and allows the normal pacemaker of the heart, usually located in the right atrium, to take over and restore a normal heart rhythm.

Early defibrillation is key to survival in cardiac arrest. For each minute that defibrillation is delayed, the chance of survival is reduced by 10%. (after 10 minutes, few people are successfully resuscitated.)

Early defibrillation can increase survival rates to greater than 50%. Rescuers should immediately begin chest compressions, and use the AED as soon as it is available and ready to use.

AEDs can be found wherever crowds of people gather- swimming pools, airports, malls, sporting arenas, schools, hotels…More and more businesses are also investing in these life-saving machines. In some communities, private AED owners are registering their AEDs with ambulance dispatch, so that they can be easily located by bystanders when needed. Make it a point to learn where the AEDs in your neighborhood or town are located- you never know when you might need one!



AEDs have been designed to be extremely “user friendly”. All you need to do as a rescuer is turn on the machine (the most important step) and listen as the machine guides you through the steps to use the AED safely and effectively. Although there are many brands of AEDs on the market, they all work in a similar fashion and are designed to be used by lay rescuers.

When an AED becomes available (i.e., when you or another rescuer have retrieved it), place it at the victim's side, closest to the rescuer who will operate it. In this way, the other rescuer can continue performing CPR until the AED is ready to analyze and deliver a shock (if needed).

There are four universal steps to using any AED. These will be highlighted in the following list of steps so they are easily recognizable.

The steps to use an AED are as follows:

1. **Turn on the machine.** This is the most important step- turning on the machine will enable the AED unit to guide you through the next steps. To turn on the AED, open the top of the carrying case and push the ON button. Note: some models will turn on automatically when you lift the lid of the carrying case.

2. **ATTACH AED pads** to the victim's bare chest. Expose the patient's chest. Dry it off if wet, shave excessive hair if possible. Use adult pads for victims who are 8 years of age or older. Peel off the adhesive backing. Place one pad on the upper right chest just below the collarbone. Place the other pad on the patient's lower left ribcage, a couple of centimeters beneath the armpit. Some pads are marked- there will be a red heart on the pad that is to be placed on the victim's left side (the heart side).
   Press pads firmly onto the patient's chest. Then attach the connecting cables to the AED unit. Note: some cables will come preconnected.

3. **Analyze the rhythm**. When the AED unit instructs you to, CLEAR the victim while the machine is analyzing the victim's heart rhythm. This means you should ensure that no one is touching the victim, including yourself. The rescuer performing chest compressions or giving breaths will need to stop at this point. Note: some AEDs will begin to analyze the victim's rhythm independently; for others, you will need to push the ANALYZE button. Analyzing the victim's rhythm will take up to 10 seconds, so don't be alarmed by this.
4. **Push to shock**. If a shock is advised, the machine will clearly state "SHOCK ADVISED, STAND CLEAR". You should ensure that no one is touching the victim, including yourself. You need to look around to make sure no one is touching the victim's body while stating "CLEAR" or some similar message that warns others a shock is to be delivered. Once you are <u>certain</u> that no one is touching the victim, push the SHOCK button. You will notice that the victim's muscles contract strongly.

- If a shock is not necessary (the rhythm is not ventricular fibrillation or pulseless ventricular tachycardia), the AED will state NO SHOCK ADVISED and tell you to resume CPR
- After approximately 5 cycles of compressions and ventilations, or 2 minutes of CPR, the AED will instruct you to repeat steps 3 and 4- analyze the rhythm and push to shock if the rhythm requires a shock and the AED instructs you to do so.
- Continue CPR alternating with analysis of the rhythm until help arrives (i.e. EMS).

Special Circumstances

There are times when using an AED may present special challenges. Here's what to do when faced with one of the following:

- **The victim has a hairy chest**- ideally, you should use a razor to shave the areas that will be covered by the AED pads. If a razor is not available and the AED machine is prompting you to CHECK PADS or CHECK ELECTRODES, try pressing down firmly on the AED pads to ensure good contact with the skin. If the machine continues to prompt you, quickly pull off the pads- this should remove enough hair to allow a new set of pads to adhere firmly to the victim's skin. Many AED machines are coming equipped with a razor in the carrying case to combat this problem. If you happen to own an AED, ensure that a razor is included in the case.

- **The victim is in water**- if the victim is in water, pull the victim to a dry area. You are not in danger of getting a shock if the victim is in water. Water is a great conductor of electricity, so if the victim is in water, the shock will be dispersed across the skin of the victim, and the victim will not receive the full dose of electrical energy required to convert them to a normal rhythm. If the victim's chest is wet, quickly dry the chest with a towel or your sleeve; however, the chest does NOT need to be completely dry. If the victim is lying in a small puddle or in snow, you can safely use the AED without moving the victim.

- **The victim has an implanted pacemaker or defibrillator**- obviously, if the victim has one of these devices, it must have failed! You will recognize these devices as a small lump under the skin on the chest, usually the upper chest on either side. Some older models may be implanted in the abdomen. They are generally about the size of a deck of cards or smaller. You will also be able to see a scar over the area. If the victim has one of these devices, avoid placing the AED pad directly over it; doing so may block delivery of the shock.

- **The victim has a medication patch on their chest**- many medications can now be delivered transdermally (through the skin). These includes pain medications, hormones, smoking cessation drugs, nitroglycerin and others. Do not place an AED pad over one of these patches. If it won't delay delivery of a shock, remove the patch and wipe the skin before applying the AED pad. These patches may cause the skin to burn under the AED pad if left in place, or they may block delivery of the shock.

Now you know what to do should you encounter one of these special circumstances.

Two Rescuer Adult CPR with an AED

We've learned a lot so far! Thus far, we've learned how to perform compressions, maintain the airway and use an AED on an adult victim. Now it's time to put it all together. The following steps outline how to perform CPR with an AED when there are two rescuers present.

When you come across a victim who is down and there are two rescuers present:

- **Rescuer 1** checks for response and breathing- tap the victim on the shoulder and ask if they are okay. At the same time, observe the victim's chest for breathing. If the victim is not breathing, or is breathing abnormally or only gasping, stay with the patient and prepare to perform the next steps.
- **Rescuer 2** activates the emergency response system and leaves to retrieve an AED.
- **Rescuer 1** checks for a carotid pulse for 5 to not more than 10 seconds. If a pulse is not felt, or the rescuer is not sure if there is a pulse, the rescuer will expose the chest (in preparation for AED use) and begin CPR, starting with chest compressions. **Rescuer 1** will continue cycles of chest compressions and ventilations with a pocket mask or bag-mask device until Rescuer 2 returns with an AED.
- **Rescuer 2** arrives with an AED and places it on the side opposite to **Rescuer 1,** who is performing chest compressions. **Rescuer 2** powers on the AED and attaches the pads to the victim's chest, also attaching the cables to the AED unit if necessary. **Rescuer 1** should continue CPR while the pads are being placed, right up until it is time to analyze the victim's heart rhythm.
- **Rescuer 2** CLEARS the victim, ensuring neither rescuer is touching the victim, and waits for the AED to ANALYZE, or pushes the ANALYZE button when prompted by the AED to do so.
- **Rescuer 2** pushes the SHOCK button if a shock is indicated, making sure that the victim is CLEAR beforehand.
- If no shock is needed, or after the victim has been shocked, **Rescuer 2** should resume chest compressions (as Rescuer 1 may be fatigued by this time) while **Rescuer 1** manages the victim's airway and delivers breaths using the face mask or bag-mask device.
- After approximately 5 cycles of CPR, or 2 minutes, the AED will state that the victim's rhythm should be ANALYZED. During analysis, which can take up to 10 seconds, **Rescuer 2** and **Rescuer 1** should switch positions, so that **Rescuer 1** CLEARS the victim, pushes SHOCK if a shock is advised, and immediately resumes chest compressions (or performs chest compressions if no shock is advised). **Rescuer 2** then takes over management of airway and breathing. Rescuers should switch positions every 2 minutes when it is time to ANALYZE the victim's heart rhythm. This will prevent rescuer fatigue and ensure that rescuers are able to provide high-quality chest compressions at the proper rate and depth. CPR and analysis with the AED should continue until EMS arrives.

CPR with an Advanced Airway

When there is no advanced airway in place, rescuers must pause compressions to deliver breaths using a face mask or bag-mask device. Once an advanced airway is in place, there is no longer a need to pause compressions to deliver breaths.

Why is this? If the chest is compressed during ventilations, most of the air (and thus oxygen) delivered would be forced out of the victim's mouth before it could enter the lungs. An advanced airway means that air is reliably delivered to the lungs, regardless of whether a rescuer is applying force to the chest at the same time a breath is being delivered.

Advanced airways include:

- Laryngeal mask airways (LMAs)
- Supraglottic airway (Combitube or King LT)
- Endotracheal (ET) tube

When an advanced airway is in place, compressions are delivered at a rate of 100-120 compressions per minute. Breaths are delivered over 1 second ***simultaneously*** at a rate of 1 breath every 6 seconds (10-12 breaths per minute).

Mouth-to-Mouth Breaths

Although it is necessary to use a pocket mask or bag-mask device to deliver breaths in the healthcare setting, there are times when these devices may not be available. Should a cardiac arrest occur at home, you would likely not hesitate to perform mouth-to-mouth breathing for a relative or loved one; you might choose to give mouth-to-mouth to a friend as well. In cases such as these, you will likely decide the benefit outweighs the risk to your own health.

To provide mouth-to-mouth breaths to an adult (or child aged 1 to 8 years):

1. Use the head tilt-chin lift to hold the victim's airway open.
2. Using the hand on the victim's forehead that is maintaining the head tilt-chin lift, pinch the victim's nose closed using the thumb and index finger.
3. Inhale a regular breath, then cover the victim's mouth with your own, creating a tight seal.
4. Give one breath over 1 second, watching to see if the chest rises.
5. If the chest doesn't rise as you give the breath, repeat the head tilt-chin lift.
6. Give a second breath over 1 second and watch for chest rise.
7. If the second breath fails to go in, go immediately to chest compressions.

You may wonder how mouth-to-mouth breathing can sustain the victim. In actual fact, your expired air contains about 17% oxygen- this is just enough oxygen to meet the victim's needs.

When providing mouth-to-mouth breathing, it is important not to provide breaths that are too forceful or too rapid. Doing so may cause air to enter the stomach rather than the lungs, which can cause gastric inflation. Gastric inflation may result in vomiting, and an unconscious victim may develop pneumonia if vomitus makes its way to the lungs. To avoid gastric inflation, give each breath slowly over 1 second and deliver just enough air to make the chest rise.

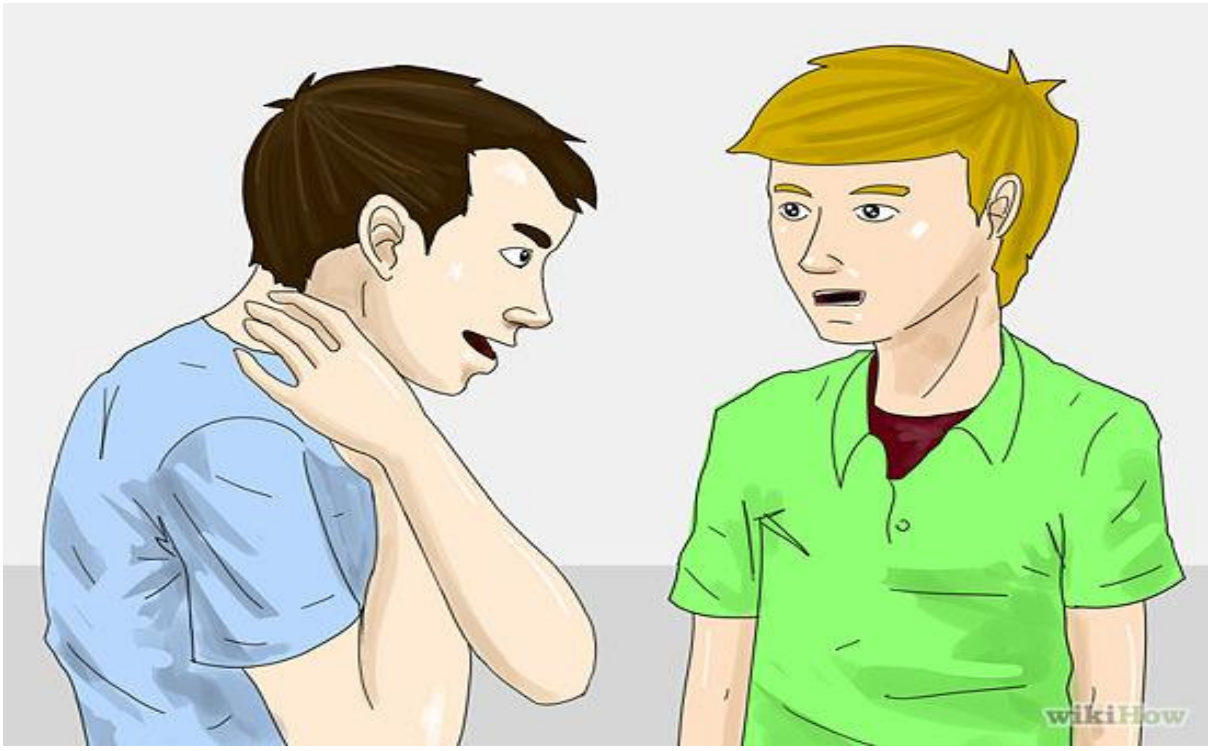
Rescue Breathing

Respiratory arrest is defined as the cessation of breathing. During respiratory arrest, as well as inadequate breathing, the victim will still have some amount of cardiac output, which you will be able to detect as a palpable pulse.

It is important to be able to recognize respiratory arrest, or impending respiratory arrest, which may be seen as slow, irregular or gasping respirations. These abnormal respirations are inadequate to support life. Respiratory arrest inevitably leads to cardiac arrest if not treated, therefore healthcare providers should intervene quickly to prevent this deterioration by providing ***rescue breathing***.

To provide rescue breathing when the victim has a pulse, give **1 breath every 5 to 6 seconds.**

Treatment of Choking



Choking in the Conscious Adult or Child (older than 1 year of age)

1. STAND (OR KNEEL) BEHIND THE VICTIM AND WRAP YOUR ARMS AROUND THE VICTIM'S WAIST.
2. PUT ONE FOOT IN BETWEEN THE VICTIM'S FEET AND ONE FOOT BEHIND YOU-THIS POSITION PROVIDES STABILITY SHOULD THE VICTIM BECOME UNCONSCIOUS AND YOU NEED TO EASE THE VICTIM TO THE GROUND.
3. PLACE YOUR FIST WITH THE THUMB SIDE IN JUST ABOVE THE VICTIM'S BELLYBUTTON AND WELL BELOW THE STERNUM (BREASTBONE).
4. GRAB YOUR FIST WITH YOUR OTHER HAND.
5. ADMINISTER ABDOMINAL THRUSTS, PULLING INWARD AND UPWARD UNTIL THE FOREIGN OBJECT COMES OUT OR THE PATIENT BECOMES UNCONSCIOUS. EACH THRUST SHOULD BE FORCEFUL, DISTINCT AND SEPARATE.

NOTE: FOR PREGNANT OR OBESE PATIENTS, GIVE CHEST THRUSTS INSTEAD OF ABDOMINAL THRUSTS.

Choking in the Unconscious Adult or Child

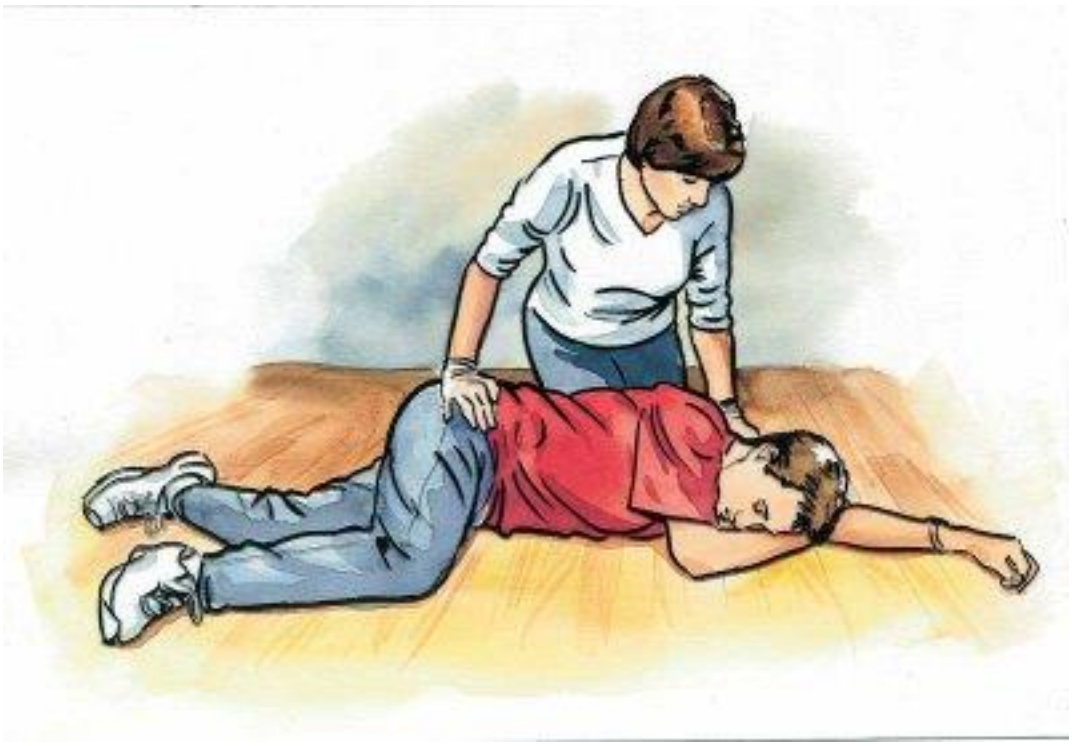
1. IF YOU ARE CARINIG FOR SOMEONE WHO IS CHOKING AND THEY LOSE CONSCIOUSNESS, LOWER THE VICTIM GENTLY TO THE GROUND.
2. ACTIVATE EMS/CALL 9-1-1
3. BEGIN CPR, STARTING WITH CHEST COMPRESSIONS- DO NOT CHECK FOR A PULSE.
4. EACH TIME YOU OPEN THE AIRWAY TO PROVIDE VENTILATIONS, OPEN THE VICTIM'S MOUTH AND CHECK FOR THE OBJECT. IF YOU CAN SEE THE OBJECT, TURN THE VICTIM's HEAD TO THE SIDE AND SWEEP IT OUT OF THE VICTIM'S MOUTH WITH YOUR INDEX FINGER. NEVER PERFORM A BLIND FINGER SWEEP- THIS MAY FORCE THE OBJECT FARTHER DOWN THE VICTIM'S AIRWAY.
5. IF YOU DO NOT SEE THE OBJECT, ATTEMPT TO PROVIDE BREATHS. IF BREATHS WILL NOT GO IN, RESUME CHEST COMPRESSIONS.
6. AFTER APPROXIMATELY 5 CYCLES OF COMPRESSIONS AND (ATTEMPTED) VENTILATIONS, ACTIVATE EMS/CALL 9-1-1 IF NOT ALREADY DONE.
7. IF THE OBSTRUCTION IS RELIEVED, CHECK RESPONSIVENESS, BREATHING AND PULSE. PROVIDE RESCUE BREATHING OR CPR AS REQUIRED. IF THE VICTIM IS RESPONSIVE, THEY SHOULD BE TAKEN TO HOSPITAL TO RULE OUT ANY INJURY CAUSED BY ABDOMINAL OR CHEST THRUSTS.

MOVING VICTIMS

It is generally not recommended that a rescuer move a victim unless there is a direct danger to the victim's (or rescuer's) life, or if it is necessary to provide care. Try to immobilize the spine and protect the head, neck, and back if it is necessary to move a victim.

Recovery position

An unconscious victim who is breathing and has a pulse should be assisted into the recovery position to protect the airway.



This position keeps the airway open, prevents aspiration of fluids into the lungs and allows fluid (such as saliva or blood that could occlude the airway) to drain from the mouth.

1. Extend the victim's arm that is closest to you above the victim's head.
2. Place the victim's leg that is farthest away from you over his other leg.
3. Place the victim's arm that is farthest from you across his chest.
4. Supporting the head and neck, roll the victim towards you.
5. Position his or her top leg so the knee props up the victim's body
6. Place the victim's hand or arm under his/ her chin to aid in keeping the airway open.

CPR CERTIFICATION SKILLS REVIEW

Skill	Adult
Check scene safety	Ensure the safety of the scene before entry
Determine patient responsiveness	Tap on the shoulder and shout, "Are you OK?" Look for gasping, abnormal breathing, or absence of chest movement.
Activate EMS – Call for help!	If alone with no mobile phone, leave the victim to activate EMS and retrieve AED before returning to victim. If another person is available, ask them to activate EMS, get an AED and return to provide assistance.
Check pulse	Check for a pulse for no more than 10 seconds in the carotid artery of the neck.
Compressions: Hard and Fast	30:2 at 100- 120/minute. Use both hands. Depth of compressions 2" minimum to 2.4" maximum.
Airway	Head tilt/chin lift. Check mouth for obstruction, foreign matter for choking victims. Use jaw thrust if suspected neck/spinal cord injury
Breathing	2 breaths over 1 second each, following 30 compressions. Watch for chest rise. When an advanced airway is present, give 1 breath every 6 seconds while continuous chest compressions are being performed.
Rescue breathing: Patient with pulse and not breathing or gasping	1 breath every 5-6 seconds, reposition airway if breaths aren't going in, recheck pulse every 2 minutes
AED	Give CPR until AED is available and charged- use AED as soon as it is available.



Standard First Aid

Welcome!

We're so excited that you've chosen us to be your learning partner in Standard First Aid. We strive to provide the best online learning experiences anywhere on the Web, and we're certain you'll find the information within this course of great value to you in your work and life.

In order to pass the course, you will need to complete the online examination. To be certain you pass on the first attempt, please read through the course at your leisure to be sure you understand the information contained within.

Topics included in this course are as follows:

- Scene Safety
- Universal Precautions
- Patient Assessment
- Control of Bleeding
- Shock
- Musculoskeletal Injuries
- Burns
- Chest Pain
- Respiratory Emergencies
- Stroke
- Seizures
- Diabetic Emergencies
- Poisoning
- Hyperthermia (Heat Exposure)
- Hypothermia
- Frostbite
- Foreign Bodies
- Traumatic Tooth Loss
- Bites and Stings
- Anaphylaxis
- Minor Cuts and Puncture Wounds

Let's get started!

Scene Safety

Imagine that you're driving along in your car, and to your horror you observe the car in front of you lose control, drive into a deep ditch at the side of the road and flip twice, landing on it's side. You have taken CPR and First Aid and you're prepared to help. What should you do first?

Scenes like these happen every day. That's why we take these courses- so that we can help at the right time in the right way. Before you attend to any victim, there is one important step you must take: *you must ensure that the scene is safe*.

Why? Scene safety is important because the last thing you would want to do in such a situation is to become a victim yourself- that would be of no help to anyone. There are several things that you should look for when determining in the scene is safe for you to enter. Look for:



- Flames or smoke
- Traffic (especially on the highway)
- Risk of an explosion
- Chemical spills/fumes
- Downed electrical wires
- Potential collapse of a building or structure
- Animals (frightened animals may bite if you attempt to render assistance to their owner)
- Potential for violence towards yourself or others
- Other dangers

Only when you have ensured that <u>you will be safe</u> should you provide first aid to a victim.

Universal Precautions



One of your major concerns when providing first aid should be to protect yourself from bloodborne pathogens. *Bloodborne pathogens* are infectious microorganism that have the ability to cause disease in humans. They are spread through direct contact with infected blood. Some of these include hepatitis C and HIV, the virus that causes AIDS. *Universal precautions* are measures that help to protect against bloodborne pathogens and assume that all victims are potentially infectious. By observing universal precautions, you are protecting yourself against exposure to potentially disease-causing microorganism present in another's blood or body fluids. To protect yourself, you should use PPE, or personal protective equipment, such as gloves. If you do not have gloves, you can improvise by using a plastic bag to cover your hands. Always follow these guidelines when providing first aid to protect yourself:

- Wash your hands, both before and after provision of first aid (you want to protect the victim from your pathogens as well, which is why you should wash your hands before providing care)
- Cover any of your own scrapes or cuts, as these can become portals of entry for microorganisms
- Avoid touching your nose, mouth or eyes while providing care and until you have washed your hands well after providing care
- Be careful to avoid injury at the site by being aware of any objects that may cut or scrape your skin

Patient Assessment

Once you have determined that the scene is safe (and only after you have ensured that this is so), you may approach the victim to perform an assessment. Assessment has three separate, but equally important, steps.

- Check responsiveness and ABC's to identify any life-threatening emergencies that must be dealt with immediately.
- Obtain a history from the victim and any bystanders.
- Perform an examination, using the head-to-toe method.

Check responsiveness and ABCs

To check responsiveness, talk to the victim to see if they respond. Shake their shoulders and yell “Are you okay?”. If they do not respond, this means they are likely unconscious and you must move quickly to assess airway, breathing and circulation.



Next, assess for breathing. Look for normal, absent or abnormal breathing (such as gasping). If responsiveness and breathing are absent or abnormal, call for help. If someone is around, ask them to activate the emergency response system (call 9-1-1) and get an AED (automated external defibrillator). If you are alone and no one responds to your call for help, use your cell phone to call 9-1-1. Quickly leave to obtain an AED if you know where one is located.

Check the victim’s pulse. If the victim is an adult, check for a pulse using the carotid artery. Use 2 or 3 fingers to locate the trachea (the windpipe), then slide these 2 fingers into the groove located between the trachea and the muscles on the side of the victim’s neck. Try to feel a pulse for 5, but not more than 10, seconds. If you cannot feel a pulse, or you’re not sure that you can feel a pulse, start CPR, beginning with compressions. If the victim is a child, you can also use the carotid artery to find a pulse, or you may also use the femoral artery. Place 2 fingers in the inner thigh, halfway between the victim’s hipbone and pubic bone and just below the crease where the victim’s leg meets the abdomen. If the victim is an infant, use the brachial artery in the upper arm. Place 2 or 3 fingers on the inside of the infant’s arm between the elbow and shoulder. In child and infant victim’s if there is no pulse, you are not sure if there is a pulse OR if the pulse is less than 60 beats per minute with signs of poor perfusion (pale color or bluish discoloration around the mouth or at the nailbeds), start CPR beginning with compressions.

When performing CPR, remember the following:

- Push HARD and FAST, 100 to 120 compressions/minute for victims of all ages
- Allow the chest to fully recoil after each compression (return to its normal position)
- Press down on the chest at least 2 inches for an adult victim; for children and infants, compress the chest to 1/3 the depth of the chest, or approximately 2 inches for most children and 1 ½ inches for infants less than 1 year of age. Use 2 hands to compress the chest of adults, 1 or 2 hands to compress the chests of children (depending on size) and 2 fingers to compress the chests of infants. When 2 rescuers are present with an infant victim, the chest is compressed using the 2 thumbs of the rescuer, while the fingers encircle the chest and provide support to the back.

- When providing breaths, use a barrier device (such as a pocket face mask) if available to prevent inhalation of expired air or other matter. If you choose to provide mouth-to-mouth breaths, use a head tilt-chin lift to open the airway and hold the airway open with your hand. With the hand you use on the forehead to hold the airway open, pinch the victim's nostrils shut. Place your mouth over the victim's mouth to create a tight seal. Take a normal breath and breathe through your mouth into the victim's mouth. Deliver each breath over 1 second and watch for chest rise. If the chest does not rise, try the head tilt-chin lift again and attempt another breath. If you cannot ventilate the victim, return immediately to chest compressions. For small children and infants, cover their mouth AND nose with your mouth. Deliver each breath over 1 second and watch for chest rise.
- Hands-only CPR can be provided when a barrier device is not available- provide continuous chest compressions at a rate of 100 to 120 compressions per minute.
- Compression-ventilation ratios for adults (1 and 2 rescuers) is 30 compressions to 2 breaths; for children and infants, the compressions to ventilation ratio is 30:2 when 1 rescuer is present, and changes to 15: 2 when a second rescuer is available.
- Avoid excessive ventilation- this may lead to gastric inflation (air in the stomach), which may cause the victim to vomit. When an unconscious victim vomits, vomited material may travel down the airway into the patient's lungs, which may cause what is known as aspiration pneumonia.
- Use an AED (automated external defibrillator as soon as it is available. Follow the steps as the AED guides you. Remember, the first and most important step when using an AED is to turn the machine ON- the AED will guide you through the next steps.

Note: This course is not designed to teach you to perform CPR. If you have not learned to perform CPR, we offer CPR with AED here. {hyperlink to the appropriate course}. We have added this material as a quick review. We recommend you take the full CPR/AED course if you have not already done so.

Obtain a history

SAMPLE History

- **S**igns and symptoms
- **A**llergies
- **M**edications
- **P**ertinent past medical history
- **L**ast oral intake
- **E**vents leading up to the incident

Assuming the victim does not require CPR, you should next obtain a history from the patient and any witnesses or bystanders. Ask the victim what happened, as well as any others who may

be able to provide important information that may determine what you do next. This information may also be important to emergency responders one they arrive on the scene. An easy way to obtain a thorough history is to use the mnemonic SAMPLE:

S- signs and symptoms- ask the victim how he/she feels. What can you notice when you look at the victim? For example, the victim may tell you that their right leg hurts- this is a symptom. You notice that the patient is holding their leg and grimacing in apparent pain- this is a sign.

A-allergy- ask the patient if they have any allergies to anything. If family members or loved ones are present, they may be able to provide you with this information on the patient's behalf.

M-medications- ask the patient if they take any medications. This should include over-the-counter medications, vitamins and supplements, as these can sometimes have unwanted side effects, such as excessive bleeding for example.

P-previous problems- ask if the patient has ever experienced a similar event. Also ask if they have any major health problems, such as heart disease, diabetes or bleeding disorders.

L-last meal taken- ask the patient when the last time was that the patient ate or drank anything.

E-event- ask for a summary of the present event. What exactly happened?

Perform an examination

Once you have obtained the victim's history, you should move on to performing a head-to-toe examination, starting at the head and working your way down to wards the feet. You should look for:

- Bruises
- Cuts/lacerations
- Bumps/lumps
- Sources of bleeding
- Obvious deformities of the arms and legs
- Painful areas

Note: If the patient is conscious, you should ask for permission (consent) before examining the victim. If the victim is unconscious, there are Good Samaritan laws that protect you legally, providing you are following the standard of care (providing care as you have learned or have been trained).

Take a close look at the victim. Assess the victim's breathing. Are they breathing very shallowly or quickly? Do they have pain when they breathe? Is their breathing too slow? Also assess their color (pale, bluish discoloration or flushing) and skin temperature. Is the skin warm and dry, or cool and clammy? This may provide important information.

During your examination, you should immediately call 9-1-1 for help if you find any life-threatening signs or symptoms:

- Chest pain or pressure
- Altered level of consciousness
- Uncontrolled bleeding
- Serious broken bones
- Difficulty breathing
- Seizures
- Signs of shock (pale, clammy and cool skin)

Note: You should not try to move the victim unless absolutely necessary, as moving the victim could cause further injury. You may need to move the victim if the scene is not safe for you and the victim (there is fire or smoke nearby, or you fear an explosion may occur). Otherwise, you should leave the victim in the same position they were in. Remember, scene safety is of utmost importance- although you may want to, you should not put your own life at risk unnecessarily- doing so may create another victim- YOU.

Control of Bleeding



Although most bleeding episodes are not life-threatening, severe bleeding can result in death if not stopped. As a first aid responder, protect yourself from bloodborne pathogens by wearing gloves if they are available- a plastic bag over your hand can be used if you don't have any gloves.

Try to identify the source of the bleeding. If blood is spurting, it is likely that an artery has been cut. Apply direct pressure to stop the bleeding. If the wound is in the victim's belly and organs are visible, do NOT try to put the organs back into the abdomen; simply cover the wound with a clean dressing, apply pressure as needed to control the bleeding and get the victim to medical care.

For other bleeding wounds, do the following:

1. Encourage the victim to lie down; if the victim is dizzy, raise the victim's feet above his/her head.
2. Do NOT remove any object that is deeply embedded in the wound.
3. If the source of blood is an extremity, raise the affected arm or leg above the level of the victim's heart. Apply pressure directly to the wound using a clean cloth or clean/sterile bandage. Hold pressure for ten minutes; do NOT peek to see if bleeding has stopped.
4. Apply a pressure bandage. If you note that there is blood coming through the pressure bandage, do not remove it- simply reinforce the bandage if you notice bleeding around the original bandage.
5. Continue to apply pressure to the wound.
6. If the victim becomes drowsy, extremely anxious, thirsty, or if his skin becomes very cool and/or clammy, you should suspect shock. If the victim is anxious, try to keep them calm and still. Raise the victim's legs above the level of their heart.
7. Get the victim to emergency care as soon as possible.

Shock

Shock is caused by inadequate blood flow to the body's organs. It can be caused by blood loss or loss of body fluids, as may occur following untreated severe vomiting or diarrhea. Symptoms of shock may include the following:

- Drowsiness
- Confusion
- Loss of consciousness
- Extreme thirst
- Pale skin color
- Rapid heart rate
- Cool, clammy skin
- Rapid breathing
- Anxiety



Victims in shock must receive emergency care as soon as possible to prevent death. Call 9-1-1. While waiting for emergency services:

1. Ensure the victim is lying down.
2. Elevate the victim's legs above the level of their heart, approximately 8 to 12 inches. This will promote blood flow to the heart.
3. Cover the victim with a blanket to keep them warm.
4. Keep the victim calm and still.

Musculoskeletal Injuries

Sprains, strains, fractures and dislocations are almost always caused by trauma. Any of these injuries may cause deformity, swelling, and severe pain. All of these injuries should be evaluated by a medical professional; however, there are some steps that should be taken in providing first aid. These steps may prevent further injury and disability:

- Call 9-1-1 if the injury is a result of major trauma.
- Assume the victim has a neck injury in any case of major trauma and avoid moving the victim until help arrives. Do not allow the victim to move his/her neck.
- Stabilize the affected body part only if doing so does not cause additional pain; do not try to force a deformed joint back into place as this may cause further damage.
- Many victims of musculoskeletal injury will assume a position of comfort- allow them to do so if this decreases their pain.
- Remember the acronym RICE when treating injured extremities (arms and legs, feet and hands):
    - *Rest* any injured joint
    - Apply a cold pack or *Ice* if available to reduce swelling and control pain
    - Apply *Compression* (an elastic bandage) for any sprains or strains
    - *Elevate* the injured body part above the heart to reduce swelling
- Do not give the victim anything by mouth until seen by a medical professional; the exception to this rule is if medical care is not available within a couple of hours, an over-the-counter pain reliever may help manage pain.
- Control bleeding if present (see section on Control of Bleeding)
- If the victim shows signs of shock (see section on Shock), have the victim lie down and wait for help.



Burns

Many burns are minor and can be treated with first aid; others burns are more serious and require medical attention. The severity of a burn can depend on many factors, including:

- The age of the victim- burns can be more severe in babies, young children and the elderly
- The size of the burn- how much of the body is affected, often described as a percentage, which may be an indicator of survivability
- The depth of the burn- how deeply the burn extends down into the subcutaneous tissue
- The location of the burn- burns located on the hands, face, groin/genitals, neck, hands and feet can have severe functional and cosmetic consequences
- The presence of other injuries- burns combined with other severe injuries may affect outcome and survivability

It is important to be able to recognize the type of burn in order to provide the correct care.

Type of Burn	Appearance & Sensation	Complications	First Aid Treatment
First Degree	Redness; Dry skin; Painful	Infection	First degree- cool the burn with cool running water for 10-20 minutes, or apply a cool wet compress. Remove rings from burned fingers as the area may swell. Apply lotion or aloe vera gel for comfort. May take an over-the-counter pain reliever if needed. Sunburns are an example of first-degree burn.
Second Degree (Partial Thickness)	Red with blisters; Moist; Painful	Infection and cellulitis; scarring and contractures; may require debridement	Treat as for a first degree burn unless the burned area is large. Seek medical help for larger burns, especially those on the hands, face, neck, groin or feet. Leave large blisters intact- do NOT

			puncture. If small blisters rupture (smaller than your fingernail) cleanse the area with mild soap and water and apply antibiotic ointment and a non-stick dressing. Seek medical care for signs of infection (increased pain, redness, swelling, oozing from the wound)
Third Degree (Full Thickness)	Stiff; dry; leathery; white or brown color; Painless	Scarring and contractures; May require debridement or amputation; large third degree burns may result in death	Call 9-1-1. If it is safe to do so, remove the victim from further contact with heat and/or smoke. Do not remove clothing that has adhered to the skin. Remove belts, jewelry or other restrictive items from the victim as the burned areas will swell very quickly. Do not immerse very large burns in cool water as this may lead to excessive loss of body heat. Elevate burned areas if possible. Cover burned areas with a clean cloth or a sheet. Monitor breathing and circulation and perform CPR if it becomes necessary.
Fourth Degree	Black; charred; Painless	Possible gangrene; Usually leads to loss of function and sometimes death	As for a full thickness burn

Smoke inhalation may be as deadly as severe burns. If a burn victim has soot around or in their mouth or nose, this indicates that the airway may be affected and breathing may become a problem. If you suspect smoke inhalation, call 9-1-1.

Chest Pain

There are many causes of chest pain, some minor and some very serious. Without a medical workup, it is often impossible to tell whether chest pain is life threatening. For this reason, if you or a companion experiences chest pain that lasts more than a few minutes, it is critical that chest pain be evaluated by medical professionals. Symptoms that should raise your suspicion of heart attack include:

- Chest pain that gets worse when you are active but decreases with rest
- Pressure or squeezing pain in the middle of your chest
- Pain that spreads to your shoulder, arm, neck or back
- Pain accompanied by dizziness, weakness, shortness of breath, nausea or vomiting.
- Pale, clammy skin (diaphoresis or sweating)

Any of these symptoms may indicate angina, pulmonary embolism (a blood clot that moves to the lung), aortic dissection or a myocardial infarction (heart attack). If you or someone you are with experiences any of these symptoms:

1. Call immediately for help (call 9-1-1). If you cannot call for help, get to medical treatment as soon as possible.
2. If you have a well-stocked first aid kit, have the victim chew an adult aspirin or 2 baby aspirins (unless allergic). This can help reduce blood clotting which is a cause of heart attack.
3. Have the victim lie or sit comfortably until help arrives, and keep the victim calm
4. If the victim becomes unresponsive, activate the emergency response system and begin CPR if indicated.

Chest pain may not be life-threatening. For example, pneumonia and costochondritis (pain in the cartilage connecting your ribs to your sternum) are common causes of chest pain. Strained chest muscles, gastric reflux, gallbladder disease, and even anxiety can cause chest pain. These syndromes and diseases may cause chest pain; however, any chest pain should be diagnosed by medical personnel to rule out one of the deadly conditions.

Respiratory Emergencies

Respiratory conditions are common in the population in every age group. For this reason, it's important to know what to do to assist someone experiencing difficulty breathing. Respiratory problems are almost always an emergency, and medical assistance will be required in almost all cases. Common causes of respiratory distress include:

- Asthma
- Emphysema
- COPD (chronic obstructive pulmonary disease, common in the elderly population)
- Respiratory infections (such as pneumonia, croup, whooping cough)
- Pneumothorax (collapsed lung, usually from trauma)
- Heart failure or severe heart disease
- Fluid around the heart or lungs (pericardial or pleural effusion)
- Anaphylaxis (life-threatening allergic reaction)
- Pulmonary thrombosis (blood clot in the lung)
- Injury to the chest, lungs or neck

Difficulty breathing may present as difficulty taking a deep breath, being short of breath or experiencing 'air hunger', the sensation of not being able to get enough air. There may also be pain with breathing in certain conditions, especially if the victim has been involved in a trauma.

Signs and symptoms of respiratory distress are often quite obvious, and may include:

- Rapid breathing
- Inability to breathe properly unless sitting upright
- Anxiety
- Altered level of consciousness (drowsiness, confusion or unconsciousness)
- Cough

- Dizziness
- Pain with breathing
- Grunting, high-pitched whistling (stridor) or wheezing when breathing
- Bluish discoloration to the lips, nailbeds or fingers
- Rapid heart rate
- Diaphoresis (sweating)
- Difficulty speaking more than a couple of words at a time
- Gasping for breath
- Fever (if the cause is infectious)
- Hemoptysis (coughing up blood)
- Rash or hives (if the victim is suffering an anaphylactic reaction)

As can be seen, respiratory emergencies can present in numerous ways. If you are administering first aid to someone experiencing a respiratory emergency, you should:

1. Call 9-1-1.
2. Loosen tight clothing that may be restricting the victim's breathing.
3. Assist the victim to use any inhalers, oxygen or other medical devices.
4. Bandage any open wounds of the neck or chest if necessary. If the victim has a 'sucking' chest wound (a chest wound that allows air to enter the chest cavity with each breath), apply plastic wrap or a plastic bag to the wound, sealing it in place EXCEPT for one side to allow air to escape. This will prevent air from entering the chest, but will also allow any trapped air to escape.
5. Do not give the person food or drink.
6. Do not put a pillow under the victim's head (this may close off the victim's airway).
7. Do not move the person if they have been injured (in case of spinal cord injury).
8. Continue to monitor the victim's breathing and heart rate. Should the victim stop breathing or lose their pulse, perform rescue breathing or CPR until help arrives.

Do not be falsely reassured if a victim with noisy breathing (wheezing, gurgles or high-pitched whistling) suddenly has quiet breathing- this may indicate that the victim is no longer moving air into the lungs and may mean that the victim is close to respiratory collapse.

Stroke

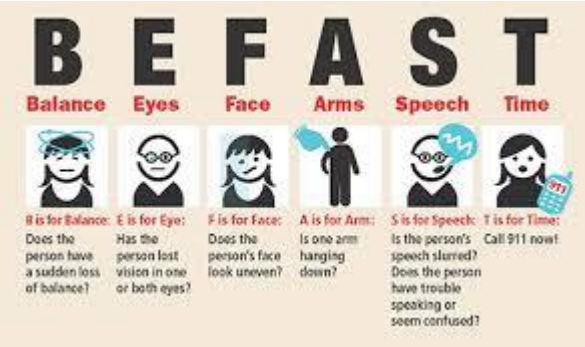
A stroke is a true medical emergency. There is a saying in medicine, "Time is Brain", which means that the longer a person waits to receive medical care after a stroke, the greater the chances of serious neurological damage or death. The brain requires oxygen and nutrients to function and cannot last long without these- brain cells begin to die within minutes when deprived of oxygen.

It is important that everyone is familiar with the signs of stroke:

- Weakness or numbness on one side of the body or face
- Facial drooping (one side of the mouth will appear to be lower than the other, smile will be crooked)
- Changes in vision (blurred vision, loss of vision or dimness)
- Difficulty speaking or difficulty "finding" words/slurred speech
- Altered gait/difficulty walking

- Severe headache (not always present)
- Dizziness or weakness

A victim of stroke may be found unconscious if the stoke is severe.



A stroke occurs for one of two reasons: a blood clot blocks blood flow to a portion of the brain (called ischemic stroke), or there is bleeding into the brain (called hemorrhagic stroke). The most important thing to note with either type of stroke is when symptoms first appeared (when the victim was last known to ne normal). This is because certain treatments for stroke are time-dependent.

If you suspect stroke:

- Call 9-1-1. This is your priority action. Be sure you tell dispatch that you suspect stroke, as this will start important preparations at the hospital.
- Keep the victim calm.
- Try to determine when the victim was last normal, or without symptoms.
- Do NOT give food or water. Some strokes may affect swallowing.
- Do NOT give aspirin. Although most strokes are ischemic in nature, some are hemorrhagic, and giving aspirin in these cases would worsen bleeding and could cause death.
- If the victim is unconscious, monitor breathing and pulse and be prepared to perform CPR if it should become necessary.

Seizures

Seizures (convulsions) are not uncommon and can be quite frightening to witness if you have never seen one before. Seizures are not a disease in and of themselves, but rather are a symptom of any number of disorders that affect the brain, including epilepsy. Seizures do not cause permanent brain damage. A seizure is a sudden surge of electrical activity in the brain which affects the way a person moves and acts. Some seizures are severe, while others are barely noticeable unless you know what you are looking for. You will not need to provide first aid unless the seizure is a grand mal seizure, in which the person loses consciousness, falls to the ground and experiences jerking and stiffening of the muscles. These are also called generalized seizures, or tonic-clonic seizures. These types of seizures usually last 1 to 3 minutes and subside on their own without treatment. Seizures that last longer than 5 minutes require medical care and monitoring.

If you are assisting a victim who is suffering a seizure:

1. Loosen and tight or restrictive clothing that may hamper breathing.

2. Do NOT try to restrain the victim or hold them down- this may result in injury to the victim.
3. Do NOT try to put anything in the victim's mouth to prevent them biting their tongue- the victim may choke on the object or injure their mouth.
4. Remove any dangerous objects/furniture from the victim's space that could cause injury
5. If there are panicked bystanders, ask them to move back and give the victim some space. Reassure them that the seizure will pass and there is no reason to panic.
6. Once the victim has stopped seizing, they may be confused and disoriented for a period of time. Provide reassurance. If the victim is semiconscious or unconscious, place the victim on his/her side to prevent choking on any secretions from the mouth, particularly if the victim bit their tongue and it is bleeding.
7. Monitor breathing and pulse until help arrives.
8. Call 9-1-1. Many seizure patients will not require transport to a hospital if the seizure is their "usual" pattern of seizure, but it is a wise idea to allow medical personnel to make that decision.

Children will sometimes experience what is known as febrile seizures when they have a high fever- treatment is the same as for an adult victim. Controlling fever will often prevent these types of seizures. Most children outgrown these types of seizures as their brain develops.

Diabetic Emergencies

Millions of people around the world suffer from diabetes, and diabetic emergencies are common. In type 1 diabetes, the body produces no insulin, and the individual must use artificial insulin to survive. In type 2 diabetes, the individual's pancreas produces insulin, but the person's body is unable to utilize it properly. Our cells require a constant supply of sugar to survive, particularly the brain. Too much sugar in the blood (hyperglycemia) and too little sugar (hypoglycemia) have different symptoms, and it is important to be able to recognize the difference between the two.

Signs and symptoms of hyperglycemia include:

- Fatigue (tiredness)
- Thirst
- Warm, dry skin
- Sweet or fruity breath
- Rapid heart rate and breathing
- Drowsiness
- Loss of consciousness (diabetic coma)

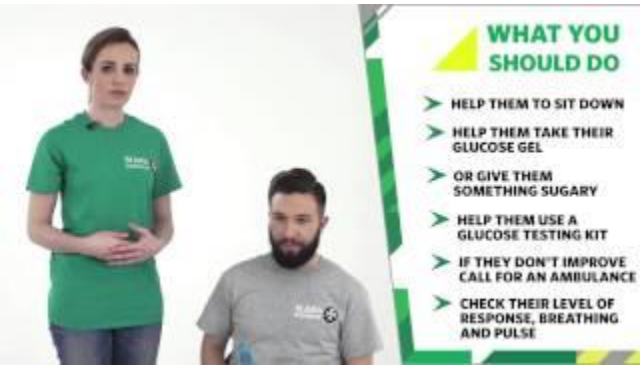
Signs and symptoms of hypoglycemia include:

- Weakness/faintness
- Hunger
- Shakiness/trembling
- Sweating
- Cold, clammy skin
- Combativeness, confusion or irrational behaviour
- Rapid heart rate
- Loss of consciousness (severe)

To treat hyperglycemia:

1. Call 9-1-1. If you know or suspect the victim is a diabetic, tell dispatch.
2. Monitor breathing and pulse and prepare to act if the person becomes unconscious.

To treat hypoglycemia:



1. Have the victim sit or lie down.
2. If they have their own source of sugar (i.e. glucose gel) have them take it; if they do not, give them food or drink containing sugar, such as soda, juice with sugar in it or candy.
3. If they show signs of improving, ask them to check their blood glucose (if they have a testing kit).
4. Give them more sugar if necessary.
5. If they do not improve quickly call 9-1-1. Look for other causes.
6. Continue to monitor breathing and pulse and prepare to act if the person becomes unconscious.

Poisoning



Poisoning can occur with almost any substance, even with medications prescribed by a healthcare provider. In fact, intentional and unintentional overdoses of medicines are much more common than poisonings by other substances. You should know the number of your regional poison control center or the United States National Poison Center at 800-222-1222 and call them before providing first aid to anyone suspected of having any kind of poisoning. When you call, have some general information about the victim readily available. In addition, if you know the medicine, product or plant the victim ingested, this will be invaluable to the expert on the phone.

Be aware that the symptoms of poisoning can be the same as many other diseases and conditions; however, there are certain signs to look for if you suspect poisoning. If you notice any of these, call for medical assistance BEFORE calling poison control:

- Altered level of consciousness (may be sleepy or hyperactive)
- Altered respiratory pattern (breathing very slowly or too quickly)
- Slurred speech
- Nausea and/or vomiting
- Chemical smell on the victim's breath
- Burns on or around the lips or in the mouth
- Seizures
- Empty pill bottles in the area
- Spilled chemical bottles

While you are waiting for medical assistance, be sure you and the victim are both safe. Move to fresh air if you suspect the victim has been breathing lethal fumes, as in carbon monoxide poisoning. Look into the victim's mouth and, with a gloved hand, remove anything you can easily reach. If the poison spilled on the victim, remove any contaminated clothing and begin to flush the area with water until help arrives, but only if you have proper protective wear, such as gloves. Do NOT make the victim vomit unless advised to do so. Monitor the victim and start CPR if that becomes necessary.

If you reach the poison control center, follow any instructions given by them. If the victim is taken to hospital, take or send any pills, bottles or plants.

Hyperthermia (Heat Exposure)

Heat related injuries include heat cramps, heat exhaustion and heat stroke. Heat stroke is the most severe of these injuries and typically happens when the victim has been performing heavy work or engaging in sports in a very hot environment. The victim usually has been sweating heavily and not replenishing fluids lost to sweating. In addition, some medications and alcohol use may predispose an individual to heat stroke. The main symptoms of heat stroke include:

- Faintness/dizziness
- Headache
- Nausea/vomiting
- Hot, dry skin or cool, clammy skin
- Rapid and weak pulse
- Weakness
- Altered level of unconsciousness (may lose consciousness)

Untreated heat exhaustion or heat stroke can cause death. Act quickly if you suspect either condition. If you suspect a person has hyperthermia, there are several things you can do to help the victim:

1. Call 9-1-1
2. Move the person to a cooler environment, out of the sun and into a shady or air-conditioned space.

3. Lie the victim down and elevate their legs and feet.
4. Remove restrictive or tight clothing.
5. Begin to actively cool the victim by fanning the person and applying cool compresses to the armpits, neck and groin areas.
6. If the victim is awake enough to drink and is not vomiting, try giving cool liquids to drink. Do NOT give the person alcohol or beverages containing caffeine. Move the person out of the sun and into a shady or air-conditioned space.
7. If the victim loses consciousness, monitor airway, breathing and circulation and prepare to perform CPR if necessary.

Hypothermia

Hypothermia is caused by prolonged exposure to the cold (cold weather or cold water).

- Shivering and loss of coordination
- Slurred speech, apathy, confusion
- Extremely slow breathing
- Loss of consciousness (severe hypothermia)

If you suspect someone has hypothermia, there are several things you can do to help the victim:

1. Call 9-1-1
2. Move the person to a warmer environment.
3. Remove any wet clothing and cover the victim with a dry blanket (or dress in dry clothing for mild cases).
4. Begin to actively warm the victim by applying hot water bottles or chemical hot packs (wrapped in towels or blankets to prevent burns) to the chest and abdominal areas (core areas). Do not worry about warming the arms and legs; warm the blood at the center (core) of the body and allow that warm blood to warm the extremities.
5. Do NOT immerse the victim in hot water- warming the victim too quickly may cause lethal heart rhythms
6. Do NOT use a heating pad or heating lamp.
7. If the victim is awake enough to drink and is not vomiting, try giving warm liquids to drink. Do NOT give the person alcohol or caffeinated beverages.
8. If the victim is unconscious, monitor breathing and pulse and prepare to perform CPR if necessary.

Frostbite

Frostbite is caused by exposure to cold temperatures for prolonged periods, which causes damage to the skin and underlying tissues.

Symptoms of frostbite may include:

- Pins and needles sensation, followed by numbness
- Pale and hard skin
- Aching, throbbing or lack of sensation in the affected area
- Red and very painful skin/tissue as unthawing occurs

- Blisters and blackened, dead tissue (severe frostbite)

Areas with poorer circulation are most prone to frostbite, such as the nose, ears, hands and feet. When frostbite extends to the blood vessels and damages them, the affected area may not recover and amputation is often necessary.



To treat frostbite:

1. Look for signs and symptoms of hypothermia and treat first (see section on Hypothermia above). Call 9-1-1 immediately if frostbite is severe or if there are signs and symptoms of hypothermia.
2. Bring the victim into a warm, dry place.
3. Remove tight jewellery if fingers are affected, or socks and boots if feet are affected. Remove any wet clothing to prevent further cooling.
4. If medical assistance is close by: wrap the affected area with dry gauze, separating fingers and toes.
5. If medical assistance is not close by, you will need to rewarm the affected areas.
6. Soak affected areas in warm water for 20 to 30 minutes (do NOT use hot water). Keep changing the water as it cools. If the victim's face is affected, apply warm compresses one after another. Pain, color changes and changes in sensation will occur as the tissue warms. Warming is complete when the affected tissue is soft again and full feeling returns.
7. Once fingers and toes have thawed, wrap them in dry gauze, being careful to separate the digits.
8. Move the injured tissue as little as possible.
9. If the frostbite is severe and affects more than one area, give warm fluids (NOT alcohol) to warm the victim and replenish fluids.
10. Do NOT break any blisters that form
11. Do NOT rewarm if there is a risk of refreezing- wait until you can get medical assistance)
12. Do NOT use direct heat to thaw damaged tissue (i.e. hair dryer, heating pad)
13. Do NOT massage the affected area

Foreign Bodies

A foreign object in any part of the body can cause pain, infection, and other problems. Most commonly, foreign bodies are found in the ear, eye, nose or skin.

Ear

Foreign bodies in the ear can lead to hearing loss, pain, and infection. The important thing to remember is to NEVER stick anything in the ear in an attempt to remove an object. Doing so may push the foreign body deeper into the ear and may result in damage to the ear. If you can see the object and grab it with a pair of tweezers, remove it. Use gravity by having the person tilt their head toward the ear with the foreign body.

Many times, an insect will crawl into an ear seeking the warmth and confined space. If you know that an insect is in the ear, you may be able to remove it by warming some baby oil and pouring it into the ear. The oil will serve to drown the insect and may allow the insect to float out of the ear. Do NOT use this method if there is any drainage from the ear that might indicate a ruptured eardrum. You should not use this method in any child who has tubes in his ears for recurrent ear infections.

Whether or not these methods are successful in removing the foreign body from the ear, a licensed practitioner should check the affected ear as soon as possible.

Eye

Foreign bodies in the eye can lead to vision loss, pain and infection. Be sure your hands are clean before you start working with the eyes. Many times, it may be possible to remove the foreign object by flushing the eye with saline solution or clean water while holding your eyelid open.

If the object is embedded in the eyeball, do NOT remove it. Instead, cover the eye loosely with a gauze pad and get immediate medical attention. An object deeply embedded in the eyeball must be removed surgically to prevent further damage to the eye.

If you flush a foreign body from the eye but the victim continues to have pain or redness, or if the victim cannot see normally, medical attention is immediately required.

Nose

Foreign bodies in the nose can lead to pain and infection. The important thing to remember is to NEVER stick anything in the nose in an attempt to remove an object. Doing so may push the foreign body deeper into the nose and may result in damage to the structures of the nose. If you can see the object and grab it with a pair of tweezers, remove it.

Instruct the victim to breathe through his mouth until the foreign body is removed. Breathing through his nose may cause the object to enter the victim's windpipe or lungs. Instead, have him blow his nose gently in an attempt to remove the object. If the object is not removed using these methods, seek medical help for professional removal.

Skin

Foreign bodies in the skin can lead to infection and pain. Before attempting to remove an object from the skin, make sure your hands are clean. If the object in the skin is wood, do NOT soak the extremity. Getting the wood wet will cause it to swell and fragment, making the removal more difficult. If enough of the foreign body is above the skin, you may be able to grasp it with tweezers and pull it out. After

removal, gently squeeze the area around the wound to encourage bleeding to wash out any fragments that might be left behind. Apply some antibiotic ointment and cover the area if it is likely to get dirty. Be sure to consult with your healthcare provider to determine if you should receive a tetanus shot.

Occasionally, the foreign body will be completely under the top layer of skin. If you feel comfortable using a sharp needle to remove the object, you must first clean the needle as well as possible using rubbing alcohol or soap and water. If you can see the point of entry for the object, use the needle to lift the skin above the object and move the point of the needle to lift the tip of the foreign body out above the level of the skin. Use the tweezers to grasp the object and pull it out. Cleanse the wound thoroughly, apply a topical antibiotic ointment, and cover the wound if it is likely to get dirty. Again, be sure to check on the status of your tetanus immunization.

Traumatic Tooth Loss

Losing a tooth due to trauma does not always mean that a tooth is lost forever. If a tooth is lost due to trauma you should:

1. Avoid touching the roots of the tooth- handle it by the crown only.
2. Avoid rubbing the tooth in an effort to clean it of debris.
3. If debris is present, gently swish the tooth around for less than 10 seconds in a bowl of lukewarm water-do NOT rinse the tooth under running water.
4. Attempt to place the tooth back in its socket. Bite down gently on a piece of gauze or moistened paper towel if it will not go all the way in.
5. If the tooth will not go back into the socket, place it in some milk or a mild saltwater solution.
6. See a dentist as soon as possible- the sooner a dentist is consulted, the better the chance of successful reimplantation.

Bites and Stings

Animal bites can be dangerous for several reasons. Animal bites can be deep and there is a high risk of infection.

1. If the bite is minor, clean the wound thoroughly with soap and water and apply a thin layer of antibiotic cream. Cover with a bandage.
2. For deeper wounds, apply pressure to stop any bleeding and seek medical attention. If medical help is not readily available, stop any bleeding, clean the wound well and apply a clean dressing until medical attention is available.

Any time the skin is broken, there is risk of infection. Signs of infection (redness, swelling, pain that is getting worse instead of better, warmth, red streaks moving out from the wound) require medical help immediately. A tetanus shot is required if tetanus is not up to date.

If there is a chance that the animal carried rabies, it is critical that medical help be sought immediately. If possible, cage the animal that bit the victim, but do not risk a second bite to yourself. Remember that unimmunized domesticated animals can also carry rabies. In many areas of the US, a large proportion of

the populations of bats and skunks are rabid. Any wild animal that is acting unusually (i.e. it lets you get close to it) may be ill with rabies.

Human Bites

The human mouth is very dirty -- and human bites can be worse than animal bites. For human bites, it is important to clean the wound very well with soap and water. Apply a thin layer of antibiotic cream and a clean bandage. Watch for signs and symptoms of infection and seek medical care if any develop.

If the wound is very deep, get health care as soon as possible. Again, if you have not had a tetanus shot in the last ten years, seek medical care to get one as soon as possible after the wound.

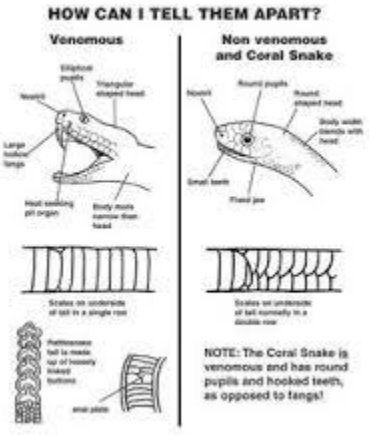
Insect Bites or Stings

Insect bites and stings can be a problem not only due to localized pain and swelling, but also because of the generalized systemic reactions that can occur. If a victim has been bit or stung:

1. Move to an area where the victim is safe from additional bites or stings.
2. Remove the stinger if it is still stuck in the skin. Do this by scraping a fingernail or other object, such as a credit card, across the skin; avoid trying to pluck the stinger out with your fingers as this may release more venom into the skin.
3. Cleanse the area with soap and water.
4. Apply an ice pack or cold cloth to help reduce swelling and pain.
5. Ibuprofen or acetaminophen may relieve the pain caused by inflammation. A topical antihistamine may help relieve pain and swelling. Calamine or Caladryl lotion may help with itching. A paste created from water and meat tenderizer or baking soda may also help reduce swelling and itching.

Snake Bite

Most snakes will avoid you if possible and will bite only when threatened or surprised. If you come across a snake, back away slowly and leave the snake alone.



If you or someone around you are bit by a snake:

1. Call 9-1-1.
2. Remain calm and try to remember what the snake looks like (color, markings, shape of pupils).
3. Do NOT try to catch the snake (and never bring a live snake to the hospital).

4. Immobilize the affected area with a splint.
5. Keep the area affected by the bite lower than the heart.
6. Keep the victim as quiet as possible to keep the venom from spreading.
7. If possible, clean the wound gently. Do not flush it with water.
8. Do NOT cut the wound or try to suck the venom from the bite. Do NOT apply ice or a tourniquet.
9. Remove any jewelry or other items that might constrict the area if swelling occurs.

Anaphylaxis

Some people are highly allergic to certain insect bites or stings. Certain foods, such as peanuts, can also cause a severe and life-threatening allergic reaction.

Anaphylaxis is a severe allergic reaction that can rapidly cause death if not immediately treated. Anaphylaxis causes a dangerous drop in blood pressure leading to shock. It can also cause swelling of the airway leading to inability to breathe. Symptoms may occur within minutes of exposure to the allergen.

Symptoms and signs may include:

- Shortness of breath
- Swelling of lips, eyes, mouth or throat
- Dizziness or faintness
- Wheezing or difficulty breathing
- Rapid heart rate
- Nausea/vomiting or diarrhea
- Hives or rash
- Flushed or pale skin
- Loss of consciousness

If you are with someone experiencing an anaphylactic reaction:

- Call 9-1-1 immediately.
- Check to see if he/she is carrying an EpiPen. If the victim is able, have them inject themselves; if they are unable, assist them to use the EpiPen. EpiPen use is simple: Remember “blue to the sky, orange to the thigh”. Hold the pen firmly with the orange side pointing down. Remove the blue cap by pulling straight up- do not bend or twist. Push the orange tip firmly into the mid-outer thigh. You will hear a click- hold firmly in place for several seconds.
- Lie the victim down with the legs and feet elevated, loosen tight clothing and keep them warm.
- Monitor the patient closely. If the patient stops breathing or loses their pulse, start CPR and continue until help arrives.



The medication contained in EpiPens is potent but wears off quickly. Even if the victim feels better after receiving the injection of epinephrine, they must go to the hospital in case symptoms return.

Minor Cuts and Puncture Wounds

Cuts and puncture wounds are common injuries. First aid for these injuries includes:

- Control bleeding by application of pressure to the area (see the section on Bleeding).
- Elevate the injury.
- Apply a cold pack to reduce swelling and pain.
- Clean the wound thoroughly with soap and water and apply a topical antibiotic cream.
- Cover the wound with a Band-Aid or clean bandage.
- Change the dressing if it becomes soiled.
- If the wound is deep or large, seek medical care to determine if stitches are required. This must be done within the first few hours after injury.
- A tetanus shot may be necessary if you have not had one in the last ten years- check with your physician.
- Watch for signs of infection (redness, swelling, drainage, or pain that is getting worse instead of better) and get treatment if you think the wound is getting infected.

Child and Infant CPR

FOLLOWS 2010 ECC/ILCOR GUIDELINES

Updated with 2015 Guidelines



Welcome!

We're excited that you've decided to take Child and Infant CPR, and that you've chosen us to direct your learning experience.

You never know when you may be called upon to perform CPR on a child or infant. All parents, grandparents, babysitters, older siblings, daycare workers, teachers and anyone else who lives or works with children should learn how to perform CPR when disaster strikes. CPR can save a child's life by providing the brain, heart and other organs with life-giving oxygen in the event of respiratory and/or cardiac arrest. Drowning, poisoning, accidents, smoke inhalation and SIDS are only a few of the emergencies that could result in death if not treated quickly.

In this course, you will learn:

- The Pediatric Chain of Survival and your place within the chain
- How to provide high quality chest compressions for children and infants
- How to initiate early use of an AED (automated external defibrillator)
- How to provide rescue breathing for pediatric victims of respiratory arrest
- How to provide CPR to children and infants when there are 2 rescuers
- How to assist child and infant victims of choking

Let's get started!

The Five Fears

Many bystanders are reluctant to perform CPR. There are many reasons why bystanders are reluctant to get involved. Let's take a look at some of these reasons so that you can understand why they have no real basis of support.

Fear of Disease: Use universal precautions when the possibility exists of coming into contact with bodily fluids. Use gloves, mask, and/or a gown when you have them available. The benefit of initiating lifesaving resuscitation in a child in cardiopulmonary arrest greatly outweighs the risk for secondary infection in the rescuer or the child. Nevertheless, use of simple infection-control measures during CPR and CPR training can reduce a very low level of risk even further (Mejicano & Maki, 1998). This fear is not as prevalent when dealing with infants and children, who are less likely to be harboring serious bloodborne pathogens.



Fear of Lawsuits: These laws state that a person acting in good faith who is rendering reasonable first aid will not be held accountable for damages to the person to whom the aid is rendered, unless gross and willful misconduct are involved. Good Samaritan laws may differ for from state to state. Implied consent means that there is an assumption that if an unconscious person were able to request care, they would do so (or their parent/guardian would do so on their behalf).



Fear of Uncertainty: People sometimes fear that they won't remember what to do. Remembering the correct number of compressions or the number of compressions to ventilations is not as important as the willingness to respond and to push hard and fast. Hands-only CPR is designed to provide simple life support. Remember, anything you do is better than doing nothing at all!



Fear of Harming the Victim: A victim in need or clinically dead can only be helped by your intervention efforts. You can't hurt someone who is dead, and any injuries you may unknowingly cause (such as injured ribs) can be dealt with in a surviving victim of respiratory or cardiac arrest. Children and



infants are surprisingly resilient!

Fear of Injury to Self: As you will learn, scene safety is of utmost importance, and checking the scene for safety should be your first action when you come across an unconscious victim, even if the victim is a child. If the scene is not safe for you to enter, you must not enter. This will only result in more victims. Under no circumstances should you risk becoming a victim. Instead, you should call for help (activate EMS).

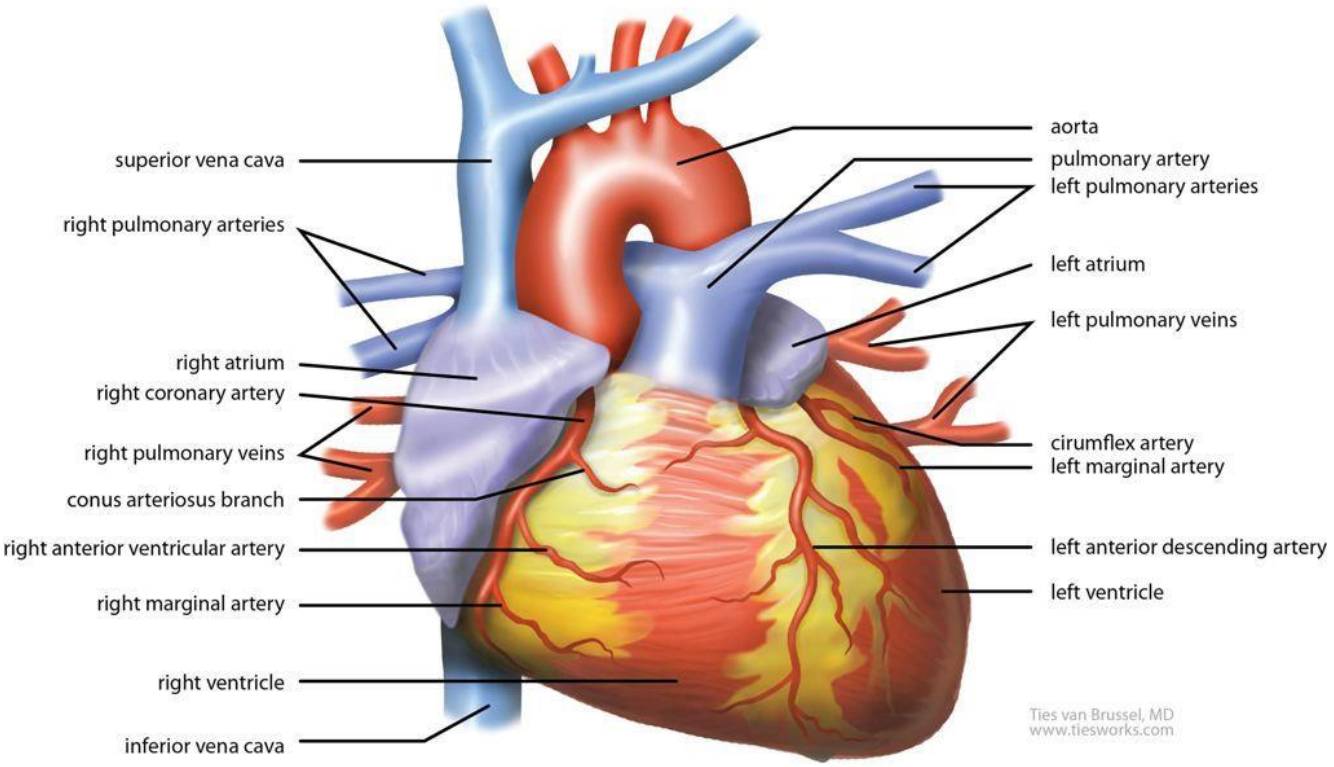


As a rescuer, you must always check to ensure the scene is safe before rendering assistance to any victim!

BASIC ANATOMY AND PHYSIOLOGY

Before you learn how to perform CPR, it's important to understand how the heart, lungs, brain and cells perform. Here is a brief review of these systems.

THE HEART



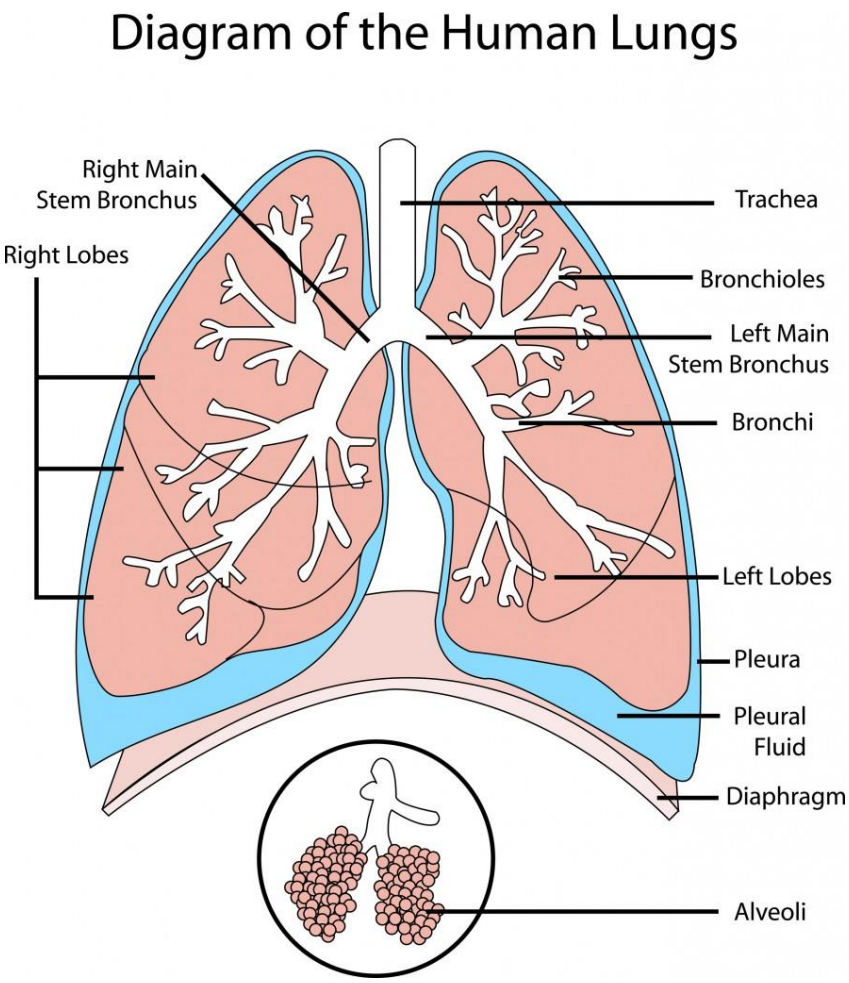
The heart consists of four chambers, the right and left atria (singular: atrium) and the right and left ventricles. The atria are located above the ventricles, as can be seen in the above diagram. The heart is a muscular organ supplied by the coronary arteries. It is located below your breastbone (sternum). In children and infants, their hearts are about the size of their fists.

The heart pumps blood that has been deoxygenated by supplying the body's tissues into the lungs, and when that blood has been oxygenated again in the lungs, it exits the lungs to the left side of the heart, where it is pumped out into the tissues once again to provide live-giving oxygen.

Your body has about 5 liters of blood, which circulate through this system approximately 3 times per minute. A child will have less circulating blood volume, depending on their age- a child of 80 pounds will have approximately half the circulating blood volume of an average-sized adult. Infants will have far less. This is why blood loss can be devastating to a child.

What is known as the cardiovascular system is composed of the heart, arteries, capillaries, and veins.

THE LUNGS



Your lungs are spongy, air-filled sacs, with one lung located on either side of the chest. The trachea, which is sometimes called the windpipe, conducts air down into the lungs through the bronchi, which are smaller tubular branches. The bronchi then divide into smaller and smaller tubules called bronchioles. Air is exchanged in the alveoli, which are tiny sacs that allow oxygen and carbon dioxide to move between the lungs and the bloodstream via tiny capillaries.

Your lungs take in oxygen to supply your body's organs and tissues. They release carbon dioxide, a waste product, into the atmosphere when you exhale.

Room air at regular atmospheric pressure contains 21% oxygen. Our bodies utilize approximately 4-6% of that oxygen and release about 17% back into the atmosphere, along with carbon dioxide, when we exhale.

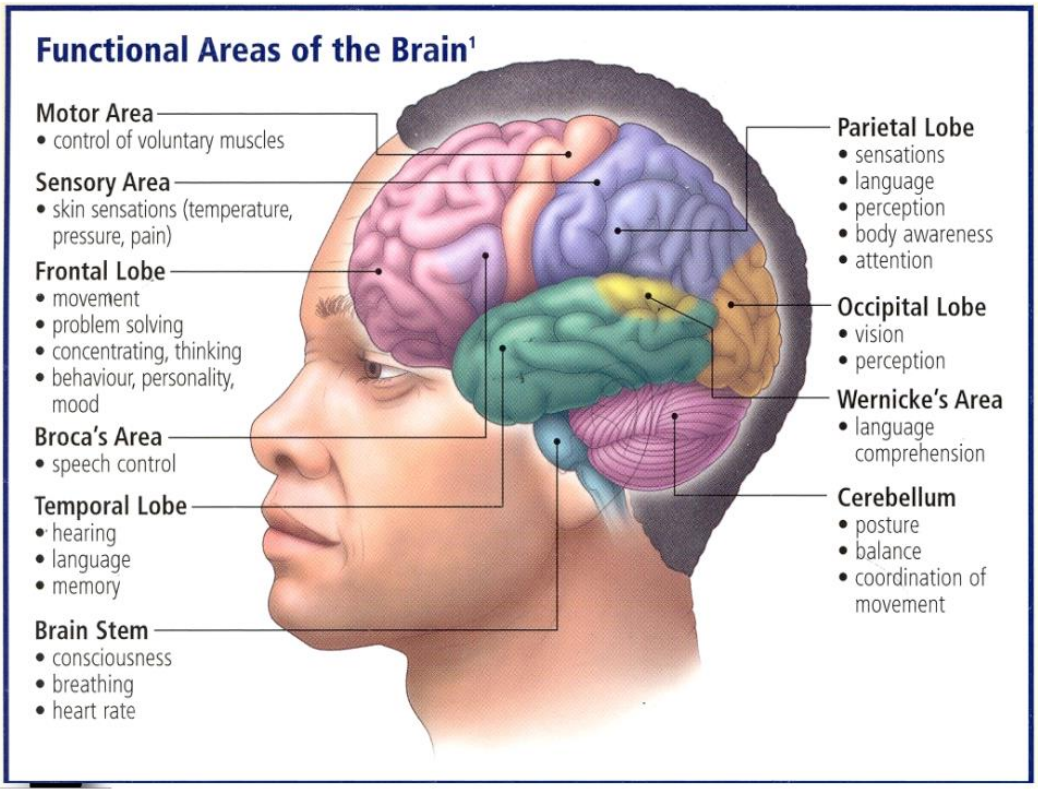
Children have higher respiratory rates than adults. Higher respirator rates lead to proportionately higher minute volumes. As a result, children may be more susceptible to agents absorbed through the pulmonary route than adults with the same exposure. Children may also respond more rapidly to such agents. Because children's airways are much smaller, and they have a shorter trachea and a relatively larger tongue size than adults, they are more susceptible to breathing and airway problems. Respiratory arrest is a common cause of cardiac arrest in children.

THE BRAIN

Your brain needs a constant supply of oxygen. Without oxygen, brain cells begin to die in 4 to 6 minutes.

On average, the adult brain weighs three pounds, and uses 20% of the body's oxygen. In children, the brain reaches 95% of adult size by the age of seven.

The medulla is located in the brain stem and controls automatic bodily functions, including consciousness and respiratory and cardiovascular function.



THE BODY'S CELLS

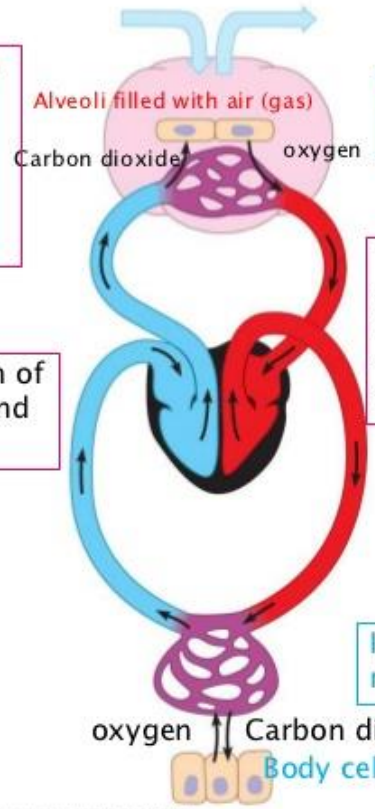
All of the body's tissues are made up of cells.

All cells require oxygen to carry out their normal functions. They also require nutrients. Oxygen deprivation, or hypoxia, will cause cells to die within a few short minutes.

The body's cells use oxygen and produce carbon dioxide as a waste product. Carbon dioxide must be eliminated through the lungs through ventilation.

Red blood cells transport oxygen throughout the body. Children have a smaller circulating blood volume; therefore, any loss of blood can affect them profoundly.

Gas exhange surface must be:
- Moist
- Thin
- Large in relation of size of body



External respiration

Blood – part of circulatory system contain red pigment – hemoglobin, to transport gasses

Process: Diffusion of gasses (oxygen and carbon dioxide

Internal respination

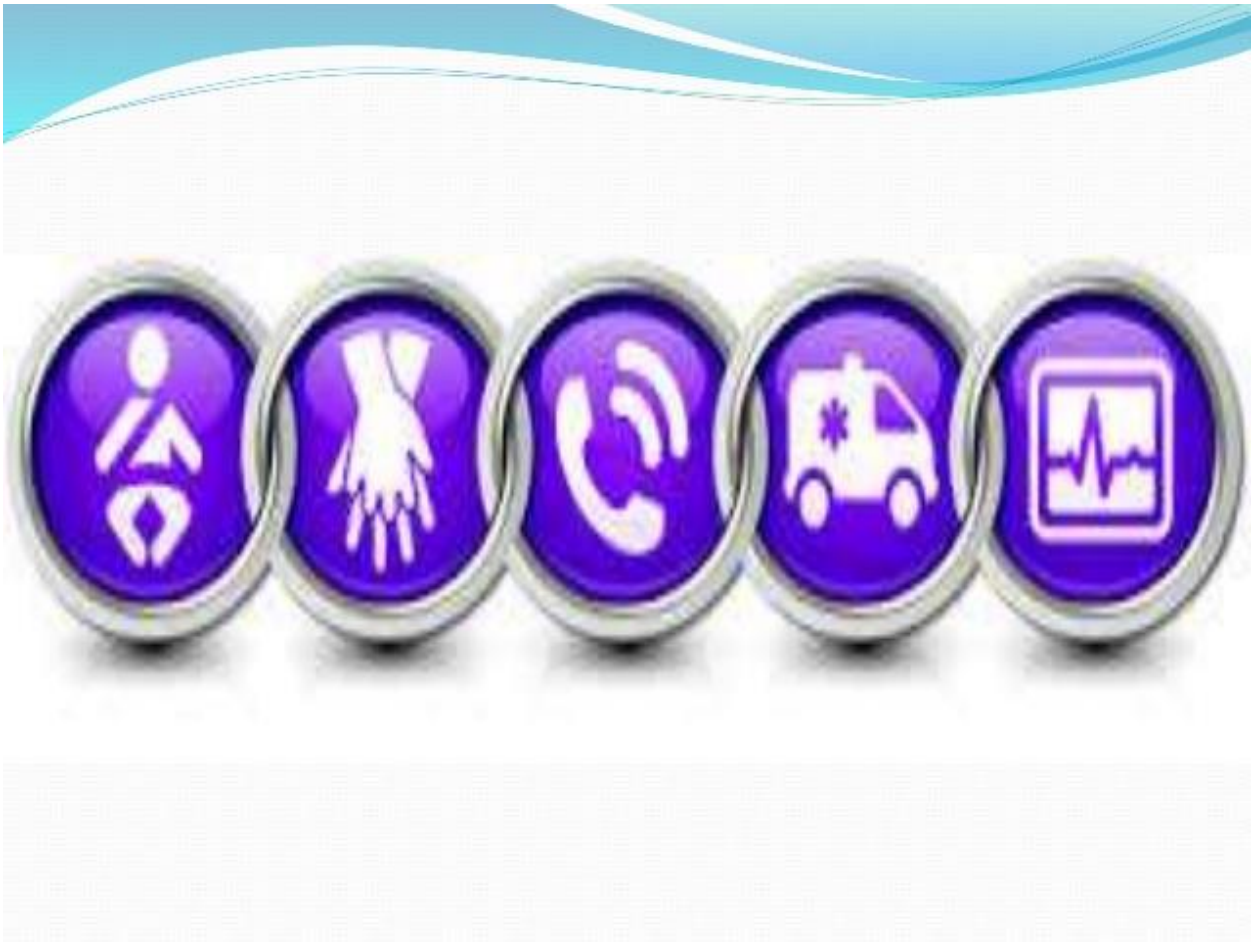
Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

The Pediatric Chain of Survival

The pediatric chain of survival can be thought of as a sequence of events that must occur in order to restore health in a child or infant victim of sudden cardiac arrest. Because children are more prone to respiratory arrest and shock, it is essential to recognize and prevent airway and breathing problems before they occur to prevent cardiac arrest and to ensure survival and full recovery. For this reason, the most important link in the pediatric chain of survival is prevention. Unintentional injuries are the number one cause of death in children. Children's lives can be saved by focusing on prevention of emergencies. Knowing CPR is important, but even more important is ensuring that CPR is never needed in the first place!

Therefore, the Pediatric Chain of Survival includes:

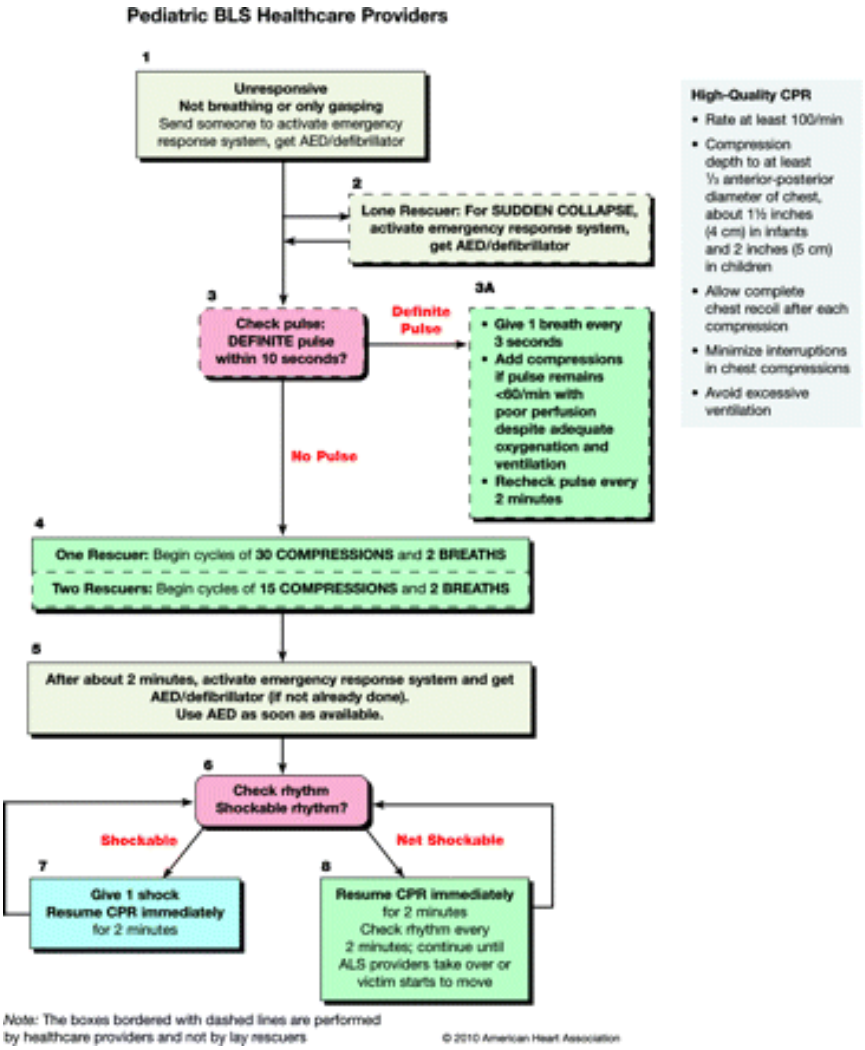
- Prevention of respiratory/cardiac arrest
- Early high-quality CPR
- Activation of the Emergency Response System
- Early advanced life support
- Comprehensive post- cardiac arrest care



THE TEAM APPROACH

In many situations, there may be only one rescuer who will carry out the normal sequence of assessments and actions. However, in many situations, there is often more than one rescuer trained and willing to help.

This is when a team approach can be used. The first rescuer should take the role of team leader and delegate tasks. One rescuer can provide compressions, one can prepare to give breaths with a face mask or mouth-to-mouth, and one can prepare the AED. By working together, the most efficient care can be given to the patient. Of course, if you are the only rescuer present, you will have to do all of these things. For this reason, the steps to performing CPR are presented in a liner sequence.



Chest Compressions

Chest compressions are the most important component of CPR. Chest compressions are an attempt to mimic the normal activity of the heart. When a rescuer presses down on a victim's chest, blood is forced out of the heart and into the arteries. When pressure on the chest is released, blood is allowed to return to the heart. A small amount of oxygen will be present in the bloodstream for several minutes after the heart ceases to beat, just enough to keep the brain alive. Compressions can keep vital organs functioning until higher level care is available.



To perform compressions on a child victim (aged 1 to puberty*):

To perform compressions on a child, place the heel of your non-dominant hand on the victim's chest between the nipples. Then place your other hand on top of the first and interlace your fingers. The heel of your hand should be pressing on the bottom two-thirds of the sternum, avoiding the xiphoid process (the small bony prominence at the very bottom of the sternum (breastbone). You should be as close to the victim's side as possible, with your knees against the victim's side (this will help prevent back injury and fatigue). Lock your elbows and press down hard, depressing the sternum at least 1/3 the depth of the chest, or approximately 2 inches (5 cm). If you find that you are pressing too deeply with two hands, use only one hand. Your shoulders should be positioned directly over your hands in a straight line. Push hard and fast 100 to 120 times per minute, counting out loud as you do so. It's important that you allow the chest to recoil (return to it's normal, relaxed position) in between compressions. If you do not allow the chest to recoil, the heart will not fill completely, which means that less blood (and therefore oxygen) will be pumped out of the heart to vital organs with the next compression.

Performing compressions is exhausting. Most people find that they become very fatigued after providing compressions for just a few minutes. When a person performing compressions becomes fatigued, there is a tendency to compress less firmly and more slowly; for this reason, it is recommended that rescuers trade off doing compressions every 2 minutes to prevent fatigue and optimize the quality of compressions. If you are alone, you will have to do the best you can- keep performing compressions until help arrives or you are physically unable to do so.

*Note that puberty, for the purposes of CPR, is defined as the presence of underarm hair in males or any breast development in a female.

One Rescuer Child CPR

If you are alone and come across a child who is down, follow the steps below. If someone else is immediately available to assist, use the 'Two Rescuer' sequence.

Stay Safe: If you come upon a child who may need CPR, look around and make sure you and the child are in a safe place. If the child is in water or on a road, try to move the child to a safer area. If you are in a safe area, do not try to move the child as he/she may have other injuries that you cannot see. Simply roll him/her over onto his back. Make sure the child is on a firm surface, in case compressions are needed.

Assess the Victim: To quickly assess the victim, shake his shoulder and yell at him. Check for breathing. If he/she is not breathing, or is not breathing normally (i.e., only gasping), shout for help.

Activate the Emergency Response System and Find an AED: Yell for help. If someone responds, tell him/her to call for help by dialing 9-1-1. If you are in an area where an AED may be available, tell him/her to go find the AED. Make sure you tell the person to return to assist you as soon as possible. If you are alone and witnessed the child collapse, call for help by dialing 9-1-1 and run to get the AED if you know where one is nearby. If you do not know where an AED is, begin CPR immediately after dialing 9-1-1.

Begin CPR

1. Check for a pulse on the side of the neck. Feel for a pulse for at least 5 seconds but NO MORE THAN 10 seconds. To check for a carotid pulse, slide 2 or 3 fingers into the groove between the traches (windpipe) and the neck muscles at the side of the neck. Alternately, you can check for a pulse in the femoral artery located in the groin. To do this, place two fingers in the inner thigh, part way between the pubic bone and the hip bone, just below the crease where the leg joins the abdomen. Remember, do not feel for a pulse for more than 10 seconds.
2. If there is no pulse or the pulse is less than 60 /minute with signs of poor perfusion* (or if you are unsure if there is a pulse), begin CPR starting with chest compressions. Provide 30 chest compressions, followed by two breaths. NOTE: If you are not comfortable giving rescue breathing and/or you do not have a mask available, do 'Compression Only' CPR.
    a. Use the heel of one hand on the lower half of the breastbone in the middle of the chest.
    b. Place the other hand on top of the first hand.

c. Straighten your arms and lock your elbows so that your body weight is over your hands.
d. The most important part of CPR is to remember to push HARD and FAST. Each compression should be one-third the depth of the chest or approximately two inches deep and the rate should be 100-120 compressions per minute.
e. Be sure to let up on the pressure on the sternum after each compression (chest recoil) so the chest can re-expand and blood can flow back into the heart. The purpose of CPR is to help the blood flow through the heart and into the rest of the vital organs; if you allow the chest to fully re-expand, more blood will flow into the heart and will be available to deliver to the rest of the body.
f. Count out loud as you do compressions. When you have done 30 compressions, try to open the victim's airway by doing a head tilt/chin lift. Note that if you are doing 'Compressions Only' CPR, you can skip this step.
    i. With your non-dominant hand, push on the victim's forehead to tilt the head back.
    ii. With your dominant hand, place your fingers under the bony part of the lower jaw and gently lift the jaw to bring the chin forward. Be sure you lift up on the bony part of the jaw and not the soft tissue under the jaw so you don't block the victim's airway. Do not use your thumb to lift the jaw. Allow the victim's mouth to remain slightly open.
    iii. If you think the person's neck may be injured, avoid the head tilt/chin lift and use the jaw thrust maneuver if you have been trained to do so. Instructions for performing a jaw thrust are provided later in the unit.
g. If you have a barrier device to use between your mouth and the child's face, use it. Although the risk of infection from performing CPR is very, very low, it is recommended to use a barrier device when providing CPR. This includes the use of face masks. Give each breath slowly – each breath should last one second. Make sure the chest rises with each breath. Repeat, giving a second breath.
h. Start another cycle of chest compressions. Remember, push HARD and FAST. Alternate chest compressions (30) and breaths (2) until help arrives.

*Signs of poor perfusion: this refers to a lack of blood flow that results in certain visible signs, including pale skin color or bluish discoloration of the skin. Fingers, earlobes, lips and nail beds may look bluish or light gray. Sometimes there may be mottling, which is a mixture of a purplish or blotchy red-blue coloring on the extremities (arms or legs).

Airway/ Breathing

Face masks provide a barrier between the rescuer and the victim. Some masks are equipped with a one-way valve that allows the rescuer's breaths to enter the victim's airway, but prevents the victim's expired air from coming in contact with the rescuer's airway. These masks also prevent contact with vomitus and blood, which could pose an infection risk to the rescuer. These masks can be purchased online for minimal cost, and can be kept virtually anywhere- in a purse, car or at home. It takes practice to learn how to use these masks effectively to provide ventilations.

Using a Face Mask:

a) Position yourself at the victim's side. If you are a lone (single) rescuer, positioning yourself at the victim's side will allow you to provide both ventilations and compressions without having to move.
b) Position the mask on the victim's face. Masks are usually triangular in shape, and you will notice that the mask has a "pointy" end- this end goes over the bridge of the victim's nose.
c) Seal the mask against the victim's face. To do this, take the hand that is closest to the top of the victim's head and place it along the edge of the mask. Some people find it easier to form a 'C' with their index finger and thumb and use these digits to grasp the mask around the base of the mouthpiece. With the thumb of your other hand, apply pressure along the bottom edge of the mask. Then place the remaining fingers of your second hand along the bony edge of the jaw and lift the jaw upwards. Open the airway by performing a head-tilt chin-lift procedure. While you lift the jaw, ensure that you are sealing the mask all the way around the outside edge of the mask to obtain a good seal against the victim's face.
d) Deliver air over 1 second, ensuring that the victim's chest rises.
e) If the victim's chest does not rise, reposition the mask and try to get a better seal. Remember, you should be lifting the victim's jaw into the mask, rather than simply pushing the mask down onto the victim's face.
f) Provide 2 ventilations over 1 second each with the mask after every 30 compressions.
g) If the victim has a pulse but is *not breathing*, provide rescue breathing by providing 1 breath every 3 to 5 seconds (12-20 breaths per minute). Check for a pulse every 2 minutes- if at any time the victim loses their pulse, start chest compressions along with ventilations at a rate of 30:2.



Mouth-to-Mouth Breaths

Although it is recommended to use a face mask to deliver breaths, there are times when these devices may not be necessary. Should a cardiac arrest occur at home, you would likely not hesitate to perform mouth-to-mouth breathing for your own child; you might choose to give mouth-to-mouth to a child in your care as well. In cases such as these, you will likely decide the benefit outweighs the risk to your own health, which is very, very small to begin with.

To provide mouth-to-mouth breaths to a child:

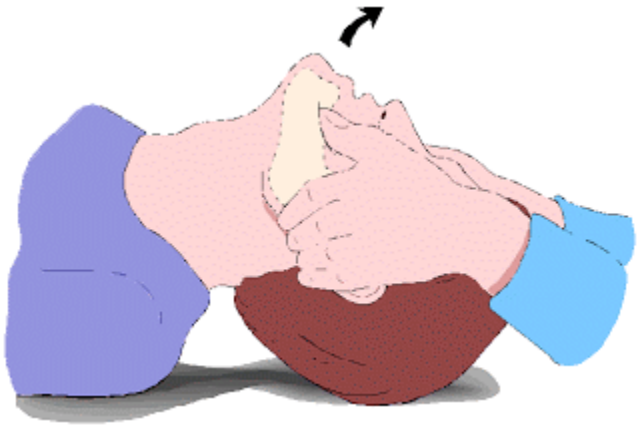
1. Use the head tilt-chin lift to hold the victim's airway open.
2. Using the hand on the victim's forehead that is maintaining the head tilt-chin lift, pinch the victim's nose closed using the thumb and index finger.
3. Inhale a regular breath, then cover the victim's mouth with your own, creating a tight seal.
4. Give one breath over 1 second, watching to see if the chest rises.
5. If the chest doesn't rise as you give the breath, repeat the head tilt-chin lift.
6. Give a second breath over 1 second and watch for chest rise.
7. If the second breath fails to go in, go immediately to chest compressions.

The Jaw Thrust Maneuver

If you suspect that a child may have a neck or spinal cord injury (i.e., the child has fallen, been in a motor vehicle accident or suffered another mechanism of injury that could result in injury to the neck or spinal cord), you should not use the head tilt-chin lift maneuver to open the victim's airway. This could further damage the neck or spinal cord. Instead, you should use the jaw thrust maneuver to open and maintain the victim's airway.

To perform this maneuver:

a) Place your hands on either side of the victim's head. Rest your elbows on the surface that the victim is laying on.
b) Put the fingers of both your hands under the angle of the victim's lower jaw and lift so that the jaw slides forward.
c) Use your thumb to push the lower lip away from you if the victim's lips close.



Using an AED

An AED, or automated external defibrillator, is a device that has the ability to detect and treat, through electrical energy, the lethal arrhythmias known as ventricular fibrillation and ventricular tachycardia.

Ventricular fibrillation is a condition in which the lower chambers of the heart, the ventricles, quiver in an unorganized fashion, which renders them incapable of pumping blood to the rest of the body. Untreated, ventricular fibrillation rapidly causes cardiac arrest. Ventricular tachycardia is a rapid rhythm originating in the ventricles. In ventricular tachycardia, the ventricles contract so quickly, albeit in a somewhat organized fashion, that inadequate blood flow is produced. Ventricular tachycardia often precedes ventricular fibrillation. Both rhythms are lethal if not treated.

An AED sends electrical energy (a 'shock') through the heart, which stuns the heart and allows the normal pacemaker of the heart, usually located in the right atrium, to take over and restore a normal heart rhythm.

Early defibrillation is key to survival in cardiac arrest. For each minute that defibrillation is delayed, the chance of survival is reduced by 10%. (after 10 minutes, few people are successfully resuscitated.)

Early defibrillation can increase survival rates to greater than 50%. Rescuers should immediately begin chest compressions, and use the AED as soon as it is available and ready to use.

AEDs can be found wherever crowds of people gather- swimming pools, airports, malls, sporting arenas, schools, hotels…more and more businesses are also investing in these life-saving machines. In some communities, private AED owners are registering their AEDs with ambulance dispatch, so that they can be easily located by bystanders when needed. Make it a point to learn where the AEDs in your neighborhood or town are located- you never know when you might need one! If you work in a place that has an AED, make sure you are familiar with the AED should you ever need to use it.



AEDs have been designed to be extremely "user friendly". All you need to do as a rescuer is turn on the machine (the most important step) and listen as the machine guides you through the steps to use the AED safely and effectively. Although there are many brands of AEDs on the market, they all work in a similar fashion and are designed to be used by lay rescuers.

When an AED becomes available (i.e., when you or another rescuer have retrieved it), place it at the child's side, closest to the rescuer who will operate it. In this way, the other rescuer can continue performing CPR until the AED is ready to analyze and deliver a shock (if needed).

There are four universal steps to using any AED. These will be highlighted in the following list of steps so they are easily recognizable.

The steps to using an AED are as follows:

1. **Turn on the machine.** This is the most important step- turning on the machine will enable the AED unit to guide you through the next steps. To turn on the AED, open the

top of the carrying case and push the ON button. Note: some models will turn on automatically when you lift the lid of the carrying case.

2. **ATTACH AED pads** to the child's bare chest. Expose the child's chest. Dry it off if wet. Choose adult pads for victims who are 8 years of age or older, and choose pediatric pads for children aged 1 year to puberty. Peel off the adhesive backing. Place one pad on the upper right chest just below the collarbone. Place the other pad on the patient's lower left ribcage, a couple of centimeters beneath the armpit. Some pads are marked- there will be a red heart on the pad that is to be placed on the victim's left side (the heart side). For defibrillators equipped with child-sized pads, there will be a picture on the front of the package to indicate whether the pads are for an adult or child.
   Press pads firmly onto the patient's chest. Then attach the connecting cables to the AED unit. Note: some cables will come preconnected.

3. **Analyze the rhythm**. If the AED unit instructs you to, CLEAR the victim while the machine is analyzing the child's heart rhythm. This means you should ensure that no one is touching the child, including yourself. The rescuer performing chest compressions or giving breaths will need to stop at this point. Note: some AEDs will begin to analyze the victim's rhythm independently; for others, you will need to push the ANALYZE button. Analyzing the victim's rhythm will take up to 10-15 seconds, so don't be alarmed by this.
4. **Push to shock**. If a shock is advised, the machine will clearly state "SHOCK ADVISED, STAND CLEAR". You should ensure that no one is touching the victim, including yourself. You need to *look around to make sure no one is touching the victim's body* while stating "CLEAR" or some similar message that warns others a shock is to be delivered. Once you are certain that no one is touching the victim, push the SHOCK button. You will notice that the victim's muscles contract strongly.

- If a shock is not necessary (the rhythm is not one that can be helped by delivery of a shock), the AED will state NO SHOCK ADVISED and tell you to resume CPR.
- After approximately 5 cycles of compressions and ventilations, or 2 minutes of CPR, the AED will instruct you to repeat steps 3 and 4- analyze the rhythm and push to shock if the rhythm requires a shock and the AED instructs you to do so. If two rescuers are present, the rescuers should switch positions each time the AED is analyzing the rhythm to prevent fatigue related to delivery of compressions.
- Continue CPR alternating with analysis of the rhythm until help arrives (i.e. EMS)

For victims younger than 8 years of age, some AEDs are modified to deliver a lower dose of electrical energy and will come with child pads, as previously mentioned. Sometimes there is a special switch or key that must be used with the child pads. But what should you do if only adult pads are available? If adult pads are the only option, use them.

In terms of pad placement, pads should not touch or overlap. If the pads are too large for a smaller child's chest, place one pad on the front of the child's chest and the other on the child's back. The idea is to "sandwich" the child's heart between the pads. Interestingly enough, research has shown that using adult pads on small children, and even infants, does not lead to permanent heart damage in most cases.

There are a few special circumstances to keep in mind when using an AED:

- **The victim is in water**- if the child is in water, pull the victim to a dry area. You are not in danger of getting a shock if the victim is in water. Water is a great conductor of electricity, so if the victim is in water, the shock will be dispersed across the skin of the victim, and the victim will not receive the full dose of electrical energy required to convert them to a normal rhythm. If the child's chest is wet, quickly dry the chest with a towel or your sleeve; however, the chest does NOT need to be completely dry. If the victim is lying in a small puddle or in snow, you can safely use the AED without moving the victim.
- **The victim has an implanted pacemaker or defibrillator**- obviously, if a child has one of these devices, it will have failed. Some children with severe congenital heart disease may have one of these devices. You will recognize these devices as a small lump under the skin on the chest, usually the upper chest on either side. They are generally about the size of a deck of cards or smaller. You will also be able to see a scar over the area. If the victim has one of these devices, avoid placing the AED pad directly over it; doing so may block delivery of the shock.

Two Rescuer CPR for Children (aged 1 year to puberty)

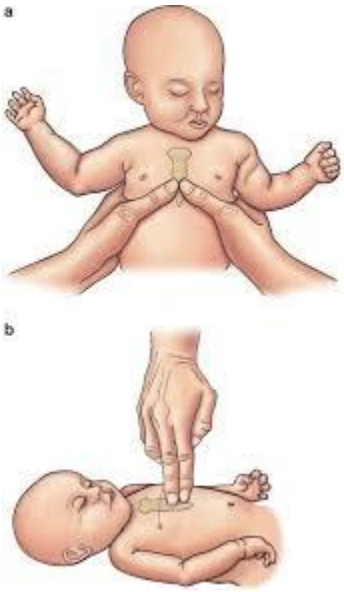
When two rescuers are present, performing CPR on a child is the same as performing CPR on an adult, except that the compression ventilation ratio when 2 rescuers are present drops to 15:2. One or two hands may be used to compress the chest to a depth of 1/3 the diameter of the chest. Remember that compressions should be done when there is no pulse present or when the child's heart rate *is less than 60 beats a minute and there are signs of poor perfusion.* Rescuers should trade off performing compressions every two minutes to avoid fatigue.

CPR for Infants (up to 12 months of age)

CPR for infants is similar to CPR for adults and children. There are a few differences as follows:

- **Checking for responsiveness**: never shake an infant as this may cause brain damage. To check for responsiveness in an infant, tap the soles of the feet while calling to the infant in a loud voice.
- **Pulse check location**- for an infant, it is easiest to check for a pulse using the brachial artery. To locate the brachial artery, place 2 or 3 fingers on the inside of the upper arm between the shoulder and elbow. Press the fingers gently for 5 to 10 seconds to feel for a pulse. Pushing too firmly may occlude the infant's pulse.
- **Depth of compressions**- compress the infant's chest to one-third the depth of the chest, or approximately 1 ½ inches.

- **Compression delivery technique**- 2 finger technique: when one rescuer is present, the chest is compressed using two fingers on the lower half of the sternum, avoiding the xiphoid process (the very end of the sternum where it narrows and comes to a point). To landmark, place 2 fingers in the center of the infant's chest, just below the nipple line. Push down on the infant's chest one-third the depth of the chest, or approximately 1 ½ inches. Allow the chest to fully recoil (return to its neutral position) in between compressions. Compressions should be delivered at a rate of 100-120 compressions/minute.
- **2 thumb-encircling hand technique**: When two rescuers are present, the compression: ventilation ratio drops to 15:2, the same as for children. To perform this technique, position yourself at the infant's feet. Place your thumbs side by side on the center of the infant's chest just below the nipple line. Encircle the infant's chest so that the fingers of both hands support the infant's back. Use your thumbs to deliver compressions at the appropriate depth and rate (100-120 compressions/minute; 1/3 of the depth of the chest or approximately 1 ½ inches). This position allows another rescuer to support airway and breathing without getting in the way of the rescuer performing chest compressions, and is therefore the preferred technique when two rescuers are present. This technique also allows for more consistent chest compressions and superior blood flow and blood pressure compared to the 2-finger technique.



- **Compression: ventilation ratio**- The compression: ventilation ratio for 1 rescuer CPR in infants is 30:2, the same as for two rescuer CPR children.
- **When to call EMS**- it is far more common for infants to experience a respiratory arrest prior to going into cardiac arrest, unless the infant has a congenital heart problem; in

adults the opposite is true- cardiac arrests are more common due to cardiovascular disease. Therefore, the goal is to intervene before the infant goes into cardiac arrest. For this reason, when to call EMS is dependent upon whether you witnessed the infant's arrest. If you did NOT witness the infant's arrest (unwitnessed arrest) and you are alone, you should provide CPR for 2 minutes prior to calling EMS and finding an AED. If you witness the arrest (i.e., the infant suddenly becomes unresponsive), you should call EMS and get an AED before returning to the child to start CPR.

One Rescuer CPR for Infants (up to 12 months of age)

If you are the lone rescuer of an infant:

1. Assess the child for responsiveness by tapping the soles of the infant's feet while calling his/her name loudly. Check for breathing- if there is no breathing or the infant is breathing abnormally or only gasping, call for help.

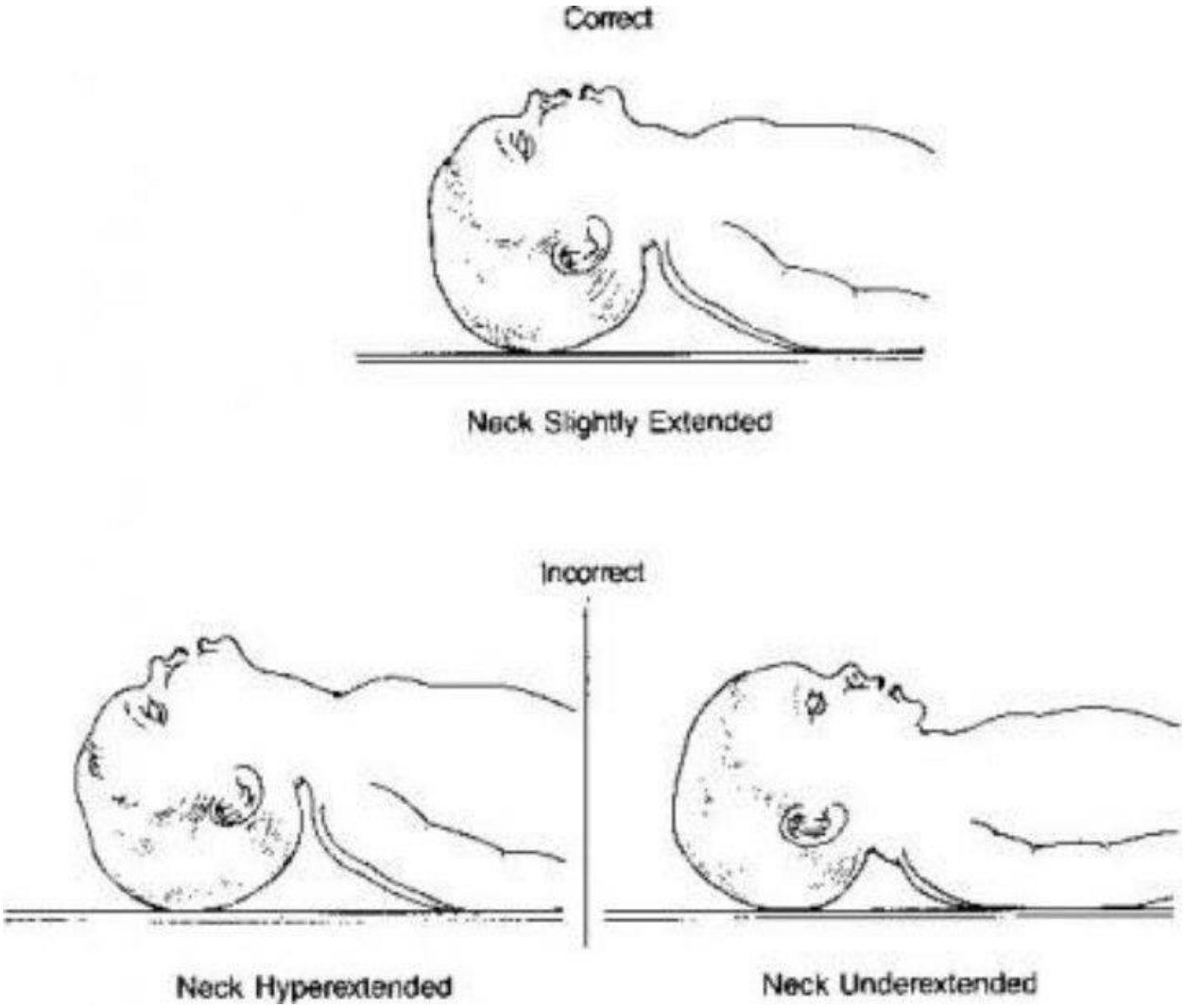
2. If someone responds to your call for help, ask them to call 9-1-1 (activate EMS) and find an AED. If you are alone and witness the infant go into arrest, perform CPR for 2 minutes before calling for help; if you come across and infant who has been "down" for an unknown period of time, call 9-1-1 first before beginning CPR.

3. Check the infant's brachial pulse for at least 5, but no more than 10, seconds. Place 2 or 3 fingers on the inside of the upper arm between the shoulder and elbow. Press the fingers down gently for 5 to 10 seconds to feel for a pulse. Remember: pushing too firmly may occlude the infant's pulse.



4. If you are not sure you can feel the pulse, the pulse is absent or the infant's heart rate is below 60 beats per minute with signs of poor perfusion (pale or bluish discoloration in the face, extremities or nail beds), start CPR, beginning with 30 compressions followed by two breaths.

a. Place the infant on his back on the ground or on a firm surface.
b. Place 2 fingers in the center of his chest just below the nipple line; do not press on the end of the breastbone.
c. Provide 100-120 chest compressions per minute to a depth of 1/3 the depth of the chest or approximately 1 ½ inches.
d. The principles of providing breaths for infants are the same as for children and adults. Use the correct sized face mask for the infant (the mask should cover the mouth and nose without extending past the chin or covering the eyes). Each breath should go in over 1 second and should cause visible chest rise. A breath should require only a small puff of air into the mouthpiece of the device to cause chest rise- avoid excessive ventilations. Lastly, perform a head tilt-chin lift maneuver to open the infant's airway; the infant's head should be placed in "sniffing position" with the infant's head tilted just enough that the nose appears to be sniffing the air. In this position, the external ear canal should be level with the top of the infant's shoulder. Avoid hyperextending the neck- you also want to avoid allowing the chin to fall down towards the neck. You can do this by placing one hand on the infant's forehead while you perform chest compressions. In this way, the infant's airway will remain open and will not close off. Placing a small towel under the infant's shoulders can help to maintain proper positioning.



5. After about two minutes of compressions (five cycles of 30 compressions and 2 breaths), leave the child to call 911 and get an AED if you know where one is (and you or someone has not already done so). Then return to the infant to continue CPR.

6. Use an AED as soon as it is available.

Two Rescuer CPR for Infants (up to 12 months of age)

If there are 2 rescuers available:

1. **Rescuer 1**- Assess the child for responsiveness by tapping the soles of the feet and calling to him/her loudly. Check for breathing- if the infant is not breathing or is breathing abnormally or only gasping, send **Rescuer 2** to activate EMS and bring the AED (if one is available).

2. **Rescuer 1**- Check the infant's brachial pulse for at least 5, but not more than 10, seconds. Place 2 or 3 fingers on the inside of the upper arm between the shoulder and elbow. Press the fingers down gently for 5 to 10 seconds to feel for a pulse. Remember: pushing too firmly may occlude the infant's pulse.

3. **Rescuer 1**- If you are not sure you can feel the pulse, the pulse is absent or the infant's heart rate is below 60 beats per minute with signs of poor perfusion (pale or bluish discoloration in the face, extremities or nail beds), start CPR, beginning with 30 compressions followed by two breaths.

    a. Place the infant on his back on the ground or on a firm surface.
    b. Place 2 fingers in the center of the infant's chest just below the nipple line; do not press on the end of the breastbone.
    c. Provide 100-120 chest compressions per minute to a depth of 1/3 the depth of the chest or approximately 1 ½ inches.
    d. The principles of providing breaths for infants are the same as for children and adults. Use the correct sized face mask for the infant (the mask should cover the mouth and nose without extending past the chin or covering the eyes). Each breath should go in over 1 second and should cause visible chest rise. A breath should require only a small puff of air into the mouthpiece of the device to cause chest rise- avoid excessive ventilations. When the second person returns, change the ratio of compressions to ventilations to 15:2.

4. **Rescuer 2** should place the thumbs of both hands on the lower half of the infant's breastbone, while being careful not to press on the end of the breastbone (xiphoid process). Put the fingers of both hands around the infant's back to provide support. Use the thumbs to depress the sternum approximately 1/3 the depth of the chest, or approximately 1 ½ inches. Compress at a rate of 100-120 compressions per minute.

5. **Rescuer 1** should provide breaths as described above.

6. Continue CPR at a ratio of 15 compressions to 2 breaths and switch roles every 2 minutes to avoid fatigue.

7. Use the AED when available.

Mouth-to-Mouth-and Nose Breathing

To provide breaths to an infant when there is no face mask available:

1. Use a head tilt-chin lift to maintain an open airway (sniffing position), being careful not to hyperextend the neck, which could block the airway.
2. Place your mouth over the victim's mouth AND NOSE to create a tight seal.
3. Blow into the infant's nose and mouth over 1 second, with just enough volume and force to cause the chest to rise with each breath. Be careful not to ventilate too forcefully, as doing so may cause lung damage.
4. If the chest does not rise, repeat the head tilt-chin lift and try to ventilate the victim again. You may need to try to provide breaths at a few different positions before you achieve airway patency (airway is in an open position).
5. If the infant is older and you cannot cover both the infant's mouth and nose, pinch the nostrils closed and place your mouth over the victim's to form a tight seal- just the same as you would do for a child.

You may wonder how mouth-to-mouth or mouth-to-mouth-and-nose breathing can sustain the victim. In actual fact, your expired air contains about 17% oxygen- this is just enough oxygen to meet the victim's needs for a brief period of time.

When providing mouth-to-mouth or mouth-to-mouth-and-nose breathing, it is important not to provide breaths that are too forceful or too rapid. Doing so may cause air to enter the stomach rather than the lungs, which can cause gastric inflation. Gastric inflation may result in vomiting, and an unconscious victim may develop pneumonia if vomitus makes its way to the lungs. To avoid gastric inflation, give each breath slowly over 1 second and deliver just enough air to make the chest rise.

Rescue Breathing

Respiratory arrest is defined as the cessation of breathing. During respiratory arrest, as well as when there is inadequate breathing, the victim will still have some amount of cardiac output, which you will be able to detect as a palpable pulse.

It is important to be able to recognize respiratory arrest, or impending respiratory arrest, which may be seen as slow, irregular or gasping respirations. These abnormal respirations are inadequate to support life. Respiratory arrest inevitably leads to cardiac arrest if not treated, therefore rescuers should intervene

quickly to prevent this deterioration by providing ***rescue breathing***. For children and infants, give one breath every 3-5 seconds (12-20 breaths per minute). Check for a pulse every 2 minutes- if the victim loses their pulse, begin chest compressions combined with breaths.

Choking in the Conscious Child (older than 1 year of age)

1. STAND (OR KNEEL) BEHIND THE VICTIM AND WRAP YOUR ARMS AROUND THE VICTIM'S WAIST.
2. IF STANDING, PUT ONE FOOT IN BETWEEN THE VICTIM'S FEET AND ONE FOOT BEHIND YOU-THIS POSITION PROVIDES STABILITY SHOULD THE VICTIM BECOME UNCONSCIOUS AND YOU NEED TO EASE THE VICTIM TO THE GROUND.
3. PLACE YOUR FIST WITH THE THUMB SIDE IN JUST ABOVE THE VICTIM'S BELLYBUTTON AND BELOW THE STERNUM (BREASTBONE).
4. GRAB YOUR FIST WITH YOUR OTHER HAND.
5. ADMINISTER ABDOMINAL THRUSTS, PULLING INWARD AND UPWARD UNTIL THE FOREIGN OBJECT COMES OUT OR THE VICTIM BECOMES UNCONSCIOUS. EACH THRUST SHOULD BE FORCEFUL, DISTINCT AND SEPARATE.

Choking in the Conscious Infant (less than 12 months of age)

1. SIT OR KNEEL WITH THE INFANT IN YOUR LAP.
2. IF NOT DIFFICULT TO DO, REMOVE THE INFANT'S CLOTHING TO EXPOSE THE INFANT'S CHEST.
3. REST THE INFANT'S BODY ON YOUR FOREARM WITH THE INFANT'S HEAD LOWER THAN THE BODY. SUPPORT THE JAW AND HEAD WITH YOUR HAND. REST YOUR FOREARM ON YOUR THIGH OR LAP TO PROVIDE SUPPORT.
4. GIVE 5 BACK BLOWS FORCEFULLY WITH THE HEEL OF YOUR HAND BETWEEN THE INFANT'S SHOULDER BLADES.
5. SUPPORT THE BACK OF THE INFANT'S HEAD WITH THE PALM OF YOUR HAND AND THE INFANT'S JAW AND HEAD WITH THE OTHER PALM WHILE TURNING HIM OVER (FACE UP) ONTO YOUR OTHER FOREARM. KEEP THE INFANT'S HEAD LOWER THAN ITS BODY. REST YOUR FOREARM ON YOUR THIGH FOR SUPPORT.
6. GIVE 5 CHEST THRUSTS, JUST AS YOU WOULD WHEN PERFORMING CHEST COMPRESSIONS IN CPR. CHEST THRUSTS SHOULD BE DELIVERED AT A RATE OF 1 CHEST THRUST PER SECOND AND SHOULD BE GIVEN WITH ENOUGH FORCE TO DISLODGE A FOREIGN BODY.
7. REPEAT BACK BLOWS/CHEST THRUSTS UNTIL THE OBJECT COMES OUT OR THE VICTIM LOSES CONSCIOUSNESS.



Choking in the Unconscious Child

1. IF YOU ARE CARINIG FOR A CHILD WHO IS CHOKING AND THEY LOSE CONSCIOUSNESS, LOWER THE VICTIM GENTLY TO THE GROUND.
2. ACTIVATE EMS/CALL 9-1-1
3. BEGIN CPR, STARTING WITH CHEST COMPRESSIONS- DO NOT CHECK FOR A PULSE.
4. EACH TIME YOU OPEN THE AIRWAY TO PROVIDE VENTILATIONS, OPEN THE VICTIM'S MOUTH AND CHECK FOR THE OBJECT. IF YOU CAN SEE THE OBJECT, TURN THE VICTIM's HEAD TO THE SIDE AND SWEEP IT OUT OF THE VICTIM'S MOUTH WITH YOUR INDEX FINGER. NEVER PERFORM A BLIND FINGER SWEEP- THIS MAY FORCE THE OBJECT FARTHER DOWN THE VICTIM'S AIRWAY.
5. IF YOU DO NOT SEE THE OBJECT, ATTEMPT TO PROVIDE BREATHS. IF BREATHS WILL NOT GO IN, RESUME CHEST COMPRESSIONS.
6. AFTER APPROXIMATELY 5 CYCLES OF COMPRESSIONS AND VENTILATIONS, ACTIVATE EMS/CALL 9-1-1 IF NOT ALREADY DONE.
7. IF THE OBSTRUCTION IS RELIEVED, CHECK RESPONSIVENESS, BREATHING AND PULSE. PROVIDE RESCUE BREATHING OR CPR AS REQUIRED. IF THE VICTIM IS RESPONSIVE, THEY SHOULD BE TAKEN TO HOSPITAL TO RULE OUT ANY INJURY CAUSED BY ABDOMINAL OR CHEST THRUSTS.

Choking in the Unconscious Infant

1. CALL FOR HELP. IF SOMEONE RESPONDS, SEND THEM TO ACTIVATE EMS/CALL 9-1-1
2. PLACE THE INFANT ON A FLAT, FIRM SURFACE.
3. BEGIN CPR, STARTING WITH COMPRESSIONS. EACH TIME YOU OPEN THE AIRWAY TO DELIVER BREATHS, LOOK TO SEE IF THE OBJECT IS VISBLE. IF THE OBJECT IS VISIBLE, TURN THE INFANT'S HEAD TO THE SIDE AND USE YOUR INDEX FINGER TO SWEEP THE OBJECT OUT OF THE INFANT'S MOUTH. NEVER PERFORM A BLIND FINGER SWEEP, AS THIS MAY FORCE THE OBJECT FARTHER DOWN INTO THE AIRWAY.
4. AFTER 5 CYCLES OF COMPRESSIONS AND VENTILATIONS (OR ATTEMPTED VENTILATIONS), ACTIVATE THE EMERGENCY RESPONSE SYSTEM IF NOT ALREADY DONE.
5. IF THE OBSTRUCTION IS RELIEVED, CHECK RESPONSIVENESS, BREATHING AND PULSE. PROVIDE RESCUE BREATHING OR CPR AS REQUIRED. IF THE VICTIM IS RESPONSIVE, THEY SHOULD BE TAKEN TO HOSPITAL TO RULE OUT ANY INJURY CAUSED BY ABDOMINAL OR CHEST THRUSTS.

CPR CERTIFICATION SKILLS REVIEW

Skill	Child	Infant
Check scene safety	Ensure the safety of the scene before entry	Ensure the safety of the scene before entry
Determine patient responsiveness	Tap on the shoulder and shout, "Are you ok?" Look for gasping, abnormal breathing, or absence of chest movement.	Tap the soles of the feet while calling the infant's name. Look for gasping, abnormal breathing, or absence of chest movement.
Activate EMS – Call for help!	If cardiac event is *witnessed*, activate EMS and get an AED (if you know where one is) before starting CPR. If cardiac event is *not witnessed*, provide 2 minutes of CPR, then leave the victim to activate EMS and get an AED. Return to the victim to resume CPR and use the AED as soon as possible.	Same as for child.
Check pulse	Use carotid or femoral artery. Check for pulse for 5 to no more than 10 seconds.	Check brachial artery in the upper arm for 5 to no more than 10 seconds.
Compressions: Hard and Fast	1 rescuer: 30:2 2 rescuers: 15:2 1 or 2 hands depending on child's size. Depth: 1/3 AP diameter of chest, or about 2"	1 rescuer: 30:2 2 rescuers: 15:2 2 fingers on breastbone below nipple line OR 2 thumbs (2 rescuers) Depth: 1/3 AP diameter of chest, or about 1 ½"
Airway	Open the airway using head tilt-chin lift-use jaw thrust if you suspect spinal cord injury.	Open the airway using head tilt-chin lift. Head should be in "sniffing" position. Use roll under shoulders to maintain proper positioning.
Breathing	2 breaths over 1 second each, following 30 compressions (or 15 compressions for 2 rescuers). Watch for chest rise. Avoid excessive ventilation. .	Same as for child
Rescue breathing: Patient with pulse	1 breath every 3-5 seconds, rechecking circulation every 2 minutes.	Same as child.

and not breathing or gasping*		
AED	For infants and children up to age 8 years, use child pads; if not available, use adult pads, don't let pads contact each other. May use anterior-posterior pad placement.	Use child attenuator pads; if not available, use adult pads, don't pads contact each other. May use anterior-posterior pad placement.

Bloodborne Pathogens

Protect Yourself!



Welcome!

In this course, you will learn:

- The definition of bloodborne pathogens
- Which bloodborne pathogens are of greatest concern
- How bloodborne pathogens can enter the body
- The importance of handwashing and PPE (personal protective equipment) in protecting against infection and reducing the spread of bloodborne pathogens
- Practices that can reduce the transmission of bloodborne pathogens
- How to remove disposable gloves safely and appropriately
- What you should do if you are exposed to a bloodborne pathogen

Once you complete the course, complete the multiple-choice exam to receive your certification card.

Let's get started!

Bloodborne Pathogens- What are they, and which ones should I be concerned about?



Bloodborne pathogens are those organisms that are carried in the blood and are capable of causing disease. Common bloodborne pathogens that cause disease in humans include human immunodeficiency virus (HIV), Hepatitis B (HBV), and Hepatitis C (HCV). These viruses are usually transmitted through exposure to blood, but you may also be exposed through other body fluids.

HIV/AIDS

Human immunodeficiency virus is one of the common bloodborne pathogens. Also called HIV, this is the virus that causes acquired immune deficiency syndrome (AIDS). Since the virus was first identified in the United States in the early 1980's, two types of HIV viruses have been identified. They are called HIV-1 and HIV-2. HIV-1 is the type most people talk about when they use the term HIV. Both types of HIV destroy T cells in the body. These T cells are critical in helping your body fight disease; when these T cells are destroyed, your body becomes 'immunocompromised' or more susceptible to other diseases.

Transmission of HIV

There are a number of different ways that HIV can be transmitted from one person to another:



- Unprotected sex with someone who has HIV will increase the risk of HIV transmission. This is particularly true if there is any contact with blood during the sexual act.
- Intravenous drug users who share needles, syringes, or other equipment used to prepare drugs are at increased risk for HIV transmission.
- Health care workers who care for an HIV patient may be at risk if they are stuck or cut with a contaminated needle, lancet or other sharp object. Although the risk is low, any worker who receives such an injury should follow the policy at their place of employment.
- Infants born to HIV infected mothers are at risk since the virus can be passed from mother to child particularly during the birth process.
- The blood supply and tissues for donation in the United States undergo many tests to ensure that they do not harbor the HIV virus; therefore, the risk of receiving contaminated blood in the U.S. is very slim. However, there is a very remote risk of infection through blood transfusions and tissue donations.
- HIV can be transmitted through medical or dental practices. The CDC and OSHA have very strict standards in place that have been established to minimize this risk. If your

doctor and dentist practice these safety standards, the risk of HIV transmission in the private office is very remote.

- Tattooing or body piercing may present a potential risk of HIV transmission; however, if equipment is properly sterilized, this risk is almost non-existent. In fact, there have been no documented cases of HIV transmission from tattooing or body piercing.

The HIV organism cannot live outside the human body. It cannot be spread by insects, saliva, tears, casual contact, sharing dishes or utensils, or breathing air around an HIV infected person.

Symptoms of HIV Infection

When a person is first infected with HIV, there may be no symptoms. Some people may have vague symptoms that mimic the flu – fever, malaise, and headache. People with HIV may appear to be healthy for months or years after infection. However, it is important that anyone who is at risk should be tested for the virus. If the test is positive, the infected person should see a doctor who specializes in treatment of HIV patients. Many treatments can be used to delay the onset of symptoms and keep an HIV patient feeling healthy for many years. The medications available today can slow the damage to the immune system. If HIV is left untreated, the infected person is at high risk for developing kidney failure, eye disorders, heart disease, and certain forms of cancer.

Despite the fact that the current medications used for HIV are effective in improving the health of the infected person, treatment must continue for the person's lifetime. A doctor who specializes in HIV treatment to prevent or delay the development of AIDS must monitor the HIV positive person.

AIDS is the stage of HIV infection when the individual's immune system can no longer fight disease. Extensive research into HIV and AIDS has led to the development of combinations of medications that allow many people with HIV to live decades before developing AIDS. However, there is currently no cure for HIV and AIDS.

HIV Prevention

HIV is primarily transmitted through unprotected sex or sharing intravenous drug devices. Therefore, HIV prevention should start with steps to eliminate those risky behaviors. First, you should know your HIV status through testing. If you are in a high-risk group, you should be tested on a regular schedule as established by your doctor. Be sure you know the HIV status of your sexual partners.

Limit your number of sexual partners and use latex condoms with all partners. Latex condoms are very effective at preventing infection with HIV and other sexually transmitted diseases. Natural condoms do NOT provide the same level of protection so be sure the condoms are latex.

If you are an IV drug user, get treatment to help you stop drug use. Many health departments have programs that can help you stop IV drug abuse. If you continue to inject drugs, always use clean needles and do not share any of the drug use equipment with anyone else. If you think you have been exposed to HIV, get medical treatment from a clinic or private physician in your area. Early treatment may prevent infection.

Hepatitis B



Hepatitis B is a bloodborne liver disease that results from infection with the Hepatitis B virus (HBV). Although it is usually spread through blood, other body fluids from a person infected with the Hepatitis B virus can also cause the disease. Hepatitis B virus infection is called "acute" when it occurs within 6 months after a person is exposed to the HBV. This short-term infection can lead to a long-term, chronic Hepatitis B infection if the disease is not treated and the virus remains in your body. Although Hepatitis B can lead to death, the good news is that there is a vaccination to prevent the disease.

Transmission of HBV

As a bloodborne pathogen, HBV is found mainly in blood, but may also be found in other body fluids. The mode of transmission for HBV is very similar to the way HIV is transmitted including:



- Unprotected sex with someone who is infected may increase the risk of HBV transmission.
- Intravenous drug users who share needles, syringes, or other equipment used to prepare drugs are at increased risk for HBV transmission.
- Health care workers may be at risk if they are stuck or cut with a contaminated needle, lancet or other sharp object.
- Infants born to infected mothers are at risk since the virus can be passed from mother to child particularly during the birth process.
- HBV can be transmitted through medical or dental practices if the doctor or dentist does not practice the safety standards recommended by the CDC and OSHA.
- Tattooing or body piercing may present a potential risk of HBV transmission; however, if equipment is properly sterilized, this risk is almost non-existent.
- Sharing personal care items such as razors, toothbrushes or other items that might harbor HBV may present a risk for infection.

As with HIV, HBV cannot be spread by insects, saliva, tears, casual contact, sharing dishes or utensils, or breathing air around an HBV infected person.

Symptoms of HBV

Once exposed to and infected by the Hepatitis B virus, symptoms will appear in six weeks to six months. Symptoms may vary by age. For example, most children under the age of 5 years may be asymptomatic with infection; as many as half of the adults over the age of 25 years will have early symptoms. Development of the chronic form of HBV infection seems to be related to the age at initial infection -- the younger the age at infection, the higher the risk for developing the chronic infection. People with chronic HBV infection may have no symptoms or may have very severe symptoms. Symptoms of HBV infection may include:

- Fever
- Fatigue and general malaise
- Loss of appetite
- Nausea and vomiting
- Abdominal pain
- Dark urine
- Jaundice (yellowing of the skin and whites of the eyes)
- Liver failure (1% of cases)
- Liver cancer (typically with chronic disease)

Prevention of HBV

Transmission of HBV can be reduced or prevented by:

- Screening of pregnant women and treatment of infants born to HBV infected mothers.
- Vaccination of all children through the age of 18 years
- Vaccination of adults who may be at increased risk for infection (healthcare workers, people with multiple sexual partners, IV drug users)

Fortunately, vaccination rates are high among infants and children. However, vaccination rates are still low in high-risk adults. Most HBV infections continue to occur in these groups.

Hepatitis C

The most common bloodborne pathogen infection in the U.S. is Hepatitis C virus (HCV). Among all of the bloodborne diseases, HCV is the one that is least likely to be transmitted through sexual contacts; however, it CAN be transmitted sexually.

Transmission of HCV

As with the other Hepatitis infections, HCV is most often transmitted through repeated exposure to infected blood. Because of this mode of transmission, the same groups are at risk for HCV infection.

- Blood transfusions with infected blood used to be the most common method of transmission. Since screening of all blood products is so rigorous, this mode of transmission is almost unheard of in the United States.
- Intravenous drug users who share needles, syringes, or other equipment used to prepare drugs are at increased risk for HCV transmission.



- Health care workers may be at risk if they are stuck or cut with a contaminated needle, lancet or other sharp object.
- Infants born to infected mothers are at risk since the virus can be passed from mother to child particularly during the birth process.
- Unprotected sex and sharing razors and toothbrushes with an infected person CAN transmit HCV, but these modes of transmission are much more uncommon.

Symptoms of HCV

The majority of people infected with HCV usually have only mild clinical symptoms. A large majority of HCV infections results in chronic disease and many of these people will have liver disease. People with HCV infections may be at risk for chronic liver disease for the rest of their lives.

Most people with new HCV infection do not have symptoms. If symptoms do occur, they may be identical to symptoms of HBV infection including:

- Fever
- Fatigue and general malaise
- Loss of appetite
- Nausea and vomiting
- Abdominal pain
- Dark urine
- Jaundice (yellowing of the skin and whites of the eyes)
- Liver failure
- Liver cancer

Transmission of Bloodborne Pathogens

In order for infection to occur, there are four conditions that must be met. These are:

- **A pathogen must be present**. No pathogen, no problem!
- **A sufficient quantity of the pathogen must be present** in order to cause disease/infection. Some pathogens are more virulent or potent than others and require fewer numbers of virus or bacteria to cause infection.
- **A person must be susceptible to the pathogen in question**. Individuals who have been vaccinated against a pathogen can still become infected as vaccinations are not always

100% effective, and effectiveness may decrease over time. There are no vaccinations for some pathogens, such as HIV.

- **The pathogen must have access to the body** via an entry site. An entry site, or *portal of entry*, may include broken skin (abrasions or punctures), mucous membranes such as the nose and mouth, or the respiratory system.

Bloodborne pathogens are primarily spread through *direct contact* (infected blood or body fluid from an infected individual enters another person's body via the correct portal of entry). *Indirect contact* occurs when an individual touches an item or surface that has been contaminated by pathogens from another person's body, and these pathogens enter the person's body via the correct portal of entry. Bloodborne pathogens cannot be spread by casual contact (shaking hands or hugging), water or food, although other pathogens that are not bloodborne can. *Droplet transmission* occurs when a person coughs or sneezes and pathogens are inhaled by another person, or are deposited on items or surfaces touched by another person. Insects and animals can also spread pathogens when they bite or sting us- this is known as *vector-borne transmission*.

Hand Hygiene

Handwashing is one of the most effective ways to prevent the spread of pathogens. Hands should be washed before and after providing care, even when gloves are worn. Although handwashing might seem like a fairly straight-forward procedure, many people do not wash their hands correctly. Hands should be washed in the following fashion:

- Wet your hands with warm running water.
- Apply soap and lather your hands by rubbing them together briskly. Don't forget under your nails, between your fingers and the backs of your hands.
- Scrub vigorously for at least 20 seconds. To time yourself, sing the "Happy Birthday" song twice.
- Rinse your hands thoroughly under warm running water.
- Dry your hands with a clean paper towel (or air dry them).

When there is no ready access to water, alcohol hand rubs may be used. Alcohol hand rubs should contain at least 60% alcohol. You should know that, although hand sanitizers containing alcohol can greatly reduce the number of pathogens on your skin, they cannot remove all pathogens.

Personal Protective Equipment (PPE)

Personal protective equipment is used to minimize exposure to bloodborne pathogens and may include gloves, gowns, masks or goggles and others. Let's look at the different types of PPE used to protect against the transmission of bloodborne pathogens:

- Goggles- protect the eyes; should fit snugly around and over eyes; eyeglasses are not a substitute for goggles
- Gloves- protect the hands
- Face shields- protect the eyes, mouth, nose and face; should cover the forehead and should wrap around the sides of the face; should also extend below the chin
- Gowns/aprons- protect clothing and skin; may be reusable or disposable; should be resistant to fluid penetration
- Masks- protect the mouth and nose; should fully cover nose and mouth; should be resistant to fluid penetration
- Respirators- protect the respiratory tract from airborne agents, such as influenza (the flu); may be half or full face; particulate respirators protect against tiny particles; full respirators have an air source; some respirators must be fit tested

Not all of these must be worn at the same time. Selection of PPE is based on the type of possible exposure. For example, when changing a patient's dressing you might wear only gloves. When dealing with a victim of a motor vehicle collision with multiple trauma, you might be required to wear gloves, a gown and a mask.

A word about gloves

It is easy to become complacent when wearing gloves, so it is important to remember that gloves are not 100% effective. You should also be mindful of protecting yourself and others. When wearing gloves, try not to touch your face (for example, to readjust goggles or a mask. Do not touch others or environmental surfaces excepting those that must be touched to provide care. Change gloves when they are heavily soiled when working with a patient, and NEVER use the same gloves to care for two different patients. Lastly, be sure to dispose of your gloves in the appropriate receptacle. Disposable gloves are meant to be discarded after a single use.

Donning and removing PPE

- PPE should be donned before coming in contact with the patient
- PPE should be removed carefully at the doorway or just outside the patient's room- respirators should be removed once you are outside the room
- Hands should be washed immediately after removing PPE

- When donning PPE, don in the following order: gown, mask, goggle/face shield, gloves (use common sense!)
- Gloves are always donned last and should extend over the cuffs of an isolation gown
- When removing PPE, remove in the following order: gloves, goggles/face shield, gown, mask/respirator

Removing gloves:

- Grasp the outside edge of one glove near the wrist
- Pull the glove down the hand and turn the glove inside out
- Hold the glove in the opposite gloved hand
- Slide an ungloved finger under the wrist edge of the remaining glove
- Pull off from the inside, creating a bag for both gloves
- Discard both gloves in the appropriate receptacle

Work Practice and Engineering Controls

Work practice controls are measures used to change the way work practices are carried out to ensure safety, whereas engineering controls are those measures used to isolate or remove hazards from the work environment.

Examples of work practice controls are:

- Removing and disposing of PPE as soon as possible after use
- Isolating areas of contamination to prevent others from being exposed
- Disposing of sharps in puncture-proof biohazard containers
- Avoiding unnecessary spraying, splashing or spattering of blood or other potentially hazardous materials
- Avoiding eating, drinking or applying cosmetics in areas where contamination might occur
- Disinfecting/cleaning soiled work areas or surfaces
- Washing hands properly and frequently

Examples of engineering controls are:

- Safe needles (self-sheathing needles)
- Biohazard containers
- Labeling of all biohazards
- Safer medical devices
- PPE

Cleanup of Spills and Equipment

It is everyone's responsibility to maintain a clean and safe environment. When equipment is soiled or there has been a spill of a potentially hazardous substance, it is important to clean and remove the biohazardous substance immediately. Know where spill kits, disinfectants and other cleanup agents are located in your workplace. When attending to spill or cleaning equipment, keep the following in mind:

- Always wear gloves and other PPE as necessary depending on the risk
- Protect others by blocking off the area with cones or ropes to prevent others from inadvertently walking through the contaminated area
- Clean up spills immediately, or as soon as is feasible under the circumstances, using the appropriate cleansers/disinfectants
- Do not pick up needles or broken glass with your gloved hands- use a broom and dustpan or other items to prevent punctures or cuts
- When using absorbent material to clean up a biohazardous material, dispose of the absorbent material in a biohazard container
- Flood the contaminated area with disinfectant (i.e. bleach and water) and allow it to stand for at least 10 minutes to kill any pathogens
- Wash uniforms between wearings and wash according to the manufacturer's instructions

What to Do if You Are Exposed

No matter how careful we are, accidents can occur. It is a serious matter to be exposed to a bloodborne pathogen, and it's important that you take the proper steps if exposure occurs.

Following an exposure, you should:

- Wash the contaminated area thoroughly with soap and water; if your mouth and nose have been exposed, flush the areas with water
- For eye exposure, irrigate the eyes with clean water or normal saline for 20 minutes- many workplaces have eye wash stations for this purpose
- Report the incident immediately to your supervisor
- See a medical professional (EMS or a physician) according to your workplace protocol
- Make a written report of the incident, including the date, time of event and the circumstances surrounding the event. Make sure to include what you did immediately after the event.

BLS for Healthcare Providers

FOLLOWS 2010 ECC/ILCOR GUIDELINES
Updated with 2015 Guidelines



Table of Contents

PART ONE
Welcome
The Five Fears that Impede the Performance of CPR
Basic Anatomy and Physiology
- The Heart
- The Lungs
- The Brain
- The Body's Cells

Cardiovascular Disease
Heart Attack
Stroke
PART TWO
Guideline Changes
The Adult Chain of Survival
The Team Approach
Chest Compressions for Adults
One Rescuer Adult CPR
Airway/ Breathing
- Using a Face Mask
- Using a Bag-Valve Device
- The Jaw Thrust Maneuver

Two Rescuer Adult CPR (without an AED)
Using an AED
- Special Circumstances

Two Rescuer Adult CPR with an AED
PART THREE
CPR for Children (age 1 year to puberty)
- The Pediatric Chain of Survival

One Rescuer CPR for Children (aged 1 year to puberty)
Two Rescuer CPR for Children (aged 1 year to puberty)
PART FOUR
CPR for Infants (up to 12 months of age)
One Rescuer CPR for Infants (up to 12 months of age)
Two Rescuer CPR for Infants (up to 12 months of age)
AED Use in Child/Infant CPR
PART FIVE
CPR with an Advanced Airway
Mouth-to-Mouth Breaths
Rescue Breathing
PART SIX
Treatment of Choking
Treatment of Choking in Victims 1 Year of Age and Older
- Choking in the Conscious Adult or Child (victims 1 year of age and older)

- Choking in the Conscious Infant
- Choking in the Unconscious Adult or Child (victims 1 year of age and older)
- Choking in the Unconscious Infant

Moving Victims

- The Recovery Position

PART SEVEN

CPR Certification Skills Review

CPR Exam Questions

Welcome!

We're excited that you've decided to take CPR, and that you've chosen us to direct your learning experience.

Despite technological advances and new scientific discoveries, heart disease remains the number one cause of death globally, with 17.3 million deaths per year; 38.7% of cases of out of hospital cardiac arrests (OHCA) are witnessed by a bystander (AHA, 2015). Learning CPR can save lives- you can save a life.

In this course, you will learn:

- The Chain of Survival and your place within the chain
- How to provide high quality chest compressions for adults, children and infants
- How to initiate early use of an AED (automated external defibrillator)
- How to provide rescue breathing for victims of respiratory arrest
- How to provide CPR with 2 rescuers
- How to assist victims of choking

Let's get started!

PART ONE

In This Section:

The Five Fears that Impede the Performance of CPR
Basic Anatomy and Physiology

- The Heart
- The Lungs
- The Brain
- The Body's Cells

Cardiovascular Disease
Heart Attack
Stroke

The Five Fears

Many bystanders are reluctant to perform CPR. There are many reasons why bystanders are reluctant to get involved. Let's take a look at some of these reasons so that you can understand why they have no real basis of support.

Fear of Disease: Use universal precautions when the possibility exists of coming into contact with bodily fluids. Use gloves, mask, and/or gown when you have them available. The benefit of initiating lifesaving resuscitation in a patient in cardiopulmonary arrest greatly outweighs the risk for secondary infection in the rescuer or the patient. Nevertheless, use of simple infection-control measures during CPR and CPR training can reduce a very low level of risk even further (Mejicano & Maki, 1998).



Fear of Lawsuits: These laws state that a person acting in good faith who is rendering reasonable first aid will not be held accountable for damages to the person to whom the aid is rendered, unless gross and willful misconduct are involved. Good Samaritan laws may differ for professional health care providers from state to state. Implied consent means that there is an assumption that if an unconscious person were able to request care, they would do so.



Fear of Uncertainty: People sometimes fear that they won't remember what to do. Remembering the correct number of compressions or the number of compressions to ventilations is not as important as the willingness to respond and to push hard and fast. Hands-only CPR is designed to provide simple life support.



Fear of Harming the Victim: A victim in need or clinically dead can only be helped by your intervention efforts. You can't hurt someone who is dead, and any injuries you may unknowingly cause (such as injured ribs) can be dealt with in a



surviving victim of cardiac arrest.

Fear of Injury to Self: As you will learn, scene safety is of utmost importance, and checking the scene for safety should be your first action when you come across an unconscious victim. If the scene is not safe for you to enter, you must not enter.

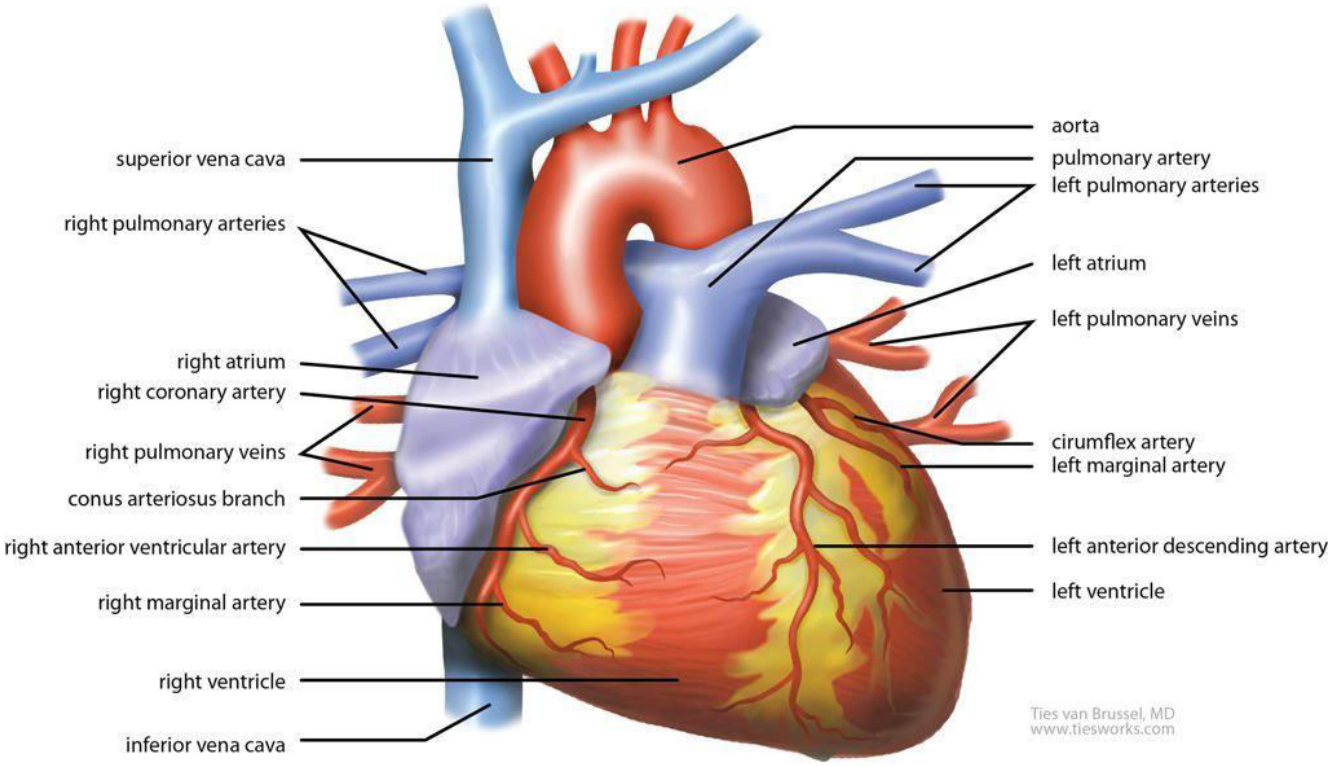
This will only result in more victims. Under no circumstances should you risk becoming a victim. Instead, you should call for help (activate EMS).



BASIC ANATOMY AND PHYSIOLOGY

Before you learn how to perform CPR, it's important to understand how the heart, lungs, brain and cells perform. Here is a brief review of these systems.

THE HEART



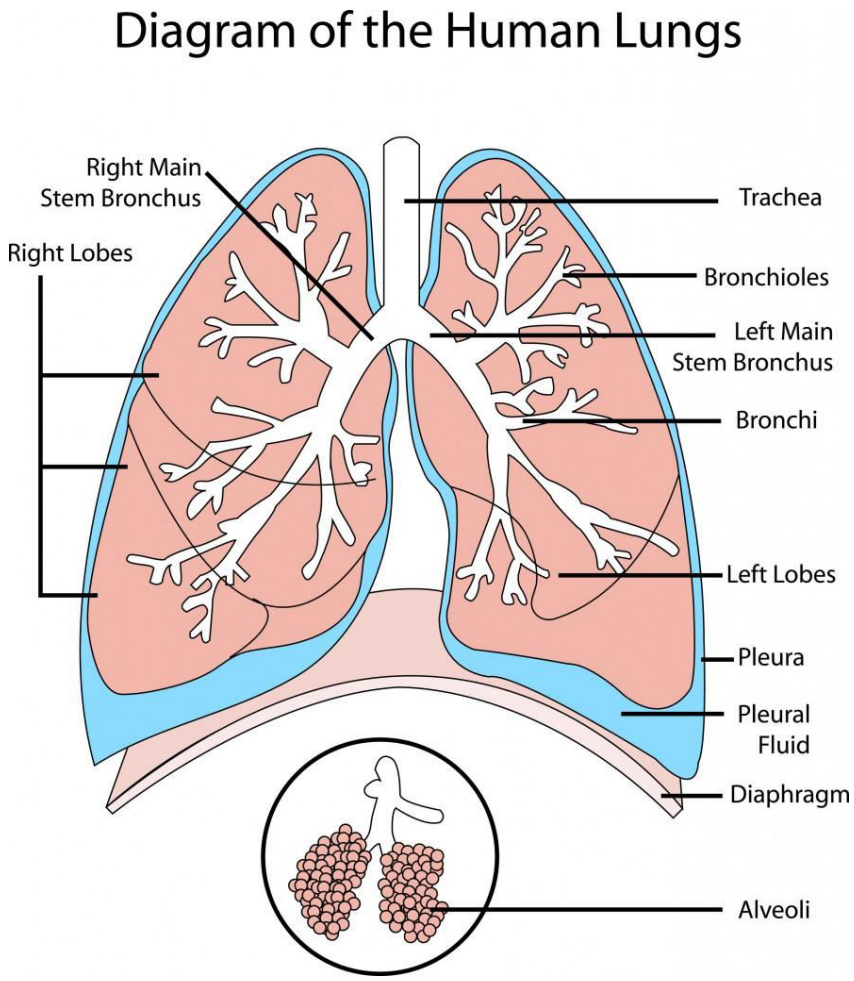
The heart consists of four chambers, the right and left atria (singular: atrium) and the right and left ventricles. The atria are located above the ventricles, as can be seen in the above diagram. The heart is a muscular organ supplied by the coronary arteries. It is located below your breastbone (sternum) and, in an adult, is approximately the size of your fist.

The heart pumps blood that has been deoxygenated by supplying the body's tissues into the lungs, and when that blood has been oxygenated again in the lungs, it exits the lungs to the left side of the heart, where it is pumped out into the tissues once again, to provide oxygen.

Your body has about 5 liters of blood, which circulate through this system approximately 3 times per minute.

What is known as the cardiovascular system is composed of the heart, arteries, capillaries, and veins.

THE LUNGS



Your lungs are spongy, air-filled sacs, with one lung located on either side of the chest. The trachea, which is sometimes called the windpipe, conducts air down into the lungs through the bronchi, which are smaller tubular branches. The bronchi then divide into smaller and smaller tubules called bronchioles. Air is exchanged in the alveoli, which are tiny sacs that allow oxygen and carbon dioxide to move between the lungs and the bloodstream via tiny capillaries.

Your lungs take in oxygen to supply your body's organs and tissues. They release carbon dioxide, a waste product, into the atmosphere when you exhale.

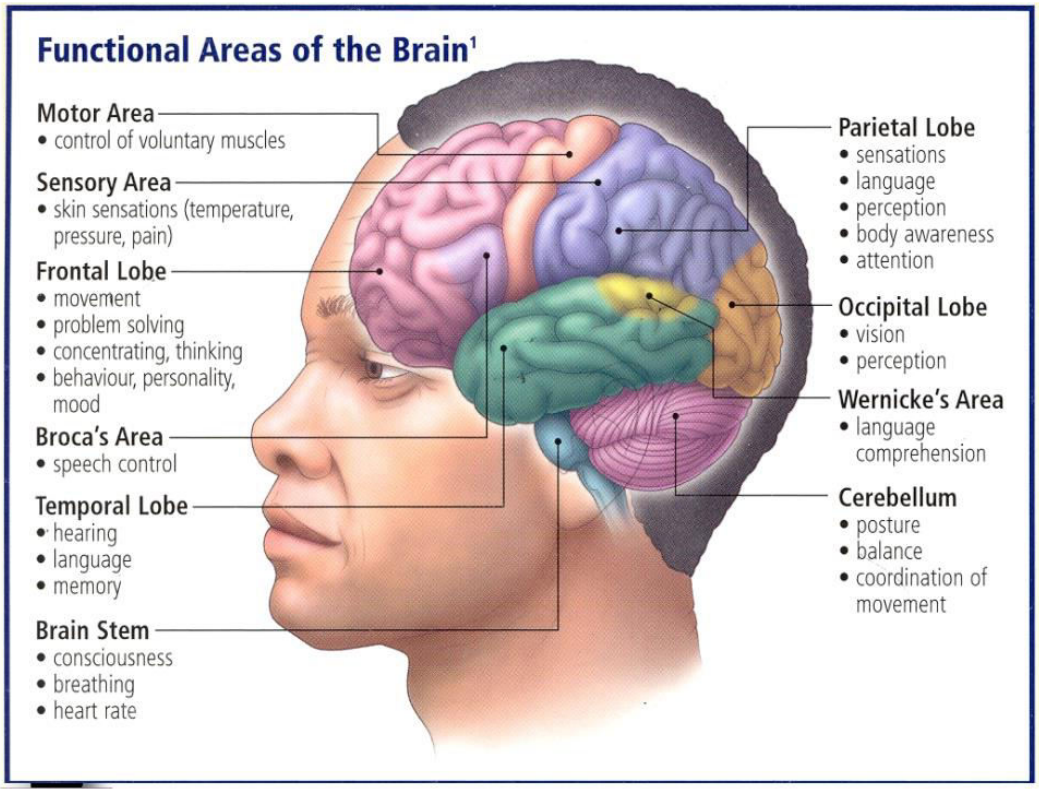
Room air at regular atmospheric pressure contains 21% oxygen. Our bodies utilize approximately 4-6% of that oxygen and release about 16% back into the atmosphere, along with carbon dioxide, when we exhale.

THE BRAIN

Your brain needs a constant supply of oxygen. Without oxygen, brain cells begin to die in 4 to 6 minutes.

On average, the brain weighs three pounds, and uses 20% of the body's oxygen.

The medulla is located in the brain stem and controls automatic bodily functions, including consciousness and respiratory and cardiovascular function.



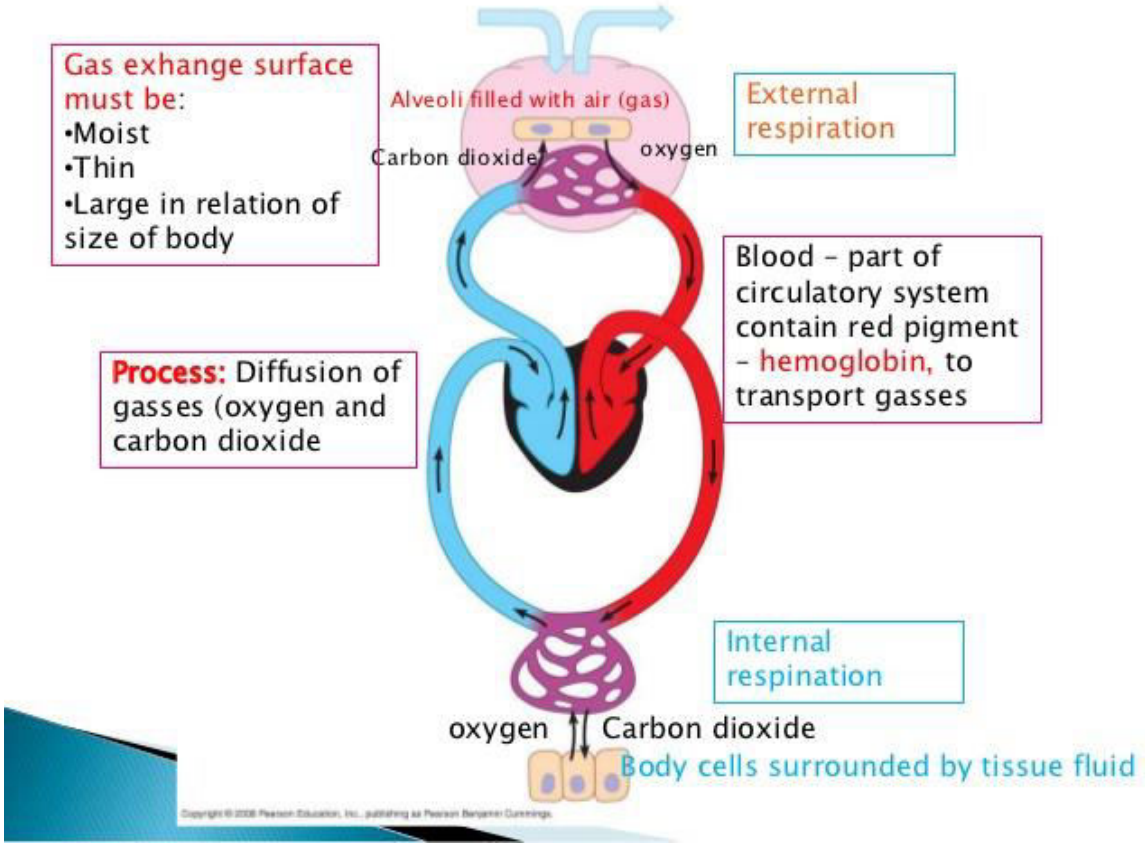
THE BODY'S CELLS

All of the body's tissues are made up of cells.

All cells require oxygen to carry out their normal functions. They also require nutrients. Oxygen deprivation, or hypoxia, will cause cells to die within a few short minutes.

The body's cells use oxygen and produce carbon dioxide as a waste product. Carbon dioxide must be eliminated through the lungs through ventilation.

Red blood cells transport oxygen throughout the body.



CARDIOVASCULAR DISEASE

As mentioned previously, cardiovascular disease is the number one cause of death worldwide. While it is important to learn what to do when cardiovascular disease leads to a heart attack or stroke, it is equally as important, if not more so, to understand how to prevent cardiovascular disease from occurring in the first place.

There are numerous factors that can increase an individual's risk of heart attack. Some of these factors can be controlled (modifiable risk factors), while others cannot (non-modifiable risk factors.

Modifiable risk factors, or factors that can be controlled to a certain degree by an individual, include:

- cigarette smoking
- hypertension
- obesity
- sedentary lifestyle (lack of exercise)
- high cholesterol
- high blood sugar (in diabetes)
- poor diet (diet high in sugar, fat)
- stress

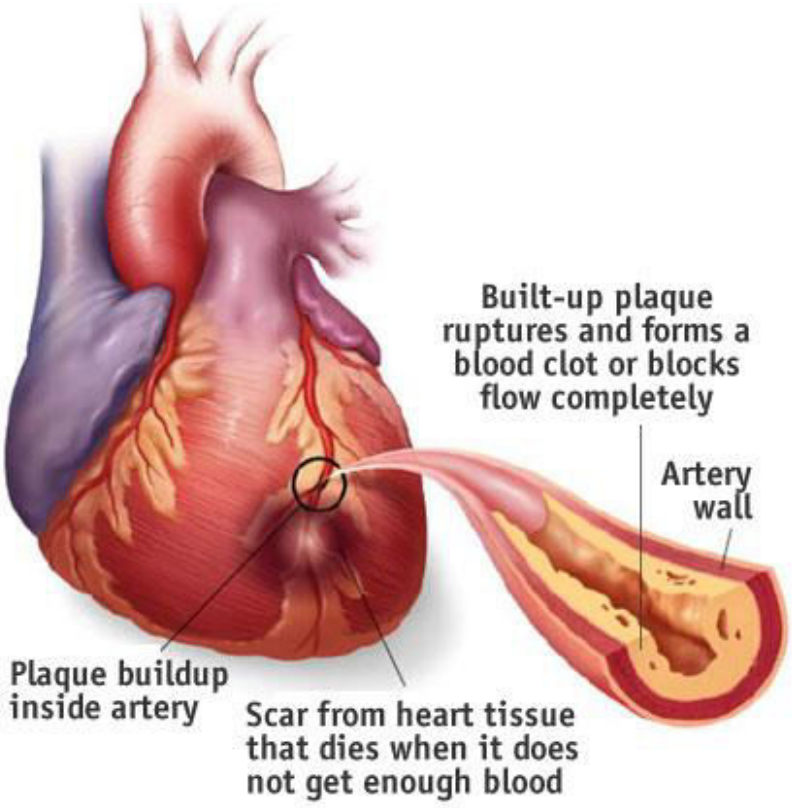
Non-modifiable risk factors, or risk factors that individuals cannot change, include:

- age
- sex
- genetics/hereditary factors (i.e. hereditary high cholesterol or hyperlipidemia)
- race (i.e., certain groups may be at higher risk for hypertension, or high blood pressure)

CARDIOVASCULAR DISEASE IS THE NUMBER ONE CAUSE OF DEATH IN THE UNITED STATES, ACCORDING TO THE CENTER FOR DISEASE CONTROL (CDC). THE CDC ESTIMATES THAT MORE THAN 650,000 AMERICANS DIE FROM CARDIOVASCULAR DISEASE (1 in 4 DEATHS) EACH YEAR IN THE UNITED STATES.

Cardiovascular disease damages the heart and blood vessels, and frequently causes heart attack and/or stroke. Americans of all ages should focus on prevention of cardiovascular disease by eating healthy foods, exercising 30 minutes (at least) each day, controlling weight, minimizing stress, consuming healthy fats and oils, and avoiding cigarette or cigar smoking.

Atherosclerosis, or hardening of the arteries, is a condition caused by the building up of plaque inside the body's arteries, the large blood vessels that carry blood away from the heart to the body's organs. Plaque is composed of fatty substances, cholesterol, fibrin (a clotting substance in the blood), calcium and cellular waste products. When plaque builds up, it can partially or totally block the flow of blood through an artery in the brain (which causes stroke), the heart (which causes a heart attack), the kidneys, the arms, the legs and other vital areas. Plaque may break off to block an artery, or a blood clot (thrombus) may form on the surface of the plaque- either of these two circumstances can lead to a heart attack or stroke.



HEART ATTACK

A heart attack often presents with one or more of the following symptoms:

- Chest tightness, pressure, or discomfort
- Nausea
- Sweating
- Shortness of breath
- Fatigue
- Weakness
- Pain in the jaw or arm
- Pallor (pale color of the skin)

Women and people with diabetes often present with atypical symptoms, such as nausea and vomiting or back pain. About one-third of patients report no chest pain at all.

WHAT CAN YOU DO?



- **Recognize** the symptoms and signs of a heart attack.
- **Call 911.**
- Don't let the patient drive him or herself to the hospital
- Keep the patient calm and seated in a position that is comfortable.
- **Offer 1 adult aspirin** or **2 baby aspirin**- aspirin should be chewed before swallowing to speed absorption (Note: do not give if the patient has already taken aspirin or if the patient has an allergy to aspirin)
- Be prepared to start CPR if it becomes necessary

STROKE

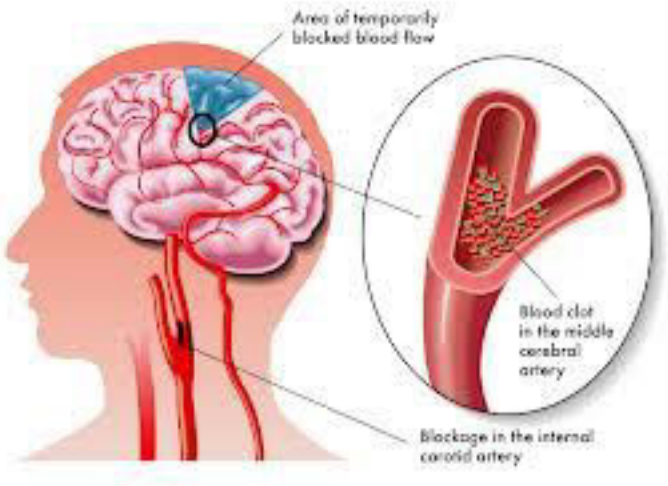
You should suspect stroke if a patient or victim presents with:

- Severe headache with no known cause
- Numbness or weakness of the face, arm or leg on one side of the body
- Drooping eyelid or mouth on one side
- Confusion or trouble understanding
- Difficulty speaking (slurred speech or difficulty finding words)
- Loss of balance or coordination
- Dizziness
- Trouble with vision (i.e. blurred or double vision, loss of vision in one eye)

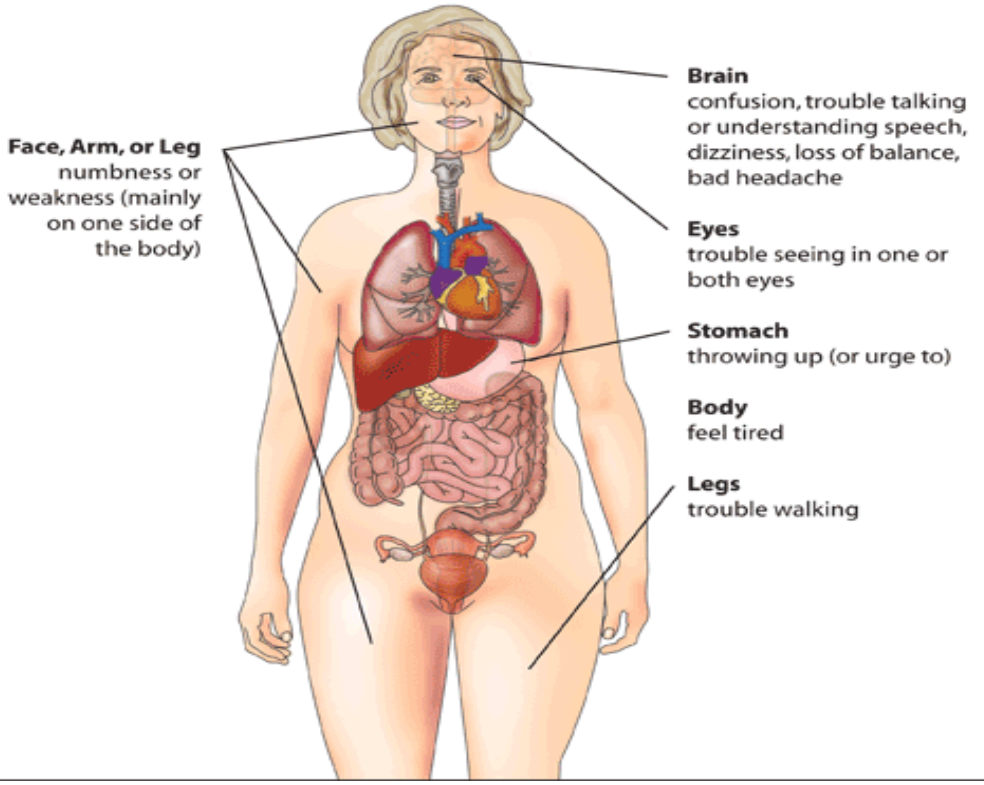
WHAT CAN YOU DO?

- **Know the signs and symptoms** of a stroke
- **Call 911**
- Don't give the patient food or drink
- Keep the patient calm and quiet
- Monitor the patient
- Be prepared to begin CPR if it becomes necessary

A stroke is caused by one of two mechanisms: it can be caused by a ruptured blood vessel in the brain (called a hemorrhagic stroke) or by a blockage in one of the arteries that causes loss of blood flow and oxygen to a part of the brain (called an ischemic stroke). Ischemic strokes are more common. Remember that "Time is brain," and act quickly. Stroke is the 3rd leading cause of death in the United States.



Signs of a Stroke



PART TWO

In This Section:

Guideline Changes
The Adult Chain of Survival
The Team Approach
Chest Compressions for Adults
One Rescuer Adult CPR
Airway/ Breathing
- Using a Face Mask
- Using a Bag-Valve Device
- The Jaw Thrust Maneuver

Two Rescuer Adult CPR (without an AED)
Using an AED
- Special Circumstances

Two Rescuer Adult CPR with an AED

Guideline Changes

For anyone who has taken CPR prior to 2010, there have been a few key changes since the 2010 Guidelines were instituted. These changes include:

- Changing from the ABCs (Airway-Breathing-Circulation) to a C-A-B sequence of steps. This was one of the major changes that took place in 2010, when research showed that decreasing the delay in beginning chest compressions lead to a higher incidence of ROSC (return of spontaneous circulation). Rescuers are now trained to provide interventions in the following order: 1) Chest Compressions, 2) Airway, and 3) Breathing. Decreasing the delay in beginning compressions increases survival rates.
- The removal of "Look, listen and feel" for breathing. It is often hard for even trained providers to identify effective breathing. It is no longer recommended to look, listen and feel to determine whether a victim is breathing. Rather, if the victim is not responsive, has no pulse, and is not breathing or is breathing in an irregular fashion (i.e. only gasping), begin CPR immediately!
- Emphasis on high-quality CPR. High-quality CPR is defined as:
    - Compression rate of 100- 120 beats per minute on all victims;
    - Compression depth of AT LEAST 2 inches in adults (no greater than 2.4 inches), and at least 1/3 the anterior-posterior diameter of the chest in children and infants (about 1.5 inches in infants and 2 inches in children);
    - Allowing the chest to completely expand (recoil) after each compression (do not lean on the chest between compressions);
    - Not interrupting CPR except to use an AED (keep interruptions in chest compressions to less than 10 seconds);
    - Avoiding excessive ventilations;
    - Above all else, PUSH HARD AND FAST!
- Since many responders (even professionals) are unable to feel or palpate a pulse quickly, the recommendation is to feel for a pulse for NO MORE THAN 10 seconds. If you can't feel a pulse or if you're not sure you can feel a pulse, begin CPR.
- A manual defibrillator is preferred to an Automatic External Defibrillator (AED) for infants. If trained first responders are not present and you don't have a manual defibrillator, the next best option is an AED with a pediatric dose attenuator. If an AED with a dose attenuator is not available, you CAN use an adult AED on an infant.

- It has been recognized that health care professionals should call for assistance when they come upon an unconscious victim, but they may also simultaneously assess breathing and check for a pulse before fully activating the emergency response system. This may encourage efficiency in assessment and response, rather than following a step-by-step response.
- Health care professionals may tailor their response to an unconscious victim by altering the response sequence to fit the situation or scenario (using an AED immediately when one is close by and the arrest is witnessed, or providing ventilations first when the cause of arrest is known to be an anoxic event (i.e. drowning).
- Rescuers should provide 1 breath every 6 seconds when an advanced airway is in place and compressions are ongoing continuously (rather than trying to remember a range of ventilations for adults, children and infants).

The Adult Chain of Survival



The Adult Chain of Survival represents a continuum of care, from early recognition of the victim in cardiac arrest to post-resuscitation care to provide the best chance of survival from cardiac arrest. In the diagram above, the first two of five steps are visualized The five links in the Adult Chain of Survival include:

- Early recognition of cardiac arrest
- Activation of the Emergency Response System
- Early CPR, with an emphasis on high-quality chest compressions
- Early defibrillation
- Comprehensive post- cardiac arrest care

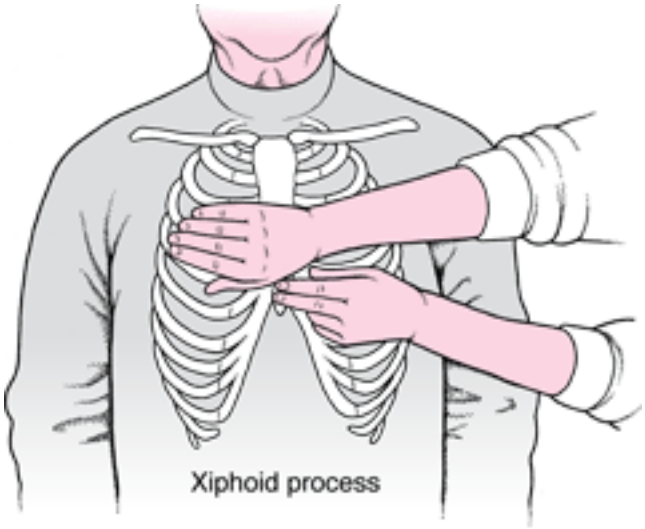
Note that in the diagram above, the first two of the five steps (early recognition of cardiac arrest and activation of the emergency response system) have been combined to acknowledge the fact that these steps often occur simultaneously when multiple rescuers are present.

THE TEAM APPROACH

In many situations, there may be only one rescuer who will carry out the normal sequence of assessments and actions. However, in many situations, there is often more than one rescuer trained and willing to help.

This is when a team approach can be used. The first rescuer should take the role of team leader and delegate tasks. One rescuer can provide compressions, one can prepare to give breaths with a bag-valve mask, and one can prepare the AED. By working together, the most efficient care can be given to the patient.

Chest Compressions for Adults





Chest compressions are the most important component of CPR. Chest compressions are an attempt to mimic the normal activity of the heart. When a rescuer presses down on a victim's chest, blood is forced out of the heart and into the arteries. When pressure on the chest is released, blood is allowed to return to the heart. A small amount of oxygen will be present in the bloodstream for several minutes after the heart ceases to beat, just enough to keep the brain alive. Compressions can keep vital organs functioning until higher level care is available.

Adult Chest Compressions

To perform compressions on an adult, place the heel of your non-dominant hand on the victim's chest between the nipples. Then place your other hand on top of the first and interlace your fingers. The heel of your hand should be pressing on the bottom two-thirds of the sternum, avoiding the xiphoid process (the small bony prominence at the very bottom of the sternum (breastbone). You

should be as close to the victim's side as possible, with your knees against the victim's side (this will help prevent back injury). Lock your elbows and press down hard, depressing the sternum at least 2 inches (5 cm), but no more than 2.4 inches (6 cm). Your shoulders should be positioned directly over your hands in a straight line. Push hard and fast 100 to 120 times per minute, counting out loud as you do so. It's important that you allow the chest to recoil (return to it's normal, relaxed position) in between compressions. If you do not allow the chest to recoil, the heart will not fill completely, which means that less blood (and therefore oxygen) will be pumped out of the heart to vital organs with the next compression.

Performing compressions is exhausting. Most people find that they become very fatigued after providing compressions for 2 or 3 minutes. When a person performing compressions becomes fatigued, there is a tendency to compress less firmly and more slowly; for this reason, it is recommended that rescuers trade off doing compressions every 2 minutes to prevent fatigue and optimize the quality of compressions. If you are alone, you will have to do the best you can- keep performing compressions until help arrives or you are physically unable to do so.

One Rescuer Adult CPR

If you are alone and come across an individual who is down, follow the steps below. If someone else is in the area to assist, use the 'Two Rescuer' sequence.

Stay Safe: If you come upon an individual who may need CPR, look around and make sure you and the victim are in a safe place. If the victim is in water or on a road, try to move the victim to a safer area. If you are in a safe area, do not try to move the victim as he/she may have other injuries that you cannot see. Simply roll him over onto his back. Make sure the victim is on a firm surface, in case compressions are needed.

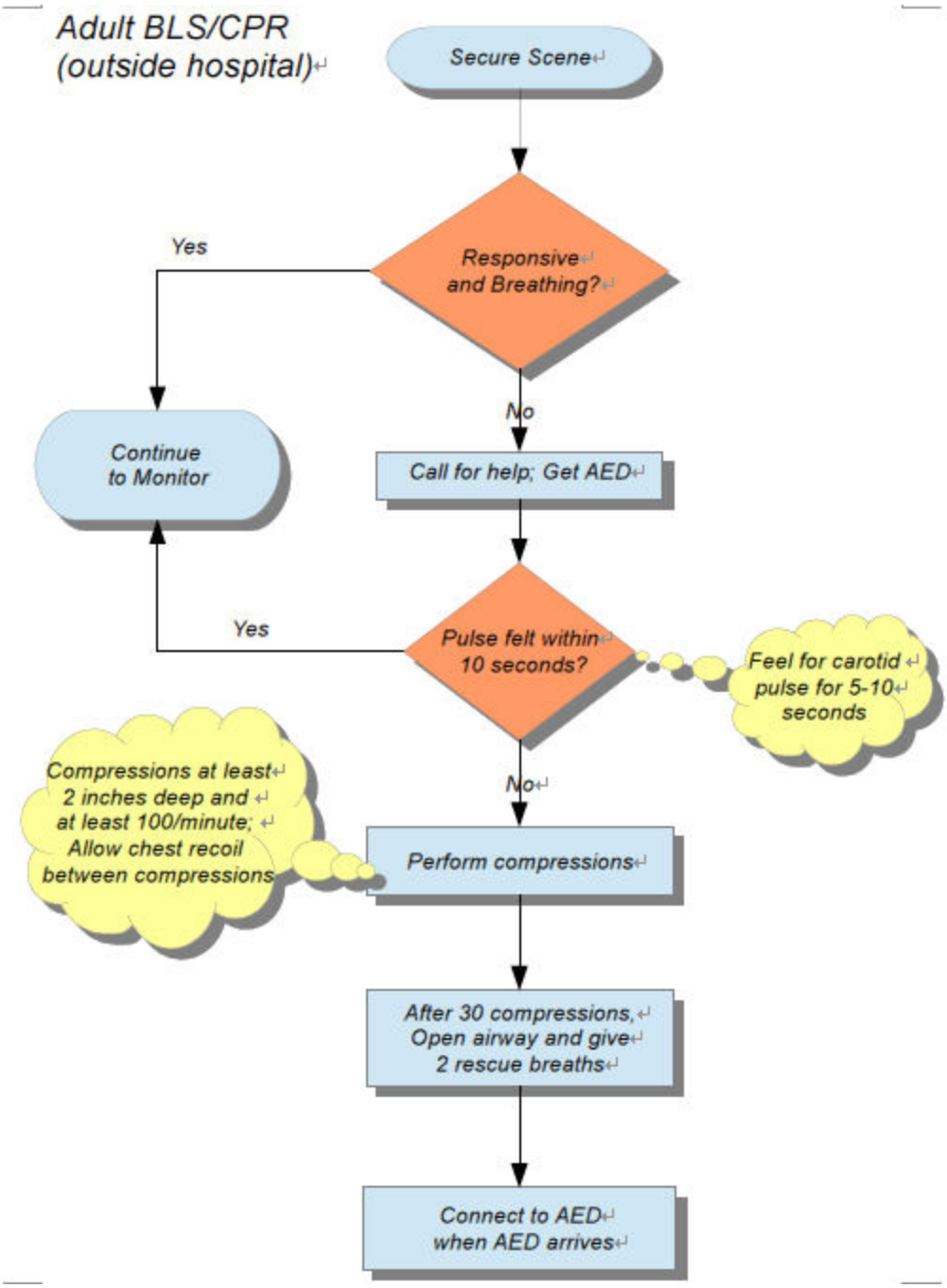
Assess the Victim: To quickly assess the victim, shake his shoulder and yell at him. Check for breathing. If he/she is not breathing, or is not breathing normally (i.e., only gasping), you must summon help.

Activate the Emergency Response System and Find an AED: Yell for help. If someone responds, tell him to call for help by dialing 9-1-1. If you are in an area where an AED may be available, tell him to go find the AED. Make sure you tell the person to return to assist you as soon as possible. If you are alone, call for help by dialing 9-1-1 and run to get the AED if you know where one is nearby. If you do not know where an AED is, begin CPR immediately.

Begin CPR

1. Check for a pulse on the side of the neck. Feel for a pulse for at least 5 seconds but NO MORE THAN 10 seconds. To check for a carotid pulse, slide 2 or 3 fingers into the groove between the windpipe and the neck muscles at the side of the neck.
2. If there is no pulse (or if you are unsure if there is a pulse), begin CPR starting with chest compressions. Provide 30 chest compressions, followed by two breaths. NOTE: If you are not comfortable giving rescue breathing and/or you do not have a mask available, do 'Compression Only' CPR.
    a. Use the heel of one hand on the lower half of the breastbone in the middle of the chest.
    b. Place the other hand on top of the first hand.
    c. Straighten your arms and lock your elbows so that your body weight is over your hands.
    d. The most important part of CPR is to remember to push HARD and FAST. Each compression should be AT LEAST two inches deep (but no more than 2.4 inches) and the rate should be 100-120 compressions per minute.

e. Be sure to let up on the pressure on the sternum after each compression (chest recoil) so the chest can re-expand and blood can flow back into the heart. The purpose of CPR is to help the blood flow through the heart and into the rest of the vital organs; if you allow the chest to re-expand, more blood will flow into the heart and will be available to deliver to the rest of the body.
f. Count out loud as you do compressions. When you have done 30 compressions, try to open the victim's airway by doing a head tilt/chin lift. Note that if you are doing 'Compressions Only' CPR, you can skip this step.
    i. With your non-dominant hand, push on the victim's forehead to tilt the head back.
    ii. With your dominant hand, place your fingers under the bony part of the lower jaw and gently lift the jaw to bring the chin forward. Be sure you lift up on the bony part of the jaw and not the soft tissue under the jaw so you don't block the victim's airway. Do not use your thumb to lift the jaw. Allow the victim's mouth to remain slightly open.
    iii. If you think the person's neck may be injured, avoid the head tilt/chin lift. Use the jaw thrust maneuver if you have been trained to do so.
g. If you have a barrier device to use between your mouth and the victim's face, use it. Although the risk of infection from performing CPR is very, very low, it is expected that healthcare workers use a barrier device when providing CPR. This includes the use of face masks or bag-mask devices (see next section). Give each breath slowly – each breath should last one second. Make sure the chest rises with each breath. Repeat, giving a second breath.
h. Start another cycle of chest compressions. Remember, push HARD and FAST. Alternate chest compressions (30) and giving breaths (2) until help arrives.


Adult BLS/CPR
(outside hospital)
Secure Scene
Responsive and Breathing?
Yes
No
Continue to Monitor
Call for help; Get AED
Pulse felt within 10 seconds?
Yes
Feel for carotid pulse for 5-10 seconds
No
Compressions at least 2 inches deep and at least 100/minute; Allow chest recoil between compressions
Perform compressions
After 30 compressions, Open airway and give 2 rescue breaths
Connect to AED when AED arrives

Airway/ Breathing

Face masks provide a barrier between the rescuer and the victim. Some masks are equipped with a one-way valve that allows the rescuer's breaths to enter the victim's airway, but prevents the victim's expired air from coming in contact with the rescuer's airway. These masks also prevent contact with vomitus and blood, which could pose an infection risk to the rescuer. It takes practice to learn how to use these masks effectively to provide ventilations.

Using a Face Mask:

a) Position yourself at the victim's side. If you are a lone (single) rescuer, positioning yourself at the victim's side will allow you to provide both ventilations and compressions without having to move.
b) Position the mask on the victim's face. Masks are usually triangular in shape, and you will notice that the mask has a "pointy" end- this end goes over the bridge of the victim's nose.
c) Seal the mask against the victim's face. To do this, take the hand that is closest to the top of the victim's head and place it along the edge of the mask. Some people find it easier to form a 'C' with their index finger and thumb and use these digits to grasp the mask around the base of the mouthpiece. With the thumb of your other hand, apply pressure along the bottom edge of the mask. Then place the remaining fingers of your second hand along the bony edge of the jaw and lift the jaw upwards. Open the airway by performing a head-tilt chin-lift procedure. While you lift the jaw, ensure that you are sealing the mask all the way around the outside edge of the mask to obtain a good seal against the victim's face.
d) Deliver air over 1 second, ensuring that the victim's chest rises.
e) If the victim's chest does not rise, reposition the mask and try to get a better seal. Remember, you should be lifting the victim's jaw into the mask, rather than simply pushing the mask down onto the victim's face.
f) Provide 2 ventilations over 1 second each with the mask after every 30 compressions.
g) If the victim has a pulse but is *not breathing*, provide rescue breathing by providing 1 breath every 5 to 6 seconds (10-12 breaths/minute). Check for a pulse every 2 minutes- if there is no pulse, start chest compressions along with ventilations at a rate of 30:2.



Using a Bag-Mask Device:

A bag-mask device (or BVM, bag-valve mask) consists of a mask attached to a reservoir bag. They are commonly used to provide positive-pressure ventilations during CPR. They can be attached to an oxygen source to provide 100% oxygenation during resuscitation. As with the face mask, it takes practice to be able to use a bag-mask device. It can also be very difficult for one person to use a bag-mask device; therefore, it is recommended that use of a bag-mask device be used only when there are two rescuers available.

The steps to using a bag-mask device are as follows:

a) Position yourself at the top of the victim's head- this allows room for the second rescuer to provide compressions.
b) Place the mask of the bag-mask device on the victim's face, using the bridge of the patient's nose as a guide to correct positioning.
c) Use the E-C clamp technique to hold the mask in the correct position while you lift the jaw to obtain an open airway. To perform the E-C clamp technique, take the index finger and thumb of your non-dominant hand and form them into a 'C' around the top of the mask. Your other fingers of the same hand are used to lift the jaw (the 'E' part of the E-C clamp technique.
d) Squeeze the bag to deliver a breath- each breath should be delivered over 1 second. Watch for chest rise. If you do not observe chest rise, you do not have a tight seal. If this occurs, reposition the mask and try again. Be careful not to overinflate the lungs- each breath should result in visible and *natural* chest rise.
e) Provide 2 ventilations after every 30 compressions. If the patient has a pulse but is not breathing, provide one breath every 5-6 seconds (10-12

breaths/minute) and check for a pulse every 2 minutes. If the victim loses their pulse, you will need to begin chest compressions.



The Jaw Thrust Maneuver

If you suspect that a victim may have a neck or spinal cord injury (i.e., the victim has fallen, been in a motor vehicle accident or suffered another mechanism of injury that could result in injury to the neck or spinal cord), you should not use the head tilt-chin lift maneuver to open the victim's airway. This could further damage the neck or spinal cord. Instead, you should use the jaw thrust maneuver to open and maintain the victim's airway.

To perform this maneuver:

a) Place your hands on either side of the victim's head. Rest your elbows on the surface that the victim is laying on.
b) Put the fingers of both your hands under the angle of the victim's lower jaw and lift so that the jaw slides forward.
c) Use your thumb to push the lower lip away from you if the victim's lips close.

Two Rescuer Adult CPR (without an AED)

If you come across a victim who is down and another person is available to help, send that person to activate the Emergency Response System and find an AED while you assess whether the victim needs CPR. When you have determined that the victim requires CPR, start CPR beginning with compressions. When the second person returns (without an AED in this case):

- You should continue chest compressions and count the compressions aloud each time you compress the victim's chest.
- The second rescuer should keep the person's airway open using the head tilt-chin lift maneuver. The second rescuer should give two rescue breaths after every 30 compressions, using a face mask or a bag-mask device.
- You and the second rescuer should switch positions/roles about every two minutes so neither one of you gets tired. If you are counting out loud, two minutes is about 5 cycles of thirty compressions and two breaths. You should change positions sooner if the person doing compressions becomes too tired to perform high-quality compressions.

Using an AED



An AED, or automated external defibrillator, is a device that has the ability to detect and treat, through electrical energy, the lethal arrhythmias known as ventricular fibrillation and ventricular tachycardia.

Ventricular fibrillation is a condition in which the lower chambers of the heart, the ventricles, quiver in an unorganized fashion, which renders them incapable of pumping blood to the rest of the body. Untreated, ventricular fibrillation rapidly causes cardiac arrest. Ventricular tachycardia is a rapid rhythm originating in the ventricles. In ventricular tachycardia, the ventricles contract so quickly, albeit in a somewhat organized fashion, that inadequate blood flow is produced. Ventricular tachycardia often precedes ventricular fibrillation. Both rhythms are lethal if not treated.

An AED sends electrical energy (a 'shock') through the heart, which stuns the heart and allows the normal pacemaker of the heart, usually located in the right atrium, to take over and restore a normal heart rhythm.

Early defibrillation is key to survival in cardiac arrest. For each minute that defibrillation is delayed, the chance of survival is reduced by 10%. (after 10 minutes, few people are successfully resuscitated.)
Early defibrillation can increase survival rates to greater than 50%. Rescuers should immediately begin chest compressions, and use the AED as soon as it is available and ready to use.

AEDs can be found wherever crowds of people gather- swimming pools, airports, malls, sporting arenas, schools, hotels…More and more businesses are also investing in these life-saving machines. In some communities, private AED owners are registering their AEDs with ambulance dispatch, so that they can be easily located by bystanders when needed. Make it a point to learn where the AEDs in your neighborhood or town are located- you never know when you might need one!



AEDs have been designed to be extremely "user friendly". All you need to do as a rescuer is turn on the machine (the most important step) and listen as the machine guides you through the steps to use the AED safely and effectively. Although there are many brands of AEDs on the market, they all work in a similar fashion and are designed to be used by lay rescuers.

When an AED becomes available (i.e., when you or another rescuer have retrieved it), place it at the victim's side, closest to the rescuer who will operate it. In this way, the other rescuer can continue performing CPR until the AED is ready to analyze and deliver a shock (if needed).

There are four universal steps to using any AED. These will be highlighted in the following list of steps so they are easily recognizable.

The steps to use an AED are as follows:

1. **Turn on the machine.** This is the most important step- turning on the machine will enable the AED unit to guide you through the next steps. To turn on the AED, open the top of the carrying case and push the ON button. Note: some models will turn on automatically when you lift the lid of the carrying case.
2. **ATTACH AED pads** to the victim's bare chest. Expose the patient's chest. Dry it off if wet, shave excessive hair if possible. Choose adult pads for victims who are 8 years of age or older. Peel off the adhesive backing. Place one pad on the upper right chest just below the collarbone. Place the other pad on the patient's lower left ribcage, a couple of centimeters beneath the armpit. Some pads are marked- there will be a red heart on the pad that is to be placed on the victim's left side (the heart side).
   Press pads firmly onto the patient's chest. Then attach the connecting cables to the AED unit. Note: some cables will come preconnected.
3. **Analyze the rhythm**. If the AED unit instructs you to, CLEAR the victim while the machine is analyzing the victim's heart rhythm. This means you should ensure that no one is touching the victim, including yourself. The rescuer performing chest compressions or giving breaths will need to stop at this point. Note: some AEDs will begin to analyze the victim's rhythm independently; for others, you will need to push the ANALYZE button. Analyzing the victim's rhythm will take up to 10 seconds, so don't be alarmed by this.
4. **Push to shock**. If a shock is advised, the machine will clearly state "SHOCK ADVISED, STAND CLEAR". You should ensure that no one is touching the victim, including yourself. You need to look around to make sure no one is touching the victim's body while stating "CLEAR" or some similar message that warns others a shock is to be delivered. Once you are certain that no one is touching the victim, push the SHOCK button. You will notice that the victim's muscles contract strongly.

- If a shock is not necessary (the rhythm is not ventricular fibrillation or pulseless ventricular tachycardia), the AED will state NO SHOCK ADVISED and tell you to resume CPR
- After approximately 5 cycles of compressions and ventilations, or 2 minutes of CPR, the AED will instruct you to repeat steps 3 and 4- analyze the rhythm and push to shock if the rhythm requires a shock and the AED instructs you to do so.
- Continue CPR alternating with analysis of the rhythm until help arrives (i.e. EMS)

Special Circumstances

There are times when using an AED may present special challenges. Here's what to do when faced with one of the following:

- **The victim has a hairy chest**- ideally, you should use a razor to shave the areas that will be covered by the AED pads. If a razor is not available and the AED machine is prompting you to CHECK PADS or CHECK ELECTRODES, try pressing down firmly on the AED pads to ensure good contact with the skin. If the machine continues to prompt you, quickly pull off the pads- this should remove enough hair to allow a new set of pads to adhere firmly to the victim's skin. Many AED machines are coming equipped with a razor in the carrying case to combat this problem. If you happen to own an AED, ensure that a razor is included in the case.

- **The victim is in water**- if the victim is in water, pull the victim to a dry area. You are not in danger of getting a shock if the victim is in water. Water is a great conductor of electricity, so if the victim is in water, the shock will be dispersed across the skin of the victim, and the victim will not receive the full dose of electrical energy required to convert them to a normal rhythm. If the victim's chest is wet, quickly dry the chest with a towel or your sleeve; however, the chest does NOT need to be completely dry. If the victim is lying in a small puddle or in snow, you can safely use the AED without moving the victim.

- **The victim has an implanted pacemaker or defibrillator**- obviously, if the victim has one of these devices, it must have failed! You will recognize these devices as a small lump under the skin on the chest, usually the upper chest on either side. Some older models may be implanted in the abdomen. They are generally about the size of a deck of cards or smaller. You will also be able to see a scar over the area. If the victim has one of these devices, avoid placing the AED pad directly over it; doing so may block delivery of the shock.

- **The victim has a medication patch on their chest**- many medications can now be delivered transdermally (through the skin). These includes pain medications, hormones, smoking cessation drugs, nitroglycerin and others. Do not place an AED pad over one of these patches. If it won't delay delivery of a shock, remove the patch and wipe the skin before applying the AED pad. These patches may cause the skin to burn under the AED pad if left in place, or they may block delivery of the shock.

Now you know what to do should you encounter one of these special circumstances.

Two Rescuer Adult CPR with an AED

We've learned a lot so far! Thus far, we've learned how to perform compressions, maintain the airway and use an AED on an adult victim. Now it's time to put it all together. The following steps outline how to perform CPR with an AED when there are two rescuers present.
When you come across a victim who is down and there are two rescuers present:

- **Rescuer 1** checks for response and breathing- tap the victim on the shoulder and ask if they are okay. At the same time, observe the victim's chest for breathing. If the victim is not breathing, or is breathing abnormally or only gasping, stay with the patient and prepare to perform the next steps.
- **Rescuer 2** activates the emergency response system and leaves to retrieve an AED.
- **Rescuer 1** checks for a carotid pulse for 5 to not more than 10 seconds. If a pulse is not felt, or the rescuer is not sure if there is a pulse, the rescuer will expose the chest (in preparation for AED use) and begin CPR, starting with chest compressions. **Rescuer 1** will continue cycles of chest compressions and ventilations with a pocket mask or bag-mask device until Rescuer 2 returns with an AED.
- **Rescuer 2** arrives with an AED and places it on the side opposite to **Rescuer 1,** who is performing chest compressions. **Rescuer 2** powers on the AED and attaches the pads to the victim's chest, also attaching the cables to the AED unit if necessary. **Rescuer 1** should continue CPR while the pads are being placed, right up until it is time to analyze the victim's heart rhythm.
- **Rescuer 2** CLEARS the victim, ensuring neither rescuer is touching the victim, and waits for the AED to ANALYZE, or pushes the ANALYZE button when prompted by the AED to do so.
- **Rescuer 2** pushes the SHOCK button if a shock is indicated, making sure that the victim is CLEAR beforehand.
- If no shock is needed, or after the victim has been shocked, **Rescuer 2** should resume chest compressions (as Rescuer 1 may be fatigued by this time) while **Rescuer 1** manages the victim's airway and delivers breaths using the face mask or bag-mask device.
- After approximately 5 cycles of CPR, or 2 minutes, the AED will state that the victim's rhythm should be ANALYZED. During analysis, which can take up to 10 seconds, **Rescuer 2** and **Rescuer 1** should switch positions, so that **Rescuer 1** CLEARS the victim, pushes SHOCK if a shock is advised, and immediately resumes chest compressions (or performs chest compressions if no shock is advised). **Rescuer 2** then takes over management of airway and breathing. Rescuers should switch positions every 2 minutes when it is time to ANALYZE the victim's heart rhythm. This will prevent rescuer fatigue and ensure that rescuers are able to provide high-quality chest compressions at the proper rate and depth. CPR and analysis with the AED should continue until EMS arrives.

PART THREE

In This Section:

CPR for Children (age 1 year to puberty)
- The Pediatric Chain of Survival

One Rescuer CPR for Children (aged 1 year to puberty)
Two Rescuer CPR for Children (aged 1 year to puberty)

CPR for Children (age 1 year to puberty)

Performing CPR on a child victim is similar to performing CPR on an adult, with a few key changes:

- Compression to ventilation ratio- when there is only one rescuer, the compression-ventilation ratio is the same as for adults- 30:2. However, when two rescuers are present, the compression-ventilation ratio changes to 15:2.

- Depth of compressions- for children, rescuers should compress to 1/3 the depth of the chest. For most children, this will be about 2 inches.

- Hands- rescuers may use one hand or two to compress the chest of a child. The number of hands used will depend on the size of the child. It does not matter how many hands you use, as long as you are compressing the chest adequately (1/3 the depth of the chest). If you find you are compressing the chest too hard (greater than 1/3 the depth of the chest, or 2 inches), simply remove one hand.

- When to call EMS- it is far more common for children to experience a respiratory arrest prior to going into cardiac arrest, unless the child has a congenital heart problem; in adults the opposite is true- cardiac arrests are more common due to cardiovascular disease. Therefore, the goal is to intervene before the child goes into cardiac arrest. For this reason, when to call EMS is dependent upon whether you witnessed the child's arrest. If you did NOT witness the child's arrest (unwitnessed arrest) and you are alone, you should provide CPR for 2 minutes prior to calling EMS and finding an AED. If you witness the arrest (i.e., the child suddenly collapses), you should call EMS and get an AED before returning to the child to start CPR.

The Pediatric Chain of Survival

The Pediatric Chain of Survival is similar to the Adult Chain of Survival, but it has an extra link. Because children are more prone to respiratory arrest and shock, it is essential to recognize airway and breathing problems before they occur to prevent cardiac arrest and ensure survival and full recovery. For this reason, an extra link in the chain of survival- prevention- has been added. Therefore, the Pediatric Chain of Survival includes:

- Prevention of respiratory/cardiac arrest
- Early high-quality CPR
- Activation of the Emergency Response System
- Early advanced life support
- Comprehensive post- cardiac arrest care

One Rescuer CPR for Children (aged 1 year to Puberty)

Note that puberty is defined as the presence of underarm or chest hair in boys and any breast development in girls.

If you are the only person available to help a child, do the following:

1. Assess the child for responsiveness by shaking their shoulders and calling their name/asking if they are okay. Check his/her breathing- if breathing is absent or abnormal (i.e., gasping), shout for help.

2. If someone answers your call for help, send that person to call EMS and get an AED if you know where one is. Note: if you witnessed the child's collapse, leave the child to activate EMS and retrieve an AED if you know where one is located.

3. Check the child's pulse for at least 5 but no more than 10 seconds. Depending on the age of the child, you might be able to feel a carotid pulse in the side of the neck. In younger children with shorter and chubbier necks, you can palpate the femoral artery in the groin area, midway between the pubic bone and the hip bone and just below the crease where the leg meets the abdomen.

4. If within 10 seconds you cannot feel a pulse, are not sure you can feel a pulse, OR the heart rate is below 60 beats a minute with signs of poor perfusion (pale skin, bluish discoloration on the face or extremities, or delayed capillary refill) begin CPR with 30 compressions followed by 2 breaths. Breaths should be delivered with a face mask or a bag-mask device in the same way as for adults- perform a head tilt-chin lift to open the airway, place the mask on the child's face using the bridge of the nose as a guide, seal the mask to the child's face and lift the jaw into the mask. Each breath should go in over 1 second and should cause visible chest rise. Avoid excessive ventilation- chest rise should appear natural and gradual rather than sudden and forceful. Masks should fit correctly and should not extend below the child's chin or cover the eyes.

5. After about 2 minutes of compressions, if someone has not already done so, you should leave the child to call 911 and get an AED (if you know where one is).

6. Use the AED as soon as possible.

Two Rescuer CPR for Children (aged 1 year to puberty)

When two rescuers are present, performing CPR on a child is the same as performing CPR on an adult, except that the compression ventilation ratio when 2 rescuers are present drops to 15:2. One or two hands may be used to compress the chest to a depth of 1/3 the diameter of the chest.

We'll discuss using an AED on a child later on. For now, we'll move on to CPR for infants.

PART FOUR

In This Section:

CPR for Infants (up to 12 months of age)
One Rescuer CPR for Infants (up to 12 months of age)
Two Rescuer CPR for Infants (up to 12 months of age)
AED Use in Child/Infant CPR

CPR for Infants (up to 12 months of age)

CPR for infants is similar to CPR for adults and children. There are a few differences as follows:

- Checking for responsiveness: never shake an infant as this may cause brain damage. To check for responsiveness in an infant, tap the soles of the feet while calling to the infant in a loud voice.

- Pulse check location- for an infant, it is easiest to check for a pulse using the brachial artery. To locate the brachial artery, place 2 or 3 fingers on the inside of the upper arm between the shoulder and elbow. Press the fingers gently for 5 to 10 seconds to feel for a pulse. Pushing too firmly may occlude the infant's pulse.

- Depth of compressions- compress the infant's chest to one-third the depth of the chest, or approximately 1 ½ inches.

- Compression delivery technique- 2 finger technique: when one rescuer is present, the chest is compressed using two fingers on the lower half of the sternum, avoiding the xiphoid process (the very end of the sternum where it narrows). To landmark, place 2 fingers in the center of the infant's chest, just below the nipple line. Push down on the infant's chest one-third the depth of the chest, or approximately 1 ½ inches. Allow the chest to fully recoil (return to its neutral position) in between compressions. Compressions should be delivered at a rate of 100-120 compressions/minute.
- 2 thumb-encircling hand technique: When two rescuers are present, the compression: ventilation ratio drops to 15:2, the same as in children. To perform this technique, position yourself at the infant's feet. Place your thumbs side by side on the center of the infant's chest just below the nipple line. Encircle the infant's chest so that the fingers of both hands support the infant's back. Use your thumbs to deliver compressions at the appropriate depth and rate (100-120 compressions/minute; 1/3 of the depth of the chest or approximately 1 ½ inches). This position allows another rescuer to support airway and breathing without getting in the way of the rescuer performing chest compressions, and is therefore the preferred technique when two rescuers are present. This technique also allows for more consistent chest compressions and superior blood flow and blood pressure compared to the 2-finger technique.

- Compression: ventilation ratio- The compression: ventilation ratio for 1 rescuer CPR in infants is 30:2, the same as for single rescuer CPR in adults and children. When two rescuers are present, the compression: ventilation ratio drops to 15:2, the same as in children.

- When to call EMS- it is far more common for infants to experience a respiratory arrest prior to going into cardiac arrest, unless the infant has a congenital heart problem; in adults the opposite is true- cardiac arrests are more common due to cardiovascular disease. Therefore, the goal is to intervene before the infant goes into cardiac arrest. For this reason, when to call EMS is dependent upon whether you witnessed the infant's arrest. If you did NOT witness the infant's arrest (unwitnessed arrest) and you are alone, you should provide CPR for 2 minutes prior to calling EMS and finding an AED. If you witness the arrest (i.e., the infant suddenly becomes unresponsive), you should call EMS and get an AED before returning to the child to start CPR.

One Rescuer CPR for Infants (up to 12 months of age)

If you are the lone rescuer of an infant:

1. Assess the child for responsiveness by tapping the soles of the infant's feet while calling his/her name loudly. Check for breathing- if there is no breathing or the infant is breathing abnormally or only gasping, call for help.

2. If someone responds to your call for help, ask them to call 9-1-1 (activate EMS) and find an AED.

3. Check the infant's brachial pulse for at least 5, but no more than 10, seconds. Place 2 or 3 fingers on the inside of the upper arm between the shoulder and elbow. Press the fingers down gently for 5 to 10 seconds to feel for a pulse. Remember: pushing too firmly may occlude the infant's pulse.

4. If you are not sure you can feel the pulse, the pulse is absent or the infant's heart rate is below 60 beats per minute with signs of poor perfusion (pale or bluish discoloration in the face, extremities or nail beds), start CPR, beginning with 30 compressions followed by two breaths.

    a. Place the infant on his back on the ground or on a firm surface.

b. Place 2 fingers in the center of his chest just below the nipple line; do not press on the end of the breastbone.
c. Provide 100-120 chest compressions per minute to a depth of 1/3 the depth of the chest or approximately 1 ½ inches.
d. The principles of providing breaths for infants are the same as for children and adults. Use the correct sized face mask or bag-mask device for the infant (the mask should cover the mouth and nose without extending past the chin or covering the eyes). Each breath should go in over 1 second and should cause visible chest rise. A breath should require only a small puff of air into the mouthpiece of the device to cause chest rise- avoid excessive ventilations. Lastly, rather than performing a head tilt-chin lift maneuver to open the infant's airway, the infant's head should be placed in "sniffing position" with the infant's head tilted just enough that the nose appears to be sniffing the air. In this position, the external ear canal should be level with the top of the infant's shoulder. Avoid hyperextending the neck- you also want to avoid allowing the chin to fall down towards the neck. You can do this by placing one hand on the infant's forehead while you perform chest compressions. In this way, the infant's airway will remain open and will not close off.

5. After about two minutes of compressions (five cycles of 30 compressions and 2 breaths), leave the child to call 911 and get an AED if you know where one is (and someone has not already done so). Then return to the infant to continue CPR.

6. Use the AED as soon as it is available.

Two Rescuer CPR for Infants (up to 12 months of age)

If there are 2 rescuers available:

1. Rescuer 1- Assess the child for responsiveness by tapping the soles of the feet and calling to him/her loudly. Check for breathing- if the infant is not breathing or is breathing abnormally or only gasping, send Rescuer 2 to activate EMS and bring the AED (if one is available).

2. Rescuer 1- Check the infant's brachial pulse for at least 5, but not more than 10, seconds. Place 2 or 3 fingers on the inside of the upper arm between the

shoulder and elbow. Press the fingers down gently for 5 to 10 seconds to feel for a pulse. Remember: pushing too firmly may occlude the infant's pulse.

3. Rescuer 1- If you are not sure you can feel the pulse, the pulse is absent or the infant's heart rate is below 60 beats per minute with signs of poor perfusion (pale or bluish discoloration in the face, extremities or nail beds), start CPR, beginning with 30 compressions followed by two breaths.

    a. Place the infant on his back on the ground or on a firm surface.
    b. Place 2 fingers in the center of the infant's chest just below the nipple line; do not press on the end of the breastbone.
    c. Provide 100-120 chest compressions per minute to a depth of 1/3 the depth of the chest or approximately 1 ½ inches.
    d. The principles of providing breaths for infants are the same as for children and adults. Use the correct sized face mask or bag-mask device for the infant (the mask should cover the mouth and nose without extending past the chin or covering the eyes). Each breath should go in over 1 second and should cause visible chest rise. A breath should require only a small puff of air into the mouthpiece of the device to cause chest rise- avoid excessive ventilations. Lastly, rather than performing a head tilt-chin lift maneuver to open the infant's airway, the infant's head should be placed in "sniffing position" with the infant's head tilted just enough that the nose appears to be sniffing the air. In this position, the external ear canal should be level with the top of the infant's shoulder. Avoid hyperextending the neck- you also want to avoid allowing the chin to fall down towards the neck. You can do this by placing one hand on the infant's forehead while you perform chest compressions. In this way, the infant's airway will remain open and will not close off.

When the second person returns, change the ratio of compressions to ventilations to 15:2.

4. Rescuer 2 should place the thumbs of both hands on the lower half of the infant's breastbone, while being careful not to press on the end of the breastbone (xiphoid process). Put the fingers of both hands around the infant's chest. Use the fingers to encircle the infant's back to provide support. Use the thumbs to depress the sternum approximately 1/3 the depth of the chest, or approximately 1 ½ inches. Compress at a rate of 100-120 compressions per minute.

5. Rescuer 1 should provide breaths as described above.

6. Continue CPR at a ratio of 15 compressions to 2 breaths and switch roles every 2 minutes to avoid fatigue.

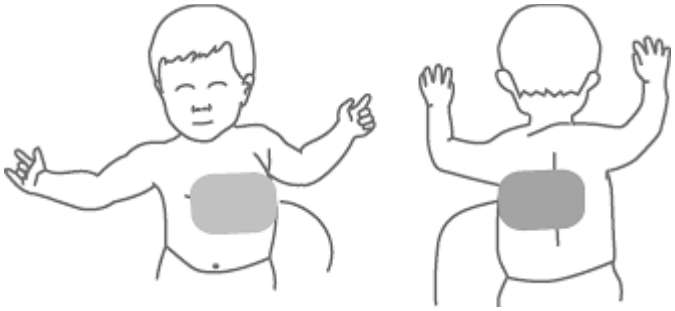
7. Use the AED when available.

AED Use in Child/ Infant CPR

As you might expect, using an AED on children and infants requires some special consideration.

Some AEDs have been modified so that they can deliver different shock dosages to adults and to infants and children. If you use an AED that is pediatric-capable, it will have special features that you will need to know about in order to operate the AED. For example, some AEDs may come with child pads, which are smaller in size. In some AEDs, there may be a key or switch that you will need to activate to use the AED on children or infants; in others you will plug the child AED pads into a separate receptacle when using the AED on infants or children. All AEDs typically come with instructions, so when all else fails, read the instructions! Turning the Machine ON will cause the AED to begin voicing instructions as well.

If the AED you are using has pads designed for use in children, use them for children under 8 years of age. If the AED does not have child pads, you may use adult pads; however, you should ensure the pads do not touch or overlap. If the pads are too large, you can use alternative placements, such as the anterior-posterior pad placement:



For infants, a manual defibrillator is preferred; however, if a manual defibrillator is not available, an AED designed for use on children and infants is the next best thing. When nothing else is available, and adult AED may be used- after all, the alternative is death, and studies have not shown that infants and children resuscitated with an adult AED suffer any permanent damage to the heart.

PART FIVE

In This Section:

CPR with an Advanced Airway
Mouth-to-Mouth Breaths
Rescue Breathing

CPR with an Advanced Airway

When there is no advanced airway in place, rescuers must pause compressions to deliver breaths using a face mask or bag-mask device. Once an advanced airway is in place, there is no longer a need to pause compressions to deliver breaths.

Why is this? If the chest is compressed during ventilations, most of the air (and thus oxygen) delivered would be forced out of the victim's mouth before it could enter the lungs. An advanced airway means that air is reliably delivered to the lungs, regardless of whether a rescuer is applying force to the chest at the same time a breath is being delivered.

Advanced airways include:

- Laryngeal mask airways (LMAs)
- Supraglottic airway (Combitube or King LT)
- Endotracheal (ET) tube

When an advanced airway is in place, compressions are delivered at a rate of 100-120 compressions per minute. Breaths are delivered over 1 second ***simultaneously*** at a rate of 1 breath every 6 seconds (10-12 breaths per minute).

Mouth-to-Mouth Breaths

Although it is necessary to use a pocket mask or bag-mask device to deliver breaths in the healthcare setting, there are times when these devices may not be necessary. Should a cardiac arrest occur at home, you would likely not hesitate to perform mouth-to-mouth breathing for a relative or loved one; you might choose to give mouth-to-mouth to a friend as well. In cases such as these, you will likely decide the benefit outweighs the risk to your own health.

To provide mouth-to-mouth breaths to an adult or child:

1. Use the head tilt-chin lift to hold the victim's airway open.
2. Using the hand on the victim's forehead that is maintaining the head tilt-chin lift, pinch the victim's nose closed using the thumb and index finger.
3. Inhale a regular breath, then cover the victim's mouth with your own, creating a tight seal.
4. Give one breath over 1 second, watching to see if the chest rises.
5. If the chest doesn't rise as you give the breath, repeat the head tilt-chin lift.
6. Give a second breath over 1 second and watch for chest rise.
7. If the second breath fails to go in, go immediately to chest compressions.

To provide breaths to an infant:

1. Use a head tilt-chin lift to maintain an open airway (sniffing position), being careful not to hyperextend the neck, which could block the airway
2. Place your mouth over the victim's mouth AND NOSE to create a tight seal.
3. Blow into the infant's nose and mouth over 1 second, with just enough volume and force to cause the chest to rise with each breath. Be careful not to ventilate too forcefully, as doing so may cause lung damage.
4. If the chest does not rise, repeat the head tilt-chin lift and try to ventilate the victim again. You may need to try to provide breaths at a few different positions before you achieve airway patency (airway is in an open position).
5. If the infant is older and you cannot cover both the infant's mouth and nose, pinch the nostrils closed and place your mouth over the victim's to form a tight seal- just the same as you would do for a child or adult.

You may wonder how mouth-to-mouth or mouth-to-mouth-and-nose breathing can sustain the victim. In actual fact, your expired air contains about 17% oxygen- this is just enough oxygen to meet the victim's needs.

When providing mouth-to-mouth or mouth-to-mouth-and-nose breathing, it is important not to provide breaths that are too forceful or too rapid. Doing so may cause air to enter the stomach rather than the lungs, which can cause gastric inflation. Gastric inflation may result in vomiting, and an unconscious victim may develop pneumonia if vomitus makes its way to the lungs. To avoid gastric inflation, give each breath slowly over 1 second and deliver just enough air to make the chest rise.

Rescue Breathing

Respiratory arrest is defined as the cessation of breathing. During respiratory arrest, as well as inadequate breathing, the victim will still have some amount of cardiac output, which you will be able to detect as a palpable pulse.

It is important to be able to recognize respiratory arrest, or impending respiratory arrest, which may be seen as slow, irregular or gasping respirations. These abnormal respirations are inadequate to support life. Respiratory arrest inevitably leads to cardiac arrest if not treated, therefore healthcare providers should intervene quickly to prevent this deterioration by providing ***rescue breathing***.

The table below provides a summary of rescue breathing guidelines for different age groups.

Rescue Breathing for Adults	Rescue Breathing for Children and Infants
1 breath every 5 to 6 seconds (10-12 breaths/minute)	1 breath every 3 to 5 seconds (12-20 breaths/minute)
Each breath should last 1 second Did the chest rise? If not, try again Be sure to check the victim's pulse every 2 minutes and begin chest compressions combined with breaths if needed	

PART SIX

In This Section:

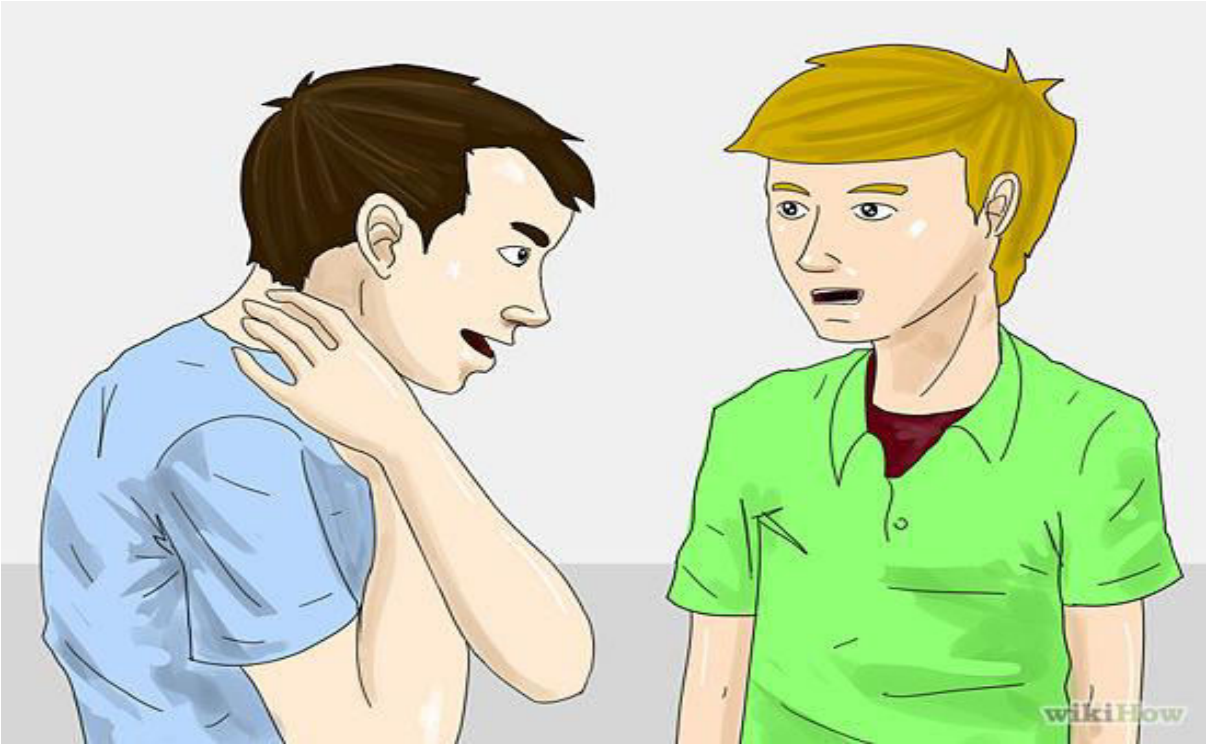
Treatment of Choking
Treatment of Choking in Victims 1 Year of Age and Older

- Choking in the Conscious Adult or Child (victims 1 year of age and older)
- Choking in the Conscious Infant
- Choking in the Unconscious Adult or Child (victims 1 year of age and older)
- Choking in the Unconscious Infant

Moving Victims

- The Recovery Position

Treatment of Choking



Choking in the Conscious Adult or Child (older than 1 year of age)

1. STAND (OR KNEEL) BEHIND THE VICTIM AND WRAP YOUR ARMS AROUND THE VICTIM'S WAIST.
2. PUT ONE FOOT IN BETWEEN THE VICTIM'S FEET AND ONE FOOT BEHIND YOU-THIS POSITION PROVIDES STABILITY SHOULD THE VICTIM BECOME UNCONSCIOUS AND YOU NEED TO EASE THE VICTIM TO THE GROUND.
3. PLACE YOUR FIST WITH THE THUMB SIDE IN JUST ABOVE THE VICTIM'S BELLYBUTTON AND BELOW THE STERNUM (BREASTBONE).
4. GRAB YOUR FIST WITH YOUR OTHER HAND.
5. ADMINISTER ABDOMINAL THRUSTS, PULLING INWARD AND UPWARD UNTIL THE FOREIGN OBJECT COMES OUT OR THE PATIENT BECOMES UNCONSCIOUS. EACH THRUST SHOULD BE FORCEFUL, DISTINCT AND SEPARATE.

Choking in the Conscious Infant

1. SIT OR KNEEL WITH THE INFANT IN YOUR LAP.
2. IF NOT DIFFICULT TO DO, REMOVE THE INFANT'S CLOTHING TO EXPOSE THE INFANT'S CHEST.

3. REST THE INFANT'S BODY ON YOUR FOREARM WITH THE INFANT'S HEAD LOWER THAN THE BODY. SUPPORT THE JAW AND HEAD WITH YOUR HAND. REST YOUR FOREARM ON YOUR THIGH OR LAP TO PROVIDE SUPPORT.
4. GIVE 5 BACK BLOWS FORCEFULLY WITH THE HEEL OF YOUR HAND BETWEEN THE INFANT'S SHOULDER BLADES.
5. SUPPORT THE BACK OF THE INFANT'S HEAD WITH THE PALM OF YOUR HAND AND THE INFANT'S JAW AND HEAD WITH THE OTHER PALM WHILE TURNING HIM OVER (FACE UP) ONTO YOUR OTHER FOREARM. KEEP THE INFANT'S HEAD LOWER THAN ITS BODY. REST YOUR FOREARM ON YOUR THIGH FOR SUPPORT.
6. GIVE 5 CHEST THRUSTS, JUST AS YOU WOULD WHEN PERFORMING CHEST COMPRESSIONS IN CPR. CHEST THRUSTS SHOULD BE DELIVERED AT A RATE OF 1 CHEST THRUST PER SECOND AND SHOULD BE GIVEN WITH ENOUGH FORCE TO DISLODGE A FOREIGN BODY.
7. REPEAT BACK BLOWS/CHEST THRUSTS UNTIL THE OBJECT COMES OUT OR THE VICTIM LOSES CONSCIOUSNESS.



NOTE: FOR PREGNANT OR OBESE PATIENTS, GIVE CHEST THRUSTS INSTEAD OF ABDOMINAL THRUSTS.

Choking in the Unconscious Adult or Child

1. IF YOU ARE CARINIG FOR SOMEONE WHO IS CHOKING AND THEY LOSE CONSCIOUSNESS, LOWER THE VICTIM GENTLY TO THE GROUND.
2. ACTIVATE EMS/CALL 9-1-1
3. BEGIN CPR< STARTING WITH CHEST COMPRESSIONS- DO NOT CHECK FOR A PULSE.
4. EACH TIME YOU OPEN THE AIRWAY TO PROVIDE VENTILATIONS, OPEN THE VICTIM'S MOUTH AND CHECK FOR THE OBJECT> IF YOU CAN SEE THE OBJECT, TURN THE VICTIM's HEAD TO THE SIDE AND SWEEP IT OUT OF THE VICTIM'S MOUTH WITH YOUR INDEX FINGER. NEVER PERFORM A BLIND FINGER SWEEP- THIS MAY FORCE THE OBJECT FARTHER DOWN THE VICTIM'S AIRWAY.
5. IF YOU DO NOT SEE THE OBJECT, ATTEMPT TO PROVIDE BREATHS. IF BREATHS WILL NOT GO IN, RESUME CHEST COMPRESSIONS.
6. AFTER APPROXIMATELY 5 CYCLES OF COMPRESSIONS AND VENTILATIONS, ACTIVATE EMS/CALL 9-1-1 IF NOT ALREADY DONE.
7. IF THE OBSTRUCTION IS RELIEVED, CHECK RESPONSIVENESS, BREATHING AND PULSE. PROVIDE RESCUE BREATHING OR CPR AS REQUIRED. IF THE VICTIM IS RESPONSIVE, THEY SHOULD BE TAKEN TO HOSPITAL TO RULE OUT ANY INJURY CAUSED BY ABDOMINAL OR CHEST THRUSTS.

Choking in the Unconscious Infant

1. CALL FOR HELP. IF SOMEONE RESPONDS, SEND THEM TO ACTIVATE EMS/CALL 9-1-1
2. PLACE THE INFANT ON A FLAT, FIRM SURFACE.
3. BEGIN CPR, STARTING WITH COMPRESSIONS. EACH TIME YOU OPEN THE AIRWAY TO DELIVER BREATHS, LOOK TO SEE IF THE OBJECT IS VISBLE. IF THE OBJECT IS VISIBLE, TURN THE INFANT'S HEAD TO THE SIDE AND USE YOUR INDEX FINGER TO SWEEP THE OBJECT OUT OF THE INFANT'S MOUTH. NEVER PERFORM A BLIND FINGER SWEEP, AS THIS MAY FORCE THE OBJECT FARTHER DOWN INTO THE AIRWAY.
4. AFTER 5 CYCLES OF COMPRESSIONS AND VENTILATIONS (OR ATTEMPTED VENTILATIONS), ACTIVATE THE EMERGENCY RESPONSE SYSTEM IF NOT ALREADY DONE.
5. IF THE OBSTRUCTION IS RELIEVED, CHECK RESPONSIVENESS, BREATHING AND PULSE. PROVIDE RESCUE BREATHING OR CPR AS REQUIRED. IF THE VICTIM IS RESPONSIVE, THEY SHOULD BE TAKEN TO HOSPITAL TO RULE OUT ANY INJURY CAUSED BY ABDOMINAL OR CHEST THRUSTS.

MOVING VICTIMS

It is generally not recommended that a rescuer move a victim unless there is a direct danger to the victim's (or rescuer's) life, or if it is necessary to provide care. Try to immobilize the spine and protect the head, neck, and back if it is necessary to move a victim.

Recovery position

An unconscious victim who is breathing and has a pulse should be assisted into the recovery position to protect the airway.



This position keeps the airway open, prevents aspiration of fluids into the lungs and allows fluid (such as saliva or blood that could occlude the airway) to drain from the mouth.

1. Extend the victim's arm that is closest to you above the victim's head.
2. Place the victim's leg that is farthest away from you over his other leg.
3. Place the victim's arm that is farthest from you across his chest.
4. Supporting the head and neck, roll the victim towards you.
5. Position his or her top leg so the knee props up the victim's body
6. Place the victim's hand or arm under his/ her chin to aid in keeping the airway open.

PART SEVEN

In This Section:

CPR Certification Skills Review
CPR Exam Questions

CPR CERTIFICATION SKILLS REVIEW

Skill	Adult	Child	Infant
Check scene safety	Ensure the safety of the scene before entry	Ensure the safety of the scene before entry	Ensure the safety of the scene before entry
Determine patient responsiveness	Tap on the shoulder and shout, "Are you OK?" Look for gasping, abnormal breathing, or absence of chest movement.	Tap on the shoulder and shout, "Are you ok?" Look for gasping, abnormal breathing, or absence of chest movement.	Tap the soles of the feet while calling the infant's name. Look for gasping, abnormal breathing, or absence of chest movement.
Activate EMS – Call for help!	If alone with no mobile phone, leave the victim to activate EMS and retrieve AED before returning to victim. If another person is available, ask them to activate EMS, get an AED and return to provide assistance.	If cardiac event is *witnessed*, follow same steps as for adults. If cardiac event is *not witnessed*, provide 2 minutes of CPR, then leave the victim to activate EMS and get an AED. Return to the victim to resume CPR and use the AED as soon as possible.	Same as for child.
Check pulse	Check for a pulse for no more than 10 seconds in the carotid artery of the neck.	Same as adult – Can also use femoral artery.	Check brachial artery in the upper arm for no more than 10 seconds.
Compressions: Hard and Fast	30:2 at 100- 120/minute. Use both hands. Depth of compressions 2" minimum to 2.4" maximum.	1 rescuer: 30:2 2 rescuers: 15:2 1 or 2 hands depending on child's size. Depth: 1/3 AP diameter of chest, or about 2"	1 rescuer: 30:2 2 rescuers: 15:2 2 fingers on breastbone below nipple line OR 2 thumbs (2 rescuers) Depth: 1/3 AP diameter of chest, or about 1 ½"
Airway	Head tilt/chin lift. Check mouth for obstruction, foreign matter. Use jaw thrust if suspected neck/spinal cord injury	Same as adult	Same as adult and child. Head should be in "sniffing" position. Use roll under shoulders to maintain proper positioning.
Breathing	2 breaths over 1 second	2 breaths over 1 second	Same as for child

	each, following 30 compressions. Watch for chest rise. When an advanced airway is present, give 1 breath every 6 seconds while continuous chest compressions are being performed.	each, following 30 compressions. Watch for chest rise. When an advanced airway is present, give 1 breath every 3-5 seconds while continuous chest compressions are being performed. .	
Rescue breathing: Patient with pulse and not breathing or gasping*	1 breath every 5 seconds, recheck airway if breaths aren't going in, recheck pulse every 2 minutes	1 breath every 3-5 seconds, rechecking circulation every 2 minutes.	Same as child.
AED	Give CPR until AED is available and charged.	For infants and children up to age 8 years, use child attenuator pads; if not available, use adult pads, don't let pads contact each other. May use anterior-posterior pad placement.	Manual defibrillator is preferred. If one is not available, use child attenuator pads; if not available, use adult pads, don't pads contact each other. May use anterior-posterior pad placement.

Table of Contents

Overview of Advanced Cardiovascular Life Support 2
Updates to ACLS in 2015 3
Chain of Survival 4
Adult BLS Chain of Survival 4
BLS and ACLS Surveys 5
The BLS Survey 5
Solo Provider Adult BLS 6
Team Adult BLS 7
Cardiac Arrest 8
Ventricular Fibrillation and Pulseless Ventricular Tachycardia 9
Ventricular Fibrillation and Pulseless Ventricular Tachycardia Algorithm 10
Pulseless Electrical Activity and Asystole 11
PEA and Asystole Algorithm 12
Respiratory Arrest 13
Airway Management 14
Return of Spontaneous Circulation (ROSC) and Post Arrest Care 16
Rapid Differential Diagnosis of Cardiac Arrest 17
Bradycardia 18
Bradycardia Algorithm 18
Tachycardia 19
Tachycardia Algorithm 20
Atrioventricular (Heart) Block 21
Acute Coronary Syndrome 22
Acute Coronary Syndrome Algorithm 23
Acute Stroke 24
Time is Brain! Stroke Time Goals for Evaluation and Therapy 25
Team Dynamics 26
Resuscitation and Life Support Medications 27

Overview of Advanced Cardiovascular Life Support

Advanced Cardiac Life Support, or ACLS, is a system of algorithms and best practice recommendations intended to provide the best outcome for patients in cardiopulmonary crisis. ACLS protocols are based on basic and clinical research, patient case studies, clinical studies, and reflect the consensus opinion of experts in the field. While the term Advanced Cardiovascular Life Support was coined by the American Heart Association, the content contained in this manual is based on the most recent guidelines published by the American Heart Association, the American College of Cardiology, the American Red Cross, and The European Society of Cardiology.

Prior to taking ACLS, it is assumed that you are proficient and currently certified in Basic Life Support (BLS). Once you become certified in ACLS, the certification is valid for two years. However, we encourage you to regularly login back in to your account to check for updates on resuscitation science advances.

This Advanced Life Support provider manual includes:

- Updates to ACLS in 2015
- Solo and Team BLS
- The ACLS Survey
- Cardiac and Respiratory Arrest
- Return of Spontaneous Circulation (ROSC) and Post Arrest Care
- Bradycardia and Tachycardia Management
- ECG Rhythm Recognition
- Atrioventricular Blocks
- Acute Coronary Syndrome and Acute Stroke Management
- Resuscitation Medications
- Team Dynamics and Systems of Care

Updates to ACLS in 2015

As we learn more about resuscitation science and medicine, physicians and researchers realize what works best and what works fastest in a critical, life-saving situation. Therefore, it is necessary to periodically update life-support techniques and algorithms. If you have previously certified in advanced cardiovascular life support, then you will probably be most interested in what has changed since the latest update in 2010. The table below also includes changes proposed since the last AHA manual was published. These changes will likely appear in future editions of the provider manual.

2015 Science Summary Table

Topic	2010	2015
Systematic Approach: BLS Assessment (name change)	• 1-2-3-4 • Check responsiveness: – Tap and shout – Scan chest for movement • Activate the emergency response system and get an AED • Circulation: Check the carotid pulse. If you cannot detect a pulse within 10 seconds, start CPR, beginning with chest compressions, immediately • Defibrillation: If indicated, deliver a shock with an AED or defibrillator	• Check responsiveness – Tap and shout • Shout for nearby help/activate emergency response system/get AED • Check breathing and pulse (simultaneously) • Defibrillation: If indicated, deliver a shock with an AED or defibrillator
Systematic Approach: Primary Assessment (name change)	• Airway • Breathing • Circulation • Differential diagnosis (H's and T's)	• Airway • Breathing • Circulation • Disability • Exposure
Systematic Approach: Secondary Assessment (new)	• NA	• SAMPLE • H's and T's
BLS: High-Quality CPR	• A rate of at least 100 chest compressions per minute • A compression depth of at least 2 inches in adults • Allowing complete chest recoil after each compression • Minimizing interruptions in compressions (10 seconds or less) • Avoiding excessive ventilation • Switching provddiac arrest with aiers about every 2 minutes to avoid fatigue	• A rate of 100 to 120 chest compressions per minute • A compression depth of at least 2 inches in adults* • Allowing complete chest recoil after each compression • Minimizing interruptions in compressions (10 seconds or less) • Avoiding excessive ventilation • Chest compression fraction of at least 60% but ideally greater than 80% • Switch compressor about every 2 minutes or sooner if fatigued • Use of audio and visual feedback devices to monitor CPR quality *When a feedback device is available, adjust to a maximum depth of 2.4 inches [6 cm]) in adolescents and adults.
ACLS: Immediate Post–Cardiac Arrest Care	• Consider therapeutic hypothermia (32°C to 34°C for 12 to 24 hours) to optimize survival and neurologic recovery in comatose patients	• Consider targeted temperature management to optimize survival and neurologic recovery in comatose patients—cool to 32°C to 36°C for at least 24 hours • Out-of-hospital cooling of patients with rapid infusion of cold IV fluids after ROSC is not recommended
ACLS: Managing the Airway	• For cardiac arrest with an advanced airway in place, ventilate once every 6 to 8 seconds	• For cardiac arrest with an advanced airway in place, ventilate once every 6 seconds
ACLS: Bradycardia	• Dopamine dosing: 2 to 10 mcg/kg per minute	• Dopamine dosing: 2 to 20 mcg/kg per minute
ACLS: ACS	• NSTEMI • Titrate O_2 saturation to ≥94%	• NSTE-ACS • Titrate O_2 saturation to ≥90%

Chain of Survival

Advanced Cardiovascular Life Support continues to emphasize the Chain of Survival. The Chain of Survival is a sequence of steps or links that, when followed to its completion, increases the likelihood that a victim of a life-threatening event will survive. The adult and pediatric chains of survival are slightly different. The person who is providing BLS is only responsible for the early links, that is, making sure the person is cared for by emergency personnel. The emphasis on **early** care is to reinforce that time is a critical factor in life support care. The 2015 standards include the concept of out of hospital care versus in-hospital care.

Adult BLS Chain of Survival (Out of Hospital)

The links of the Adult ACLS Chain of Survival are:



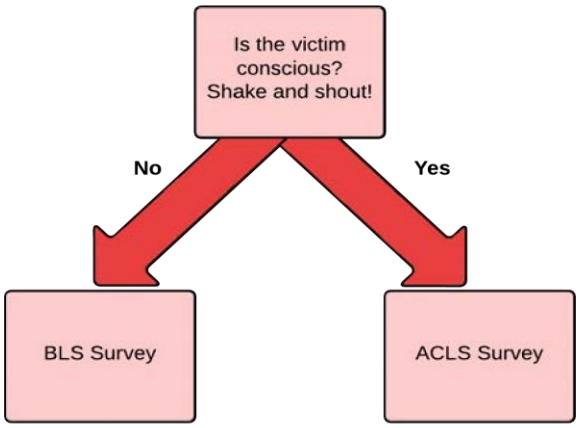
- EMS – Early activation of EMS
- CPR – Early administration of High-quality CPR
- AED – Early use of an AED
- ACLS – Early Advanced Cardiovascular Life Support
- Post-Arrest Care – Transport victim to the hospital

Adult BLS Chain of Survival (In Hospital)

This new chain should include: Primary Providers/ Code Team/ Cath Lab/ Post Arrest Care

BLS and ACLS Surveys

ACLS draws heavily on Basic Life Support (BLS). In fact, it is assumed that all people who are pursuing ACLS will be competent in the techniques of BLS—so much so that it is considered a prerequisite to ACLS.



The first step in any resuscitation is to make sure the rescuers (you!) and the victim are safe. Therefore, if your victim is in the middle of the highway or in a burning building, the first step is to move the victim to safety.

Assuming you and the victim are in a safe location, the next step is to assess whether the patient is responsive

*If patient **is not** responsive, move to BLS survey*

*If patient **is** responsive, move to ACLS survey*

The BLS Survey

The BLS Survey	
1. Responsive?	Shake and Shout! Don't be afraid to make noise. Check for effective breathing for 5 to 10 seconds.
2. Activate EMS/Call Code	In the hospital, you can call a "code" and send someone to get a defibrillator. In the community, call 911 and send for an AED
3. Circulation	Simultaneously, check the carotid pulse for no more than 10 seconds while checking for respiratory effort. If no pulse, begin high quality CPR.
4. Defibrillation	If there is a shockable rhythm, pulseless ventricular tachycardia or ventricular fibrillation, provide a shock

Adult BLS is slightly different if there is one provider (solo) or more than one provider (team) present. The difference between solo provider BLS and team BLS is that responsibilities are shared when more than one person is present. These will be detailed in Solo and Team Adult BLS.

For healthcare providers, the difference between a witnessed cardiac arrest and a victim who is found down is the order of the initial steps.

- ***If you are alone and witness a victim suddenly collapse:*** Assume cardiac arrest with a shockable rhythm. If you can get an AED quickly, you may activate EMS, leave the victim to get an AED, provide CPR for 2 minutes, and use the AED.
- ***If you are alone and find an unresponsive adult***: Tailor response to the prospective cause of injury.
    - If you suspect cardiac arrest: Activate EMS, get AED, 2 min of CPR, use AED
    - If you suspect asphyxia: 2 min of CPR, Activate EMS, get AED, use AED

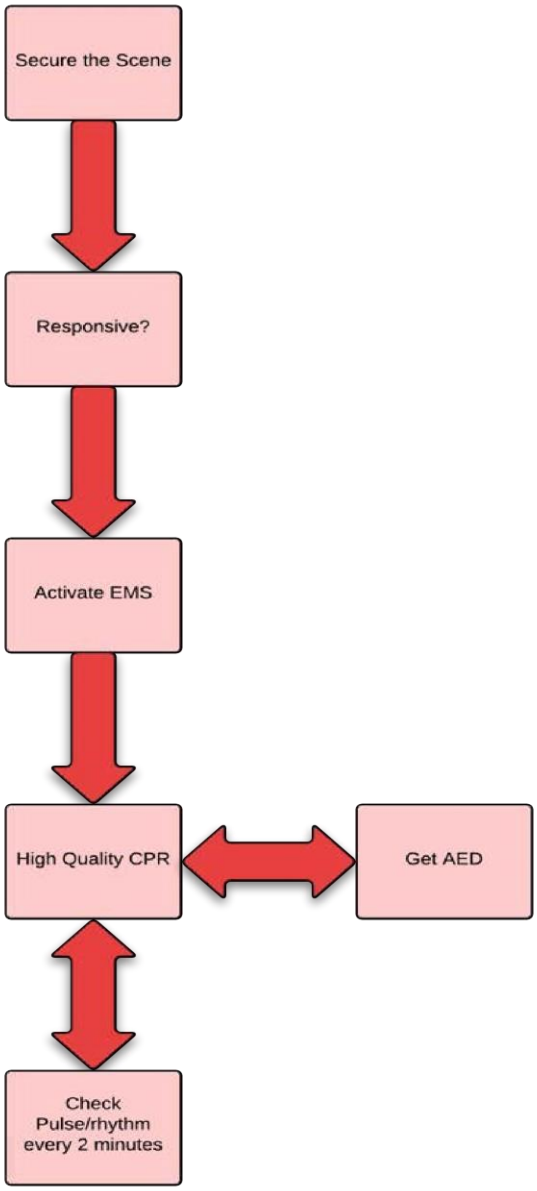
- Always make sure that you are safe and the victim is safe before you start BLS.

- Check to see if the victim is responsive. Shake and shout! Is the victim breathing effectively? Does the victim have a pulse in the carotid artery?

- If you witnessed the victim suddenly collapse, assume cardiac arrest with a shockable rhythm. If you can get an AED quickly, you may activate EMS, leave the victim to get an AED, CPR for 2 minutes, and use AED.

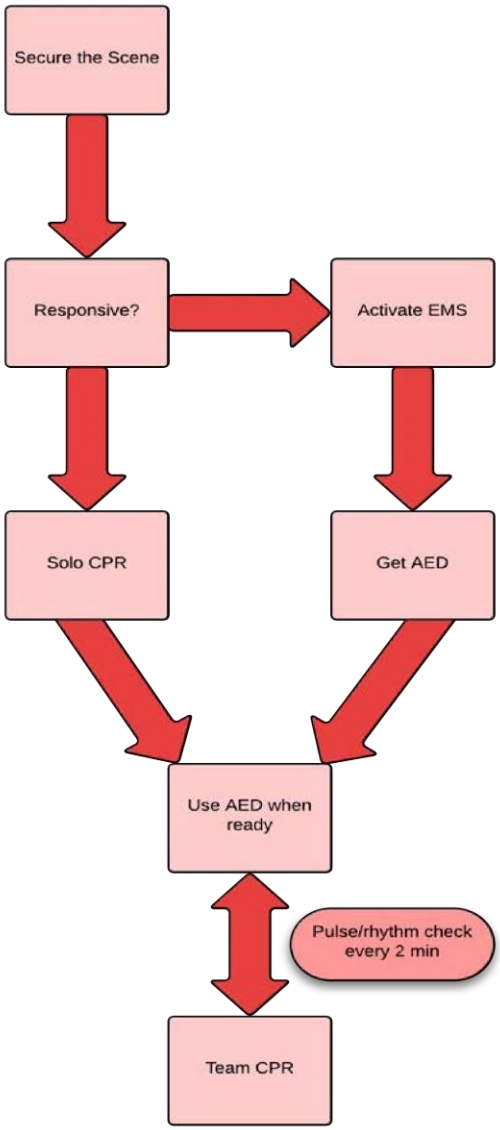
- If you find an unresponsive adult, tailor response to the presumed cause of injury.
  - If you suspect cardiac arrest: Activate EMS, get AED, 2 min of CPR, use AED
  - If you suspect asphyxia: 2 min of CPR, Activate EMS, get AED, use AED

- High Quality CPR includes
  - Fast and deep compressions, 100 to 120 compressions per minute
  - Two inches deep, complete rebound
    - If you can provide respiration, 2 breaths for 30 comps
    - If you cannot provide respiration, just give chest comps



- Check for a pulse and cardiac rhythm every two minutes. Follow directions on the AED. After providing a shock, immediately resume CPR. Keep going until EMS arrives or the victim regains circulation.

- Always make sure that your team is safe and the victim is safe before you start BLS.



- Check to see if the victim is responsive. Shake and shout! Is the victim breathing effectively? Does the victim have a pulse in the carotid artery?

- One provider activates EMS and retrieves an AED. The other provider(s) stays with the victim.

- Provide High Quality CPR includes
  - Fast and deep compressions, 100 to 120 compressions per minute
  - Two inches deep, complete rebound
    - If you can provide breaths, 2 breaths for 30 comps
    - If you cannot provide breaths, just give chest comps

- The provider who retrieved the AED applies the AED and follows directions given by the device. The provider that stayed with the victim provides CPR until the AED is ready.

- Check for a pulse and cardiac rhythm every two minutes. Follow directions on the AED. If a shock is indicated, clear everyone and administer a shock. After providing a shock, immediately resume Team CPR.

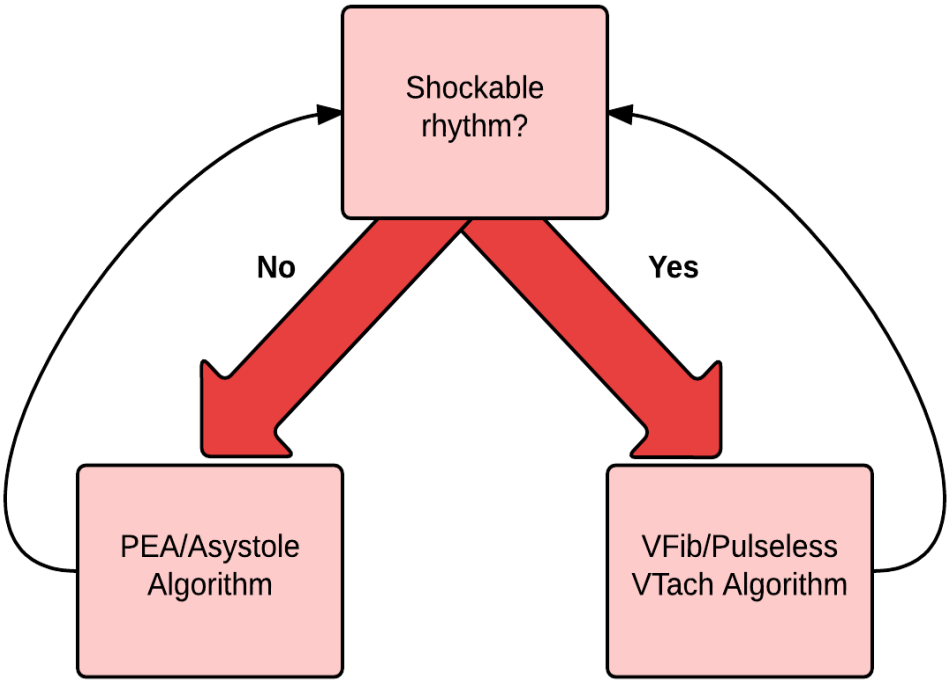
- In Team CPR, the provider giving chest compressions changes every 2 minutes

- Keep going until EMS arrives or the victim regains spontaneous circulation.

Cardiac Arrest

Cardiac arrest is the sudden sensation cessation of blood flow to the tissues in brain the results from a heart that is not pumping effectively. Four rhythms may occur during cardiac arrest: ventricular fibrillation, pulseless ventricular tachycardia, pulseless electrical activity, and asystole. The primary intervention for ventricular fibrillation and pulseless ventricular tachycardia is unsynchronized cardioversion, more commonly known as a "shock." The primary intervention for pulseless electrical activity and asystole is pharmacological, beginning with the administration of epinephrine.

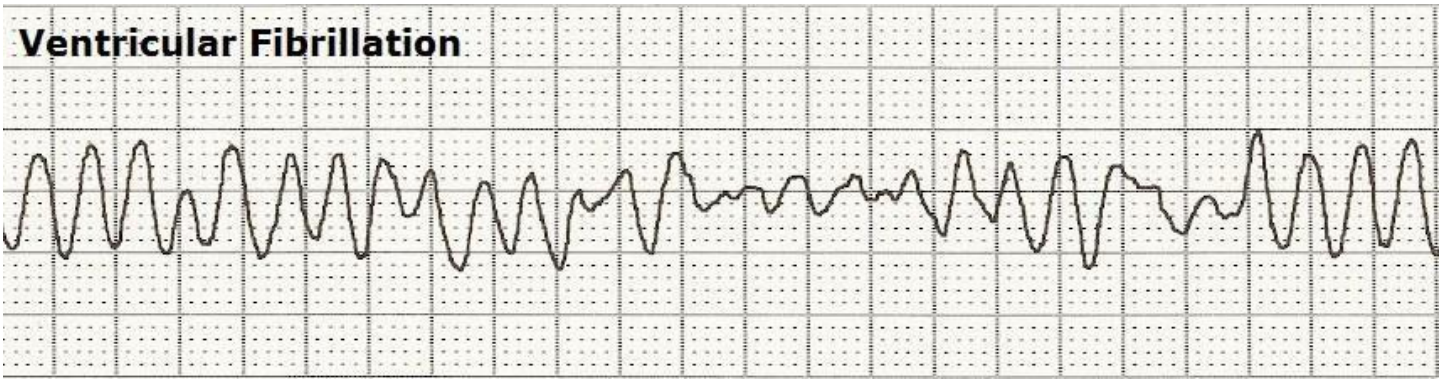
While ACLS provides algorithms for each of these cardiac arrest rhythms, in the real world a patient may move between these rhythms during a single instance of cardiac arrest. Therefore, the provider must be able to accurately assess and adapt to changing circumstances. After every 2 minutes of CPR, check for a pulse and check the cardiac rhythm. If the rhythm has switched from shockable or to shockable, then switch algorithms.



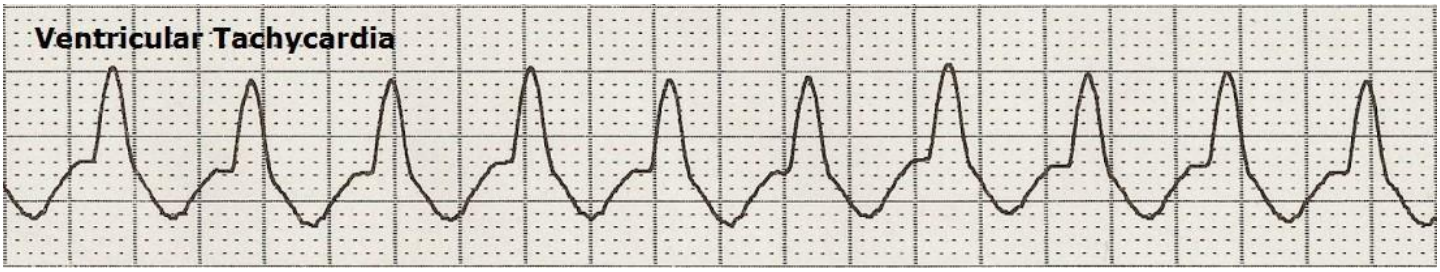
Ventricular Fibrillation and Pulseless Ventricular Tachycardia

In ventricular fibrillation or pulseless ventricular tachycardia, the heart's conduction system exhibits a disordered rhythm that can sometimes be corrected by applying energy to it. This energy may come in the form of an automated external defibrillator (AED) defibrillator paddles, or defibrillator pads. VFib and VTach are treated with unsynchronized cardioversion, since there is no way for the defibrillator to decipher the disordered waveform. In fact, it is important **not** to provide synchronized shock for these rhythms.

Ventricular fibrillation is recognized by a disordered waveform, appearing as rapid peaks and valleys as shown in this ECG rhythm strip:



Ventricular tachycardia may provide waveform similar to any other tachycardia; however, the biggest difference in cardiac arrest is that the patient will not have a pulse and, consequently, will be unconscious and unresponsive. Two examples of ventricular tachycardia are shown in this ECG rhythm strips. The first is narrow complex tachycardia and the second is wide complex tachycardia:





Ventricular Fibrillation and Pulseless Ventricular Tachycardia Algorithm

- Once you have determined that a patient has a shockable rhythm, immediately provide an unsynchronized shock.

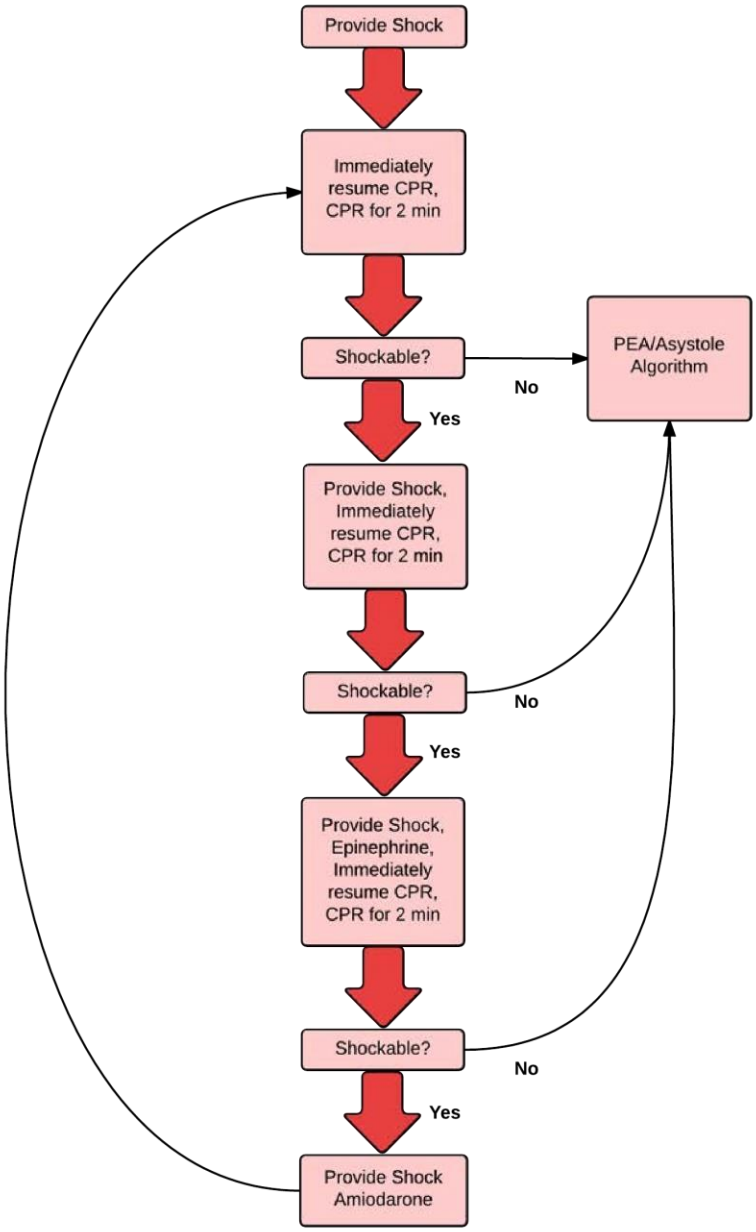
- If you are using **biphasic** energy, use recommended settings on the device. If you do not know what that setting is, use the highest available setting, (120 to 200 J).

- If you are using a **monophasic** energy source, administer 360 J.

- Resume CPR immediately after a shock. Minimize interruptions of chest compressions. Provide 2 rescue breaths for each 30 compressions.

- Epinephrine (1 mg IV/IO) is given every 3 to 5 minutes (two 2 minute cycles of CPR)

- Vasopressin (40 units IV/IO) can be used instead of the first or second dose of epinephrine



- Amiodarone (IV/IO)
  - First dose 300 mg
  - Second dose 150 mg

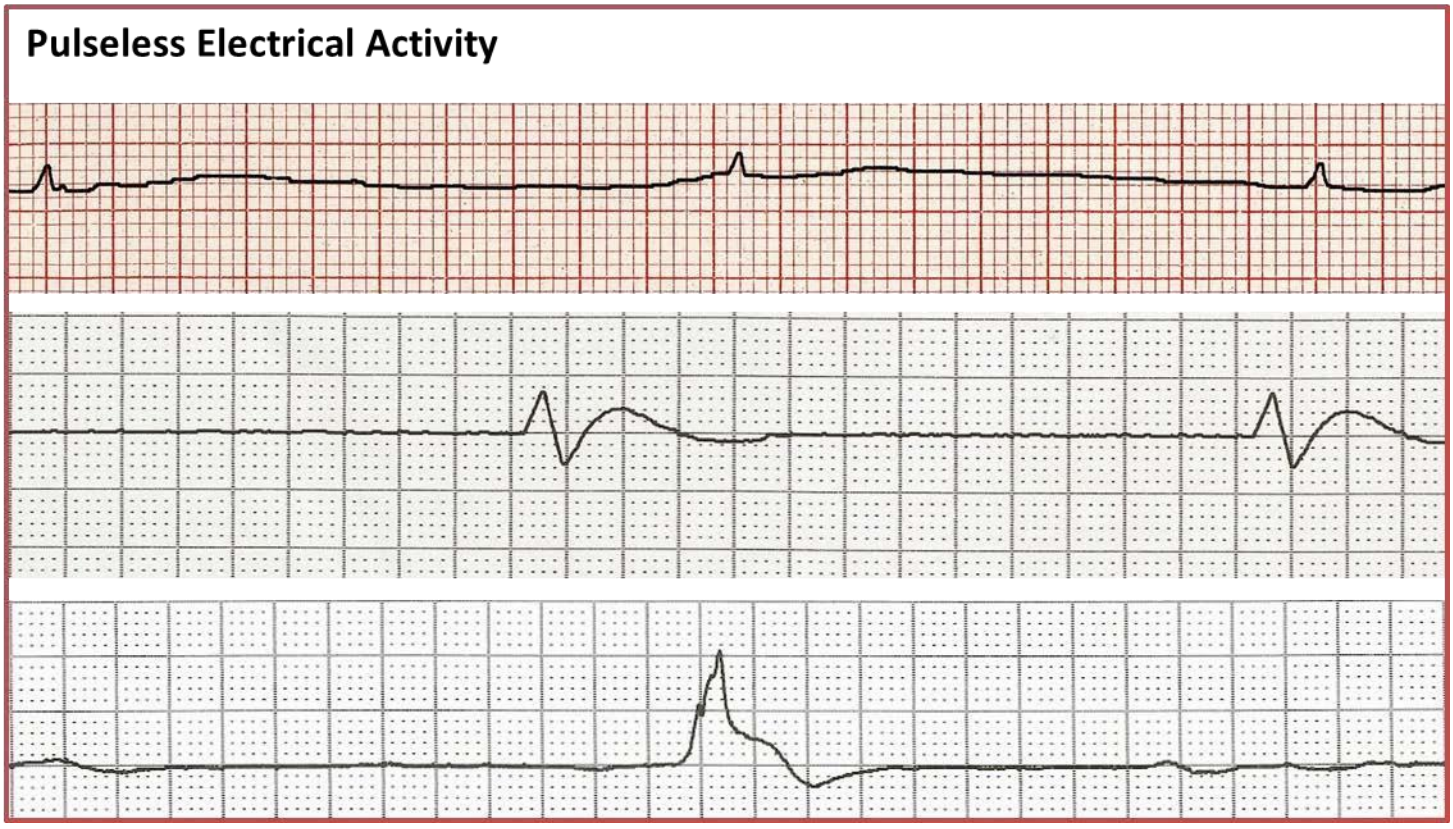
☐Lidocaine may replace amiodarone when amiodarone is not available.
  - First dose: 1-1.5 mg/kg IV
  - Second dose: 0.5-0.75 mg/kg IV every 5 to 10 min

☐If the arrest rhythm is no longer shockable, move to PEA/Asystole algorithm

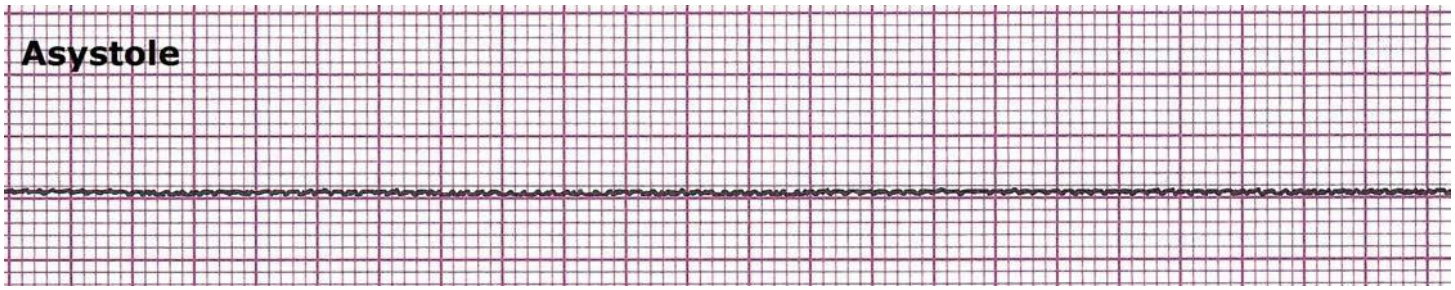
☐If the patient regains consciousness, move to ROSC algorithm for him and him

Pulseless Electrical Activity and Asystole

Pulseless electrical activity or PEA is a cardiac rhythm that does not create a palpable pulse is even though it should. A PEA rhythm can be almost any rhythm except ventricular fibrillation (incl. torsade de pointes) or pulseless ventricular tachycardia.



Asystole is the "flatline" on the ECG monitor. It represents a lack of electrical activity in the heart. It is critically important not to confuse true asystole with disconnected leads or an inappropriate gain setting on an in-hospital defibrillator. Asystole may also masquerade as a very fine ventricular fibrillation. If the ECG device is optimized and is functioning properly, a flatline rhythm is diagnosed as asystole. Note that asystole is also the rhythm one would expect from a person who has died. Consider halting ACLS efforts in people who have had prolonged asystole.



It is inappropriate to provide a shock to pulseless electrical activity or asystole. Cardiac function can only be recovered in PEA or asystole through the administration of medications.

☐ As long as the patient is in PEA or asystole, the rhythm is not shockable.

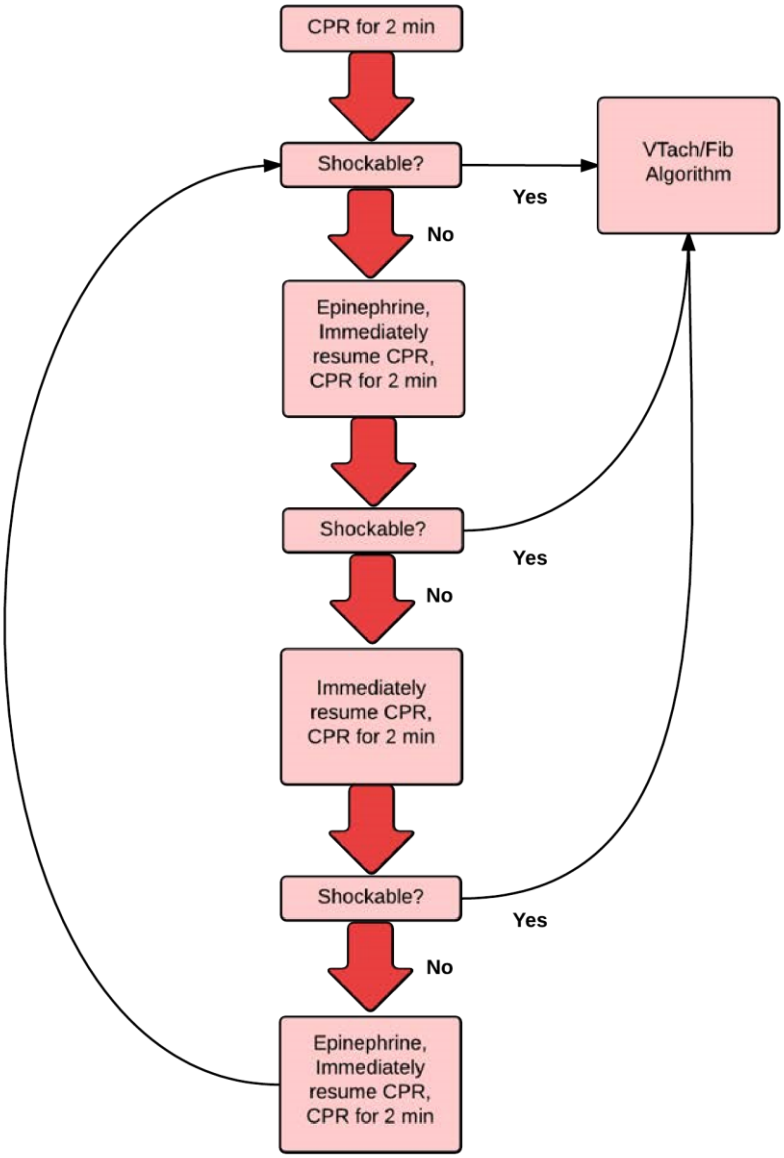
☐ Chest compressions/high-quality CPR should be interrupted as little as possible during resuscitation.

☐ After 2 min. of high-quality CPR, give 1 mg of epinephrine IV/IO. Consider advanced airway and waveform capnography.

☐ Remember, chest compressions are a means of artificial circulation, which should deliver the epinephrine to the heart. Without chest compressions, epinephrine is not likely to be effective.

☐ Chest compressions should be continued while epinephrine is administered.

☐ Rhythm checks every 2 min.



☐Epinephrine (1 mg IV/IO) is given every 3 to 5 minutes (after two, 2 minute cycles of CPR)

☐Vasopressin (40 units IV/IO) can be used instead of the first or second dose of epinephrine

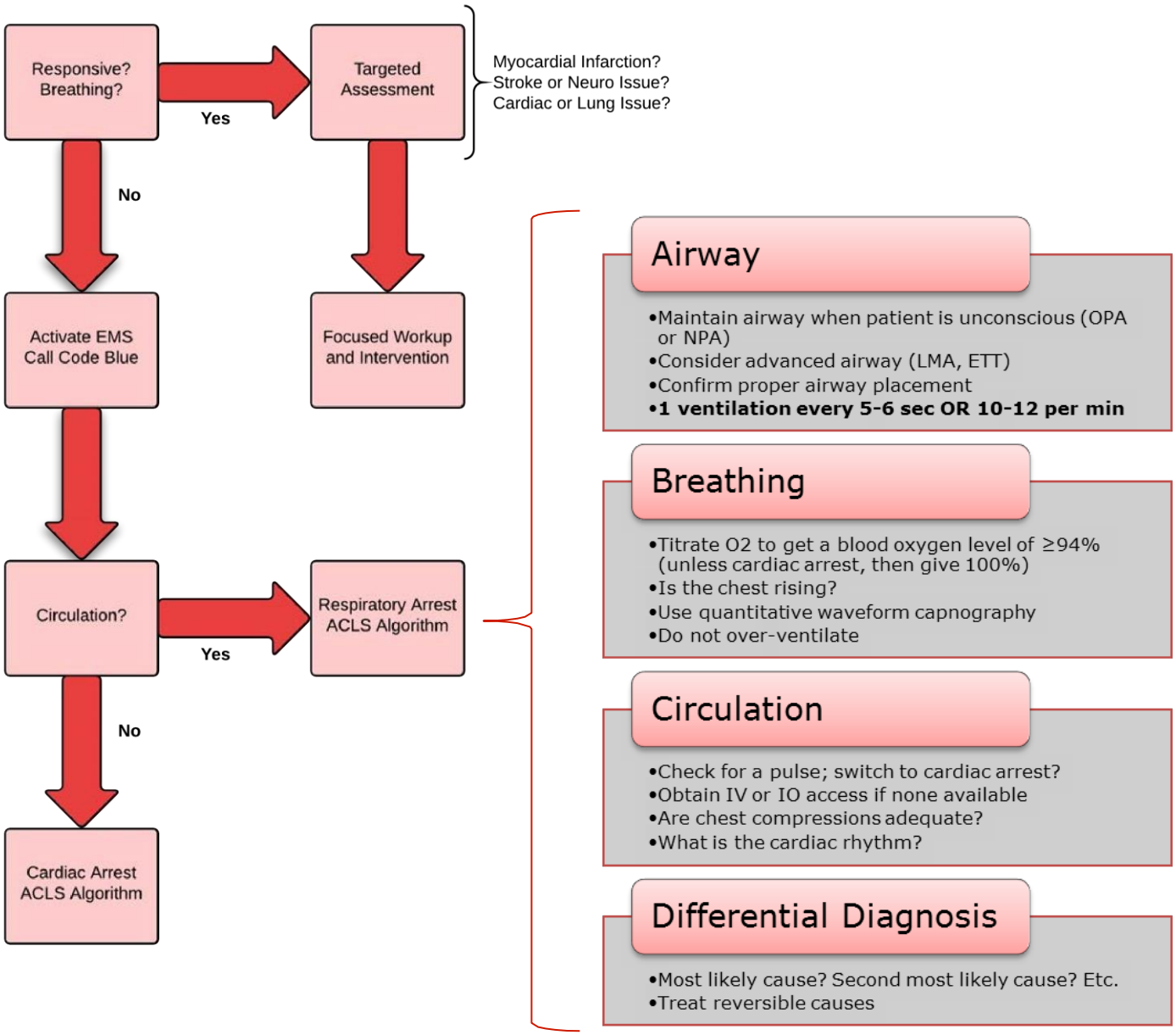
☐If the arrest rhythm becomes shockable, move to VFib/Pulseless VTach algorithm

☐If the patient regains circulation, move to ROSC algorithm

Respiratory Arrest

While cardiac arrest is more common in adults than respiratory arrest, there are times when patients will have a pulse but are not breathing or not breathing *effectively* (e.g., agonal breathing). A person who has a pulse but is not breathing effectively is in respiratory arrest.

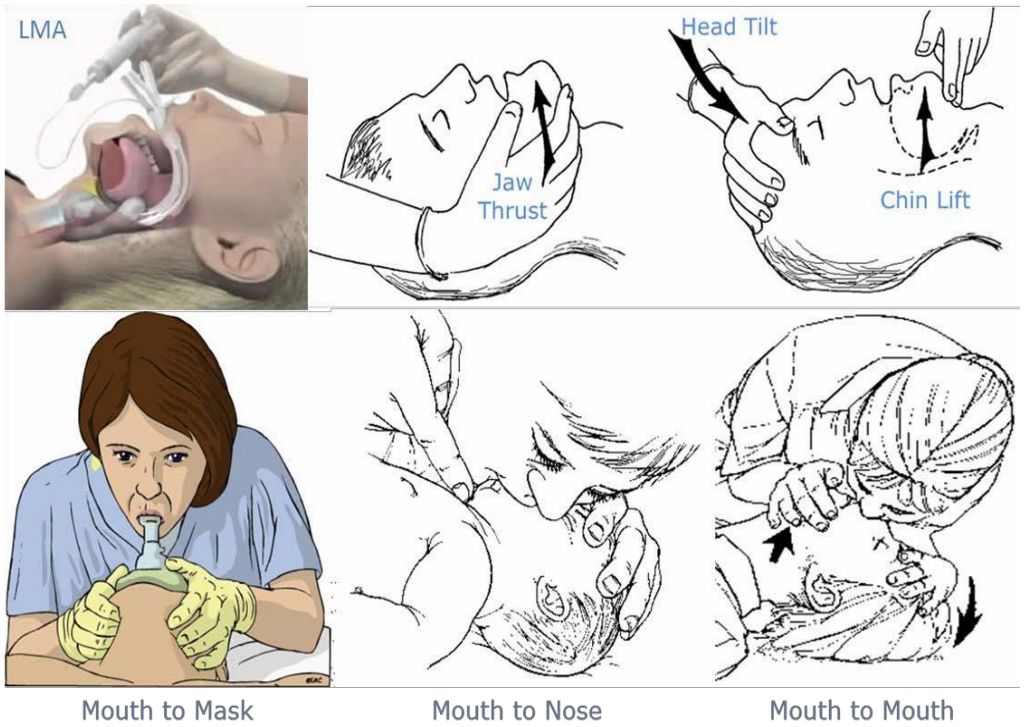
When you encounter a patient in need, you will not know he or she is in respiratory arrest, so perform a BLS assessment:



Airway Management

In ACLS, the term airway is used to refer both to the pathway between the lungs and the outside world and victim in the devices that help keep that airway open. The simplest way to "manage an airway" is the head tilt-chin lift, which creates the straightest passageway for air to flow into the lungs. As if the victim may have experienced head or neck trauma, airway





management should include a jaw thrust, which leaves the head and neck unmoved, but which opens up the airway. Current guidelines recommend that ventilations for a victim with pulses should be delivered every 5 to 6 seconds. It is critical that the provider avoid delivering excessive ventilation which can lead to less effective circulation and cerebral vasoconstriction.

If one is to use a pocket mask or a bag mask to perform ventilations, it is important to make a tight seal with a mask on the victim's face. Proper use of these masks may require the rescuer to use one or even two hands to secure the mask to the victim's face.

A nasopharyngeal airway, which extends from the nose to the pharynx, can be used in both conscious and unconscious patients. An oropharyngeal airway can only be used in unconscious patients because it may stimulate the gag reflex.

Advanced airways such as endotracheal tubes (ET tubes) and laryngeal mask airways (LMAs) usually require specialized training, but are useful in-hospital resuscitations (especially LMAs).

While nasopharyngeal and oropharyngeal airways are basic airways, they do require a bit of preparation and skill to use correctly.

Inserting an Oropharyngeal Airway

- Select an airway that is the correct size for the patient
  - Too big and it will damage the throat
  - Too small and it will press the tongue into the airway
- Place the device at the side of the patient's face. Choose the device that extends from the corner of the mouth to the earlobe
- Clear the mouth or blood or secretions with suction, if possible.
- Insert the device so that the point is toward the roof of the mouth or parallel to the teeth
  - Do not press the tongue back into the throat
- Once the device is almost fully inserted, turn it so that the tongue is cupped by the interior curve of the device.

Inserting an Nasopharyngeal Airway

- Select an airway that is the correct size for the patient
- Place the device at the side of the patient's face. Choose the device that extends from the tip of the nose to the earlobe. Use the largest diameter device that will fit.
- Lubricate the airway with a water-soluble lubricant
- Insert the device slowly, straight into the face (not toward the brain!)
- It should feel snug; do not force the device. If it feels stuck, remove it and try the other nostril.

Tips on Suctioning

- Adequate suctioning usually requires negative pressures of -80 to -120 mmHg. Wallmounted suction can deliver this, but portable devices may not.
- When suctioning the oropharynx, do not insert the catheter too deeply. Extend the catheter to the maximum safe depth and suction as you withdraw.
- When suctioning an endotracheal tube, remember that the tube is within the trachea and you may be suctioning near the bronchi/lung. Therefore sterile technique should be used.
- Each suction attempt should be for no longer than 10 seconds. Prior to suctioning, give a brief period of 100% oxygen—remember that the patient will get no oxygen during suctioning.
- Monitor vital signs during suctioning and stop suctioning immediately if the patient experiences hypoxemia (O_2 sats <94%), has a new arrhythmia, or becomes cyanotic.

Return of Spontaneous Circulation (ROSC) and Post Arrest Care

☐ The patient who has been successfully resuscitated will regain spontaneous circulation.

☐ You can detect spontaneous circulation by feeling a palpable pulse at the carotid artery.

☐ Even after Return of Spontaneous Circulation (ROSC), the patient still needs close attention and support. The patient is at risk for reentering cardiac arrest at any time. Therefore, the patient should be moved to an intensive care unit.

➢ Titrate the patient's blood oxygen levels to ≥94%

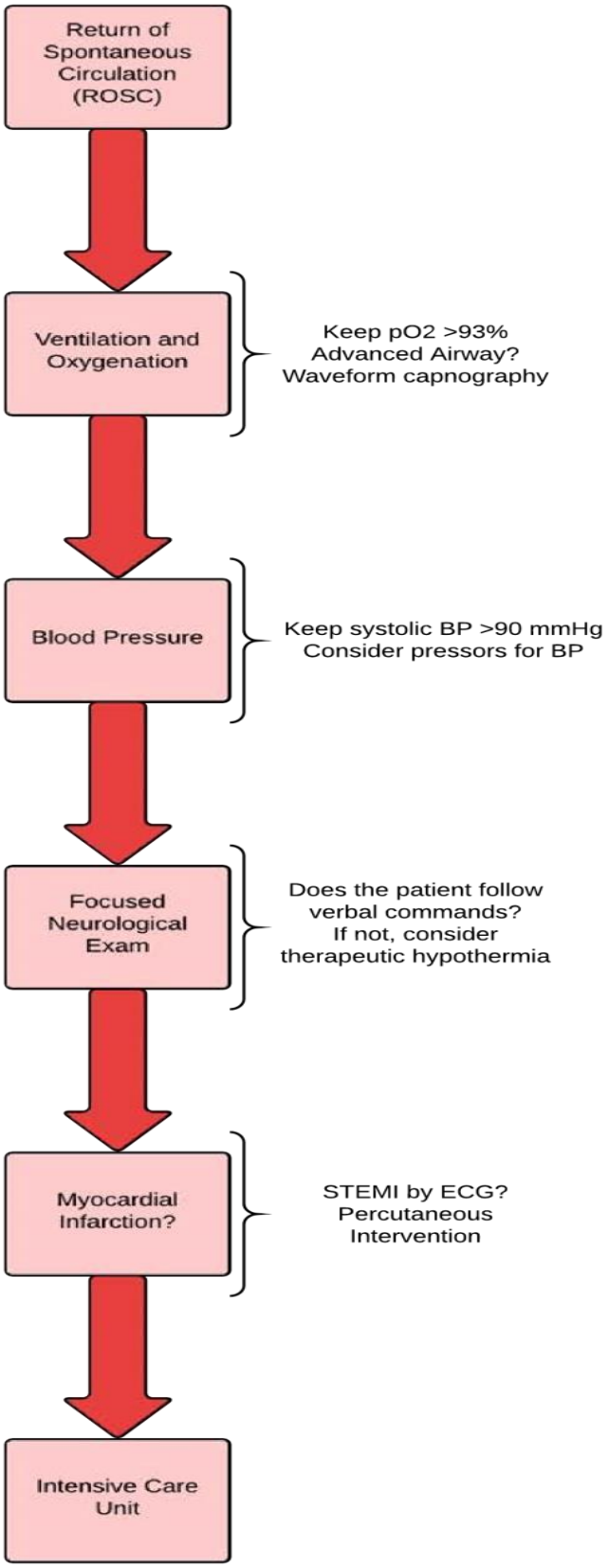
☐ Does the person need an advanced airway? If so, it should be placed.

☐ Add quantitative waveform capnography

☐ Titrate the patient's systolic blood pressure to at least 90 mmHg. this may require the addition of fluids and/or vasopressors ○ Epinephrine IV 0.1-0.5 mcg/kg/min ○ Dopamine IV 5-10 mcg/kg/min ○ Norepinephrine IV 0.1-0.5 mcg/kg/min ○ 1-2 liters of fluid resuscitation

➢ Does the person follow verbal commands? If not, there may be neurological compromise. Consider inducing therapeutic hypothermia with 4°C fluids during fluid resuscitation.

☐ Does the person have signs of myocardial infarction by ECG? Move to ACS algorithm.



Rapid Differential Diagnosis of Cardiac Arrest

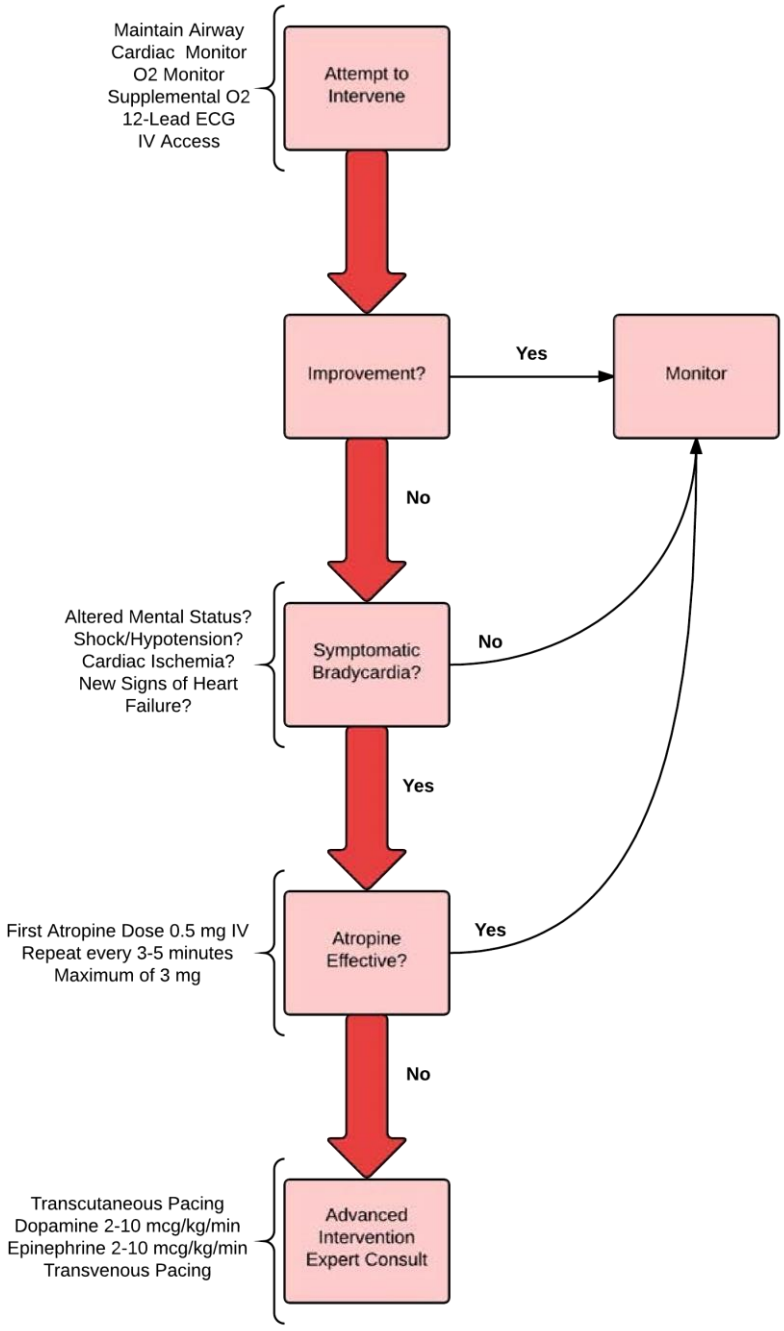
Many different disease processes and traumatic events can cause cardiac arrest, but in an emergency, it is important to be able to rapidly consider and eliminate or treat the most typical causes of cardiac arrest. To facilitate remembering the main, reversible causes of cardiac arrest, they can be organized as the Hs and the Ts.

The Hs	Symptoms/Signs/Tests	Intervention
Hypovolemia	Rapid heart rate, narrow QRS complex,	Fluid resuscitation
Hypoxia	Decreased heart rate	Airway management, oxygen
Hydrogen Ion (Acidosis)	Low amplitude QRS complex	Hyperventilation, sodium bicarb
Hypoglycemia	Fingerstick glucose testing	IV Dextrose
Hypokalemia	Flat T waves, pathological U wave	IV Magnesium
Hyperkalemia	Peaked T waves, wide QRS complex	Calcium chloride, sodium bicarb, insulin/glucose, hemodialysis
Hypothermia	History of cold exposure	Rewarming blankets/fluids

The Ts	Symptoms/Signs/Tests	Intervention
Tension Pneumothorax	Slow heart rate, narrow QRS complex, acute dyspnea, history of chest trauma	Thoracotomy, needle decompression
Tamponade (Cardiac)	Rapid heart rate and narrow QRS complex	Pericardiocentesis
Toxins	Variable, prolonged QT interval, neuro deficits	Antidote/antivenom (toxin-specific)
Thrombosis (pulmonary)	Rapid heart rate, narrow QRS complex	Fibrinolytics, embolectomy
Thrombosis (coronary)	ST segment elevation/depression, abnormal T waves	Fibrinolytics, Percutaneous intervention

Bradycardia

Bradycardia Algorithm



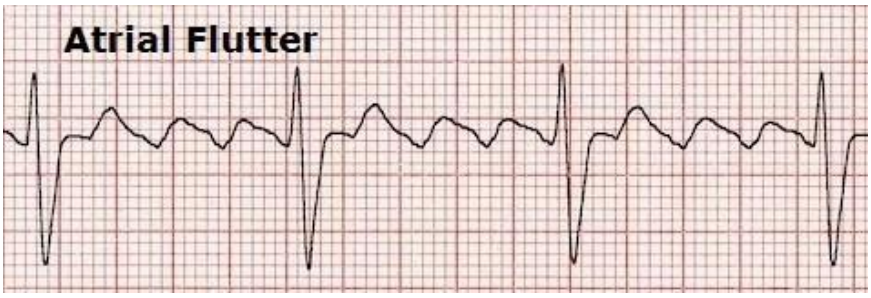
- ☐ Bradycardia is any heart rate less than 60 bpm. In practice, however, bradycardia is only a concern if it is unusual or abnormal for the patient or causing symptoms.
- ☐ New cases of bradycardia should be evaluated, but most will not require specific treatment.
- ☐ Evaluation of bradycardia includes cardiac and blood oxygen monitoring and a 12 lead ECG if available.
- ☐ Bradycardia may be treated by providing supplemental oxygen and supporting the patient's airway if needed.
- ☐ Unstable bradycardia (i.e., an abnormally slow heart rate that causes altered mental status, hypotension, symptoms of shock, cardiac chest pain, or new signs and symptoms of heart failure) should be treated immediately.
- ☐ Unstable bradycardia is first treated with intravenous atropine at a dose of 0.5 mg. Additional doses can be given every 3 to 5 min. up to a maximum of 3 mg. Pulseless bradycardia is considered PEA.
- ☐ If atropine is unsuccessful in treating symptomatic, unstable bradycardia, consider transcutaneous pacing, dopamine or epinephrine infusion, or transvenous pacing. An intensive or cardiologist may need to be consulted for these interventions and the patient may need to be moved to the intensive care unit

Tachycardia

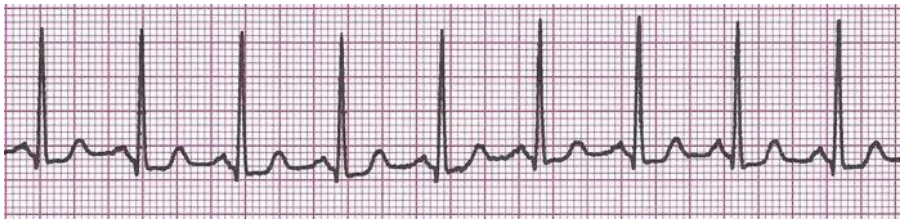
Atrial fibrillation is the most common arrhythmia. It is diagnosed by electrocardiogram, specifically the RR intervals follow no repetitive pattern. Some leads may show P waves while most leads do not. Atrial contraction rates may exceed 300 bpm. The ventricular rate often range is between 100 to 180 bpm. The pulse may be "irregularly irregular."



Atrial flutter is a cardiac arrhythmia that generates rapid, regular atrial depolarizations at a rate of about 300 bpm. This often translates to a regular ventricular rate of 150 bpm, but may be far less if there is a 3:1 or 4:1 conduction. By electrocardiogram, or atrial flutter is recognized by a sawtooth pattern sometimes called F waves. These waves are most notable in leads II, III, and aVF.



Narrow QRS complex tachycardias include several different tachyarrhythmias. A narrow QRS complex tachycardia is distinguished by a QRS complex of less than 120 ms. One of the more common narrow complex tachycardias is supraventricular tachycardia, shown below.



Wide complex tachycardias are difficult to distinguish from ventricular tachycardia. Ventricular tachycardia leading to cardiac arrest should be treated using the ventricular tachycardia algorithm. A wide complex tachycardia in a conscious person should be treated using the tachycardia algorithm.

Tachycardia Algorithm

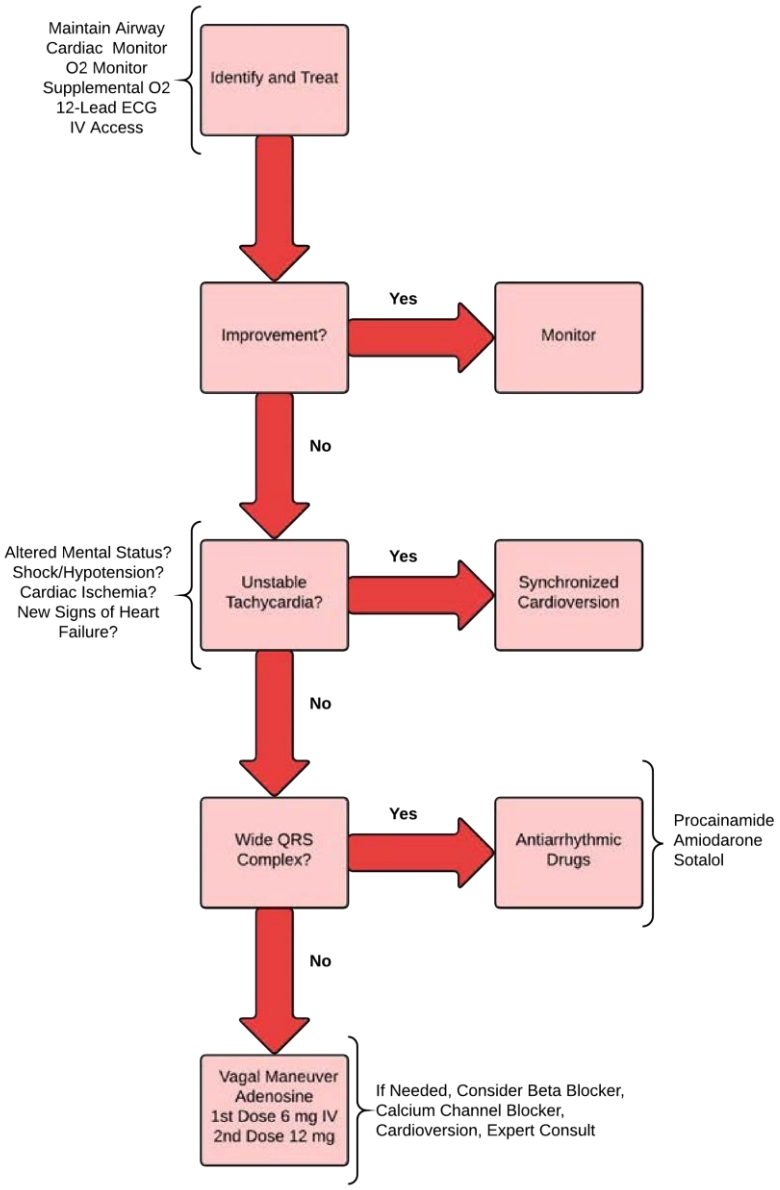
- ☐ Tachycardia is any heart rate greater than 100 bpm. In practice, however, tachycardia is usually only a concern if it is <150 bpm or causing symptoms.

- ☐ New cases of tachycardia should be evaluated with cardiac and blood oxygen monitoring and a 12 lead ECG if available.

- ☐ Tachycardia may be treated by providing supplemental oxygen, supporting the patient's airway if needed, vagal maneuvers, and IV adenosine.

- ☐ Adenosine IV rapid push
  - First dose: 6 mg
  - Second dose: 12 mg

- ☐ Unstable tachycardia (i.e., an abnormally slow heart rate that causes altered mental status, hypotension, symptoms of shock, cardiac chest pain, or new signs and symptoms of heart failure) should be treated with synchronized cardioversion or adenosine. Consider beta-blocker or calcium channel blocker.



☐ Wide QRS tachycardia may require antiarrhythmic drugs.

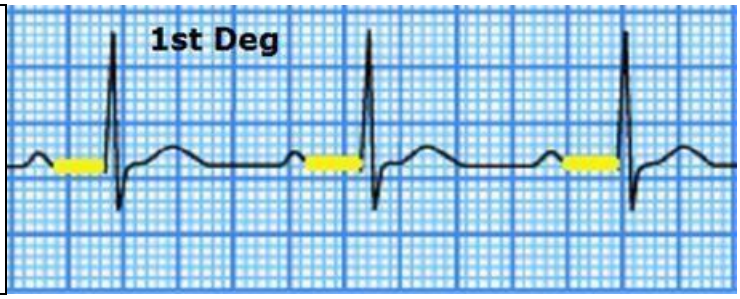
Procainamide	Amiodarone	Sotalol
20-50 mg IV, stop if hypotensive or max dose of 17 mg/kg. Maintenance infusion at 1-4 mg/min.	150 mg IV over 10 minutes with second dose for recurrent VT. Maintenance at 1 mg/min for 6 h	100 mg (1.5 mg/kg) over 5 min.

Atrioventricular (Heart) Block

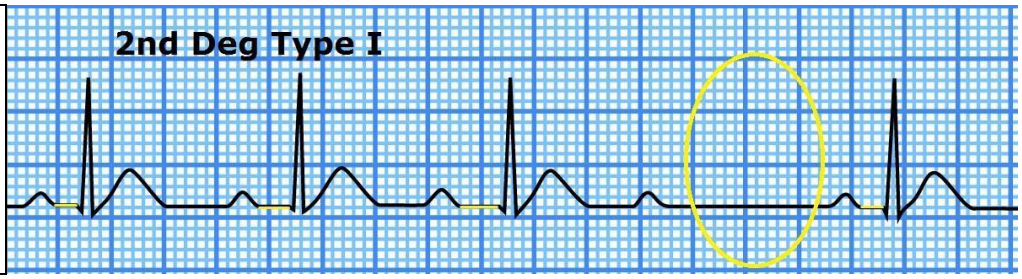
Atrioventricular block or heart block is a failure of the heart's electrical system to properly coordinate conduction. There are four main types of atrioventricular block: first degree, second degree type I, second degree type II, and third degree heart block. The types of second degree heart block are referred to as Mobitz type I and Mobitz type II. Second degree heart block Mobitz type I is also known as the Wenckebach phenomenon.

Atrioventricular blocks may be acute or chronic. Chronic heart block may be treated with pacemaker devices. From the perspective of ACLS assessment and intervention, heart block is important because it can cause hemodynamic instability and can evolve into cardiac arrest. In ACLS, heart block is often treated as a bradyarrhythmia.

The PR interval is a consistent size, but longer or larger than it should be in first degree heart block.



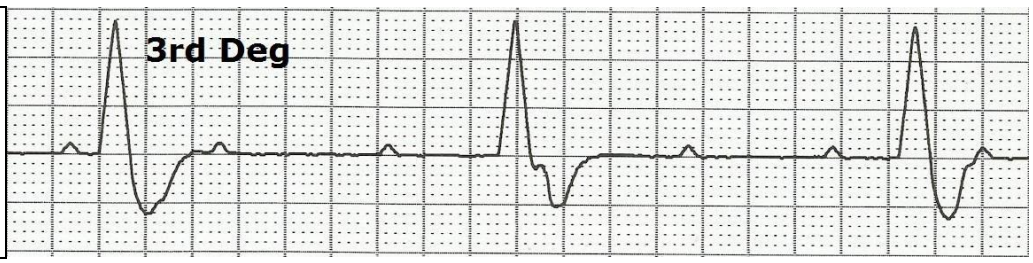
The PR interval increases in size until a QRS complexes dropped, resulting in missed "beat."



A QRS wave will occasionally drop, though the PR interval is the same size.



Complete dissociation between P waves and the QRS complex. No atrial impulses reach the ventricle.



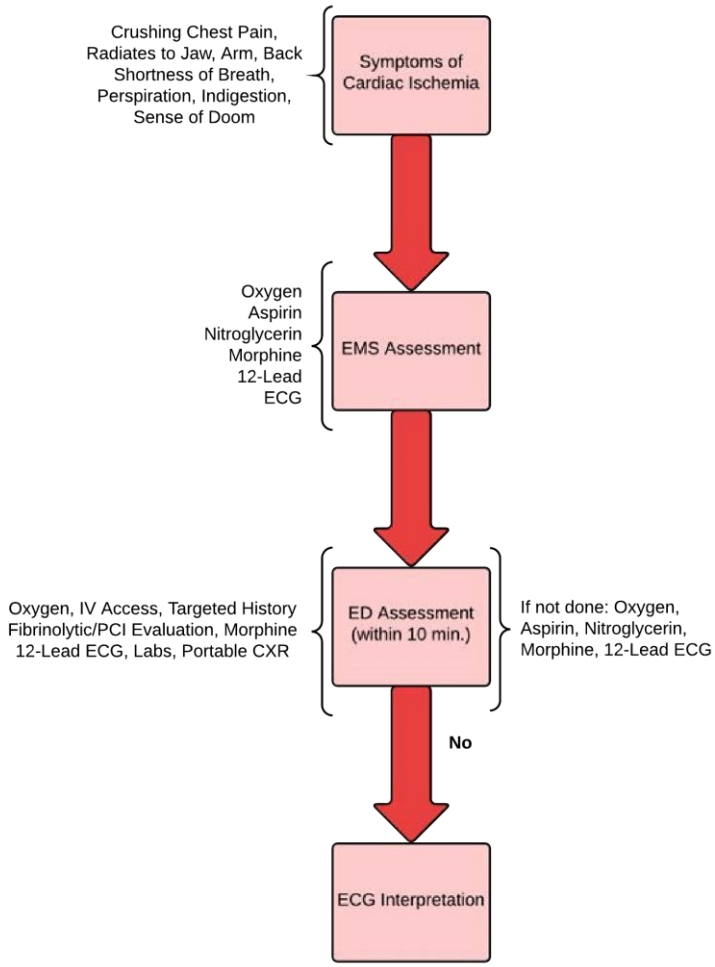
Acute Coronary Syndrome

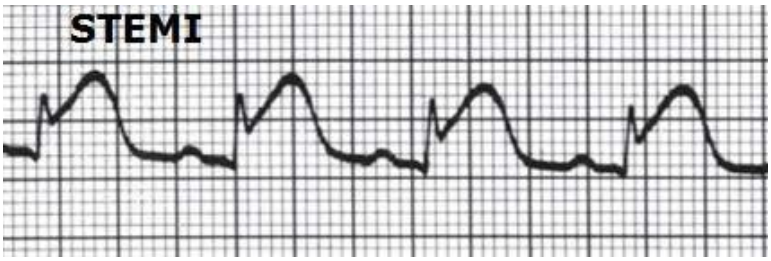
- Acute coronary syndrome or ACS is a spectrum of signs and symptoms ranging from angina to myocardial infarction. ACS includes ST segment elevation myocardial infarction (STEMI) non- ST segment elevation myocardial infarction (NSTEMI), and unstable angina.

- Cardiac chest pain (any new chest discomfort) should be evaluated promptly. This includes high degree of suspicion by individuals in the community, prompt action by EMS personnel, rapid assessment in the emergency department, and definitive treatment.

- People with symptoms of cardiac ischemia should be given oxygen, aspirin (if not allergic), nitroglycerin, and possibly morphine. Obtain a 12 lead ECG ASAP.

- The patient should be assessed in the ED within 10 min. of arrival. Draw and send labs (e.g., cardiac enzymes, coags), Obtain IV access. Give statin (if not contraindicated). Obtain chest Xray.





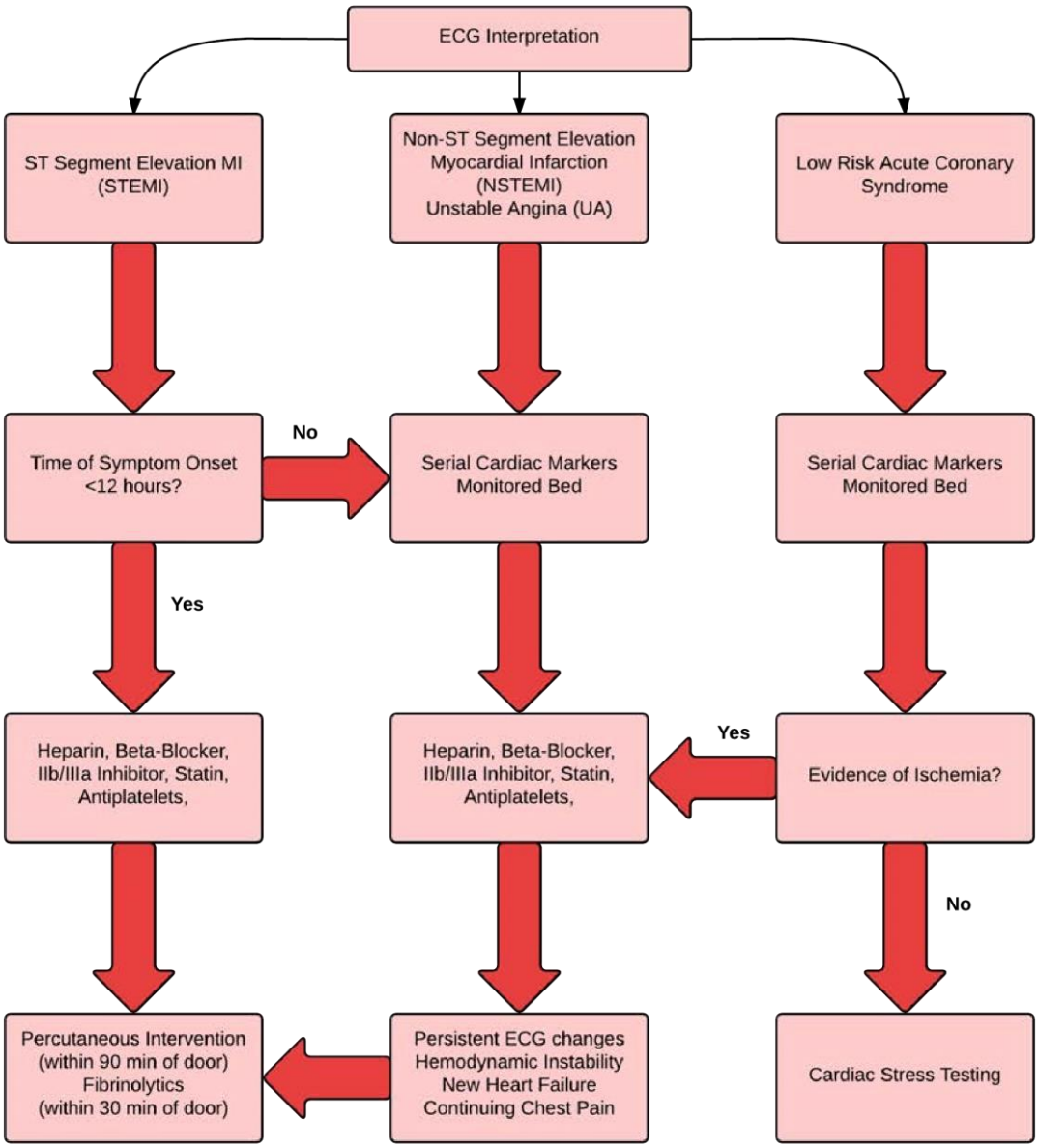
The results of the ECG will be the primary guidance for how the patient with possible cardiac chest pain is managed. The ECG diagnosis of acute coronary syndrome can be complex. STEMI is recognized by ST segment elevation with/without pathological Q waves.

- NSTEMI can be a more challenging electrocardiographic diagnosis. It may result in ST segment depression, "flipped" T waves (T wave flattening or inversion), peaked T waves, U wave inversion, and bundle branch block. The electrocardiographic of diagnosis of an NSTEMI is beyond the scope of ACLS.

☐ Unstable angina is new onset cardiac chest pain without ECG changes, angina that occurs at rest and lasts for more than 20 min., and/or angina that has become rapidly and progressively worse.

☐ STEMI and NSTEMI patients will have elevated cardiac markers in the blood (e.g. troponins) several hours after the acute event. People with unstable angina will not have elevated cardiac markers.

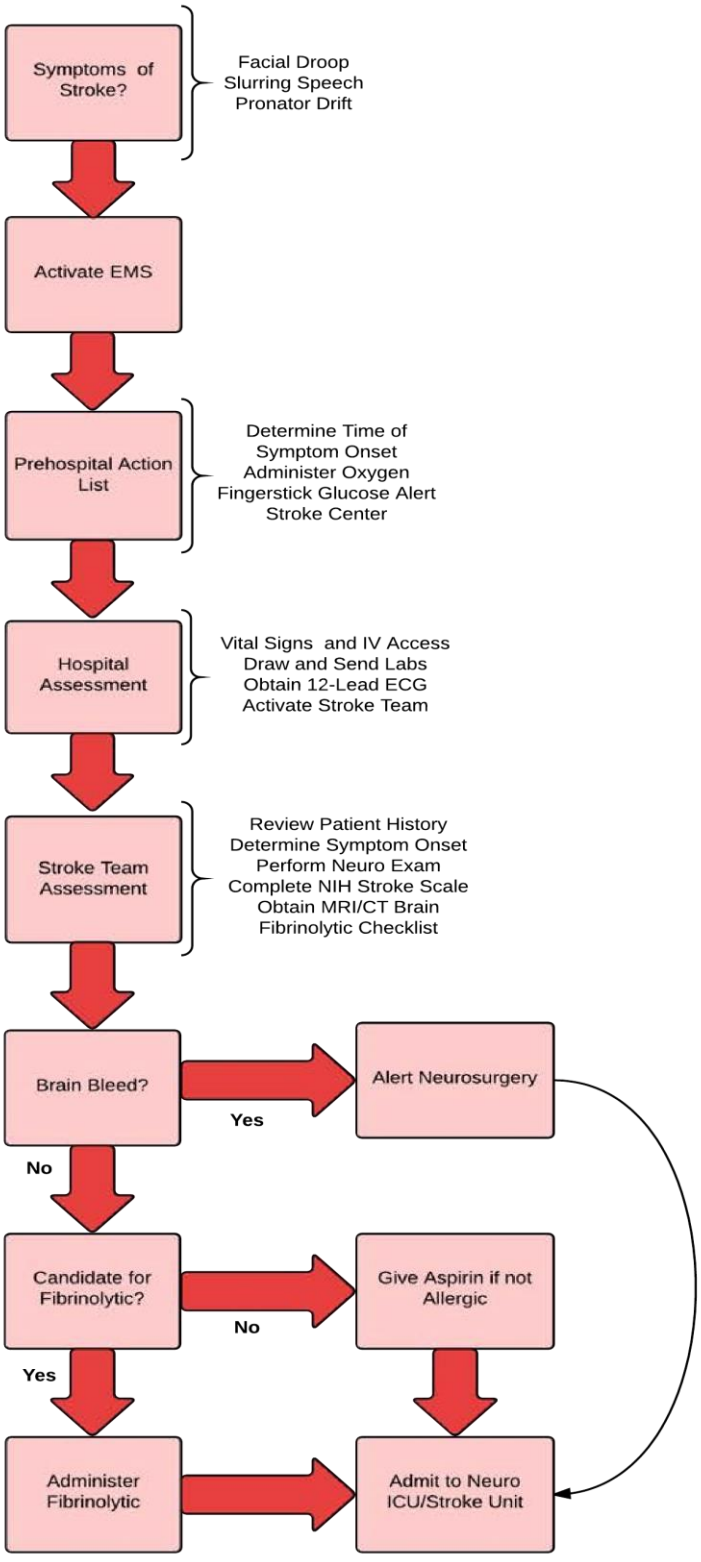
Acute Coronary Syndrome Algorithm



- ☐ STEMI patients should be treated per hospital protocol. This may include anti-platelet drug(s), anticoagulation, a beta-blocker, an ACE inhibitor, a statin, and either PCI or a fibrinolytic.
- ☐ NSTEMI is treated with medical therapy as above without a PCI or fibrinolytic, unless they do not improve with medical therapy.
- ☐ Patients with unstable angina are admitted and monitored for evidence of MI. Patients who do not "rule in" (develop MI) can undergo cardiac stress testing the next day or as an outpatient.

Acute Stroke

The Cincinnati Prehospital Stroke Scale	
Task	**Abnormal**
Ask the patient to smile	Face droops on one side
Ask the patient to say *"you can't teach an old dog new tricks"*	Speech slurs
Ask patient to close eyes and extend both arms inward (pronator drift) straight out, palms up	One arm drifts lower and turns
If one of these tasks is abnormal, stroke probability is 72% If all three tasks are abnormal, stroke probability is >85%	



☐ The EMS team should take patients with suspected stroke to a stroke center. While in transit, the EMS team should try to determine the time at which the patient was last normal, which is considered the onset of symptoms. EMS administer oxygen via nasal cannula or face mask, obtain a fingerstick glucose measurement, and alert the stroke center.

☐ Within 10 min. of the patient's arrival at the hospital, personnel should assess the patient. They should obtain vital signs and IV access, draw and send labs (e.g. coags), obtain a 12-lead ECG, order CT, and perform a general assessment.

☐ Within 25 min. of the patient's arrival at the hospital, the stroke team should determine symptom onset, perform and narrow exam including the NIH stroke scale or equivalent, perform the fibrinolytic checklist, have the results of the CT scan of the brain.

☐ Within 45 min. of the patient's arrival at the hospital, the CT scan of the brain should be read for the presence of ischemic or hemorrhagic stroke.

☐ Within 60 min. of the patient's arrival at the hospital, fibrinolytic therapy should be administered in cases of ischemic stroke if the patient is a candidate. If the patient with an ischemic stroke is not a candidate for fibrinolytic, administer aspirin if the patient is not allergic. If the patient is having a hemorrhagic stroke, neurosurgery should be consulted.

☐ Within three hours of the patient's arrival at the hospital, the patient should be moved to the neurology/neurosurgery intensive care unit, stroke unit, or med/surg intensive care unit.

Time is Brain!Stroke Time Goals for Evaluation and Therapy

In people who are candidates for fibrinolytics, the goal is to administer the agent within 3 hours of the onset of symptoms.

Fibrinolytic Checklist	
Inclusion Criteria	**Exclusion Criteria**
Ischemic stroke with neurological deficit	Stroke/ head trauma in last 3 months
Onset of symptoms <3 hours	History of brain
Age >18 years old	Brain tumor, arteriovenous malformation, or aneurysm
	Brain or spine surgery in last
	Arterial line or blood draw in last week
	Possible subarachnoid hemorrhage
	Systolic ≥ 185 mmHg or diastolic ≥ 110 mmHg consistently
	Serum glucose <50 mg/dL (<2.8 mmol/L)
	Currently bleeding internally or bleeding diathesis
	Platelet <100,000/mm (if known)
	INR >1.7 or PT >15 seconds (if known)
	Elevated aPTT (if known)
	Currently taking anticoagulants
	Hemorrhage on CT
	Stroke includes > 1/3 of entire cerebrum
	Relative Exclusion Criteria
	Minor neurologic deficits
	Rapidly improving neurologic deficits
	Major surgery/serious trauma in last 2 weeks
	Gastrointestinal/urinary tract bleeding in last 3 weeks
	Heart attack in last 3 months
	Seizure with stroke with postictal period
	Pregnant

In select individuals, the window for fibrinolytics can be extended to 4.5 hours.

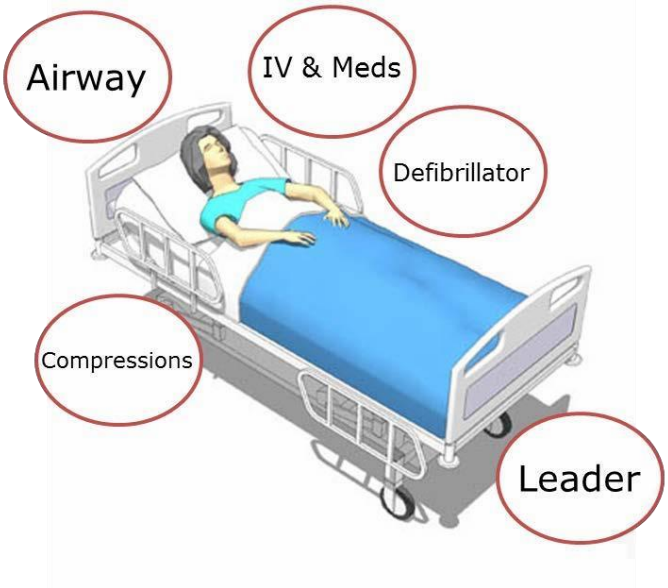
Fibrinolytic Checklist for 3 to 4.5 Hours from Symptom Onset	
Meets all criteria for fibrinolytic use at 3 hours plus these criteria:	
Inclusion Criteria	**Exclusion Criteria**
Ischemic stroke with neurological deficit	Currently taking anticoagulants
Onset of symptoms 3 to 4.5 hours	Severe stroke; NIH Stroke Scale score >25
Age 18 to 79 years old	Previous ischemic stroke and diabetes



Team Dynamics

The 2015 edition of the AHA ACLS guidelines continues to highlight the importance of effective team dynamics during resuscitation. ACLS in the hospital will be performed by several providers. These individuals must provide coordinated, organized care. Providers must organize themselves rapidly and efficiently. The AHA recommends establishing a Team Leader and several Team Members.
The Team Leader is usually a physician, ideally the provider with the most experience in leading ACLS codes. Resuscitation demands mutual respect, knowledge sharing, and constructive criticism, after the code.

Team Leader Responsibilities	Team Member Responsibilities
Usually stands at the foot of the bed	Stands in a position dictated by role
Competent in all ACLS duties	Competent in specific role (at least)
Directs Team Members in a professional, calm voice	Responds with eye contact and voice affirmation
Assigns roles	Clearly states when he/she cannot perform a role
Listens for confirmation from Team Member	Informs Team Leader when task is complete
Ask for ideas from Team Members when needed	Openly share suggestions if it does not disrupt flow
Critiques Team Performance after code	Provides constructive feedback after code
Documents resuscitation in patient chart	Provides information for documentation as needed



When performing a resuscitation, the Team Leader and Team Members should assort themselves around the patient so they can be maximally effective and have sufficient room to perform their role.

Resuscitation and Life Support Medications

Drug	Use(s)	Dosage/Route	Contraindications/Warnings
Adenosine	Supraventricular Tachycardia; May be used for other tachycardia rhythms	First dose: 6 mg IV push Second dose: 12 mg IV	Second or third degree heart block; tachycardia due to poisoning
Amiodarone	Pulseless ventricular tachycardia Ventricular fibrillation	First dose: 300 mg bolus Second dose: 150 mg Max: 2.2 grams/day	Second or third degree heart block; hypotension may result with rapid infusion or multiple doses
Atropine	Symptomatic bradycardia *(No longer recommended for PEA or asystole)*	0.5 mg IV every 3-5 min Max: 3 mg max dose Higher doses may be required for organophosphate poisoning	Doses < 0.5 mg may worsen bradycardia
	Toxins, poisons, and overdoses	Up to 2 to 4 mg until symptoms resolve	
Dopamine	Symptomatic bradycardia (if atropine fails) Pressor for shock/hypotension	2 to 20 mcg/kg IV per min Titrate to blood pressure response	Do not mix with sodium bicarbonate or alkalis/bases Correct hypovolemia before administration
Epinephrine	Cardiac arrest Anaphylaxis Symptomatic bradycardia instead of dopamine	1 mg (1:10000) IV OR 2-2.5 mg (1:1000) ETT q3 to 5 min Then infuse 0.1-0.5 mcg/kg/min titrated to blood pressure response	Cocaine-induced ventricular tachycardia May increase oxygen demand
	Symptomatic bradycardia (if atropine fails) Pressor for hypotension	2-10 mcg/minute infusion titrated to blood pressure response	
Lidocaine	Cardiac arrest VFib/VTach	First dose: 1-1.5 mg/kg IV Second dose: 0.5-0.75 mg/kg IV every 5 to 10 min Max: 3 mg/kg Infuse 1-4 mg IV per min	Wide complex bradycardia Should not be used in cases of acute myocardial infarction Observe for signs of toxicity
	Wide complex tachycardia	Wide complex tachycardia with pulse: 0.5-1.5 mg/kg IV; may repeat twice at half dose in 5-10 minutes to total of 3mg/kg; followed with infusion of 1-4 mg per minute infusion	Wide complex bradycardia
Magnesium Sulfate	Torsades de pointes	Pulseless Torsades: 1-2 gram IV bolus Torsades with a pulse: 1-2 gram IV over 5-60 minutes followed by infusion at 0.5-1 gram per hour IV	Rapid bolus may cause hypotension and bradycardia; Can also be used to reverse digitalis poisoning
	Hypomagnesemia with cardiac arrest	1-2 gram IV bolus	
Vasopressin	Ventricular fibrillation Pulseless ventricular tachycardia Asystole PEA	40 units IV instead of epinephrine	Deliver through central line Peripheral IV administration can cause tissue necrosis
	Shock	Infuse 0.02-0.04 units/min IV	

Table of Contents

Updates to PALS in 2015 3
PALS Systematic Approach 4
First Impression 4
Evaluate-Identify-Intervene 5
Primary Assessment 6
Airway 6
Breathing 7
Circulation 8
Disability 9
Exposure 9
Secondary Assessment and Diagnostic Tests 10
Respiratory Distress/Failure 11
Causes of Respiratory Distress 12
Cardiac Arrest 14
Ventricular Fibrillation and Pulseless Ventricular Tachycardia 14
PEA and Asystole 15
Rapid Differential Diagnosis of Cardiac Arrest 16
Shock 17
Fluid Resuscitation 18
Return of Spontaneous Consciousness (ROSC) and Post Arrest Care 19
Postresuscitation Management 20
Transport to Tertiary Care Center 21
Bradycardia 22
Tachycardia 23
Tachycardia with Pulse and Poor Perfusion 24
Tachycardia with Pulse and Good Perfusion 25
PALS Tools 26
Broselow Pediatric Emergency Tape System 26
PALS Airways 26
Intraosseus Access 27
Team Dynamics/Systems of Care 28

ECG Rhythms 29
Atrioventricular (Heart) Block 29
Pulseless Electrical Activity and Asystole 30
Ventricular Fibrillation and Pulseless Ventricular Tachycardia 31
Tachyarrhythmias 32
Resuscitation and Life Support Medications 33

Updates to PALS in 2015

As we learn more about resuscitation science and medicine, physicians and researchers realize what works best and what works fastest in a critical, life-saving situation. Therefore, it is necessary to periodically update life-support techniques and algorithms. If you have previously certified in pediatric advanced life support, then you will probably be most interested in what has changed since the latest update in 2015. The table below also includes changes proposed since the last AHA manual was published.

Updates to the 2015 Guidelines		
Intervention	2015 Guideline	2010 Guideline
Volume for children with febrile illness	Restrictive volumes of isotonic crystalloids	Aggressive volume resuscitation
Atropine for emergency tracheal intubation	Controversial for neonates; no minimum dose	Routine premedication prior to intubation
Arterial blood pressure monitoring	If in place, may be useful to adjust CPR	No guideline
Amiodarone and Lidocaine	Acceptable for shock-refractory VFib or Pulseless VTach	No guideline
Therapeutic hypothermia	Fever should be avoided after ROSC but use of therapeutic hypothermia is controversial	Therapeutic hypothermia should be used
Blood Pressure	Fluids and vasoactive agents to maintain systolic blood pressure above the 5th percentile for age	No guideline
Compressions	100 to 120 per minute	At least 100 per minute

PALS Systematic Approach

The PALS systematic approach is an algorithm that can be applied to every injured or critically ill child.

First Impression

The first step is to determine if the child is in imminent danger of death, specifically cardiac arrest or respiratory failure. The PALS systematic assessment starts with a quick, first impression. The provider or rescuer makes it very quick assessment about the child's condition.

Is the child in imminent danger of death? Is there time to evaluate the child to identify and treat possible causes for the current illness? Is the child conscious? Is she breathing? What is her color?

Conscious?	Responsive? Irritable?
Breathing?	Loud/Increased Work of Breathing? Gasping?
Color?	Cyanotic? Pale? Mottled? Flushed?

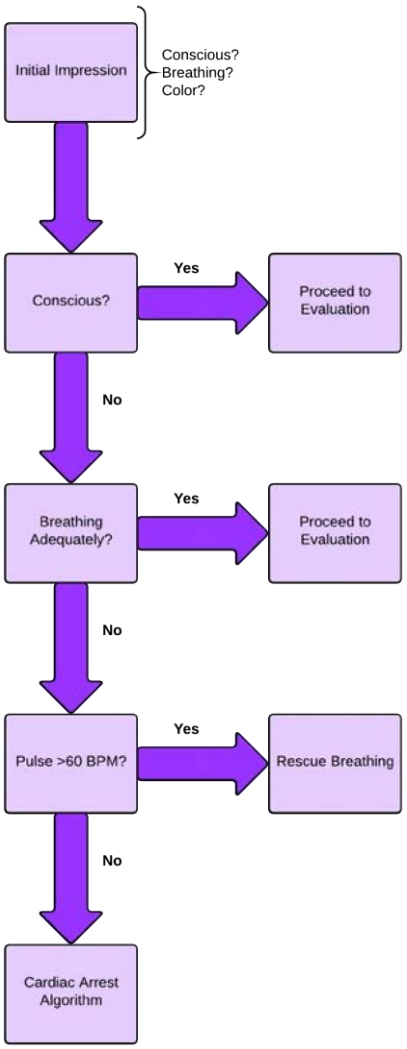
➢➢ A conscious child who is breathing effectively can be managed in the next steps of PALS, Evaluate-Identify-Intervene.

➢➢ A unconscious child who is breathing effectively can be managed in the next steps of PALS, Evaluate-Identify-Intervene.

➢➢ A child who is not breathing adequately but who has a pulse >60 BPM should be treated with rescue breathing.

➢➢ A child who has a pulse <60 BPM should be treated with CPR and according to the cardiac arrest algorithm.

➢➢ A child who has a pulse <60 BPM should be treated with CPR and according to the cardiac arrest algorithm.

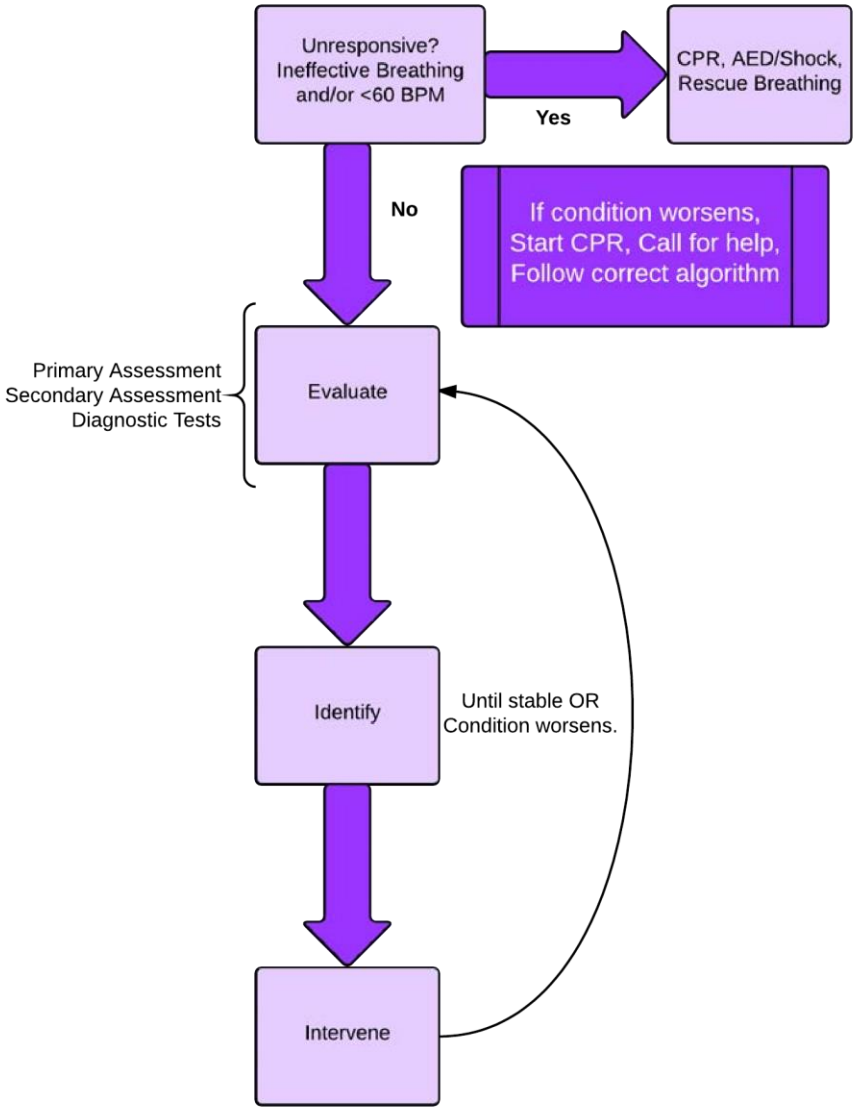


Evaluate-Identify-Intervene

- Assuming that the child does not need CPR, rescue breathing, or defibrillation, the next step in this systematic approach in PALS is a circular construct that includes evaluation, identification, and intervention.

- The provider will evaluate, identify, and intervene as many times as necessary until the child either stabilizes or her condition worsens, requiring CPR and other lifesaving measures.

- "Evaluate" pertains to evaluation of the child's illness, but also to the success or failure of the intervention.



- If the child's condition worsens at any point, revert to CPR and emergency interventions as needed.

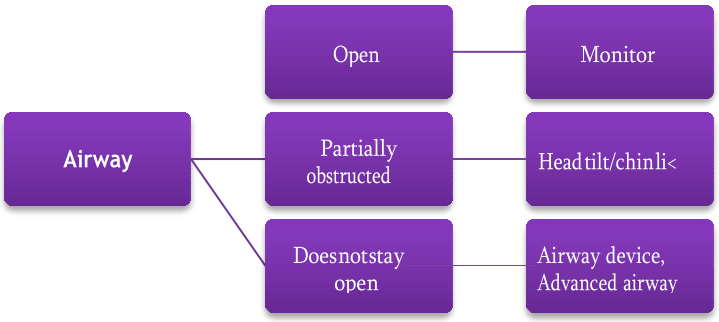
- After Spontaneous Return of Circulation (ROSC), use the evaluate–identify–intervene sequence.

- The evaluate phase of the sequence includes Primary Assessment, Secondary Assessment, and Diagnostic Tests that are helpful in pediatric life support situations.

Primary Assessment

Primary Assessment follows ABCDE: Airway, Breathing, Circulation, Disability, Exposure.

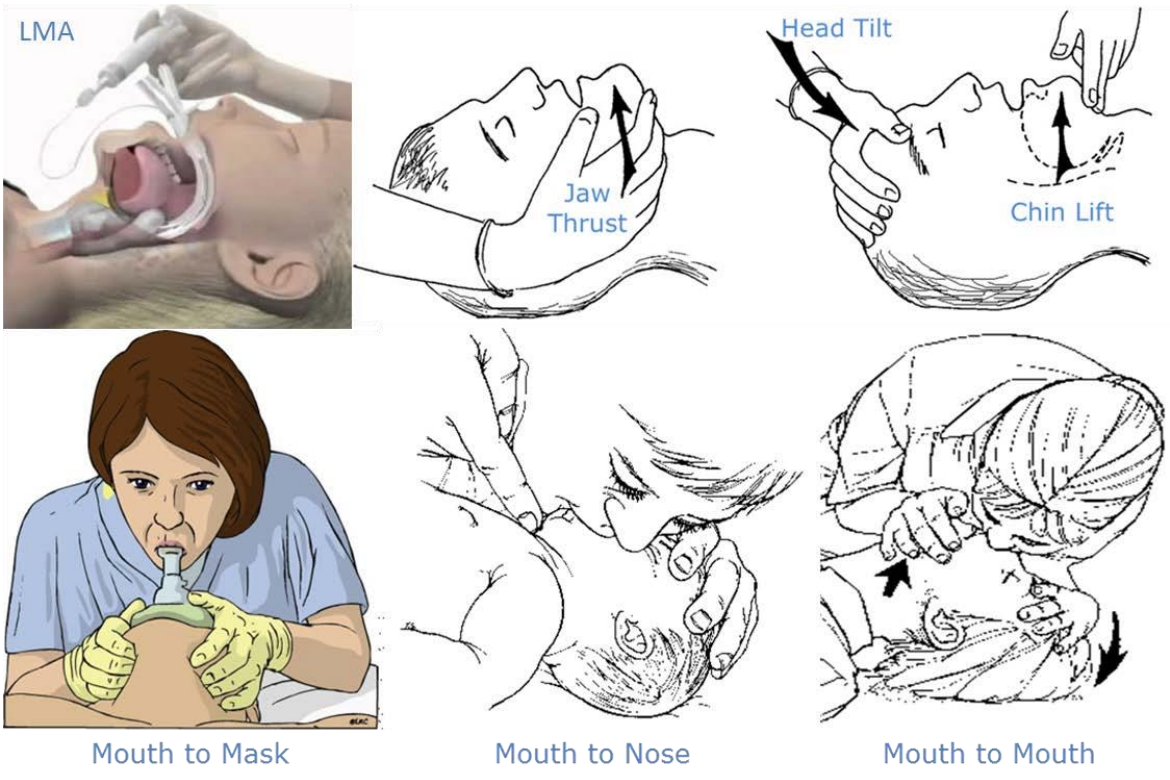
➢➢ While CPR currently uses the C-A-B approach or compressions, airway, breathing, the Primary Assessment in PALS still begins with Airway.



➢➢ If the child airway is open, you may move onto the next step. However, if the airway is likely to become compromised, you may consider a basic or advanced airway.

➢➢ Often, in unresponsive patient or in someone who has a decreased level of consciousness, the airway will be partially obstructed. This instruction does not come from a foreign object, but rather from the tissues in the upper airway. You can improve a partially obstructed airway by performing a head tilt and chin lift. If there is suspected trauma to the cervical spine, use a jaw thrust instead.

➢➢ A blocked airway would usually requires a basic or advanced airway.



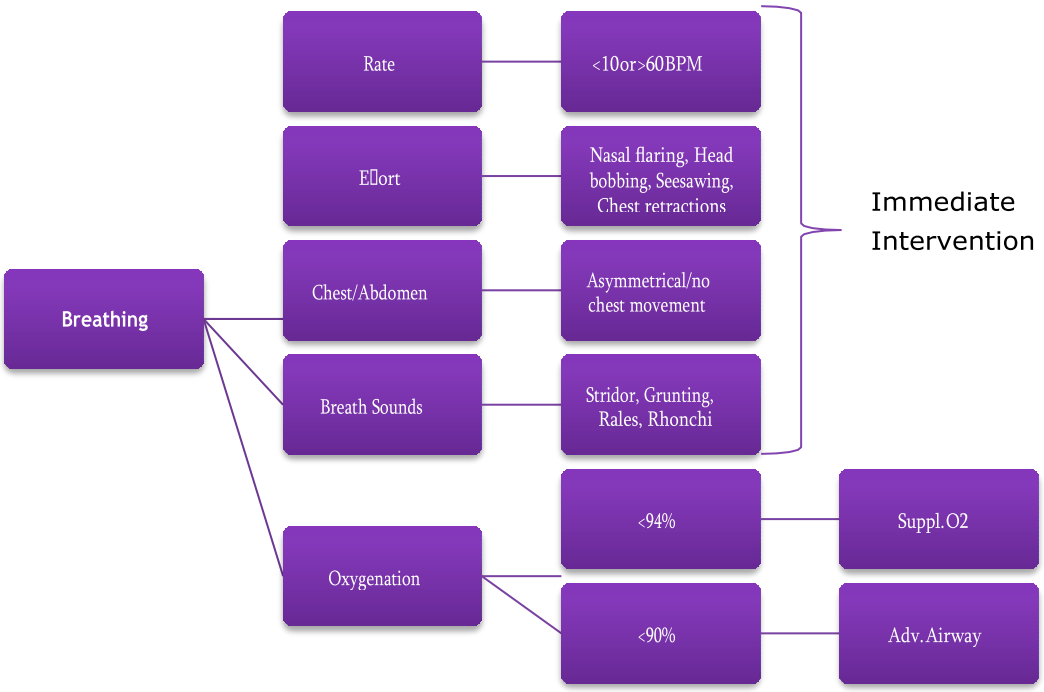
The evaluation of breathing include several signs including breathing rate, breathing effort, motion of the chest and abdomen, breath sounds, and blood oxygenation levels. Normal breathing rates vary by age and are shown in the table. The breathing rate higher or lower than the normal range indicates the need for intervention.

Normal Respiratory Rate by Age		
Age	**Range**	**Rate (BPM)**
Infant	0-12 months	30-60
Toddler	1-3 years	24-40
Preschooler	4-5 years	22-34
School Age	6-12 years	18-30
Adolescent	13-18 years	12-16

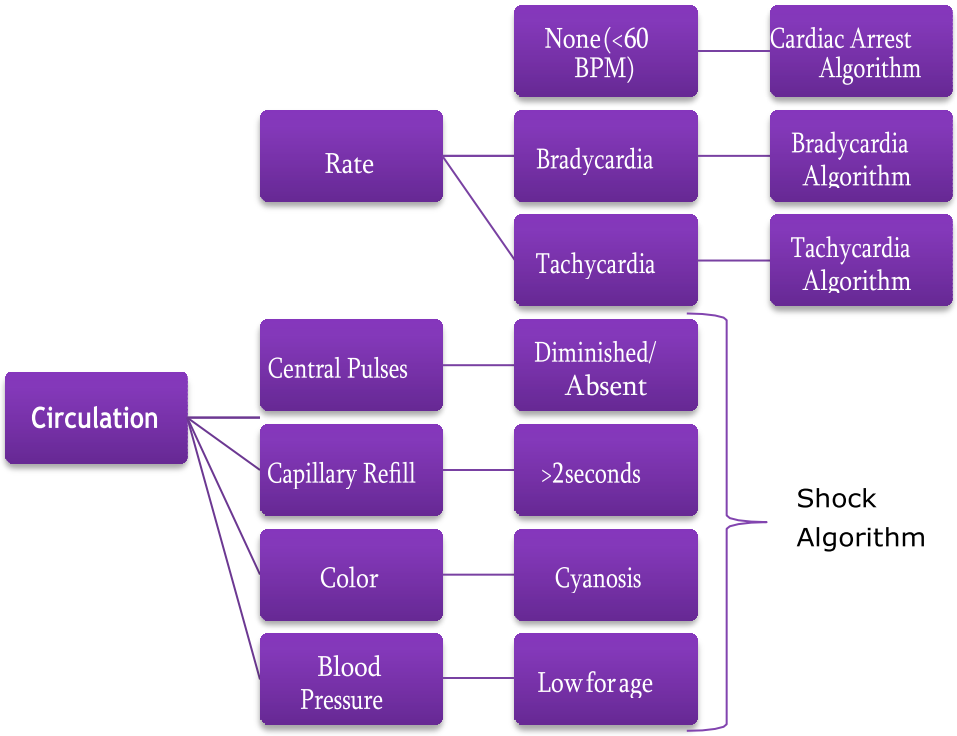
Nasal flaring, head bobbing, seesawing, and chest retractions are all signs of increased effort of breathing. The chest may show labored movement (e.g., using the chest accessory muscles), asymmetrical movement, or no movement at all.

Stridor is a high-pitched breath sounds, usually heard on inspiration, that usually indicates a blockage in the upper airway. Rales or crackles often indicate fluid in the lower airway. Rhonchi are coarse rattling sounds usually caused by fluid in the bronchi.

Blood oxygen saturation below 90% indicate that an advanced airway, such as an endotracheal tube, is needed. Blood oxygenation can be 100% during cardiopulmonary arrest but should be titrated to between 94 and 99% after ROSC or in non-acute situations.



A heart rate that is either too fast or too slow can be problematic. In children, heart rate less than 60 bpm is equivalent to cardiac arrest. Diminished central pulses, such as in the carotid, brachial, or femoral arteries, indicate shock. The same is true for capillary refill the takes longer than 2 seconds to return, cyanosis, and blood pressure that is lower than normal for the child's age. Bradycardia and tachycardia that are interfering with circulation and causing a loss of consciousness should be treated as cardiac arrest or shock, rather than as a bradycardia or tachycardia.



Normal Heart Rate by Age	
Range	**Rate (BPM)**
0-3 months	80-205
4 months - 2 years	75-190
2-10 years	60-140
Over 10 years	50-100

Normal Blood Pressure by Age			Low Systolic Blood Pressure by Age
Age	**Systolic Range**	**Diastolic Range**	
1 Day	60-76	30-45	<60
4 Days	67-84	35-53	<60
1 month	73-94	36-56	<70
1-3 months	78-103	44-65	<70
4-6 months	82-105	46-68	<70
7-12 months	67-104	20-60	<72
2-6 years	70-106	25-65	<70 + (2 X age)
7-14 years	79-115	38-78	<70 + (2 X age)
15-18 years	93-131	45-85	<90

Rapidly assess disability using the AVPU paradigm: Alert, Verbal, Pain, Unresponsive.

AVPU paradigm		
A	Alert	Appropriate, normal activity for the child's age and usual status
V	Verbal	Responds only to voice
P	Pain	Responds only to pain
U	Unresponsive	Does not respond to stimuli, even pain

A more thorough assessment would be the Pediatric Glasgow Coma Scale.

Pediatric Glasgow Coma Scale			
Response	**Score**	**Verbal Child**	**Pre-verbal Child**
Eye Opening	4 3 2 1	Spontaneously To verbal command To pain None	Spontaneously To speech To pain None
Verbal Response	5 4 3 2 1	Oriented and talking Confused but talking Inappropriate words Sounds only None	Cooing and babbling Crying and irritable Crying with pain only Moaning with pain only None
Motor Response	6 5 4 3 2 1	Obeys commands Localizes with pain Flexion and withdrawal Abnormal flexion Abnormal extension None	Spontaneous movement Withdraws when touched Withdraws with pain Abnormal flexion Abnormal extension None
Mild: 13-15 Moderate: 9-12 Severe: 3-8			

Exposure is included in the primary assessment to remind the provider to look for causes of injury or illness that may not be readily apparent. To do this, the child's clothes need to be removed in a ordered and systematic fashion. During the removal, the provider should look for signs of discomfort or distress that may point to an injury in that region.

The provider should look for and treat, at a minimum, hypothermia, hemorrhage, local and/or systemic infection, fractures, petechiae, bruising or hematoma.

Secondary Assessment and Diagnostic Tests

When a child is experiencing an acutely life-threatening event, such as cardiopulmonary failure, it is appropriate to treat the child with CPR and the appropriate arrest algorithm.

When a child has a condition that may soon become life-threatening or if something does "not feel right", continue using the Primary Assessment sequence of Evaluate-Identify-Intervene. If at any time the child's condition worsens, treat the child with CPR and the appropriate arrest algorithm.

Use SAMPLE in Secondary Assessment		
S	Signs/Symptoms	Fever Decreased intake Vomiting/Diarrhea Bleeding Shortness of breath Altered mental status Fussiness/Agitation
A	Allergies	Medication allergy Environmental allergy Food allergy
M	Medications	Prescribed Over-the-counter New meds? Last dose?
P	Past History	Birth history Chronic health issues Immunization status Surgical history
L	Last Meal	Breast/bottle/solid? When? What? How much? New foods?
E	Events/Exposures	History of present illness Onset/time course

When a child is ill but does not likely have a life-threatening condition, you may proceed to the Secondary Assessment. The Secondary Assessment includes a focus history and focused physical examination looking for things that might cause respiratory or cardiovascular compromise.

The focused physical examination may be quite similar to the Exposure phase of the Primary Assessment, but will be guided by the data that the provider collects during the focused history. The focused history will also help determine which diagnostic tests should be ordered.

Key Diagnostic Tests Used in PALS		
Test/Study	**Identifies**	**Possible Intervention**
Arterial Blood Gas (ABG)	Hypoxemia	Increase Oxygenation
	Hypercarbia	Increase Ventilation
	Acidosis	Increase Ventilation
	Alkalosis	Reduce Ventilation
Arterial Lactate	Metabolic acidosis, Tissue hypoxia	Shock Algorithm
Central Venous Oxygen Saturation	Poor O2 delivery (SVO2 <70%)	Shock Algorithm
Central Venous Pressure	Heart contractility, others	Vasopressors, Shock Algorithm
Chest X-ray	Respiratory conditions	Specific to cause, Respiratory Algorithm
Echocardiogram	Heart anatomy and function	Specific to cause
Electrocardiogram	Rhythm Disturbances	Specific to cause
Peak Expiratory Flow Rate	Respiratory conditions	Specific to cause, Respiratory Algorithm
Venous Blood Gas (VBG)	Acidosis	Increase Ventilation
	Alkalosis	Reduce Ventilation

Respiratory Distress/Failure

Cardiac arrest in children can occur secondary to respiratory failure, hypotensive shock, or sudden ventricular arrhythmia. In most pediatric cases, however, respiratory failure, shock, and even ventricular arrhythmia are preceded by a milder form of cardiovascular compromise. For example, respiratory failure is usually preceded by some sort of respiratory distress. In fact, respiratory distress is the most common cause of respiratory failure and cardiac arrest in children. As you may expect, outcomes are better if one can intervene during respiratory distress rather than respiratory failure.



Signs and Symptoms of Worsening Respiratory Distress,				
Sign/Sx	**Mild**	**Moderate**	**Severe**	**Verge of Arrest**
Accessory Muscles Use	No	Yes	Marked	Seesawing
Activity	Walking, talking	Talking, will sit	No activity, infant will not feed	Drowsy
Alert	Slightly agitated	Agitated	Markedly agitated	Lethargic
O_2 Sat.	>95%	91 to 95%	<90%	<90%
$PaCO_2$	<45 mmHg	<45 mmHg	<45 mmHg	<45 mmHg
PaO_2	Normal	>60 mmHg	<60 mmHg ± Cyanosis	<60 mmHg + Cyanosis
Pulse	Normal	100-200 BPM	>200 BPM	<100 BPM
Respiratory Rate	Increased	Increased	Markedly Increased	Increased or Decreased
Speaking?	Sentences	Phrases	Words	Not talking
Wheeze	Audible	Loud	Very Loud	None

Causes of Respiratory Distress

Respiratory distress/failure is divided into four main etiologies for the purposes of PALS: upper airway, lower airway, lung tissue disease, and disordered control of breathing.

Respiratory Distress Identification and Management	
Type of Respiratory Problem	**Possible Causes**
Upper Airway	Anaphylaxis Croup Foreign body aspiration
Lower Airway	Asthma Bronchiolitis
Lung Tissue Disorder	Pneumonia Pulmonary edema
Disordered Control of Breathing	Increased intracranial pressure Neuromuscular disease Toxic poisoning

Respiratory Distress, Key Signs and Symptoms				
	Upper Airway Obstruction	**Lower Airway Obstruction**	**Lung Disease**	**Disordered Control of Breathing**
Air Movement	Decreased			Unchanged or decreased
Airway	May or may not be fully patent in respiratory distress. Not patent in respiratory failure.			
Breath Sounds	Cough, hoarseness, stridor	Wheezing	Diminished breath sounds, grunting, crackles	Unchanged
Heart Rate	Increased in respiratory distress Decompensates rapidly to bradycardia as respiratory failure ensues			
Skin Color and Temperature	Pale, cool, and clammy in respiratory distress Decompensates rapidly to cyanosis as respiratory failure ensues			Varies
Level of Consciousness	Agitation in respiratory distress Decompensates rapidly to decreased mentation, lethargy, and LOC as respiratory failure ensues			
Respiratory Rate and Effort	Increased in respiratory distress Decompensates rapidly in respiratory failure			Varies

Respiratory Distress Management

Respiratory Distress Management by Type and Cause		
Type	**Possible Causes**	**Treatment**
Upper Airway Obstruction	Anaphylaxis	Epinephrine Albuterol nebulizer Watch for and treat airway compromise, advanced airway as needed Watch for and treat shock
	Croup	Humidified oxygen Dexamethasone Nebulized epinephrine for moderate to severe croup Keep O2 sat >90%, advanced airway as needed
	Foreign body aspiration	**Do not perform a blind finger sweep**, remove foreign object if visible Infant <1 year old: Back slaps/chest thrusts Child >1 year old: Abdominal thrusts
Lower Airway Obstruction	Asthma	Nebulized epinephrine or albuterol Keep O2 sat >90%, advanced airway or non-invasive positive pressure ventilation as needed Corticosteroids PO or IV as needed Nebulized ipratropium Magnesium sulfate slow IV (moderate to severe asthma) Terbutaline SQ or IV (impending respiratory failure)
	Bronchiolitis	Oral and nasal suctioning Keep O2 sat >90%, advanced airway as needed Nebulized epinephrine or albuterol
Lung Disease	Pneumonia	Empiric antibiotics and narrow antibiotic spectrum based on culture results Nebulized albuterol for wheezing Reduce the work of breathing and metabolic demand Keep O2 sat >90%, advanced airway as needed Continuous positive airway pressure (CPAP)
	Pulmonary edema	Reduce the work of breathing and metabolic demand Keep O2 sat >90%, advanced airway as needed Diuretics if cardiogenic CPAP
Disordered Control of Breathing	Increased intracranial pressure	Pediatric neurological/neurosurgery consult Hyperventilation as directed Use medications (e.g., mannitol) as directed
	Neuromuscular disease	Identify and treat underlying disease CPAP or ETT and mechanical ventilation as needed
	Toxic poisoning	Identify toxin/poison Call Poison Control: 1.800.222.1222 Administer antidote/anti-venom when possible Maintain patent airway, advanced airway as needed Provide suctioning

Cardiac Arrest

Cardiac arrest occurs when the heart does not supply blood to the tissues. Strictly speaking, cardiac arrest occurs because of an electrical problem (i.e., arrhythmia). Shock (i.e., too little blood pressure/volume) and respiratory failure may lead to cardiopulmonary failure and hypoxic arrest.

Ventricular Fibrillation and Pulseless Ventricular Tachycardia

➢➢ Ventricular fibrillation and pulseless ventricular tachycardia are shockable rhythms.

➢➢ The first shock energy is 2 J/kg.

➢➢ The second shock energy (and all subsequent shocks) is 4 J/kg.

➢➢ All subsequent shocks are 4 J/kg or greater.

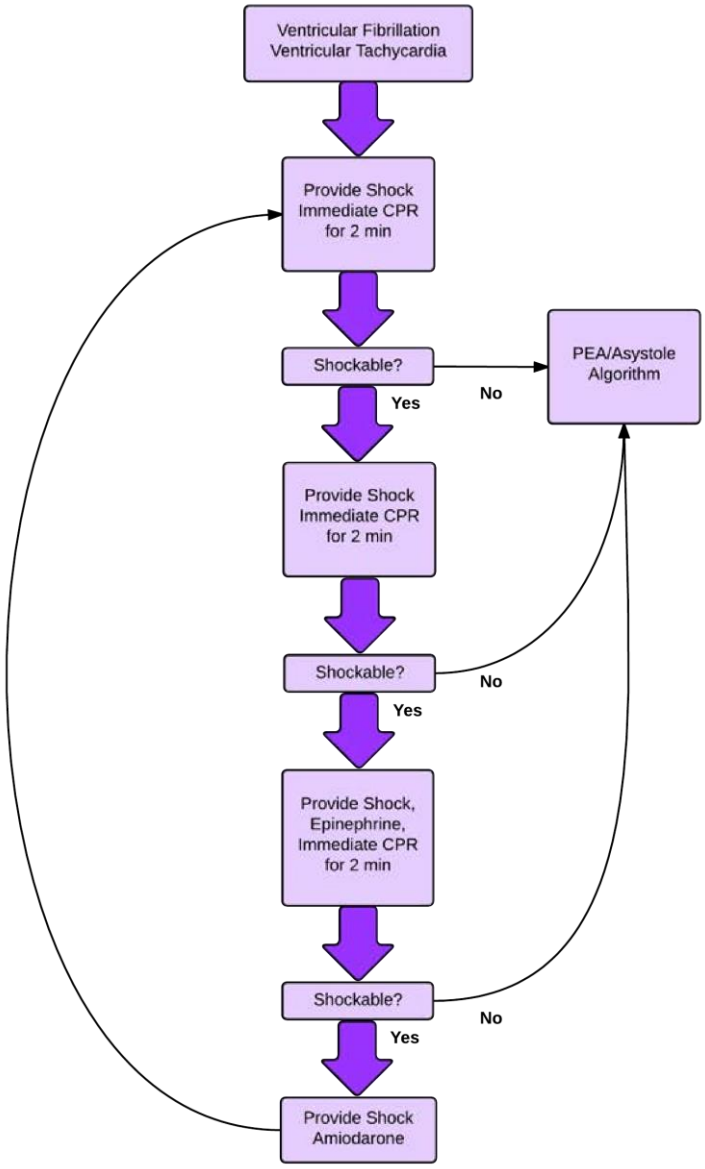
➢➢ The maximum energy is 10 J/kg or the adult dose (200 J for biphasic, 360 J for monophasic).

➢➢ Epinephrine (0.01 mg/kg IV/IO) is given every 3 to 5 minutes (two 2 minute cycles of CPR).

➢➢ Amiodarone (IV/IO)
- 5 mg/kg bolus
- Can be given three times total

➢➢ Lidocaine may be used instead of Amiodarone

If the arrest rhythm is no longer shockable, move to PEA/Asystole algorithm.



➢➢ If the patient regains consciousness, move to ROSC algorithm.

PEA and Asystole

➢➢ As long as the patient is in PEA or asystole, the rhythm is not shockable.

➢➢ Chest compressions/high-quality CPR should be interrupted as little as possible during resuscitation.

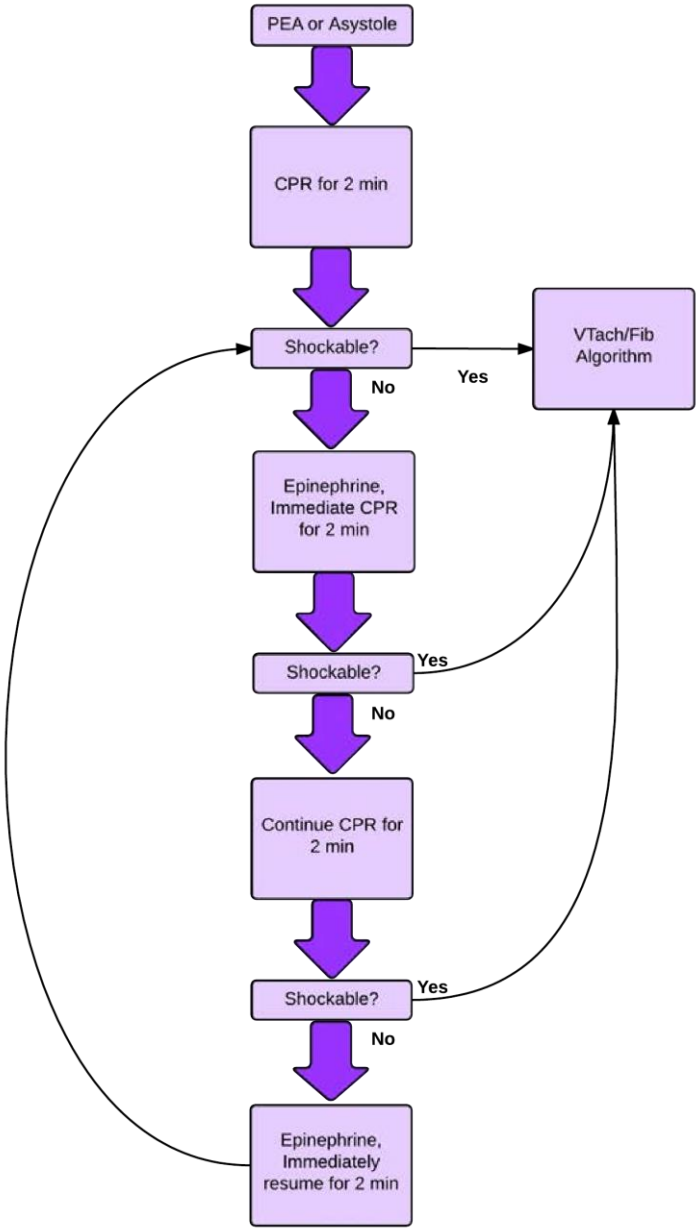
➢➢ After 2 min. of high-quality CPR, give 0.01 mg/kg epinephrine IV/IO every 3 to 5 minutes (two 2 minute cycles of CPR).

➢➢ Remember, chest compressions are a means of artificial circulation, which should deliver the epinephrine to the heart. Without chest compressions, epinephrine is not likely to be effective.

➢➢ Chest compressions should be continued while epinephrine is administered.

➢➢ Rhythm checks every 2 min.

➢➢ Look for and treat reversible causes (Hs and Ts).



➢➢ If the arrest rhythm becomes shockable, move to VFib/Pulseless VTach algorithm.

➢➢ If the patient regains circulation, move to ROSC algorithm.

Rapid Differential Diagnosis of Cardiac Arrest

Many different disease processes and traumatic events can cause cardiac arrest, but in an emergency, it is important to be able to rapidly consider and eliminate or treat the most typical causes of cardiac arrest. To facilitate remembering the main, reversible causes of cardiac arrest, they can be organized as the Hs and the Ts.

The Hs	Symptoms/Signs/Tests	Intervention
Hypovolemia	Rapid heart rate, narrow QRS complex,	Fluid resuscitation
Hypoxia	Decreased heart rate	Airway management, oxygen
Hydrogen Ion (Acidosis)	Low amplitude QRS complex	Hyperventilation, sodium bicarb
Hypoglycemia	Fingerstick glucose testing	IV Dextrose
Hypokalemia	Flat T waves, pathological U wave	IV Magnesium
Hyperkalemia	Peaked T waves, wide QRS complex	Calcium chloride, sodium bicarb, insulin/glucose, hemodialysis
Hypothermia	History of cold exposure	Rewarming blankets/fluids

The Ts	Symptoms/Signs/Tests	Intervention
Tension Pneumothorax	Slow heart rate, narrow QRS complex, acute dyspnea, history of chest trauma	Thoracotomy, needle decompression
Tamponade (Cardiac)	Rapid heart rate and narrow QRS complex	Pericardiocentesis
Toxins	Variable, prolonged QT interval, neuro deficits	Antidote/antivenom (toxin-specific)
Thrombosis (pulmonary)	Rapid heart rate, narrow QRS complex	Fibrinolytics, embolectomy
Thrombosis (coronary)	ST segment elevation/depression, abnormal T waves	Fibrinolytics, Percutaneous intervention

Shock

The goals of shock management include:

- Improving blood oxygenation
- Easing oxygen demand
- Improving volume and fluid distribution
- Normalizing electrolyte and metabolic disturbances

Shock Identification and Management	
Treatment Goal	**Key Intervention (s)**
Improving blood oxygenation	Supplemental O2 via face mask/non-rebreather Mechanical ventilation through advanced airway Packed red blood cells
Easing oxygen demand	Reduce fever Treat pain Treat anxiety
Normalizing electrolyte and metabolic disturbances	Treat imbalances promptly IV electrolytes for deficiencies Ventilatory settings for acidosis/alkalosis Glucose for hypoglycemia
Improving volume and fluid distribution	Treatment depends on type of shock

Types of Shock, Signs and Symptoms				
	Hypovolemic	**Distributive**	**Cardiogenic**	**Obstructive**
Mechanism	Too little volume	Volume distributed to tissues	Heart problem	Cardiac outflow impediment
Potential Causes	Vomiting/Diarrhea Hemorrhage DKA Burns Poor Fluid Intake	Sepsis Head/Spine Injury Anaphylaxis	Congenital Heart Dz Poisoning Myocarditis Cardiomyopathy Arrhythmia	Cardiac Tamponade Tension Pneumo Congenital Heart Dz Pulmonary Embolus
Preload	Decreased	Normal or Decreased	Varies	Varies
Contractility	Normal/Increased	Normal or Decreased	Decreased	Normal
Afterload	Increased	Varies	Increased	Increased
Respiratory Rate and Effort	Increased rate No increased effort	Increased rate +/- Increased effort	Markedly increased effort	Markedly increased effort
Breath Sounds	Normal	+/- Rales	Rales and grunting	
Systolic BP	May be normal (compensated), but soon compromised without intervention			
Pulse Pressure	Narrow	Varies	Narrow	Narrow;
Heart Rate	Increased	Increased	Increased	Increased Distant heart sounds
Peripheral Pulses	Weak	Bounding or Weak	Weak or absent Jugular vein distention	Weak
Capillary Refill	Delayed	Varies	Delayed	Delayed
Urine Output	Decreased			
Consciousness	Irritable and anxious, early. Altered mental status, later.			

Fluid resuscitation in PALS depends on the weight of the child and the severity of the situation. While dehydration and shock are separate entities, the symptoms of dehydration can help the provider to assess the level of fluid deficit and to track the effects of fluid resuscitation. In the current guidelines, the clinician must fully evaluate the child with febrile illness since aggressive fluid resuscitation with isotonic crystalloid solution may not be indicated.

Signs and Symptoms of Dehydration			
Category	**Deficit ml/kg (% body wt.)**		**Signs/Sx**
	Infants	**Adolescents**	
Mild	50 (5%)	30 (3%)	Slightly dry buccal mucosa, increased thirst, slightly decreased urine output
Moderate	100 (10%)	50–60 (5–6%)	Dry buccal mucosa, tachycardia, little or no urine output, lethargy, sunken eyes and fontanelles, loss of skin turgor
Severe	150 (15%)	70–90 (7–9%)	Same as moderate plus a rapid, thready pulse; no tears; cyanosis; rapid breathing; delayed capillary refill; hypotension; mottled skin; coma

Interventions by Shock Type		
Broad Type	**Specific Type**	**Management**
Hypovolemic	Hemorrhagic	Fluid resuscitation, packed red blood cells
	Non-hemorrhagic	Fluid resuscitation
Distributive	Septic	*Septic Shock Algorithm*
	Anaphylactic	Epinephrine IM, fluid resuscitation
	Neurogenic	Fluid resuscitation, pressors
Cardiogenic	Bradyarrhythmia	*Bradycardia Algorithm*
	Tachyarrhythmia	*Tachycardia Algorithm*
	Heart Disease	Fluid resuscitation, pressors, expert consult
Obstructive	Ductus Arteriosis	PGE1 (alprostadil), expert consult
	Tension Pneumo	Needle decompression, tube thoracostomy
	Tamponade	Pericardiocentesis
	Pulmonary Embolism	Fluid resuscitation, fibrinolytics, expert consult

Fluid Resuscitation			
Broad Type	**Specific Type**	**Volume**	**Rate**
Hypovolemic	Hemorrhagic	3 ml of crystalloid for each ml blood lost	Over 5-10 min
	Non-hemorrhagic	20 ml/kg bolus, repeat as needed	Over 5-10 min
	Diabetic Ketoacidosis	10-20 ml/kg bolus, repeat as needed	Over 60 min
Distributive	All types	20 ml/kg bolus, repeat as needed	Over 5-10 min
Cardiogenic	All types	5-10 ml/kg bolus, repeat as needed	Over 10-20 min
Obstructive	Tamponade	20 ml/kg bolus	Over 5-10 min
	Pulmonary Embolism	20 ml/kg bolus, repeat as needed	Over 5-10 min

Return of Spontaneous Consciousness (ROSC) and Post Arrest Care

- In a successful resuscitation, there will be a spontaneous return of circulation.

- You can detect spontaneous circulation by feeling a palpable pulse at the carotid or femoral artery in children and the brachial artery in infants up to 1 year.

- Even after Return of Spontaneous Circulation (ROSC), the patient still needs close attention and support. The patient is at risk for reentering cardiac arrest at any time. Therefore, the patient should be moved to an intensive care unit.

- Titrate the patient's blood oxygen to between 94% and 99%. Wean down supplemental oxygen for blood oxygenation of 100%.

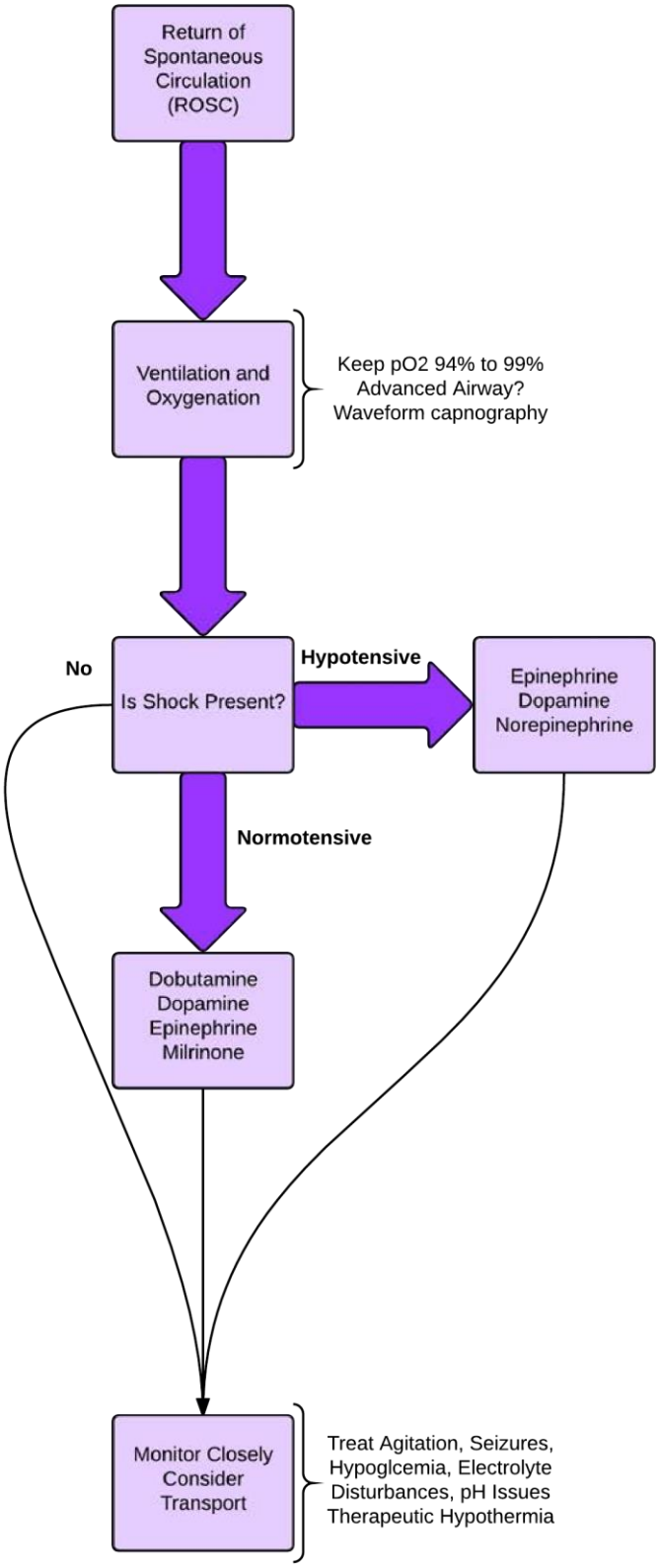
- Does the person need an advanced airway? If so, it should be placed. Also, apply quantitative waveform capnography, if available.

- Is the patient in shock? If not, monitor and move to supportive measures. If shock is present, determine if it is hypotensive or normotensive.

- Identify and treat causes (Hs and Ts). Fluid resuscitation according to cause of shock. Consider vasopressors.

- Hypotensive Shock
  - Epinephrine IV 0.1-1.0 mcg/kg/min
  - Dopamine IV 2-20 mcg/kg/min
  - Norepinephrine IV 0.1-2 mcg/kg/min

- Normotensive Shock
  - Dobutamine 2-20 mcg/kg/min
  - Dopamine IV 2-20 mcg/kg/min
  - Epinephrine IV 0.1-1.0 mcg/kg/min
  - 50 mcg/kg IV over 10-60 minutes as loading dose, then 0.25-0.75 mcg/kg/minute IV infusion as maintenance dose



Postresuscitation Management

The child is still in a delicate condition. All major organ systems should be assessed and supported. Maintenance fluids should be given. If the child has been resuscitated in the community or at a hospital without pediatric intensive care facilities, arrange to have the child moved to an appropriate pediatric hospital.

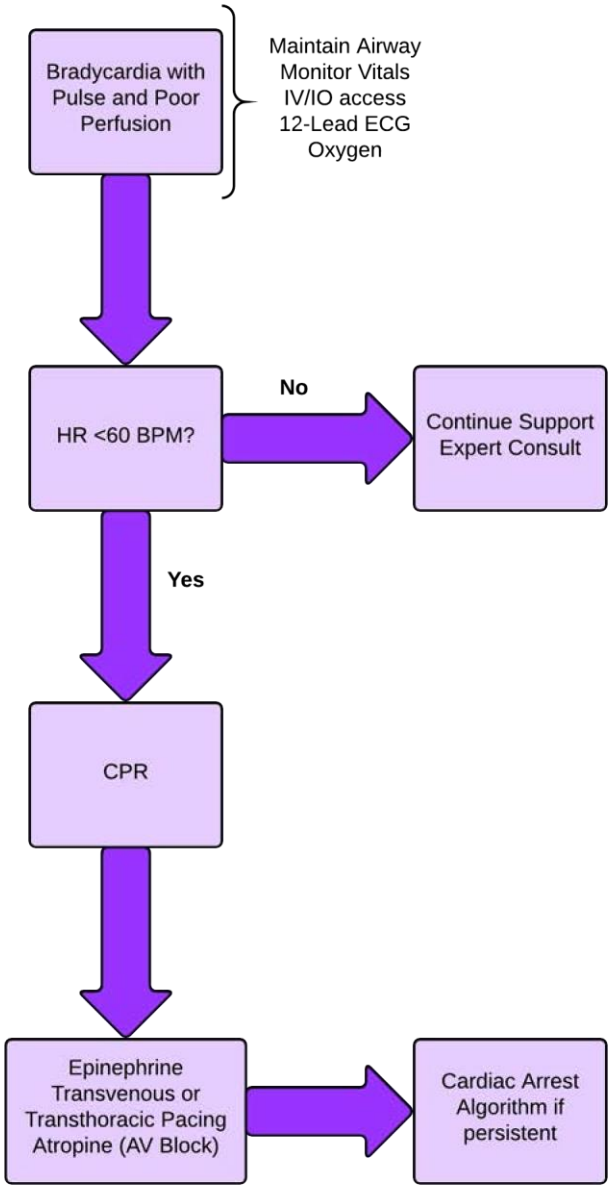
Fluid Maintenance	
Body Weight (kg)	**Hourly Maintainence Fluid Rate**
<10 kg	4 mL/kg/hour
10-20 kg	40mL/hour + 2 mL/kg/hour for each kg >10
>20 kg	60mL/hour + 1 mL/kg/hour for each kg >20

Postresuscitation Management Priorities		
System	**Priority**	**Intervention**
Respiratory	Maintain oxygenation	Titrate oxygen to maintain O2 sat: 94%-99%
	Maintain ventilation	Intubate and use ventilator if needed
	Monitor vital signs	Pulse oximetry, pO2, resp. rate, end tidal CO2
	Testing	CXR, ABGs
	Control pain/anxiety	Fentanyl or morphine as needed
Cardiovascular	Testing	Heart rate, blood pressure, CVP and cardiac output, blood gases, hemoglobin/hematocrit, blood glucose, electrolytes, BUN, calcium, creatinine, ECG
	Maintain fluid volume	Use the Shock Algorithm or maintenance fluids
	Treat arrhythmias	Use drugs or electrical therapy (Bradycardia or Tachycardia Algorithms)
Neurologic	Testing	Avoid fever, do not re-warm a hypothermic patient unless the hypothermia is deleterious, consider therapeutic hypothermia if child remains comatose after resuscitation, neurologic exam, pupillary light reaction, blood glucose, electrolytes, calcium, lumbar puncture if child is stable to rule out CNS infection
	Intracranial Pressure	Support oxygenation, ventilation and cardiac output Elevate head of bed unless blood pressure is low Consider IV mannitol for increased ICP
	Seizure Precautions and Treatment	Treat seizures per protocol, consider metabolic/toxic causes and treat
Renal	Monitor urine output	Insert urinary catheter Urine output, infants and children: > 1 ml/kg/h Urine output, adolescents: > 30 ml/h,
	Testing	Urine glucose, lactate, BUN, creatinine, electrolytes, urinalysis, fluids as tolerated, correct acidosis/alkalosis with ventilation (not sodium bicarb)
Gastrointestinal	Nasogastric tube	Maintain NG tube to low suction, watch for bleeding
	Testing	Liver function tests, amylase, lipase, abdominal ultrasound and/or CT
Hematologic	Testing	Hemoglobin/Hematocrit/Platelets, PT, PTT, INR, fibrinogen and fibrin split products, type and screen
	Consider blood therapy	If fluid resuscitation inadequate: Tranfuse packed red blood cells Active bleeding/low platelets: Tranfuse platelets Active bleeding/abnormal coags: Tranfuse fresh frozen plasma

- Prepare for Transport
  - Identify nearest tertiary pediatric facility with resources to care for condition
  - Follow hospital transport protocol
  - Provide medications/fluids/blood products for use during transport
- Coordinate with Tertiary Pediatric Facility
  - Contact the specific receiving provider
  - Resuscitation Team Leader should "present" the patient to receiving provider
- Determine Mode of Transportation
  - Ground ambulance
    - Inexpensive and available in most weather conditions
    - Takes longer
  - Helicopter
    - Faster than ground ambulance
    - More expensive than ground ambulance
    - Weather limited
  - Fixed wing aircraft
    - Best long distances/unstable child.
    - Expensive
    - Also requires ground ambulance on both ends to trip
- Prepare the Child and Family
  - Inform the family of treatments
  - Inform the family of plan
  - Obtain consent for transport
  - Answer questions and provide comfort to the child and family
- Prepare Documentation
  - Send copy of chart including labs and studies with the child
  - Send contact information for all pending tests/studies
- Use Precautions
  - Universal precautions
  - Isolation specific to probable pathogen
  - Obtain cultures if indicated
  - Give empirical antibiotics if infection suspected

Bradycardia

Bradycardia is a common cause of hypoxemia and respiratory failure in infants and children. Bradycardia is a slower than normal heart rate. Since the normal heart rate in children varies, the provider must take into account the normal values for the child's age. A heart rate less than 60 beats per minute in a child under 11 years old is worrisome for cardiac arrest (unless congenital bradycardia is present). In fact, pulseless bradycardia defines cardiac arrest.



➢➢ If bradycardia interferes with tissue perfusion, maintain the child's airway and monitor vital signs. Obtain intravenous or intraosseous access. Obtain a 12 lead ECG and provide supplemental oxygen.

➢➢ If the above interventions help, continue to support the patient and consult an expert regarding additional management.

➢➢ If the heart rate is still less than 60 bpm despite the above interventions, begin to treat with CPR.

➢➢ If the child is still experiencing bradycardia, administer epinephrine IV/IO (0.01 mg/kg). May repeat every 3-5 minutes.

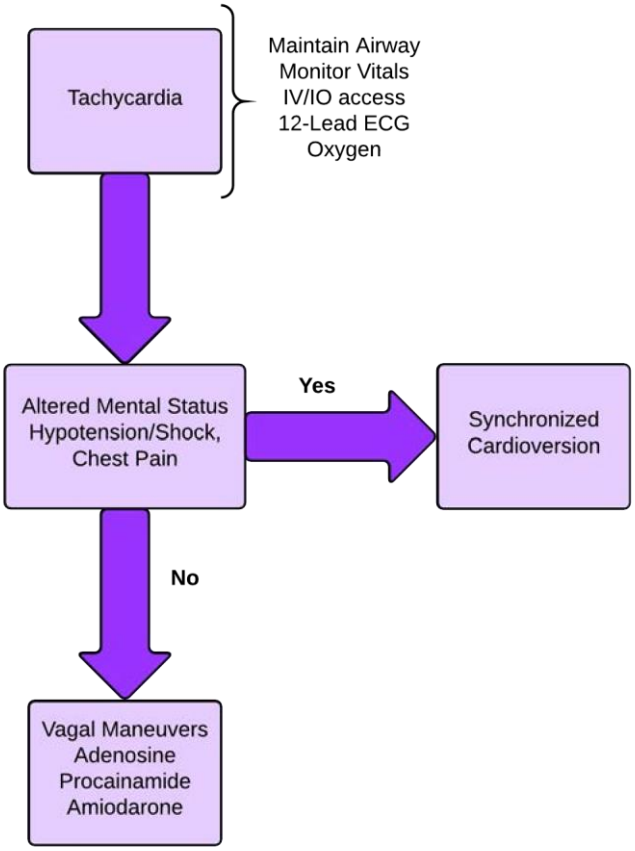
➢➢ Atropine can be given at a dose of 0.02 mg/kg up to two times.

- Min Dose: 0.1 mg.
- Max Dose: 0.5 mg.

➢➢ Consider transvenous or transthoracic pacing if available. You may need to move to the cardiac arrest algorithm if the bradycardia persists despite interventions.

Tachycardia

Tachycardia is a faster than normal heart rate. Since the normal heart rate in children varies, the provider must take into account the normal values for the child's age. Pulseless tachycardia is cardiac arrest.



➢➢ During tachycardia, maintain the child's airway and monitor vital signs. Obtain intravenous or intraosseous access. Access. Obtain a 12 lead ECG and provide supplemental oxygen.

➢➢ If the tachycardia is causing a decreased level of consciousness, hypotension or shock, or significant chest pain, move directly to synchronized cardioversion.

➢➢ If the tachycardia is not causing a decreased level of consciousness, hypotension or shock, or significant chest pain, you may attempt vagal maneuvers, first.

- Cooperative children can participate in a Valsalva maneuver by blowing through a narrow straw
- Carotid sinus massage may be effective in older children. Tachycardia is a slower than normal heart rate.
- A vagal maneuvers for an infant or small child is to place ice on the face for 15 to 20 seconds
- Ocular pressure may injure the child and should be avoided

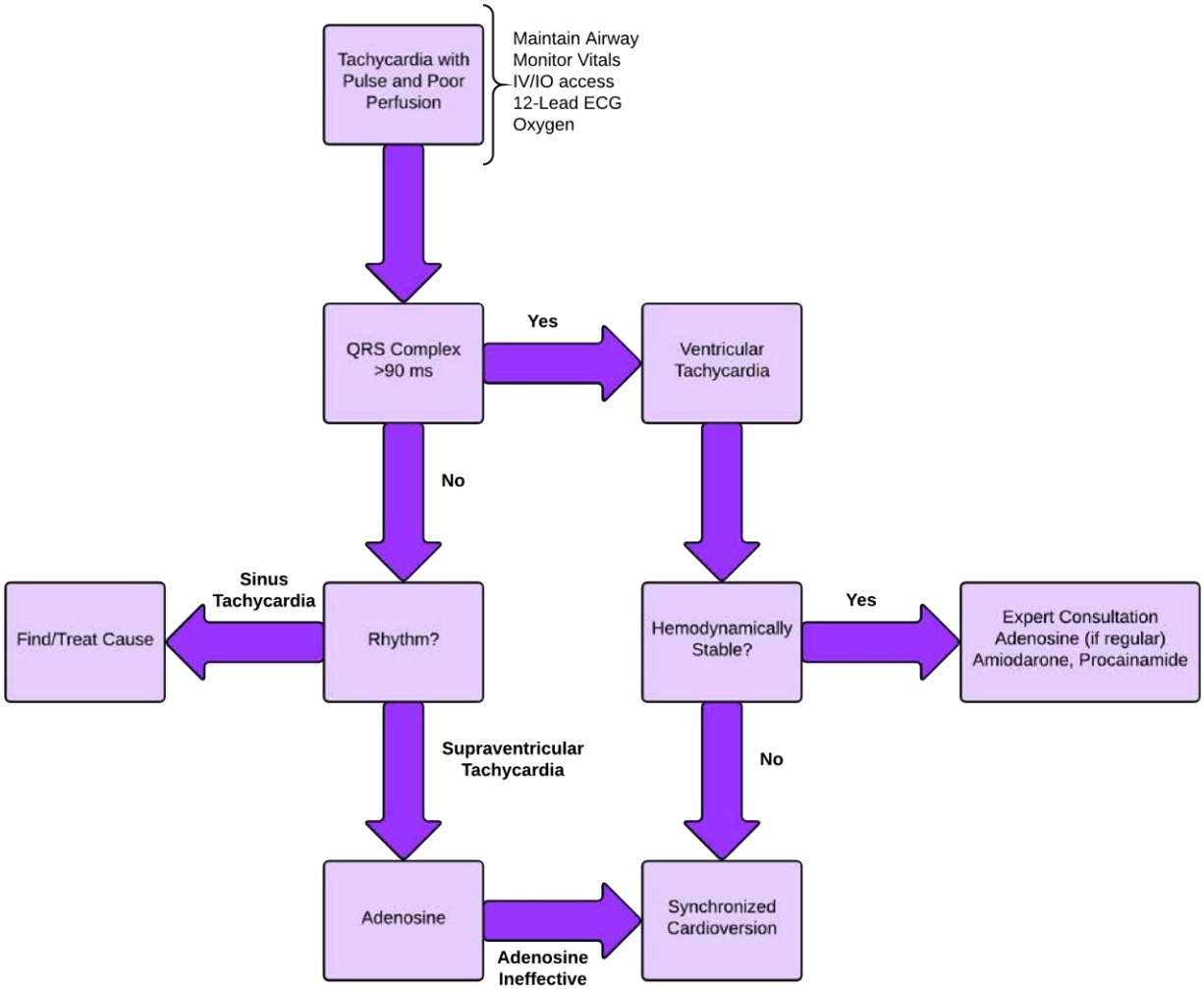
➢➢ If vagal maneuvers fail, you may use

- Adenosine: 0.1 mg/kg IV push to a max of 6 mg, followed by 0.2 mg/kg IV push to a max of 12 mg
- Procainamide: 15 mg/kg over 30-60 min
- Amiodarone: 5mg/kg over 20-60 min to a max of 300 mg

Tachycardia with Pulse and Poor Perfusion

It is important to determine if the tachycardia is narrow complex or wide complex. A QRS complex that is longer than 90 ms is wide QRS complex tachycardia. This should be considered possible ventricular tachycardia. If the child is not hemodynamically stable then provide cardioversion immediately.

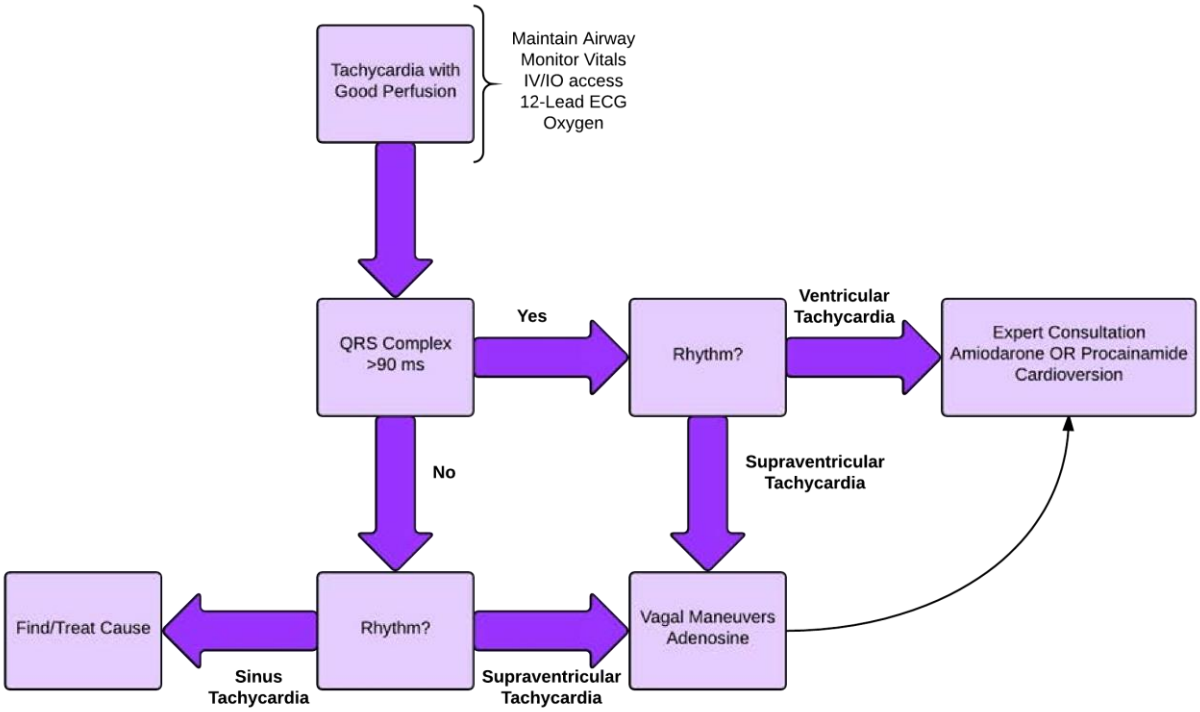
- If the wide QRS complex has a regular rhythm, then you can supply synchronized cardioversion at 100 J.
- If the wide QRS complex is irregular, this is ventricular tachycardia and should be treated with unsynchronized cardioversion (i.e. shock) immediately.
- Narrow complex tachycardia may be sinus tachycardia or supraventricular tachycardia.
- Sinus tachycardia has many causes; the precise cause should be identified and treated.
- Supraventricular tachycardia can be treated with 0.1 mg/kg adenosine IV push to a max of 6 mg. If the first dose is unsuccessful, follow it with 0.2 mg/kg adenosine IV push to a max of 12 mg. If adenosine is unsuccessful, proceed to synchronized cardioversion.
- Narrow complex supraventricular tachycardia with a regular rhythm is treated with 50-100 J of synchronized cardioversion energy.
- Narrow complex supraventricular tachycardia with an irregular rhythm is treated with 120-200 J of synchronized cardioversion energy.



Tachycardia with Pulse and Good Perfusion

Again, it is important to determine if the tachycardia is narrow complex or wide complex. A QRS complex that is longer than 90 ms is wide QRS complex tachycardia.

- Narrow complex tachycardia may be sinus tachycardia or supraventricular tachycardia.
- Wide complex tachycardia may be supraventricular tachycardia or ventricular tachycardia.
- Wide QRS complex tachycardia with good perfusion can be treated with amiodarone OR procainamide (not both). Expert consultation is recommended.
- Wide QRS complex is irregular, this is ventricular tachycardia and should be treated with unsynchronized cardioversion (i.e. shock) immediately.
- Both wide and narrow supraventricular tachycardia with good perfusion can be treated with vagal maneuvers and adenosine by rapid bolus. If adenosine is unsuccessful, proceed to synchronized cardioversion.
- Narrow complex supraventricular tachycardia with a regular rhythm is treated with 50-100 J of synchronized cardioversion energy.
- Narrow complex supraventricular tachycardia with an irregular rhythm is treated with 120-200 J of synchronized cardioversion energy.



ECG Characteristics of Tachyarrhythmias		
Sinus Tachycardia	**Supraventricular tachycardia**	**Ventricular tachycardia**
Narrow QRS complex P waves normal PR interval constant R-R interval may be variable	Narrow or wide QRS complex P waves absent or abnormal R-R interval may be constant	Wide QRS complex P waves may not be present/seen QRS complexes may be uniform or variable

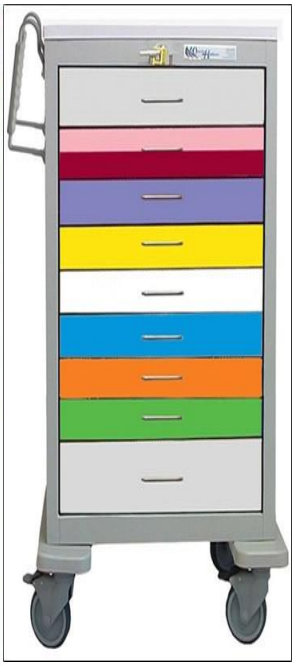
PALS Tools

Broselow Pediatric Emergency Tape System

A variety of tools is available for use in PALS, each with a size adapted to the child's size. The most commonly used system for correlating tools to the size of a child is the Broselow Pediatric Emergency Tape System. The provider can quickly measure the length/height of the child using color-coded tape. The resuscitation then uses tools (and in some hospitals, medications) proportional to the child's size. The medication cart or crash cart is stocked using the color coding system.

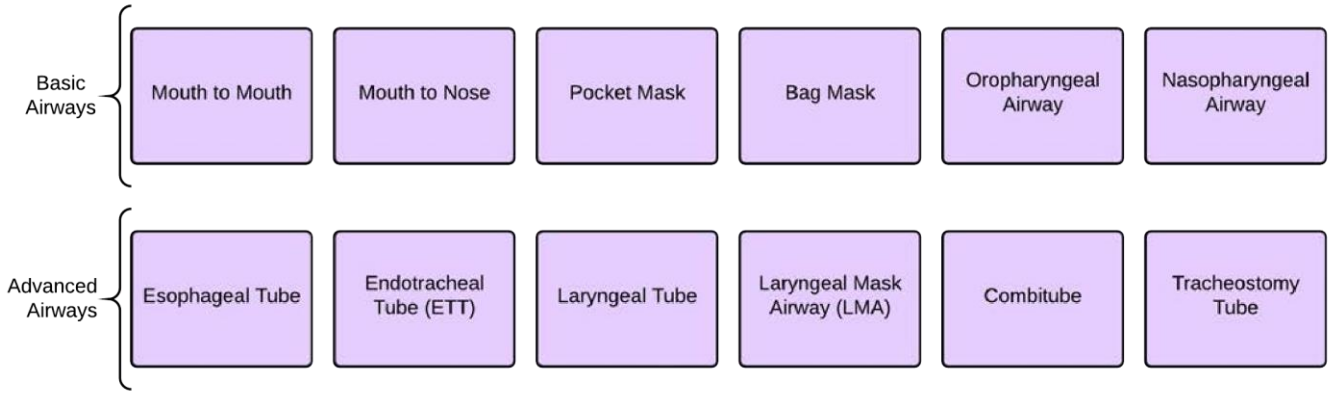
Equipment	Newborn/ Small infant (3-5 kg)	Infant (6-9 kg)	Toddler (10-11 kg)	Small Child (12-14 kg)	Child (15-18 kg)	Child (19-22 kg)	Large Child (24-30 kg)	Adult (≥32 kg)
Resuscitation bag	Infant	Child	Child	Child	Child	Child	Child/adult	Adult
O_2 mask	Newborn	Newborn	Pediatric	Pediatric	Pediatric	Pediatric	Adult	Adult
Oral airway	Infant/small child	Infant/small child	Small child	Child	Child	Child/small adult	Child/small adult	Medium adult
Laryngoscope blade (size)	0-1 straight	1 straight	1 straight	2 straight	2 straight or curved	2 straight or curved	2-3 straight or curved	3 straight or curved
Tracheal tube (mm)	Premature infant 2.5 Term infant 3.0-3.5 uncuffed	3.5 uncuffed	4.0 uncuffed	4.5 uncuffed	5.0 uncuffed	5.5 uncuffed	6.0 cuffed	6.5 cuffed
Tracheal tube length (cm at lip)	10-10.5	10-10.5	11-12	12.5-13.5	14-15	15.5-16.5	17-18	18.5-19.5
Stylet (F)	6	6	6	6	6	14	14	14
Suction catheter (F)	6-8	8	8-10	10	10	10	10	12
BP cuff	Newborn/ infant	Newborn/ infant	Infant/child	Child	Child	Child	Child/adult	Adult
IV catheter (G)	22-24	22-24	20-24	18-22	18-22	18-20	18-20	16-20
Butterfly (G)	23-25	23-25	23-25	21-23	21-23	21-23	21-22	18-21
Nasogastric tube (F)	5-8	5-8	8-10	10	10-12	12-14	14-18	18
Urinary catheter (F)	5-8	5-8	8-10	10	10-12	10-12	12	12
Defibrillation/ cardioversion external paddles	Infant paddles	Infant paddles until 1 yr or 10 kg	Adult paddles when ≥1 yr or ≥10 kg	Adult paddles	Adult paddles	Adult paddles	Adult paddles	Adult paddles
Chest tube (F)	10-12	10-12	16-20	20-24	20-24	24-32	28-32	32-40

*Adapted from the Broselow Pediatric Resuscitation Tape, with permission from Armstrong Medical Industries, Lincolnshire, Ill. Modified from Hazinski MF, ed.



PALS Airways

Basic airways do not require specialist training; however, some proficiency is needed for oropharyngeal and nasopharyngeal airway placement. Proper bag mask technique requires a tight seal between the mask and the child's face.



Intraosseus Access

➢➢ Intraosseus access is an acceptable alternative to IV access in children because the bones are softer and the marrow can be accessed quickly and reliably in emergencies.

➢➢ IO access also permits chest compressions to continue without interruption (arm IV placement is sometimes more difficult during chest compressions).

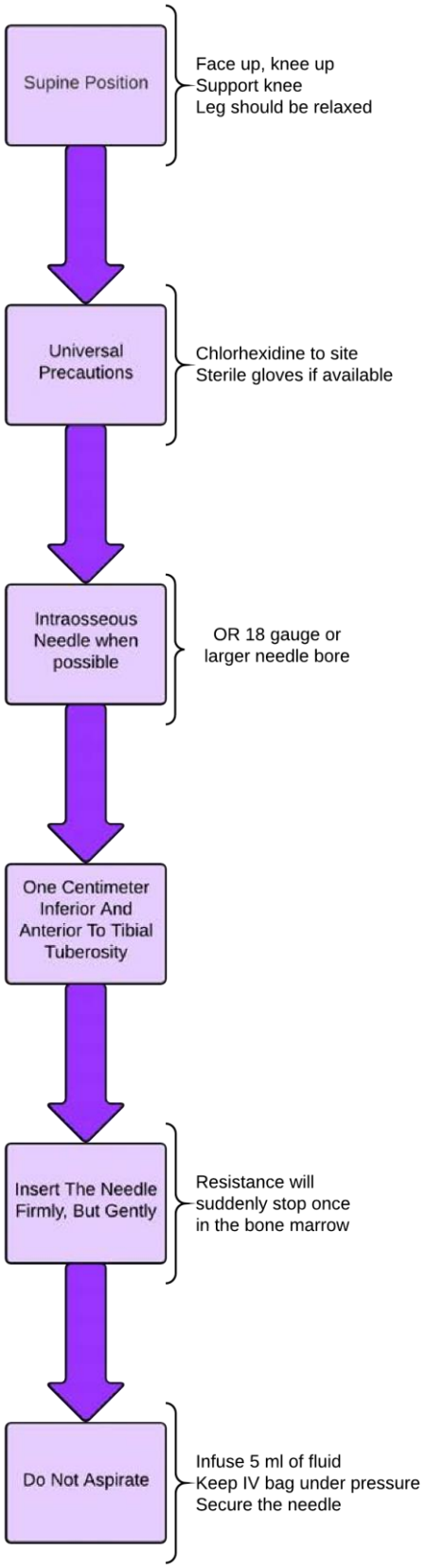
➢➢ IO access can be obtained in the:
- Proximal tibia
- Distal tibia
- Distal femur
- Anterior superior iliac spine

➢➢ An algorithm for obtaining IO access in the proximal tibia is shown.

➢➢ Avoid IO access in fractured bones, near infection, or in the same bone after a failed access attempt.

➢➢ After reaching the bone's interior, do not aspirate and immediately flush with 5 ml of fluid.

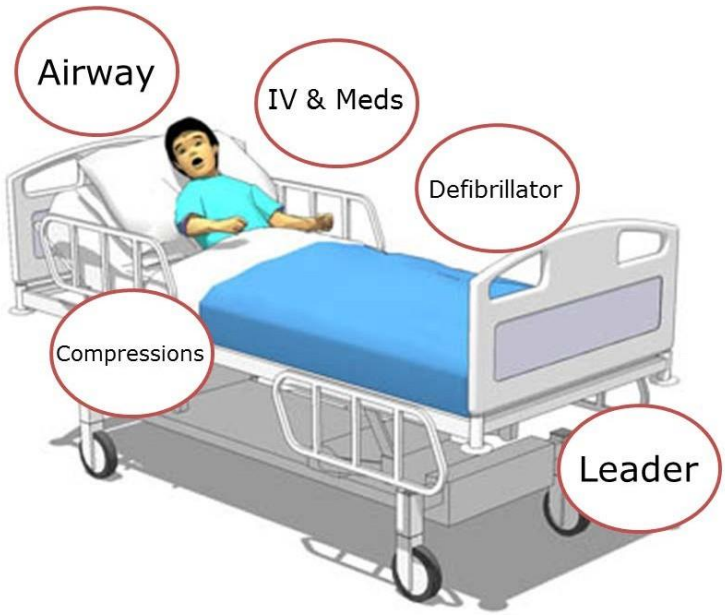
➢➢ Once the resuscitation is successful, replace the IO access with large bore IV access or central line as soon as possible (<24 hours) to avoid infection.



Team Dynamics/Systems of Care

The 2015 edition of the AHA ACLS guidelines highlights the importance of effective team dynamics during resuscitation. ACLS in the hospital will be performed by several providers. These individuals must provide coordinated, organized care. Providers must organize themselves rapidly and efficiently. The AHA recommends establishing a Team Leader and several Team Members. The Team Leader is usually a physician, ideally the provider with the most experience in leading ACLS codes. Resuscitation demands mutual respect, knowledge sharing, and constructive criticism, after the code.

Team Leader Responsibilities	Team Member Responsibilities
Usually stands at the foot of the bed	Stands in a position dictated by role
Competent in all ACLS duties	Competent in specific role (at least)
Directs Team Members in a professional, calm voice	Responds with eye contact and voice affirmation
Assigns roles	Clearly states when he/she cannot perform a role
Listens for confirmation from Team Member	Informs Team Leader when task is complete
Ask for ideas from Team Members when needed	Openly share suggestions if it does not disrupt flow
Critiques Team Performance after code	Provides constructive feedback after code
Documents resuscitation in patient chart	Provides information for documentation as needed



When performing a resuscitation, the Team Leader and Team Members should assort themselves around the patient so they can be maximally effective and have sufficient room to perform the tasks of their role.

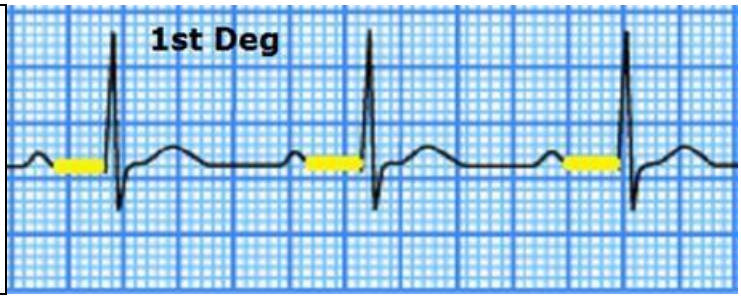
ECG Rhythms

Atrioventricular (Heart) Block

Atrioventricular block or heart block is a failure of the heart's electrical system to properly coordinate conduction. There are four main types of atrioventricular block: first degree, second degree type I, second degree type II, and third degree heart block. The types of second degree heart block are referred to as Mobitz type I and Mobitz type II. Second degree heart block Mobitz type I is also known as the Wenckebach phenomenon.Heart block is important because it can cause hemodynamic instability and can evolve into cardiac arrest.

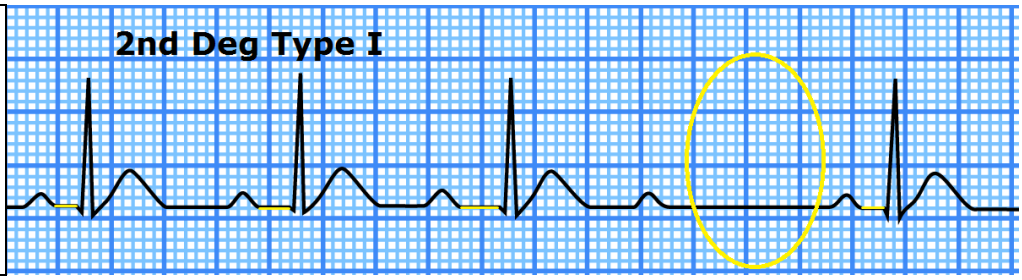
First degree atrioventricular block

The PR interval is a consistent size, but longer or larger than it should be in first degree heart block.



Second degree atrioventricular block, Mobitz type I (Wenckebach)

The PR interval increases in size until a QRS complexes dropped, resulting in missed "beat."



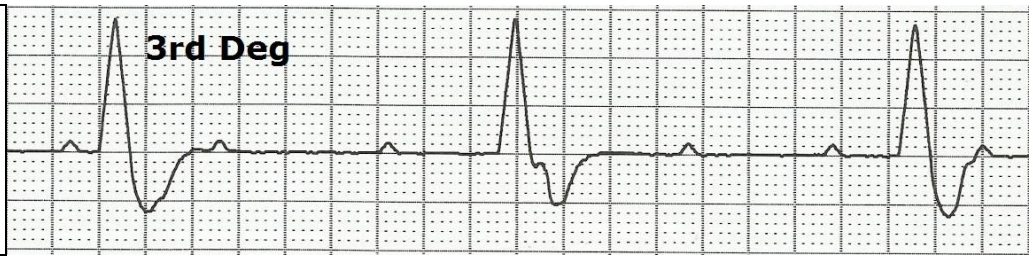
Second degree atrioventricular block, Mobitz type II

A QRS wave will occasionally drop, though the PR interval is the same size.



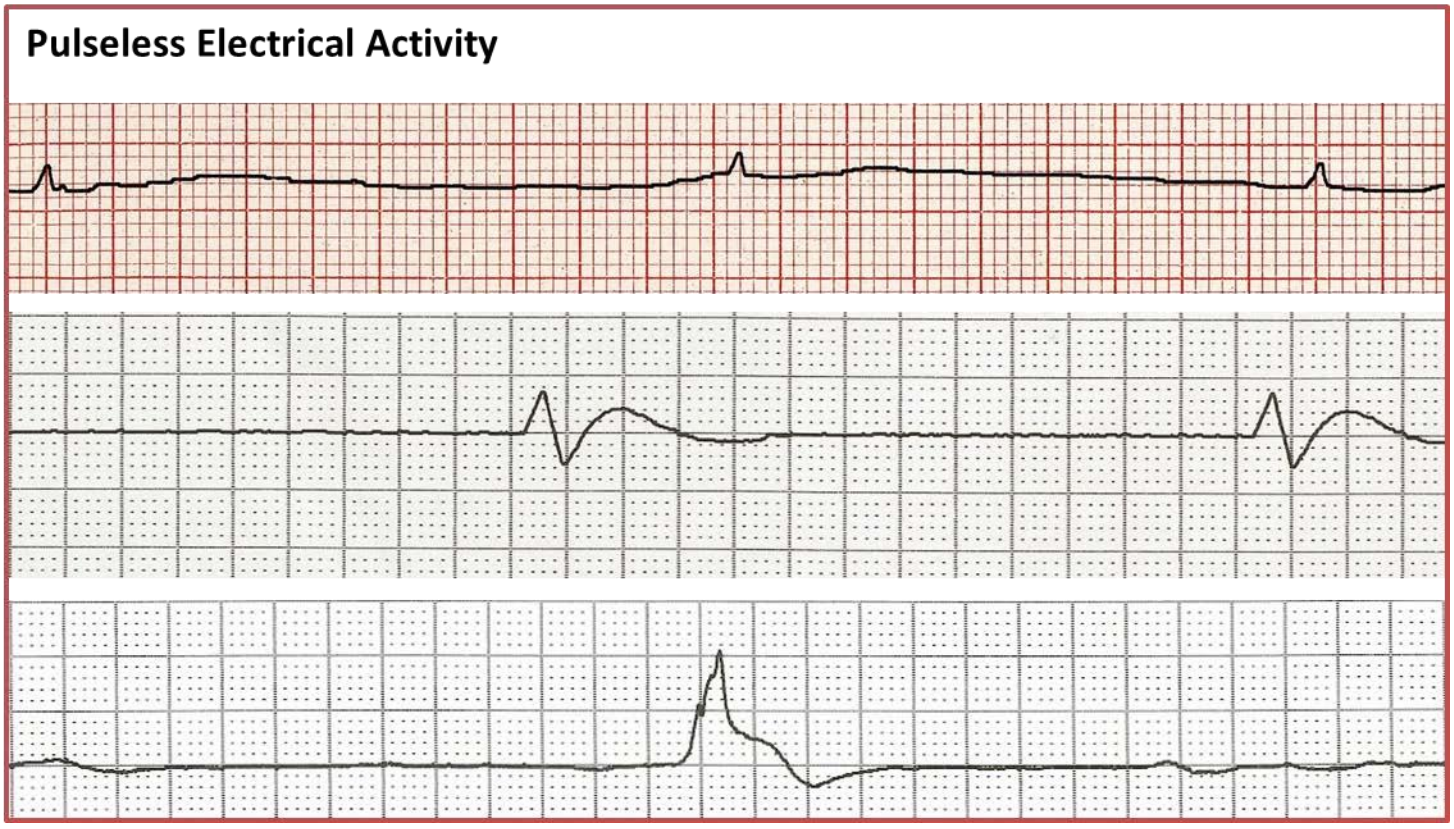
Third degree (complete) atrioventricular block

Complete dissociation between P waves and the QRS complex. No atrial impulses reach the ventricle.

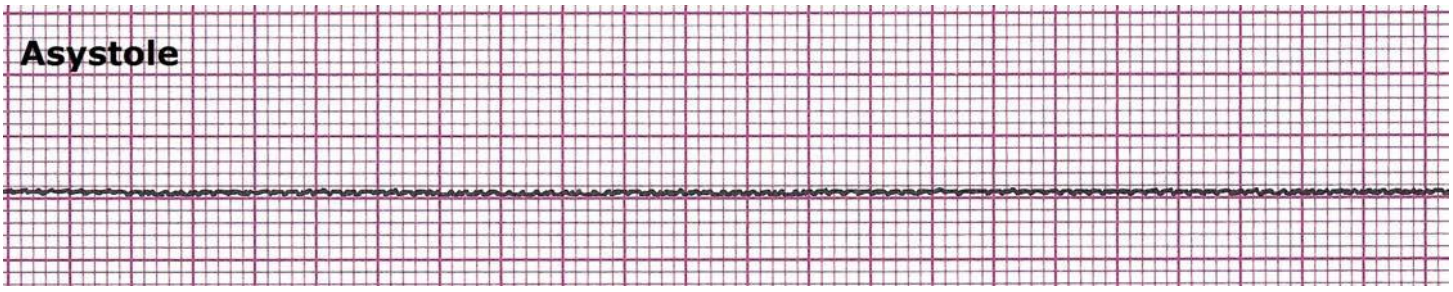


Pulseless Electrical Activity and Asystole

Pulseless electrical activity or PEA is a cardiac rhythm that does not create a palpable pulse is even though it should. A PEA rhythm can be almost any rhythm except ventricular fibrillation (incl. torsade de pointes) or pulseless ventricular tachycardia.



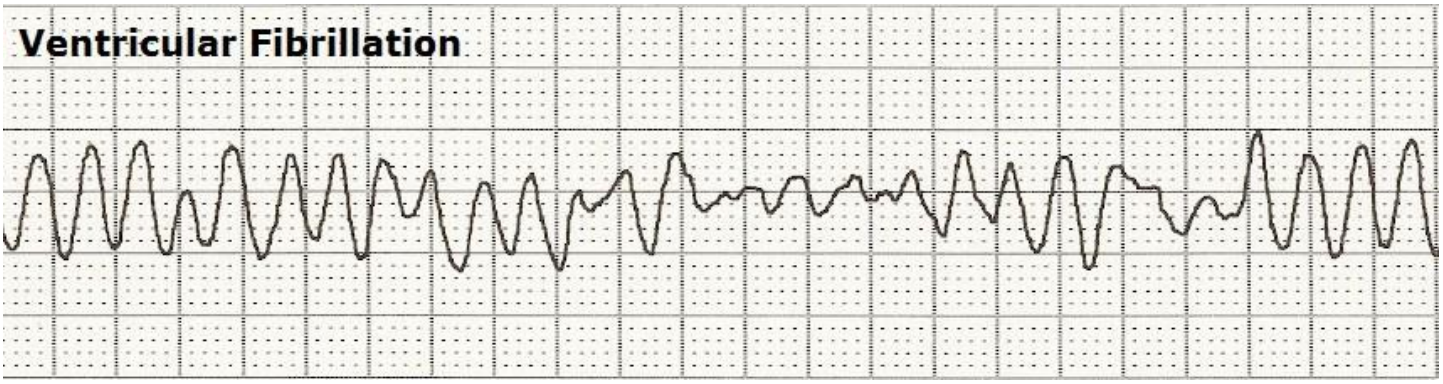
Asystole is the "flatline" on the ECG monitor. It represents a lack of electrical activity in the heart. It is critically important not to confuse true asystole with disconnected leads or an inappropriate gain setting on an in-hospital defibrillator. Asystole may also masquerade as a very fine ventricular fibrillation. If the ECG device is optimized and is functioning properly, a flatline rhythm is diagnosed as asystole. Note that asystole is also the rhythm one would expect from a person who has died. Consider halting PALS efforts in people who have had prolonged asystole.



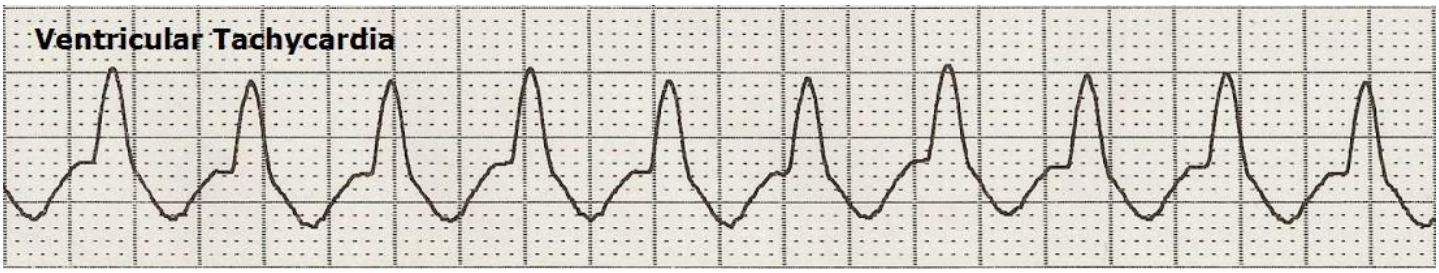
It is inappropriate to provide a shock to pulseless electrical activity or asystole. Cardiac function can only be recovered in PEA or asystole through the administration of medications.

In ventricular fibrillation or pulseless ventricular tachycardia, the heart's conduction system exhibits a disordered rhythm that can sometimes be corrected by applying energy to it. This energy may come in the form of an automated external defibrillator (AED) defibrillator paddles, or defibrillator pads. VFib and VTach are treated with unsynchronized cardioversion, since there is no way for the defibrillator to decipher the disordered waveform. In fact, it is important **not** to provide synchronized shock for these rhythms.

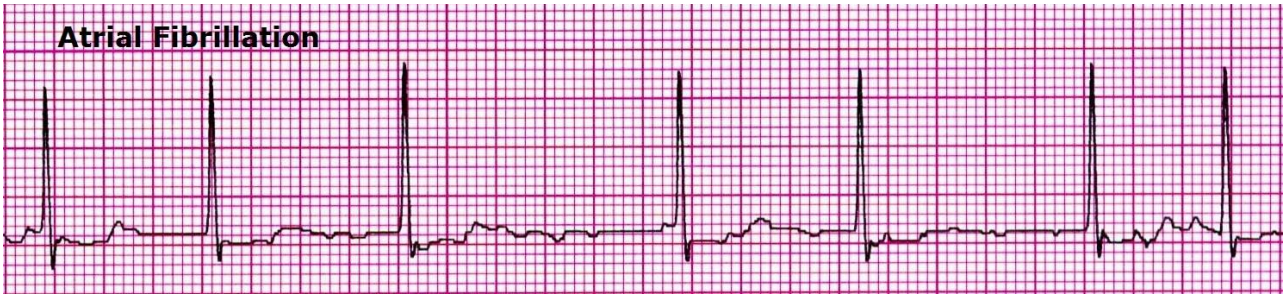
Ventricular fibrillation is recognized by a disordered waveform, appearing as rapid peaks and valleys as shown in this ECG rhythm strip:



Ventricular tachycardia may provide waveform similar to any other tachycardia; however, the biggest difference in cardiac arrest is that the patient will not have a pulse and, consequently, will be unconscious and unresponsive. Two examples of ventricular tachycardia are shown in this ECG rhythm strips. The first is narrow complex tachycardia and the second is wide complex tachycardia:



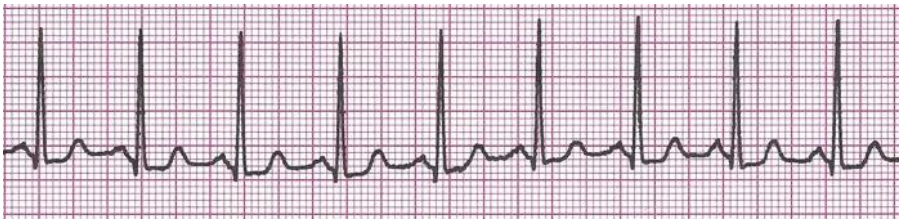
Atrial fibrillation is the most common arrhythmia. It is diagnosed by electrocardiogram, specifically the RR intervals follow no repetitive pattern. Some leads may show P waves while most leads do not. Atrial contraction rates may exceed 300 bpm. The ventricular rate often range is between 100 to 180 bpm. The pulse may be "irregularly irregular."



Atrial flutter is a cardiac arrhythmia that generates rapid, regular atrial depolarizations at a rate of about 300 bpm. This often translates to a regular ventricular rate of 150 bpm, but may be far less if there is a 3:1 or 4:1 conduction. By electrocardiogram, or atrial flutter is recognized by a sawtooth pattern sometimes called F waves. These waves are most notable in leads II, III, and aVF.



Narrow QRS complex tachycardias include several different tachyarrhythmias. A narrow QRS complex tachycardia is distinguished by a QRS complex of less than 90 ms. One of the more common narrow complex tachycardias is supraventricular tachycardia, shown below. The heart rate can exceed 220 bpm in infants and 180 bpm in children.



Wide complex tachycardias are difficult to distinguish from ventricular tachycardia. Ventricular tachycardia leading to cardiac arrest should be treated using the ventricular tachycardia algorithm. A wide complex tachycardia in a conscious child should be treated using the tachycardia algorithm. Tissue perfusion will dictate which algorithm to use.

Resuscitation and Life Support Medications

Drug	Use(s)	Dosage/Route	Contraindications/ Warnings
Adenosine	Supraventricular Tachycardia	First dose: 0.1 mg/kg IV Push Max 1st Dose: 6 mg. First dose: 0.2 mg/kg IV Push Max 1st Dose: 12 mg.	Second or third degree heart block
Albumin	Shock, trauma, burns	0.5 to 1 g/kg IV Push	Blood product
Albuterol	Asthma, bronchospasm, hyperkalemia	Weight <20kg: 2.5 mg Weight >20kg: 5 mg	Caution in tachyarrhythmias
Alprostadil	Maintain ductus arteriosis	First Dose: 0.05 to 0.1 mcg/kg/min Maintenance: 0.01 to 0.05 mcg/kg/min	Tissue sloughing in extravasation
Amiodarone	Supraventricular Tachycardia, Ventricular Tachycardia with Pulse	5 mg/kg over 20-60 min Max:300 mg max	Second or third degree heart block
	Ventricular Tachycardia Ventricular Fibrillation	5 mg/kg rapid bolus to 300 mg max Max:300 mg max	
Atropine	Symptomatic bradycardia	0.02 mg/kg IV (May give twice) Max dose: 0.5 mg 0.04-0.06 mg/kg via ETT	Dose < 0.5 mg may worsen bradycardia Do not use in glaucoma, tachycardia
	Toxins, poisons, and overdoses	0.02-0.05 mg/kg every 20-30 min	
Calcium chloride	Hypocalcemia, hyperkalemia, Calcium channel blocker overdose	20 mg/kg IV over 30-60 min	Monitor ECG and BP Contraindicated in digoxin toxicity or hypercalcemia
	PLUS Cardiac arrest	20 mg/kg IV bolus into central line	
Dexamethasone	Croup, Asthma	First Dose: 0.6mg/kg Max Dose: 16 mg	Can be given PO/IM/IV
Dextrose	Hypoglycemia	0.5 to 1 g/kg	Follow glucose
Diphenhydramine	Anaphylaxis after epinephrine	1 to 2 mg/kg every 4 to 6 h Max Dose: 50 mg	Use with caution in glaucoma, ulcer, hyperthyroidism
Dobutamine	Ventricular dysfunction	2 to 20 mcg/kg per min	Do not mix w/ sodium bicarb
Dopamine	Ventricular dysfunction, Cardiogenic or distributive shock	2 to 20 mcg/kg per min Titrate to response	Do not mix w/ sodium bicarb
Epinephrine	Anaphylaxis	0.01 mg/kg q15 min Max: 0.3mg	Make sure to distinguish and account for 1:1000 and 1:10000 concentrations.
	Asthma		
	Croup	0.25 ml via nebulizer	
	Cardiac Arrest	0.01 mg/kg (1:10000) IV or 0.1 mg/kg (1:1000) ETT q3-5 min	
	Shock	0.1 to 1 mcg/kg per min IV	
	Symptomatic Bradycardia	0.01 mg/kg IV q3-5 min Max: 1 mg	
	Toxins Overdose	0.01 mg/kg (1:10000) IV Max Dose: 1 mg	Avoid in cocaine-induced ventricular tachycardia
Etomidate	Sedation for intubation	0.2 to 0.4 mg/kg IV over 30-60 s Max Dose: 20 mg	Avoid in septic shock
Furosemide	Pulmonary Edema Diuresis	1 mg/kg IV/IM Max Dose: 20 mg	Monitor potassium
Hydrocortisone	Adrenal Insufficiency	2 mg/kg IV bolus Max Dose: 100 mg	Use with caution in infection
Inamrinone	Myocardial Dysfunction Cardiogenic Shock CHF	Loading: 0.75-1 mg/kg IV over 5-10 min. May repeat twice Max Dose: 3 mg/kg Maintenance 5-10 mcg/kg/min	Monitor ECG, oxygen, BP
Ipratropium	Asthma	250 to 500 mcg neb q 20 min	
Lidocaine	V Fib and Pulseless VT Wide Complex Tachycardia	1 mg/kg IV bolus Then 20 to 50 mcg/kg per min	Contraindicated for wide complex bradycardia
	Rapid Sequence Intubation	1-2 mg/kg IV	
Magnesium Sulfate	Asthma	25-50mg/kg IV over 15-30 min	Rapid bolus may cause hypotension and bradycardia
	Pulseless Torsades de pointes	25-50 mg/kg IV bolus	
	Ventricular Tachycardia w/ pulses	25-50 mg/kg over 10-20 m	

Drug	Use(s)	Dosage/Route	Contraindications/ Warnings
Methylprednisolone	Asthma Anaphylactic shock	Loading: 2 mg/kg IV (up to 60mg) Maintenance: 0.5 mg/kg q 6 h	Anaphylaxis possible
Milrinone	Cardiogenic shock Post-surgery CHF	Loading: 50 mcg/kg IV over 10-60 m Maintenance: 0.25-0.75 mcg/kg/min	Watch in hypovolemia
Naloxone	Narcotic reversal	Total Reversal: 0.1 mg/kg IV q 2 min Max Dose: 2 mg Partial Reversal: 1-5 mcg/kg IV	Consider airway before use
Nitroglycerine	CHF Cardiogenic Shock	Begin: 0.25-0.5 mcg/kg/min Titrate: q 15-20 minutes Max Dose: 10 mcg/kg/min	Watch for hypotension in hypovolemia
Nitroprusside	Cardiogenic Shock Hypertension	Begin: 0.3 to 1 mcg/kg/min Max Dose: 8 mcg/kg/min	Check thiocyanate and cyanide levels
Norepinephrine	Shock	0.1 to 2 mcg/kg/min Titrate to target blood pressure	Extravasation leads to tissue necrosis Give via central line
Procainamide	Atrial Flutter Supraventricular Tachycardia; Ventricular Tachycardia w/ Pulse	Loading: 15 mg/kg over 30-60 min	Follow QT int., BP Consider expert consultation
Sodium Bicarb	Severe Metabolic Acidosis Hyperkalemia	1 mEq/kg slow IV bolus Max: 50 mEq	Support ventilation *Not recommended in cardiac arrest*
Terbutaline	Asthma Hyperkalemia	10 mcg/kg SQ q 10-15 min until IV access 0.1-10 mcg/kg/min IV	Use with caution in hypokalemia
Vasopressin	Cardiac Arrest Septic Shock	0.4-1 unit/kg IV bolus Max Dose: 40 units	Check distal pulses Water intoxication Extravasation causes tissue necrosis



Neonatal Resuscitation

A Life-Saving Guide for Newborns

Contents

Neonatal Resuscitation: An Overview 2

Transitioning from Fetus to Neonate 2

- Normal Transition: A Birth 2
- Problems with Transition 3
- Predicting the Need for Resuscitation 3

Neonatal Assessment 4

- The Apgar Score 4
  - The Apgar Score Is Not Used to Direct Resuscitation 5
- Assessing the Need for Neonatal Resuscitation 5
  - Staying with the Mother 6
  - Further Evaluation 6
- Pulse Oximetry 6

Neonatal Resuscitation Techniques 7

- Supplemental Oxygen 8
- Positive Pressure Ventilation 8
  - Positive Pressure Ventilation with Bag-Mask Devices 8
  - Effective Positive Pressure Ventilation 9
- Laryngeal Mask Airway 9
- Intubation 10
- Chest Compressions 11
- Epinephrine 11
- Volume Expansion 11

Resuscitation Tools 12

Factors That May Complicate Resuscitation 12

- Airway Obstructions 12
  - Choanal Atresia 12
  - Robin Syndrome 12
- Pulmonary Complications 1
- Impaired Respiratory Drive 1
- Cardiac Abnormalities 1

Post-Resuscitation Care 2

Resuscitating Preterm Babies 3

Ethical Considerations 4

References 5

Neonatal Resuscitation: An Overview

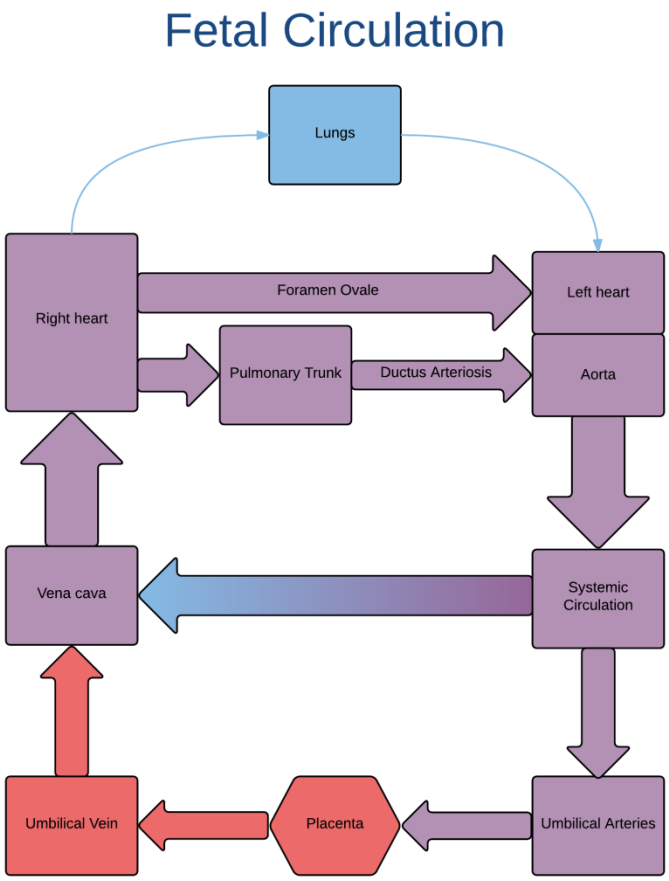
Nine out of ten neonates successfully transition from fetus to newborn. Unfortunately, the remaining 10% of neonates require assistance from medical providers. Less than 10% of these newborn babies (approximately 1% overall) will require neonatal resuscitation. On the other hand, almost one-quarter of neonatal deaths occur because of birth asphyxia, according to the medical journal *The Lancet* (Lawn, Cousens, & Zupan, 2005). Therefore, practitioners who are adept at neonatal resuscitation can save the lives of newborn babies.

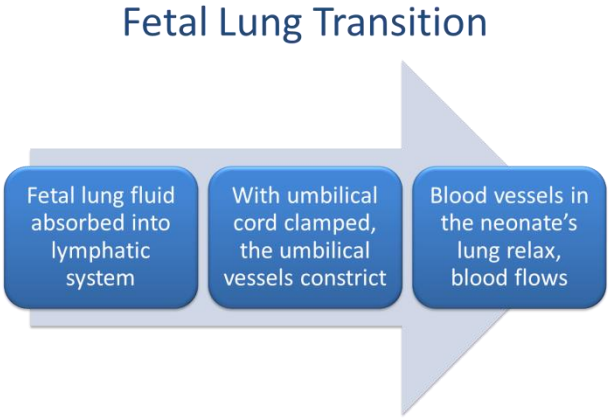
Transitioning from Fetus to Neonate

Normal Transition: A Birth

The transition from fetus to neonate, commonly called birth, is one of the most profound changes that human will ever undergo. During this transition, the fetus ceases to receive blood, oxygen, and nutrients from the maternal circulation and must derive those things from the outside world. Before birth, the neonate's lungs and gastrointestinal system are not functional.

The neonate's lungs, once filled with amniotic and fetal lung fluid in utero, become filled with air upon birth. The neonate's circulatory system must change from one that receives blood flow through the umbilicus, to a closed system driven by his own heart. In other words, the newborn baby must begin to breathe on its own and his heart must pump blood to his body. The blood will then circulate to the gastrointestinal system, as it will throughout his life.





In utero, the fetus' lungs are filled with amniotic fluid and fetal lung fluid. The blood vessels surrounding the alveoli of the fetal lung are constricted, with little blood flow. When the fetus is born, fluid within the lungs is rapidly absorbed as oxygen fills the lungs. At this time, blood flow increases in the lungs so that oxygen can be readily absorbed and then distributed to the rest of the body. For blood flow to increase, a few things must happen…

Problems with Transition

As with any complex biological process, sometimes things can go wrong. The transition from fetal to neonatal life may not occur smoothly. Any one (or more) of the things that must change during the transition may not actually change. Perhaps the neonate has air in the lungs, but the blood vessels surrounding the alveoli do not relax. Thus, the newborn is ventilated but the circulatory system is not receiving oxygen through the alveoli. This is a problem of the **pulmonary circulation**. The circulatory system may not transition from fetal to neonate; the problem is with the **systemic circulation**. The most common cause of problems with transition, however, occurs from problems with **pulmonary ventilation**. Despite the neonate's efforts, the newborn's lungs do not fill with air and remain full of amniotic/fetal fluid. The newborn will attempt to preserve blood flow to the heart and brain as much as possible, but sustained problems with transition will lead to brain damage, myocardial infarction, and/or death.

If anything goes wrong in the transition from receiving oxygen via the placenta to receiving oxygen from the lungs, the baby's ventilation will be compromised. In this case, a couple of things could happen:

1) The baby could attempt to breathe and then endure primary apnea. The reduced heart rate that occurs in this situation can be reversed with tactile stimulation.

2) With secondary apnea, the heart rate continues to drop, and blood pressure decreases as well. In this case, assisted ventilation must be employed because tactile stimulation is not enough to help the baby recover.

Predicting the Need for Resuscitation

There are some clues that can indicate that a baby may require resuscitation. We have already seen that babies born *prematurely* are at higher risk for requiring resuscitation. Babies born to obese mothers have also been shown to be at higher risk for requiring ventilation (Khalak, Cummings, & Dexter, 2015). Muscle tone is another indicator of health related to ventilation. With good muscle tone, the extremities are flexed, whereas *flaccid extremities* can indicate poor ventilation. The following table includes factors that increase the risk of a difficult child birth.

Mother	Fetus/Neonate	Peri-partum
<16 years or >40 years of age	Prematurity or Postmaturity	Prolapsed cord
Lower socioeconomic status	Macrosomia	Utero-placental bleeding
Smoking, alcohol/drug abuse	Intrauterine growth retardation	Breech presentation
Chronic/untreated medical conditions (e.g., diabetes, preeclampsia)	Multiple gestation	Chorioamnionitis
Worrisome obstetric/gestational issues (e.g., PROM, placenta previa)	Congenital anomalies	Meconium-stained amniotic fluid

PROM = Premature rupture of membranes

Neonatal Assessment

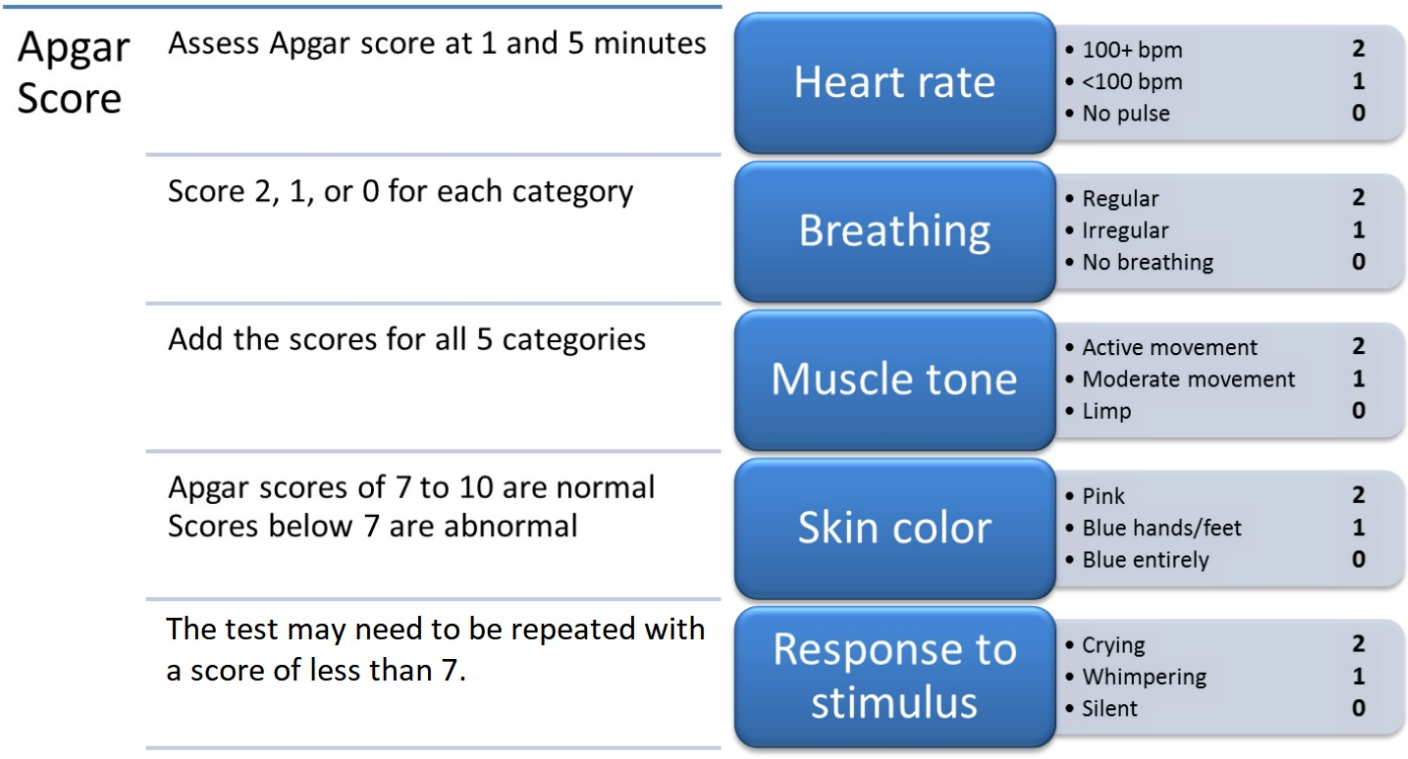
The Apgar Score

The Apgar score is a standardized score that provides information on the condition of the baby when the baby is born. The score quantifies 5 key factors: color, heart rate, reflex irritability, muscle tone, and respiration. Each factor is given a score of 0, 1, or 2. A 0 indicates a problem, a 2 indicates normal, healthy activity and a score of 1 indicates an intermediate state, neither healthy nor completely problematic. The 5 scores at each time interval are added for a total Apgar score. Apgar scores are routinely assessed at 1 and 5 minutes after birth. The test may need to be repeated with a score of less than 7.

A score 7 or above is normal, while a score below 7 indicates distress. A 5-minute Apgar score at or below 3 strongly suggests that neonatal death is imminent (Iliodromiti, Mackay, Smith, Pell, & Nelson, 2014). Despite this, **Apgar scores are not used to direct resuscitation efforts**. They simply provide a measure of the neonate's health status.

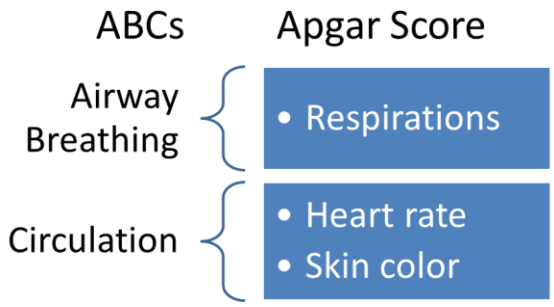
To calculate heart rate, count the number of heartbeats that occur in 6 seconds. Because one minute is 10 times more than 6 seconds, you can multiple the number of heartbeats that occur in 6 seconds by 10 to determine the beats per minute (often referred to as bpm), which is the standard metric for heart rate.

Acceptable stimuli are drying the baby and suctioning the airway, which will be sufficient to stimulate most neonates. If these do not elicit a response, one may try flicking in the feet and/or rubbing the baby's back. Do not stimulate the baby too vigorously and never shake a baby. Do not spend too long trying to induce a response through stimulation. Instead, focus on other interventions.



The Apgar Score Is Not Used to Direct Resuscitation

The Apgar score is a virtually universal instrument to assess newborn health and viability. You may wonder why it is not used to direct resuscitation efforts in neonates. The reason is simple: neonatal resuscitation, for those who need it, must begin before the first assignment of an Apgar score. While 60 seconds after birth does not seem like a long time, every second matters in neonatal resuscitation. Instead of Apgar score, healthcare professionals will focus on three factors at the time of birth: airway, breathing, and circulation.



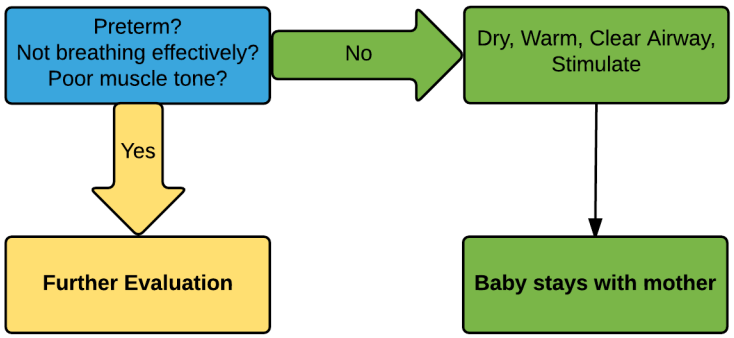
While adult resuscitation guidelines have moved to CAB, neonatal resuscitation still follows the ABCs because most neonates who need resuscitation require airway and breathing support, and less often circulatory support. The reverse is true in adults.

Assessing the Need for Neonatal Resuscitation

The initial assessment of every newborn baby, performed at the moment of birth, includes three questions:

1. Is the baby born at term?
2. Is the baby breathing/crying?
3. Does the baby have good muscle tone?

If the answer to all three of these questions is yes, the baby may stay with the mother. If the answer to any of these 3 questions is no, proceed to further evaluation.



Staying with the Mother

Staying with the mother does not necessarily mean placing the baby and the mother's arms immediately. Every neonate requires a few steps before initial mother child bonding can occur.

1. ***Warm the baby*** – To reduce any further heat loss, dry the baby and remove any wet linens. The baby may be put under a radiant warmer to reduce heat loss, but not be put under towels or blankets. Monitor the baby's temperature to ensure that the baby does not overheat.
2. ***Open the baby's airway*** – Put the baby in the "sniffing" position on its back or side, and be careful not to extend the neck too much or too little.
3. ***Clear the baby's airway*** – How precisely you clear the airway depends on whether the baby's skin has meconium on it, as well as the baby's activity level.
    a. *If meconium is present* – clear the baby's mouth and nose and dry the baby, stimulate it, and reposition it.
    b. *If meconium is absent* – check to see if the baby is vigorous, meaning that the baby has a heart rate over 100 bpm, good muscle tone, and is making respiratory efforts.
        i. *If the baby is vigorous* – behave as if meconium were present
        ii. *If the baby is not vigorous* - suction the baby's mouth and trachea gently

Further Evaluation

The first step in "further evaluation" is the same 3 steps as above:

1. ***Warm the baby***
2. ***Open the baby's airway***
3. ***Clear the baby's airway***

All babies should be immediately assessed for airway patency and proper ventilation. Once a baby is born, a vigorous cry usually means that a baby is breathing, but breathing can also be observed by watching a baby's chest. If the baby is *not breathing*, he or she will need resuscitation.

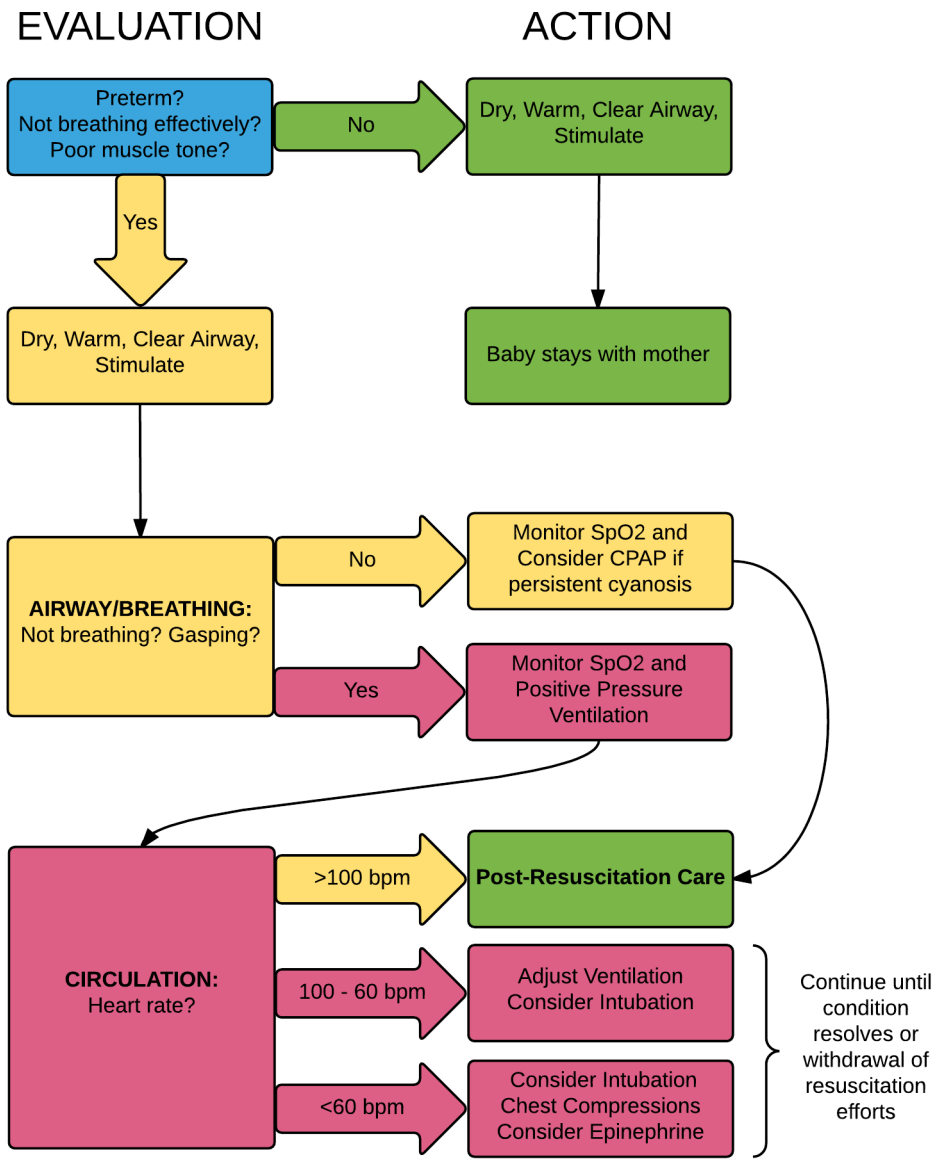
Pulse Oximetry

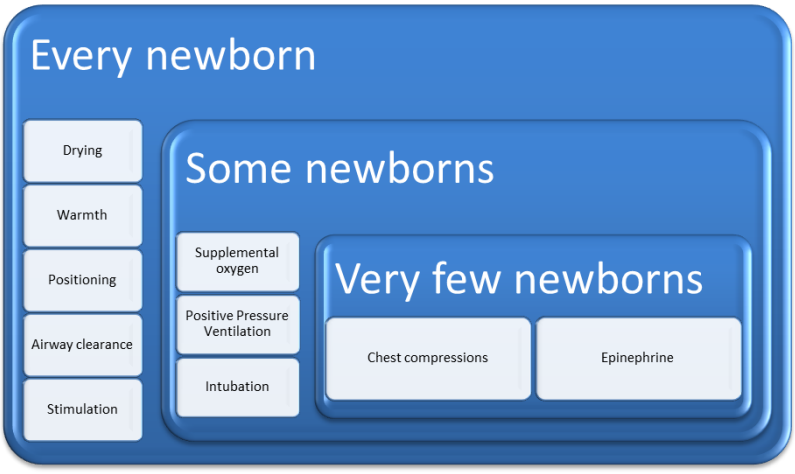
If parts of the baby's body appear blue, rather than pink, it could indicate cyanosis, or low oxygen in the blood. To confirm cyanosis, you can use a pulse oximeter. An oximeter provides the percentage of oxygenation in the blood (specifically hemoglobin). Place the oximeter in the baby's **right** hand or wrist to get a preductal (i.e., before the ductus arteriosus) measure of oxygen saturation of the blood. The right arm receives blood from the aorta prior to the ductus arteriosus. Importantly, you do not want oximetry to get in the way of resuscitation.

Target Oxygen Saturation over Time after Birth	
1 min.	60-65%
2 min.	65-70%
3 min.	70-75%
4 min.	75-80%
5 min.	80-85%
10 min.	85-95%

A pulse oximeter is used when there is persistent cyanosis, when supplemental oxygen is given, and/or positive pressure ventilation is required for more than a few breaths. Normal oxygen saturation values vary by the age of the neonate, measured in minutes, as presented in the table.

Neonatal Resuscitation Techniques





As previously described, very few newborn babies will require chest compressions or epinephrine. In fact, most neonates will respond to simple warming and airway suctioning. The goal of neonatal resuscitation is to act promptly, but with a measured, reasonable approach.

As such, it is helpful to think about the interventions that neonates require most often to least often.

Supplemental Oxygen

The percent of oxygen that should be used during resuscitation depends on whether the baby made it to term. Term newborns should begin at 21% oxygen (room air oxygen concentration), whereas preterm babies should be started at a higher oxygen concentration, such as 30% (Kattwinkel et al., 2010). Unfortunately, high concentrations of oxygen are toxic to lung tissue, especially in preterm neonates. Oxygen concentrations exceeded room air (21%) must be used judiciously, since there is a trade-off between lung tissue damage and resuscitation efforts.

Positive Pressure Ventilation

Positive pressure ventilation may be appropriate in these circumstances to increase heart rate:

- if apnea and gasping is occurring
- if the heart rate is below 100 beats per minute
- if there is persistent cyanosis

Positive Pressure Ventilation with Bag-Mask Devices

Positive pressure ventilation can be achieved with different types of bag-mask devices, which have different relative advantages and disadvantages.

Self-Inflating Bags

Pros

- Oxygen fills these bags spontaneously with a simple squeeze of the bag.
- These bags remain inflated on their own.
- Using these bags does not require a compressed gas source.

Cons

- These bags require a tight seal to inflate the newborn's lungs.
- These bags need an oxygen reservoir.
- These bags cannot deliver positive airway pressure continuously
- These bags need an integral pressure gauge.

Flow-Inflating Bags

Pros

- These bags can administer positive airway pressure continuously.
- Pressure and inflation can be regulated with a valve.

Cons

- These bags require a tight seal to inflate the newborn's lungs.
- These bags need a compressed gas source.
- There are a number of scenarios in which these bags do not work, including the valve being too far open, the port not being occluded, or the gauge missing.

T-Piece Resuscitators

Pros

- These resuscitators can administer positive airway pressure continuously.

Cons

- These resuscitators require a tight seal to inflate the newborn's lungs.
- These resuscitators require a compressed gas source.

Effective Positive Pressure Ventilation

The process of bag mask ventilation in neonatal resuscitation is the same regardless of the device chosen:

Suction: Mucus or secretions should be suction from the nose and mouth before starting positive pressure ventilation and as needed throughout the procedure.

Position: Proper positioning is key to effective positive pressure ventilation. The baby's neck should be in a neutral position, not constricted but also not hyperextended. In other words, the oropharynx should be placed at a 90° angle.



Seal: there must be an airtight seal between the rim of the mask and the neonate's face. Likewise, the size of the mask used should be appropriate to the size of the baby. Proper technique is important for achieving and maintaining a tight seal.

It is also important to make sure that the initial breaths administered with positive pressure ventilation provide enough pressure to inflate a newborn's lungs. On the other hand, it is important not to use excessive volume or pressure as this can cause barotrauma, or trauma to the lungs due to excessive pressures.

Positive pressure ventilation has generally been effective if the baby makes bilateral breath sounds and demonstrates chest movement. If the baby's heart rate rises above 100 breaths per minute, the baby begins breathing on its own, and experiences improved oxygen saturation, positive pressure ventilation can probably be stopped.

If positive pressure ventilation is not working, there are things to check, including the position of the mask you may be using and the position of the airway. Increasing pressure and the suction on mouth or nose are other strategies to improve the effects of positive pressure ventilation. Research has also shown that applying surfactant, which is a substance that reduces surface tension, through a catheter, can improve positive airway pressure and minimize the requirement of mechanical ventilation (Gopel et al., 2011).

Laryngeal Mask Airway

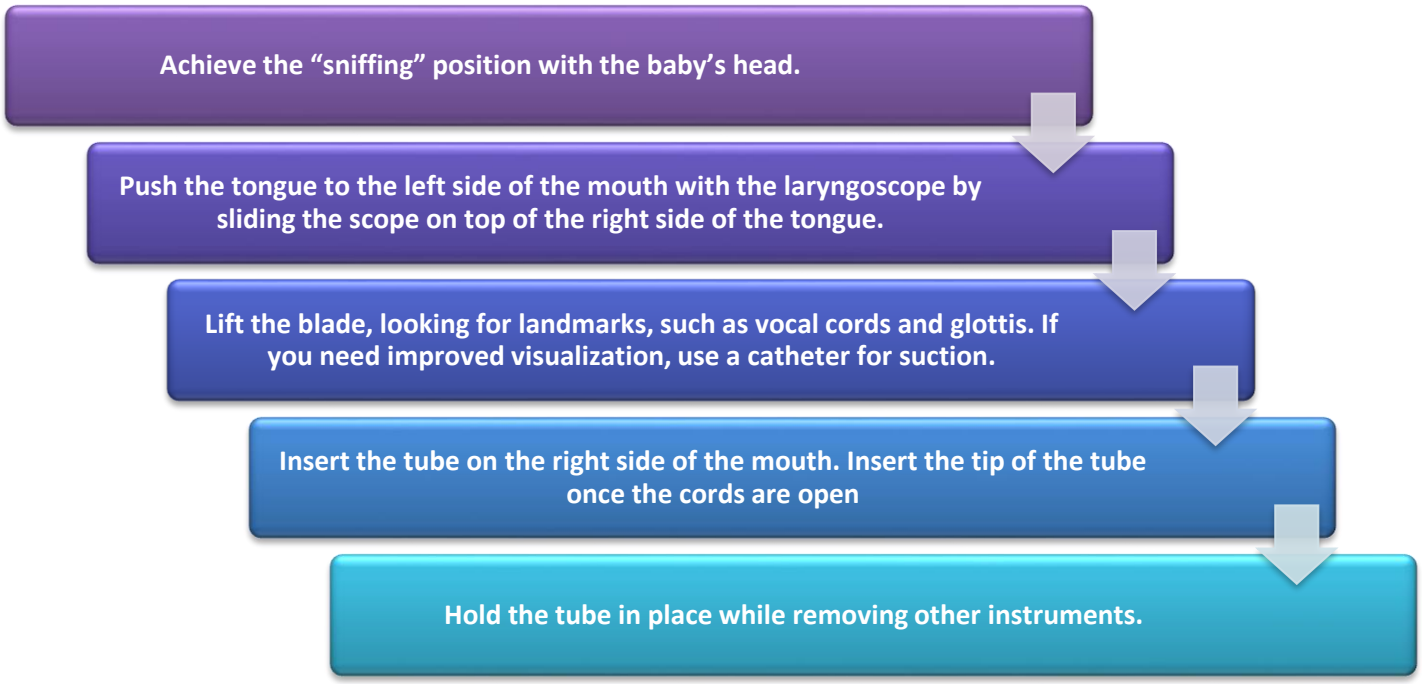
You may want to consider placing a laryngeal mask airway if the ventilation mask is not working due to malformations of the newborn's face or upper airway. Another time to think about using a laryngeal mask is when the facemask is not achieving positive-pressure ventilation and intubation is not feasible. However, there are a number of shortcomings associated with laryngeal masks that should also be considered before initiating their use, which include:

- It is not a long-term option for ventilation.
- Air leaks can occur, minimizing pressure delivered to the lungs.
- The mask does not provide a way to suction meconium out of the airway.
- Laryngeal masks are too big for preterm babies born before about 32 weeks of gestational age.

Intubation

Successful intubation requires a specifically trained professional, and one of these individuals should always be present at delivery, in case intubation is necessary. Intubation aims to:

- Improve how well ventilation is working
- Suction the trachea if needed
- Improve chest compression and ventilation coordination



Key things to keep in mind during intubation include:

- The laryngoscope should be held in the left hand
- The process should be completed within approximately 30 seconds.
- Blade size for the laryngoscope depends on whether the baby was at term
  - If at term, blade No. 1 is appropriate
  - If preterm, blade No. 0 is appropriate
  - If very preterm, blade No. 00 is appropriate
- Tube diameter size depends on the baby's weight
  - If the baby is less than 1000 grams, the 2.5 mm tube should be used.
  - If the baby is 1000-2000 grams, the 3.0 mm tube should be used.
  - If the baby is 2000-3000 grams, the 3.5 mm tube should be used.
  - If the baby is more than 3000 grams, the 3.5 or 4.0 mm tube should be used.

A successful intubation is signaled by:

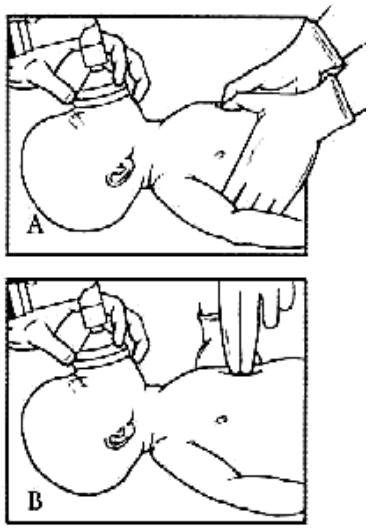
- An improvement in vital signs
- Carbon dioxide being exhaled or vapor in the tube during exhalation
- The presence of breathing signs
- Chest movement during breaths

The placement of the intubation tube can also be confirmed by visualizing the tube between the vocal cords, and x-rays can be used to confirm the chest placement of the tube.

Chest Compressions

Chest compressions are not often required during the resuscitation of newborns; however, if a baby's heart rate has not risen above 60 beats per minute after 30 seconds of positive-pressure ventilation, chest compressions should be administered. Chest compressions increase the pressure within the thoracic cavity by compressing the heart against the spine, thereby reducing the volume within that space. The effect is that blood should circulate to important organs of the body.

The two-thumb technique is generally the best way to perform chest compressions (Panel A). The thumb technique is recommended in neonates because it generates higher systolic and coronary perfusion pressures (Saini, Gupta, Kumar, Bhalla, & Kaur, 2012). You can locate the area where compressions should be performed by finding the xiphoid along the lower part of the rib cage. You can then place your thumbs above the xiphoid, on the sternum. During the compressions, you will want to ensure that chest movement occurs, and your thumb remains in contact with the chest. Compress to a depth of about one third the diameter of the baby's chest. Release all the pressure during the relaxation phase of compression, and the release should last longer in time than the downward compression. There should be 120 movements that occur each minute – 30 breaths and 90 compressions, with 3 compressions for each breath (Hemway, Christman, & Perlman, 2013). Finger compressions (Panel B) are acceptable, but less effective, generally.



After about a minute of chest compressions, check the baby's heart rate.

- If the heart is still beating at a rate below 60 beats per minute, the baby should be intubated to continue ventilation.
- If the heart rate is between 60 and 100 beats per minute, stop the chest compressions and continue ventilation at 40 to 60 breaths per minute.
- If the heart rate is over 100 beats per minute, you can stop compressions and taper back on ventilation as the baby begins to breathe on his own.

Epinephrine

Epinephrine, or adrenaline, increases blood pressure and stimulates the heart. Epinephrine should be rapidly administered to a newborn through the umbilical vein if the heart rate has stayed below 60 beats per minute after 30 seconds of assisted ventilation. Epinephrine can also be delivered via an endotracheal route, but it is not the preferred method because epinephrine does not absorb into the baby's system as well with this method as when epinephrine is delivered through the umbilical vein. Newborns should be given doses of epinephrine ranging from 0.1 to 0.3 mL/kg of a 1:10,000 concentration solution, or 0.1 mg/mL. After epinephrine is used, chest compressions and ventilations should be resumed for an additional minute. Fewer than 1% of babies who need to be resuscitated also need epinephrine.

Volume Expansion

If a newborn is not responding to resuscitation and seems to be in shock, volume expansion can be undertaken with a volume expander. Volume expansion can be achieved with normal saline (0.9% NaCl) or Ringer's lactate. If severe anemia is present, Type O, Rh-factor-negative blood is also acceptable. The initial dose of a volume expander is 10 mL/Kg of infant weight. This can be followed by a second dose, if needed.

Resuscitation Tools

There are a number of things that should be prepared ahead of every single birth, to ensure efficient resuscitation procedures are implemented if resuscitation is required. The things to prepare include:

- A stethoscope
- A laryngoscope with blades
- A bulb syringe
- A meconium aspirator
- A suction catheter
- A preheat warmer
- Warming blankets or towels
- Warming pad
- A free-flow oxygen device
- An air-oxygen blender
- A pulse oximeter
- A pulse oximeter probe
- A positive-pressure ventilation device
- A feeding tube
- Endotracheal tubes
- Stylets
- A laryngeal mask
- Epinephrine
- Saline
- Documentation materials
- A transport incubator
- Plastic bags or wraps

Factors That May Complicate Resuscitation

Airway Obstructions

If resuscitation does not seem to be working, there are some special considerations that should be assessed. In many cases, complication relates to a constricted or blocked airway such as laryngeal webs, cystic hygroma, or congenital goiter. Practically speaking, the airway obstruction is usually in the nasal pharynx (e.g., choanal atresia) or the oral pharynx (e.g., Robin syndrome).

Choanal Atresia

Babies do not normally breathe through their mouths unless they are crying. In a way, they can be considered obligate nose breathers. In the case of choanal atresia, however, the nasal airway is not fully patent (open). This means that the baby can only breathe effectively through crying or with assistance. One clue to the existence of choanal atresia is the presence of meconium or mucus is in the nasal airway. A suction catheter gently applied through the nares into the posterior pharynx can test for this condition. If the catheter cannot pass so that it is visible in the oral pharynx, you can assume that choanal atresia exists and an oral airway will be necessary.

Robin Syndrome

Robin syndrome or Pierre Robin syndrome is the co-existence of three congenital abnormalities:

- A small or underdeveloped lower jaw (micrognathia/retrognathia)
- Cleft palate
- Upper respiratory breathing obstruction



These congenital abnormalities essentially result in a blocked oral pharynx (as opposed to a blocked nasal pharynx present in choanal atresia. Intubation through the mouth is quite difficult in a child with Robin syndrome. Putting the baby on its stomach can push the tongue forward and open the airway. If that action is not adequate to improve the condition, a catheter can be used to open the airway.

Pulmonary Complications

The neonate, and especially the premature infant, can develop one or more problems in the lungs that complicate neonatal resuscitation. In the very premature infant, the lungs either cannot support respiration and oxygenation or can only do so marginally. Artificial surfactant can help considerably in these cases by reducing surface tension in the alveoli and reducing pressures required to ventilate the lungs.

Another form of lung malformation is pulmonary hypoplasia. In pulmonary hypoplasia (which is more common in fetuses exposed to insufficient amounts of amniotic fluid during gestation), the lungs have simply not formed during fetal development. Less severe cases of pulmonary hypoplasia can be effectively treated with long-term intensive care, but children with severe cases of pulmonary hypoplasia often do not survive the neonatal period.

Some of the more common causes of impaired lung function can be reversed with timely bedside or surgical procedures, assuming they are detected in the early neonatal period. For example, many babies who require neonatal resuscitation are born with a pneumothorax or develop one during resuscitation (particularly ventilation). In pneumothorax is the presence of air in the pleural space, between the chest wall and the outside of the lungs. A pneumothorax causes substantial respiratory distress and is diagnosed through trans illumination of the chest cavity, the absence of lung sounds of one of the chest, or a portable chest x-ray if needed. A pneumothorax can be treated with needle thoracostomy where the placement of a catheter to evacuate the air in the pleural space.

Pleural effusions and congenital diaphragmatic hernias are rare, but potentially treatable causes of poor lung function in the neonate. A pleural effusion is treated in much the same way as a pneumothorax, releasing fluid instead of air. A baby with congenital diaphragmatic hernia is usually diagnosed by ultrasound prior to delivery. However in women who have not had routine prenatal screenings, the hernia may go undiagnosed until delivery. The baby can be stabilized with separate tubes in the trachea and stomach until pediatric surgery can repair the hernia.

Impaired Respiratory Drive

Women who received opioid analgesics during delivery or women who are actively intoxicated with illicit opioids may deliver infants with substantial levels of opioids in their systems. In these cases, the problem with respiration is not an impaired airway or a pulmonary problem, but the drive to breathe is depressed. When this occurs, the baby can be ventilated until the opioids had been metabolized. Naloxone, an opioid antagonist, should be avoided in babies of women with opioid abuse problems or on methadone treatment because the drug can cause withdrawal seizures in the neonate.

Cardiac Abnormalities

Several types of congenital heart malformation can interfere with circulation, but few of them manifest in the newly born infant. Providers may consider a congenital heart problem after ventilation has proved fruitless. This requires specialist diagnostic and management skills that are outside the purview of neonatal resuscitation.

Post-Resuscitation Care

Once the newborn has been successfully resuscitated, the baby is moved to post-resuscitation care. Post-resuscitation care is considered separately because infants who have required resuscitation are at risk of developing complications from the resuscitation and during the period after resuscitation (Frazier & Werthammer, 2007). As such, neonates who require resuscitation are usually moved to the neonatal intensive care unit for close monitoring.

Blood pressure: Hypotension is the most likely cardiovascular result of resuscitation. Monitoring heart rate and blood pressure are the best ways to determine if hypotension is an issue for newborns who have been resuscitation. Volume replacement and inotrope administration are relevant interventions in the case of hypotension.

Electrolytes: Hyponatremia and hypocalcemia are common in recently resuscitated newborns. Electrolyte abnormalities are diagnosed/monitored with chemistry panels. Standard treatment is to reverse deficits with intravenous supplementation.

Metabolic acidosis: Poor cardiac output and/or hypoxemia can cause a buildup of lactic acid, which leads to metabolic acidosis. When possible, acidosis (acidemia) should be treated with increased ventilation (drawing off carbon dioxide from the lungs) Sodium bicarbonate can be given in cases of extreme or persistent metabolic acidosis, but it should be used with extreme caution since it is caustic, irritates blood vessels, and can actually decrease pH in cells.

Blood glucose: Hypoglycemia is a concern in the post-resuscitation period. Even in a healthy newborn, plasma glucose concentration will fall during the first two hours after delivery, usually to a value as low as (but usually not below) 40 mg/dL (2.2 mmol/L). Within 4 to 6 hours of birth, blood glucose should stabilize between 45 and 80 mg/dL (2.5 and 4.4 mmol/L) (Cornblath et al., 2000). In general, newborns within the first 48 hours of birth should be able to maintain circulating blood glucose above 50 mg/dL (2.8 mmol/L) (Stanley et al., 2015).

Central nervous system function: Seizures, apnea, and other neurological issues can result from resuscitation. Therapeutic hypothermia and anticonvulsants are potential interventions for brain disturbances resulting from resuscitation.

Pulmonary function: A number of lung complications can arise because of resuscitation. These complications include pulmonary hypertension, meconium aspiration syndrome, pneumonia, pneumothorax, transient tachypnea, and surfactant deficiency (especially in premature infants). Maintaining proper oxygenation and ventilation, delaying feedings, using antibiotics, taking x-rays, and using surfactant therapy are all interventions that can help with specific lung complications.

Feeding difficulties: Providers should look for ileus, gastrointestinal bleeding, or poor suckling/swallowing. Delaying feedings and providing intravenous fluids and parenteral nutrition are potential ways to intervene with these issues.

Renal function: Acute tubular necrosis is the most common kidney complication resulting from resuscitation. This condition can be identified by monitoring urine output and serum electrolytes.

Infection and blood cell counts: Complete blood cell counts (CBCs) can be used to diagnose anemia (low red blood cell count), thrombocytopenia (low platelet count), and infection (elevated white blood cell count, usually with elevated body temperature). Pneumonia is the likely culprit of infection in the neonate, either from aspiration that occurred during resuscitation or from infection that is present congenitally. Provider should also be aware of the possibility of sepsis.

Resuscitating Preterm Babies

Preterm babies are at increased risk for requiring resuscitation. There are a number of reason for this group's vulnerability, including that they lose heat quickly, they are quite vulnerable to the changes in oxygen levels, their vital organs, such as the brain and lungs are immature, they are more susceptible to infection, and they have a lower blood volume, which makes blood loss more problematic for them.

When preparing for a preterm birth, those responsible for resuscitation should compile extra resources and personnel. Having additional equipment for warming the baby is important, as is a compressed air source, an oxygen blender, and a pulse oximeter. These extra tools will be useful for the following reasons:

- Because preterm babies lose heat quickly, a number of different mechanisms to reduce heat loss should be employed. The room temperature should be increased, and a radiant warmer should be preheated. Having a warming pad, a polyethylene wrap, and a warmed transport incubator are other strategies to consider for preterm babies.

- Because preterm babies are more susceptible to changes in oxygen concentration, increasing their oxygen levels needs to occur at a slower rate than would occur with normal babies of term. The oximeter and blender can therefore be used to achieve optimal oxygen saturations during right after resuscitation.

The actual process of ventilation for preterm babies should follow the same protocol as positive-pressure ventilation for term babies.

There are a number of precautions you can take during the resuscitation of preterm babies to reduce the chances that the baby endures brain injury. These precautions including avoiding the Trendelenburg position, high airway pressures, as well as intravenous fluid that enters too rapidly or has high ionic concentrations. Generally treating the baby with care and gently altering ventilation can also reduce the risk of brain injury.

Post-resuscitation care is the same as it is for less premature neonates; however, assessments and treatments should be more frequent and every maneuver should be done even more gently in the extremely premature neonate (e.g., ventilation, feedings, IV infusions)

Ethical Considerations

There are no specific legal or ethical guidelines regarding when to attempt resuscitation or when to cease resuscitation protocols. These decisions are largely determined collaboratively by healthcare professionals and the family of the patient. The parents are generally deferred to for decisions regarding the health of the baby. Accordingly, research has shown that the majority of healthcare professionals initiate resuscitation or refrain from resuscitation according to the wishes of the parents, though 98% of healthcare professionals have reported initiating resuscitation when the parents are unsure of which option to pursue (Peerzada, Richardson, & Burns, 2004). However, after 10 minutes of no heart rate, the discontinuation of resuscitation efforts should be seriously considered.

When dealing with neonates who do not survive, it is important to remember the following:

- The ethical issues involved with resuscitation are the same for any human of any age—neonates are no different even though they have only been alive for a very short period.
- Some neonates will be born with problems that are not compatible with life such as incomplete gestation, ultra-low birth weight, and/or congenital abnormalities.
- In some cases, resuscitation may only be able to prolong life temporarily, prolong suffering, or result in a viable infant with massive, permanent disabilities. It may be acceptable, in these cases, to withhold resuscitation efforts. Specific examples of cases in which it is appropriate to withhold resuscitation include:
    - Anencephaly
    - Lethal genetic abnormality
    - Marked, ongoing disability, usually a case that would not be considered a “meaningful” life
    - Gestational age of less than 23 weeks
    - Birth weight of less than 400 g (fetal weight estimates may be wrong by ± 20%)
- When possible, parents of fetuses at very high risk should be engaged in end of life discussions before delivery. Providers should consider and respect the wishes of parents who have been fully informed. On the other hand, providers have a legal obligation to provide care if, in their expert opinion, the neonate has a reasonable chance of surviving and acceptable risk of morbidity, even if this is against the parents’ wishes.
- Difficult or borderline cases should include discussions with physicians, nurses, social workers, and medical ethicists, though parental wishes should be strongly considered.
- Words matter. The loss of a neonate should be treated like the loss of any other child—a somber, important moment. Be empathic and clear with parents of the deceased. Avoid phrases such as
    - “You are young. You can always have another baby.”
    - “At least you did not know this baby yet.”
    - “It was for the best.”

References

AAP/AHA. (2011). Neonatal Resuscitation. 6th Edition.

Cornblath, M., Hawdon, J. M., Williams, A. F., Aynsley-Green, A., Ward-Platt, M. P., Schwartz, R., & Kalhan, S. C. (2000). Controversies regarding definition of neonatal hypoglycemia: suggested operational thresholds. *Pediatrics, 105*(5), 1141-1145.

Frazier, M. D., & Werthammer, J. (2007). Post-resuscitation complications in term neonates. *J Perinatol, 27*(2), 82-84. doi: 10.1038/sj.jp.7211644

Gopel, W., Kribs, A., Ziegler, A., Laux, R., Hoehn, T., Wieg, C., . . . Herting, E. (2011). Avoidance of mechanical ventilation by surfactant treatment of spontaneously breathing preterm infants (AMV): an open-label, randomised, controlled trial. *Lancet, 378*(9803), 1627-1634. doi: 10.1016/s0140-6736(11)60986-0

Hemway, R. J., Christman, C., & Perlman, J. (2013). The 3:1 is superior to a 15:2 ratio in a newborn manikin model in terms of quality of chest compressions and number of ventilations. *Arch Dis Child Fetal Neonatal Ed, 98*(1), F42-45. doi: 10.1136/archdischild-2011-301334

Iliodromiti, S., Mackay, D. F., Smith, G. C., Pell, J. P., & Nelson, S. M. (2014). Apgar score and the risk of cause-specific infant mortality: a population-based cohort study. *Lancet, 384*(9956), 1749-1755. doi: 10.1016/s0140-6736(14)61135-1

Kattwinkel, J., Perlman, J. M., Aziz, K., Colby, C., Fairchild, K., Gallagher, J., . . . Zaichkin, J. (2010). Part 15: neonatal resuscitation: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. *Circulation, 122*(18 Suppl 3), S909-919. doi: 10.1161/circulationaha.110.971119

Khalak, R., Cummings, J., & Dexter, S. (2015). Maternal obesity: significance on the preterm neonate. *Int J Obes (Lond)*. doi: 10.1038/ijo.2015.107

Lawn, J. E., Cousens, S., & Zupan, J. (2005). 4 million neonatal deaths: when? Where? Why? *Lancet, 365*(9462), 891-900. doi: 10.1016/s0140-6736(05)71048-5

Peerzada, J. M., Richardson, D. K., & Burns, J. P. (2004). Delivery room decision-making at the threshold of viability. *J Pediatr, 145*(4), 492-498. doi: 10.1016/j.jpeds.2004.06.018

Saini, S. S., Gupta, N., Kumar, P., Bhalla, A. K., & Kaur, H. (2012). A comparison of two-fingers technique and two-thumbs encircling hands technique of chest compression in neonates. *J Perinatol, 32*(9), 690-694. doi: 10.1038/jp.2011.167

Stanley, C. A., Rozance, P. J., Thornton, P. S., De Leon, D. D., Harris, D., Haymond, M. W., . . . Wolfsdorf, J. I. (2015). Re-evaluating "transitional neonatal hypoglycemia": mechanism and implications for management. *J Pediatr, 166*(6), 1520-1525 e1521. doi: 10.1016/j.jpeds.2015.02.045

This Neonatal Resuscitation Course complies with the sixth edition of the *Neonatal Resuscitation* textbook by the American Heart Association and the American Academy of Pediatrics (AAP/AHA, 2011). Other sources used to compile this guide are listed in the References section. Cover image is attributed to ceejayoz.

Copyright 2015. All rights reserved.